SGA 5 — Exposé I

Dualizing Complexes

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Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the one developed in HARTSHORNE [H] for the Zariski topology and coherent sheaves: the latter served, to a very large extent, as its model. But, whereas in the case of "continuous" coefficients the existence of dualizing complexes presents no difficulties, in the case of "discrete" coefficients, on the other hand, it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See however Deligne's biduality theorem (SGA $4^{1/2}$, Finiteness Theorem 4.3.).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, notably the interpretation of dualizing functors as "exchangers of functors."

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to a translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1.), which asserts that, on a "good" regular prescheme X, for example on an excellent noetherian regular prescheme of characteristic zero[1], the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Paragraph 4 gives the applications of the theory to the local duality theorem. The latter expresses that, on a "good" prescheme X endowed with a closed point x whose residue field is separably closed, and for a constructible sheaf F, one has a perfect pairing between $\underline{\mathrm{H}}^i_x(F')$ and $\underline{\mathrm{H}}^{-i}_x(F)$, where F' is the dual of F, i.e. $F' = \mathrm{R}\,\underline{\mathrm{Hom}}(F, K_X)$, where K_X is a dualizing complex.

Finally, in paragraph 5, one proves directly that on a regular prescheme of dimension 1 the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing (n prime to the residual characteristics).

1 Definition and formal properties of dualizing complexes

We fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, we denote by A_X the constant sheaf on $X_{\text{ét}}$ with value A, and by $\mathrm{D}(X)$ the derived category of the category of A_X-modules. We denote, on the other hand, by $\mathrm{D}_c(X)$ the full triangulated subcategory of $\mathrm{D}(X)$ formed by the complexes F such that $\mathrm{H}^i(F)$ is a constructible A_X-module for every i. We put $\mathrm{D}^*_c(X) = \mathrm{D}^*(X) \cap \mathrm{D}_c(X)$, where $* = \varnothing, +, -,$ or b.

[1] This last restriction will become unnecessary once one has resolution of singularities à la Hironaka for these preschemes, and the "purity theorem."

Definition 1.1

An A_X-module I is said to be *quasi-injective* if one has

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every *constructible* A_X-module F and every $i > 0$.

Remarks 1.2

a) Contrary to what happens in the case of "continuous" coefficients (cf. [H] II.7.17), a quasi-injective sheaf is not injective in general. For example, let $X = \mathrm{Spec}(\mathrm{R})$ and $A = \mathbb{Z}/2\mathbb{Z}$: the sheaf A_X is quasi-injective, but is not injective, nor even of finite injective dimension, since $\mathrm{H}^*(X; A_X) = \mathrm{H}^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$ is a polynomial algebra on one generator of dimension 1.

b) Let I be a quasi-injective A_X-module. It is false in general that the relation $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F, is valid for every F. Indeed, take $A = \mathbb{F}_\ell$. Let k be a field, $\overline{k}$ a separable closure of k, and $f : \overline{x} = \mathrm{Spec}(\overline{k}) \to X = \mathrm{Spec}(k)$ the canonical morphism, and suppose that, for every connected étale covering $X' \to X$, there exists a connected étale covering $X'' \to X'$ such that $[X'' : X']$ is divisible by ℓ (take, for example, $k = \mathbb{Q}$). Consider the exact sequence

$$(*) \qquad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow ε. In view of the hypothesis, the section of $\mathrm{Ext}^1(F, A_X)$ defined by $(*)$ does not vanish above any U étale over X; hence $\underline{\mathrm{Ext}}^1(F, A_X)_{\overline{x}} \neq 0$. But one checks trivially that A_X is quasi-injective.

Proposition 1.3 (cf. [H] 1.7.6)

Let X be a prescheme and $N \in \mathbb{Z}$. The following conditions on $K \in \mathrm{ob}\, \mathrm{D}^+(X)$ are equivalent:

(i) K is isomorphic, in $\mathrm{D}(X)$, to a bounded complex I such that I^i is quasi-injective for every i and zero for $i > N$;

(ii) for every $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ such that $\mathrm{H}^i(F) = 0$ for $i < 0$, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$;

(iii) for every constructible A_X-module F, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$.

Proof. $(i) \Rightarrow (ii)$. Let $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ be such that $\mathrm{H}^i(F) = 0$ for $i < 0$, and let $I \in \mathrm{ob}\, \mathrm{D}^+(X)$ be bounded and such that I^i is quasi-injective for every i and zero for $i > N$; we show that $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > N$. Arguing by induction on the number of indices i such that $\mathrm{H}^i \neq 0$, one is reduced, by dévissage, to the case where $N = 0$ and I is concentrated in degree 0. One then has a biregular spectral sequence

$$\mathrm{E}_2^{pq} = \underline{\mathrm{Ext}}^p(\mathrm{H}^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since I is quasi-injective, $\mathrm{E}_2^{pq} = 0$ for $p > 0$ and every q, and hence one has $\mathrm{E}_2^{0i} = \underline{\mathrm{Hom}}(\mathrm{H}^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I)$, which gives the conclusion.

$(ii) \Rightarrow (iii)$ is trivial.

$(iii) \Rightarrow (i)$ We may suppose that K is bounded below and has injective components. The hypothesis, applied to $F = A_X$, implies that $\mathrm{H}^i(K) = 0$ for $i > N$. We may therefore replace K by an isomorphic complex K', bounded, and such that K'^i is injective for $i < N$ and zero for $i > N$. It is then observed that K'^N is quasi-injective, which completes the proof.

Remark 1.3.1

The writer does not know whether (1.3) is still true when, in condition (ii), the hypothesis "$F \in \operatorname{ob} \mathrm{D}_c^b(X)$" is replaced by "$F \in \operatorname{ob} \mathrm{D}_c^+(X)$".

Definition 1.4

The *quasi-injective dimension* of K, denoted $\dim . q. inj(K)$, is the lower bound (in $\overline{\mathrm{R}}$) of the integers N satisfying the equivalent conditions of (1.3).

If K is of finite quasi-injective dimension, the functor $\mathrm{R}\,\underline{\mathrm{Hom}}(\quad, K) : \mathrm{D}(X) \to \mathrm{D}(X)$ induces a functor $\mathrm{D}_c^b(X) \to \mathrm{D}^b(X)$.

We shall see below that if X is smooth of dimension d over a field k and n is prime to the characteristic of k, then $\dim . q. inj(A_X) = 2d$.

The example of (1.2 a)) shows that a complex of finite quasi-injective dimension is not necessarily of finite injective dimension. Nevertheless one has:

Proposition 1.5

Let X be a locally noetherian prescheme and $K \in \operatorname{ob} \mathrm{D}^+(X)$. Suppose that there exists an integer N such that $\mathrm{cd}_{\mathbf{n}}(X) \leqslant N$, where $\mathbf{n}$ is the set of prime divisors of n (notation of SGAA X 1). Then, for $N' \in \mathbb{Z}$, one has

$$\dim . q. inj(K) \leqslant N' \quad \Longrightarrow \quad \dim . inj(K) \leqslant N + N'.$$

We shall use, in the proof, the following

Lemma 1.5.1

Let C be an abelian category satisfying (AB5) and possessing a family of generators $(U_i)_{i \in I}$. Let $K \in \operatorname{ob} \mathrm{D}^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:

(i) $\dim . inj(K) \leqslant N$;

(ii) for every $i \in I$ and every quotient F of U_i, one has

$$\operatorname{Ext}^n(F, K) = 0 \qquad \text{for } n > N.$$

Proof. It suffices to prove $(ii) \Rightarrow (i)$. By a well-known argument (cf. for example [H] 1.7.6), one is reduced to showing that if $K' \in \operatorname{ob}(C)$ is such that $\operatorname{Ext}^1(F, K') = 0$ every time F is a quotient of a U_i, then K' is injective. In other words, one has to prove that, if every morphism from a subobject

V of a U_i into K' extends to U_i, then K' is injective. But this follows immediately from the proof of Lemma 1, p. 136 of (Tohoku).

Proof of (1.5). Suppose $\dim . q. inj(K) \leqslant N'$ and prove $\dim . inj(K) \leqslant N + N'$. Since the category of A_X-modules admits as generators the $A_{U,X}$ where U is étale of finite type over X, it suffices, by (1.5.1), to prove that $\mathrm{Ext}^i(X; F, K) = 0$ for $i > N + N'$ when F is a quotient of such an $A_{U,X}$, hence constructible (SGAA IX 2.9). Consider the spectral sequence

$$\mathrm{E}_2^{pq} = \mathrm{H}^p(X; \underline{\mathrm{Ext}}^q(F, K)) \Rightarrow \mathrm{Ext}^*(X; F, K).$$

The hypothesis $\mathrm{cd}_{\mathbf{n}}(X) \leqslant N$ (resp. $\dim . q. inj(K) \leqslant N'$) implies $\mathrm{E}_2^{pq} = 0$ for $p > N$ (resp. $q > N'$). Hence $\mathrm{E}_2^{pq} = 0$ for $p + q > N + N'$, whence the conclusion.

Remark 1.5.2

The hypothesis of (1.5) is verified (with $N = 2d$) if X is a prescheme of finite type and dimension d over a separably closed field k, with n prime to $\mathrm{char}(k)$ (SGAA X 4).

Proposition 1.6

Let $f : X \to Y$ be a separated quasi-finite morphism of noetherian preschemes, and let $K \in \mathrm{ob}\,\mathrm{D}^+(Y)$. Then, for $N \in \mathbb{Z}$, one has

$$\dim . q. inj(K) \leqslant N \quad \Longrightarrow \quad \dim . q. inj(\mathrm{R}^! f(K)) \leqslant N.$$

Proof. According to the "Main theorem" (EGA IV 8.12.6), there exists a factorization of f as $f = uf'$, where f' is an open immersion and u is a finite morphism. We have $\mathrm{R}^! f = \mathrm{R}^! f' \,\mathrm{R}^! u$, which reduces us to examining separately the case of an open immersion and that of a finite morphism. If f is an open immersion, the assertion is trivial (since every constructible sheaf on X is then induced by a constructible sheaf on Y). Suppose therefore that f is a finite morphism, and let F be a constructible A_X-module. We then have the duality isomorphisms (SGAA XVIII 3.1.9.6)

$$f_* \underline{\mathrm{Ext}}^i(F, \mathrm{R}^! f(K)) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(f_* F, K).$$

Since $f_*(F)$ is constructible (SGAA IX 2.14), it follows that, if $\dim . q. inj(K) \leqslant N$, then $f_* \underline{\mathrm{Ext}}^i(F, \mathrm{R}^! f(K)) = 0$ for $i > N$, hence $\underline{\mathrm{Ext}}^i(F, \mathrm{R}^! f(K)) = 0$ for $i > N$ (SGAA VIII 5.5), as required.

Let now X be a prescheme, and let $K \in \mathrm{ob}\,\mathrm{D}^+(X)$. Denote by $\underline{D}_K$ the functor $\mathrm{R}\,\underline{\mathrm{Hom}}(\quad, K)$. We shall define, for $F \in \mathrm{ob}\,\mathrm{D}(X)$, a functorial homomorphism $F \to \underline{D}_K \underline{D}_K F$ (cf. [H] V.1.2). The construction which follows would be valid more generally on a ringed topos. We may first suppose that K is formed of injectives, which allows us to "remove the R". The identity morphism $\underline{\mathrm{Hom}}(F, K) \to \underline{\mathrm{Hom}}(F, K)$ defines, by the formula dear to Cartan, a homomorphism $F \otimes \underline{\mathrm{Hom}}(F, K) \to K$, whence, by a second application of the said formula, a homomorphism $F \to \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}(F, K), K)$, which is the announced homomorphism, and which we baptize the canonical homomorphism. We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or again that F is *reflexive* relative to K, if the canonical homomorphism $F \to \underline{D}_K \underline{D}_K F$ is an isomorphism.

From now on, unless expressly mentioned otherwise, all preschemes considered will be assumed lo

Definition 1.7

Let X be a prescheme. A complex $K \in \operatorname{ob} \mathrm{D}^+(X)$ is said to be *dualizing* if the following conditions are satisfied:

(i) K is of finite quasi-injective dimension;

(ii) for every $F \in \operatorname{ob} \mathrm{D}_c^-(X)$, one has $\underline{D}_K(F) \in \operatorname{ob} \mathrm{D}_c^+(X)$;

(iii) every $F \in \operatorname{ob} \mathrm{D}_c^b(X)$ is reflexive relative to K.

Remark 1.8

a) By a standard dévissage (truncating F and using the fact that $\mathrm{D}_c(X)$ is triangulated), one sees that condition (ii) is equivalent to

(ii bis): for every constructible A_X-module F and every $i \in \mathbb{Z}$, $\underline{\mathrm{Ext}}^i(F, K)$ is constructible.

b) Condition (ii bis) implies $K \in \operatorname{ob} \mathrm{D}_c^+(X)$. Contrary to the example of coherent sheaves (EGA 0_{III} 12.3.3), the converse is perhaps not automatically verified. It is however verified (see no. 3) in the "good" cases, for example when one has on X resolution of singularities and purity (in particular if X is an excellent prescheme of characteristic zero).

c) By dévissage on F, one sees that condition (iii) is equivalent to (iii bis): every constructible A_X-module F is reflexive relative to K.

d) If $K \in \operatorname{ob} \mathrm{D}^+(X)$ satisfies (i) and (ii) (resp. is dualizing), the functor $\underline{D}_K$ induces a functor (resp. an equivalence of categories) $(\mathrm{D}_c^b(X))^\circ \to \mathrm{D}_c^b(X)$.

e) If $K \in \operatorname{ob} \mathrm{D}^+(X)$ is of finite *injective* dimension (cf. 1.5.2) and satisfies (ii), then, for every $F \in \operatorname{ob} \mathrm{D}_c(X)$, one has $\underline{D}_K(F) \in \mathrm{D}_c(X)$, and $\underline{D}_K$ induces a functor $(\mathrm{D}_c^+(X))^\circ \to \mathrm{D}_c^-(X)$.

If moreover K is dualizing, every $F \in \operatorname{ob} \mathrm{D}_c(X)$ is reflexive relative to K, and $\underline{D}_K$ induces equivalences of categories

$$(\mathrm{D}_c(X))^\circ \xrightarrow{\approx} \mathrm{D}_c(X), \quad (\mathrm{D}_c^-(X))^\circ \xrightarrow{\approx} \mathrm{D}_c^+(X), \quad (\mathrm{D}_c^+(X))^\circ \xrightarrow{\approx} \mathrm{D}_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K be of finite injective dimension.

We now turn to the "exchange formulas." We shall not return to the notion of the tor-dimension of a complex, see SGA A, Exposé XVII, 4.1.9, and also SGA 6, I, 5. Retain only the following point: if X is a prescheme and $G \in \operatorname{ob} \mathrm{D}^-(X)$, the condition $\underline{\mathrm{Tor}}_i(F, G) = 0$ for $i > N$ and all A_X-modules F is equivalent to the same condition with F constructible, by SGA A, Exposé IX, 2.9 and the commutation of tensor product with filtered inductive limits. Thus there is no need to introduce a notion of "quasi-tor-dimension."

Lemma 1.9, cf. [H] II.4.3.

a) Let X be a prescheme and let $F, G \in \operatorname{ob} \mathrm{D}_c^-(X)$. One has $F \otimes^{\mathrm{L}} G \in \operatorname{ob} \mathrm{D}_c(X)$ if $F \in \operatorname{ob} \mathrm{D}_c^-(X)$, or if G has finite tor-dimension.

b) Let $f : X \to Y$ be a morphism of preschemes and let $G \in \operatorname{ob} \mathrm{D}_c^-(Y)$. Then $f^*(G) \in \operatorname{ob} \mathrm{D}_c(X)$.

Proof. Part (b) follows trivially from SGA A, Exposé IX, 2.4. We prove (a). Either hypothesis on (F, G) implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\mathrm{Tor}}_{-p}(\mathrm{H}^{q_1}(F), \mathrm{H}^{q_2}(G)) \Rightarrow \mathrm{H}^*(F \otimes^{\mathrm{L}} G)$$

is biregular. Hence one is reduced to the case where F and G are concentrated in degree 0, that is, are constructible A_X-modules. The question is local, so one may suppose X noetherian. Then F admits, by SGA A, Exposé IX, 2.7, a left resolution F' in which the F'^i are of the form $A_{U,X}$ with U étale and of finite type over X. We have

$$F \otimes^{\mathrm{L}} G \xleftarrow{\sim} F' \otimes G,$$

and the result follows since the $A_{U,X} \otimes G$ are constructible. □

Notation 1.10. If X is a prescheme, we denote by $\mathrm{D}^b(X)_{\mathrm{torf}}$, respectively $\mathrm{D}^b(X)_{\mathrm{qinjf}}$, the full subcategory of $\mathrm{D}(X)$ consisting of objects of finite tor-dimension, respectively of finite quasi-injective dimension.

Proposition 1.11, cf. [H] V.2.6. Let X be a prescheme and $K \in \operatorname{ob} \mathrm{D}^+(X)$.

a) For $F, G \in \operatorname{ob} \mathrm{D}^-(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism
$$\underline{D}_K(F \otimes^{\mathrm{L}} G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, \underline{D}_K(G)).$$

b) If K satisfies conditions (i) and (ii) of Definition 1.7, then
$$G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \Longrightarrow \underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}}$$
and $\underline{D}_K(G)$ satisfies condition 1.7(ii).

c) If K is dualizing, then for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$ with $\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism
$$\underline{D}_K(F \otimes^{\mathrm{L}} \underline{D}_K(G)) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G).$$
Moreover,
$$\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \Longrightarrow G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}},$$
and G satisfies condition 1.7(ii).

d) If K is dualizing and has finite injective dimension, then the implications in (b) and (c) are equivalences.

Proof. Part (a) is precisely Cartan's adjunction formula, SGA 6, I, 7.4.

For (b), let $G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$. If F is a variable constructible A_X-module, then the cohomology objects of $F \otimes^{\mathrm{L}} G$ are constructible, by Lemma 1.9, and vanish outside two fixed bounds independent of F. The same therefore holds for the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$, and (a) gives the assertion.

For (c), the first assertion follows from (a) applied to $\underline{D}_K(G)$ in place of G, together with the reflexivity of G. The second follows by a similar argument to (b).

For (d), prove for instance the converse implication of (b). Suppose that $\underline{D}_K(G) \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}}$ and satisfies condition 1.7(ii). Then, as F varies through constructible modules, the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$ are constructible and vanish outside two fixed bounds. The same is consequently true for the cohomology objects of $\underline{D}_K \underline{D}_K(F \otimes^{\mathrm{L}} G)$; but

$$F \otimes^{\mathrm{L}} G \xrightarrow{\sim} \underline{D}_K \underline{D}_K(F \otimes^{\mathrm{L}} G)$$

because K is dualizing and of finite injective dimension, cf. 1.8(d). Hence G has finite tor-dimension. The converse implication in (c) is proved analogously. □

Remarks 1.11.1.

a) The editor does not know whether assertion (d) of Proposition 1.11 is valid under the sole hypothesis that K be dualizing.

b) One expresses the preceding proposition pictorially by saying that the dualizing functor $\underline{D}_K$ exchanges the functors $\otimes^{\mathrm{L}}$ and $\mathrm{R}\,\underline{\mathrm{Hom}}$, as well as the notions of finite tor-dimension and finite quasi-injective dimension. The exchange is, however, fully satisfactory only when K has finite injective dimension.

Proposition 1.12. Let Y be a noetherian prescheme, let $f : X \to Y$ be a morphism of finite type, and let $K \in \operatorname{ob} \mathrm{D}^+(Y)$. Suppose that n is prime to the residual characteristics of Y, and put

$$K_X = \mathrm{R}^! f(K_Y), \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y},$$

in the notation of SGA A, Exposé XVIII, 3.1.

a) For $F \in \operatorname{ob} \mathrm{D}^-(X)$ there is a functorial isomorphism

$$\mathrm{R}f_* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R}f_*(F). \tag{i}$$

If, moreover, K_X and K_Y are dualizing,[2] then for $F \in \operatorname{ob} \mathrm{D}_c^b(X)$ there is a functorial isomorphism

$$\mathrm{R}f_! \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R}f_*(F). \tag{ii}$$

b) For $F \in \mathrm{D}_c^-(Y)$ there is a functorial isomorphism[3]

$$\mathrm{R}^! f\, \underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_X f^*(F). \tag{i}$$

If, moreover, K_X and K_Y are dualizing, then for $F \in \mathrm{D}_c^b(Y)$ there is a functorial isomorphism

$$f^* \underline{D}_Y F \xrightarrow{\sim} \underline{D}_X \mathrm{R}^! f\, F. \tag{ii}$$

[2]We shall see below that under fairly general conditions K_Y dualizing implies K_X dualizing.

[3]If the functor $\mathrm{R}^! f$ has finite cohomological dimension, as for example when Y is of finite type over a field (SGA A, Exposé X, 4.3 and Exposé XVIII, 3.1.7), the isomorphism (b)(i) is valid for $F \in \operatorname{ob} \mathrm{D}_c(Y)$.

Proof. The isomorphism (a)(i) is just the duality isomorphism of SGA A, Exposé XVIII, 3.1.9.6.

The isomorphism (a)(ii) is obtained by applying $\underline{D}_Y$ to both sides of (a)(i), written with argument $\underline{D}_X F$, and using the finiteness theorem SGA A, Exposé XVII, 5.3.6, which asserts that $\mathrm{R}f_*(\underline{D}_X(F)) \in \mathrm{ob}\,\mathrm{D}_c(Y)$.

The isomorphism (b)(i) is the induction formula of SGA A, Exposé XVIII, 3.1.12.2.

The isomorphism (b)(ii) is obtained by applying $\underline{D}_X$ to both sides of (b)(i), written with argument $\underline{D}_Y F$, and using 1.9(b). □

Scholium 1.12.1. In figurative language, the dualizing functors $\underline{D}_X$ and $\underline{D}_Y$ exchange the usual and unusual direct images, $\mathrm{R}f_*$ and $\mathrm{R}f_!$, and also the usual and unusual inverse images, f^* and $\mathrm{R}^!f$.

Corollary 1.13. Under the hypotheses of Proposition 1.12, suppose in addition that f is proper, and let $F \in \mathrm{ob}\,\mathrm{D}^-(X)$. If (F, K_X) is bidualizing, then $(\mathrm{R}f_*F, K_Y)$ is bidualizing. Conversely, if f is finite and $(\mathrm{R}f_*F, K_Y)$ is bidualizing, then (F, K_X) is bidualizing.

Proof. Suppose (F, K_X) is bidualizing. Applying $\mathrm{R}f_*$ to the canonical isomorphism $F \xrightarrow{\sim} \underline{D}_X^2 F$, one obtains

$$\begin{aligned} \mathrm{R}f_*F &\xrightarrow{\sim} \mathrm{R}f_*\underline{D}_X\underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y\mathrm{R}f_*\underline{D}_X F \qquad \text{by (a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y\underline{D}_Y\mathrm{R}f_*F \qquad \text{by (a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this proves the first assertion.

Now suppose f finite and (f_*F, K_Y) bidualizing, since $\mathrm{R}f_* = f_*$ for f finite. We prove that the canonical morphism $F \to \underline{D}_X^2 F$ is an isomorphism. By Lemma 1.14(a) below, it suffices to show that the induced morphism

$$f_*F \longrightarrow f_*\underline{D}_X^2 F$$

is an isomorphism. But by applying (a)(i) twice, one convinces oneself that this morphism is precisely the canonical isomorphism

$$f_*F \longrightarrow \underline{D}_Y^2 f_*F,$$

which completes the proof. □

Lemma 1.14. Let $f : X \to Y$ be a finite morphism of preschemes.

a) For a morphism $u : F \to G$ in $\mathrm{D}(X)$, u is an isomorphism if and only if f_*u is an isomorphism.

b) For $F \in \mathrm{ob}\,\mathrm{D}(X)$, one has $F \in \mathrm{ob}\,\mathrm{D}_c(X)$ if and only if $f_*F \in \mathrm{ob}\,\mathrm{D}_c(Y)$.

Proof. For (a), since f_* is exact, one may restrict to the case where F and G are concentrated in degree zero. The assertion then follows immediately from the computation of the fibres of f_*F and f_*G, SGA A, Exposé VIII, 5.5.

For (b), again one may restrict to the case where F is concentrated in degree zero. Necessity is known, SGA A, Exposé IX, 2.14. We prove sufficiency: f_*F constructible implies F constructible.

The question is local upstairs, hence a fortiori downstairs, so we may suppose Y and hence X affine. By EGA IV, 17.16.6, there exists a finite family (Y_i) of affine subpreschemes of Y, pairwise disjoint and with union Y, such that, if $X_i = X \times_Y Y_i$, the induced morphism $f_i : X_i \to Y_i$ factors as

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{v_i} Y_i,$$

where u_i is finite radicial and surjective and v_i is finite étale. Put $F|_{X_i} = F_i$. By the base-change theorem for finite morphisms, SGA A, Exposé VIII, 5.5, one has

$$f_{i*}(F_i) \xrightarrow{\sim} (f_* F)|_{Y_i},$$

and by SGA A, Exposé IX, 2.8, one is reduced to proving the assertion for (f_i, F_i). Put $F'_i = u_{i*}(F_i)$, so that $f_{i*}(F_i) = v_{i*}(F'_i)$. It is clear from the definition of constructibility, respectively follows from SGA A, Exposé VIII, 1.1, that F_i, respectively F'_i, is constructible if and only if F'_i, respectively $v_{i*}(F'_i)$, is constructible. This proves the claim. □

Corollary 1.15. Let $f : X \to Y$ be a separated quasi-finite morphism of noetherian preschemes. Let $K_Y \in \operatorname{ob} \mathrm{D}^+(Y)$, and put, as in Proposition 1.12,

$$K_X = \mathrm{R}^! f(K_Y), \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y}.$$

If K_Y is dualizing, then K_X is dualizing.

Proof. We already know, by Proposition 1.6, that if K_Y has finite quasi-injective dimension, then the same is true of K_X. We show that if K_Y satisfies condition (ii), respectively condition (iii), of Definition 1.7, then K_X satisfies the same condition.

By the Main Theorem, EGA IV, 8.12.6, f factors as $f = f'i$, where i is an open immersion and f' is finite. Since

$$\mathrm{R}^! f \simeq \mathrm{R}^! i \, \mathrm{R}^! f',$$

one must treat separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since a constructible sheaf on X is induced by a constructible sheaf on Y, and formation of $\mathrm{R}\,\underline{\mathrm{Hom}}$ commutes with restriction to an open.

Suppose then that f is finite. It follows from Corollary 1.13 that if K_Y satisfies (iii), then K_X satisfies (iii), taking account of the finiteness theorem SGA A, Exposé IX, 2.14. It remains to prove that if K_Y satisfies (ii), then K_X satisfies (ii). Let $F \in \operatorname{ob} \mathrm{D}_c^-(X)$. We must show that $\underline{D}_X(F) \in \operatorname{ob} \mathrm{D}_c^+(X)$. By the finiteness theorem, $f_* F \in \operatorname{ob} \mathrm{D}_c^-(Y)$. Hence, by the hypothesis on K_Y,

$$\underline{D}_Y f_* F \in \operatorname{ob} \mathrm{D}_c^+(Y).$$

But

$$\underline{D}_Y f_* F \xrightarrow{\sim} f_* \underline{D}_X(F)$$

by Proposition 1.12(a)(i), and Lemma 1.14(b) completes the proof. □

2. Uniqueness of the dualizing complex

In this section we suppose that A is local, that is,

$$A = \mathbb{Z}/\ell^\nu \mathbb{Z},$$

where ℓ is prime.

Theorem 2.1, cf. [H] V.3.1. Let X be a connected locally noetherian prescheme, and let K be a dualizing complex on X in the sense of Definition 1.7. Let also $K' \in \operatorname{ob} \mathrm{D}_c^+(X)$. Then K' is dualizing if and only if there exist an invertible A_X-module L and an integer r, necessarily determined up to unique isomorphism, such that

$$K' \xrightarrow{\sim} K \otimes^{\mathrm{L}} L[r].$$

Proof. Let L be an invertible A_X-module and r an integer, and put $L^\vee = \underline{\mathrm{Hom}}(L, A_X)$. For $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$ there are canonical isomorphisms

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F \otimes^{\mathrm{L}} L^\vee[-r], G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G \otimes^{\mathrm{L}} L[r]) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \otimes^{\mathrm{L}} L[r]. \tag{*}$$

It follows immediately that, if $K' \simeq K \otimes^{\mathrm{L}} L[r]$, then K' is dualizing.

Moreover, if we write $\underline{D}_X = \underline{D}$ and $\underline{D}_{K'} = \underline{D}'$, then for $F \in \operatorname{ob} \mathrm{D}(X)$ the isomorphism $(*)$ gives

$$\underline{D}'F \xrightarrow{\sim} \underline{D}F \otimes^{\mathrm{L}} L^\vee[-r].$$

Since $\underline{D}K \simeq A_X$, one obtains

$$\underline{D}'K \xrightarrow{\sim} L^\vee[-r],$$

which shows that $L^\vee[r]$, hence L and r, are determined essentially uniquely by K'.

Conversely, suppose that K' is dualizing, and put

$$P = \underline{D}'K = \underline{D}'\underline{D}A_X = \mathrm{R}\,\underline{\mathrm{Hom}}(K, K').$$

We shall prove:

$$\begin{aligned} &(\mathrm{a}) \qquad K' \xrightarrow{\sim} K \otimes^{\mathrm{L}} P, \\ &(\mathrm{b}) \qquad P \xrightarrow{\sim} L[r] \quad \text{for an invertible } A_X\text{-module } L. \end{aligned}$$

This will complete the proof of the theorem.

First observe that, by Proposition 1.9(b)(ii), for $F \in \operatorname{ob} \mathrm{D}_c(X)$ there is an isomorphism

$$F \otimes^{\mathrm{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}F. \tag{**}$$

Applying $(**)$ to $F = K$ gives

$$K \otimes^{\mathrm{L}} P \xrightarrow{\sim} \underline{D}'\underline{D}K \xrightarrow{\sim} \underline{D}'A_X = K',$$

which proves (a).

Applying $(**)$ to

$$F = P' = \underline{D}\underline{D}'A_X$$

gives

$$P' \otimes^{\mathrm{L}} P = \underline{D}'\underline{D}\underline{D}\underline{D}'A_X \xrightarrow{\sim} \underline{D}'\underline{D}'A_X \xrightarrow{\sim} A_X.$$

Thus (b) follows from the following lemma. $\square$

Lemma 2.2, cf. [H] V.3.3. Let X be a connected locally noetherian prescheme, and let $P, P' \in \operatorname{ob} \mathrm{D}^-_c(X)$ be such that

$$P \otimes^{\mathrm{L}} P' \xrightarrow{\sim} A_X.$$

Then there exist an invertible A_X-module L and an integer r such that

$$P \xrightarrow{\sim} L[r], \qquad P' \xrightarrow{\sim} L^\vee[-r].$$

Proof. The proof has two steps.

First suppose that $X = \operatorname{Spec}(k)$, with k separably closed. One merely repeats the easy proof given in [H]. The essential point is this: by EGA 0_{I}, 5.4.3, if M and M' are finite-type A-modules such that

$$M \otimes M' \xrightarrow{\sim} A,$$

then M and M' are invertible and $M' \simeq M^\vee$. This is where the hypothesis that A is local is used.

We now pass to the general case. By the special case just treated, for every $x \in X$ there exists an integer $r(x)$ such that

$$P_{\overline{x}} \simeq A_{\overline{x}}[r(x)], \qquad P'_{\overline{x}} \simeq A_{\overline{x}}[-r(x)].$$

Equivalently, $\mathrm{H}^i(P_{\overline{x}}) = 0$, respectively $\mathrm{H}^i(P'_{\overline{x}}) = 0$, except for $i = -r(x)$, respectively $i = r(x)$, and the corresponding nonzero cohomology modules are isomorphic to $A_{\overline{x}}$. It remains to show that the function $x \mapsto r(x)$ is locally constant.

Assume this for the moment. Since X is connected, r is constant, say equal to r_0. Then

$$\mathrm{H}^i(P) = 0 \ (i \neq -r_0), \qquad \mathrm{H}^i(P') = 0 \ (i \neq r_0),$$

and

$$\mathrm{H}^{-r_0}(P) \otimes \mathrm{H}^{r_0}(P') \xrightarrow{\sim} A.$$

Put $L = \mathrm{H}^{-r_0}(P)$ and $L' = \mathrm{H}^{r_0}(P')$. Let $u : \overline{x} \to \overline{y}$ be a specialization morphism, and let

$$u^* : L_{\overline{y}} \to L_{\overline{x}}, \qquad u'^* : L'_{\overline{y}} \to L'_{\overline{x}}$$

be the corresponding specialization morphisms. There is a commutative square

$$\begin{array}{ccc} L_{\overline{y}} \otimes L'_{\overline{y}} & \xrightarrow{\sim} & A_{\overline{y}} \\ \downarrow u^* \otimes u'^* & & \downarrow \\ L_{\overline{x}} \otimes L'_{\overline{x}} & \xrightarrow{\sim} & A_{\overline{x}}. \end{array} \tag{*}$$

Choose a basis e, respectively f, of $L_{\overline{x}}$, respectively $L_{\overline{y}}$, and let e', respectively f', be the dual basis of $L'_{\overline{x}}$, respectively $L'_{\overline{y}}$, meaning that $e \otimes e'$ and $f \otimes f'$ map to 1. If $u^*(f) = \lambda e$ and $u'^*(f') = \lambda' e'$, the commutativity of $(*)$ gives $\lambda\lambda' = 1$. Hence u^* is an isomorphism, and u'^* is the contragredient isomorphism. By SGA A, Exposé IX, 2.13, it follows that L and L' are locally constant, hence of finite presentation as A_X-modules. Consequently, for every $x \in X$ the canonical map

$$\underline{\mathrm{Hom}}(L, F)_{\overline{x}} \longrightarrow \mathrm{Hom}(L_{\overline{x}}, F_{\overline{x}})$$

is an isomorphism for every A_X-module F. Since the geometric fibres of L are invertible, L itself is invertible. Similarly L' is invertible, and $L' \simeq L^\vee$. Hence

$$P \simeq L[-r_0], \qquad P' \simeq L^\vee[r_0].$$

It remains only to prove that r is locally constant. For $m \in \mathbb{Z}$, the condition $r(x) = m$ is equivalent to

$$\mathrm{H}^m(P')_{\overline{x}} \neq 0.$$

Thus, by SGA A, Exposé IX, 2.4, the function r is locally constructible. Therefore, to verify that it is locally constant, EGA IV, 1.10.1, allows one to suppose that X is local. Let x be its closed point and U the open complement.

We prove that r is constant. By induction on the dimension of X, we may already suppose that $r(y)$ is constant and equal to r_0 for $y \in U$; the point is to prove $r(x) = r_0$.

Assume the contrary, and derive a contradiction. By hypothesis

$$P|_U \xrightarrow{\sim} L[r_0]$$

with L an invertible A_U-module. After translating degrees we may suppose $r_0 = 0$. On the other hand $r(x) \neq 0$; write $r(x) = a > 0$. Let

$$i : U \to X, \qquad j : x \to X$$

be the canonical inclusions. Consider the exact sequence

$$0 \longrightarrow i_!(P|_U) \longrightarrow P \longrightarrow j_*j^*P \longrightarrow 0. \tag{**}$$

By hypothesis the only nonzero cohomology objects of P are $\mathrm{H}^0(P)$ and $\mathrm{H}^{-a}(P)$, and

$$\mathrm{H}^0(P) \xrightarrow{\sim} i_!L.$$

It follows that $(**)$ splits naturally, via the map $P \to \mathrm{H}^0(P)$, hence

$$P \xrightarrow{\sim} i_!(P|_U) \oplus j_*j^*P.$$

Therefore

$$\begin{aligned} P \otimes^{\mathrm{L}} P' &\xrightarrow{\sim} i_!(P|_U) \otimes^{\mathrm{L}} P' \oplus j_*j^*P \otimes^{\mathrm{L}} P' \\ &\xrightarrow{\sim} i_!\big(P|_U \otimes^{\mathrm{L}} P'|_U\big) \oplus j_*\big(j^*P \otimes^{\mathrm{L}} j^*P'\big) \\ &\xrightarrow{\sim} i_!A_U \oplus j_*A_x, \end{aligned}$$

which is impossible, since $i_!A_U \oplus j_*A_x$ is plainly not isomorphic to A_X; already the degree-zero cohomology sheaves are not isomorphic. This proves Lemma 2.2, and hence Theorem 2.1. $\square$

3. Existence of dualizing complexes

3.1. Preliminaries

As was said in the introduction, the object of this section is to prove that, on a "good" regular prescheme X, the sheaf A_X is dualizing, where n is assumed prime to the residual characteristics.

In fact, one would like this to hold on excellent regular preschemes of equal characteristic; but there is no doubt that the proof given below will extend to them once resolution of singularities and the purity theorem are available in that case.

We first present the various ingredients which will be used.

3.1.1. Singular locus. Let X be a prescheme, locally noetherian. We shall say that X satisfies condition *(reg)* if the following equivalent properties hold (EGA IV, 6.12.3):

(i) for every sub-prescheme Y of X, the set $\mathrm{Reg}(Y)$ of regular points of Y is open in Y;

(ii) for every closed integral sub-prescheme Y of X, the set $\mathrm{Reg}(Y)$ contains a non-empty open subset of Y.

The condition (reg) is satisfied if X is locally of finite type over a complete noetherian local ring, for example over a field (EGA IV, 6.12.8), or over a Dedekind ring whose field of fractions has characteristic zero, for example over $\mathbb{Z}$ (EGA IV, 6.12.6); more generally, it is satisfied if X is excellent (EGA IV, 7.8.6).

3.1.2. Dévissage of constructible sheaves. Let X be a noetherian prescheme, and let C be a strictly full subcategory of the category $\mathrm{Cons}(X)$ of constructible A_X-modules. Suppose that C is stable under direct factors and extensions. The following conditions are equivalent:

(i) $C = \mathrm{Cons}(X)$;

(ii) C contains sheaves of the form $i_!(F)$, where $i : Y \to X$ is the immersion of an integral sub-prescheme and F is locally constant on Y;

(iii) C contains sheaves of the form $f_* i_!(A'_U)$, where $i : U \to Y$ is an open subset of an integral prescheme, $f : Y \to X$ is a finite morphism such that the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$, for ℓ a prime divisor of n.

If X satisfies (reg) (3.1.1), these conditions are again equivalent to the following:

(i bis) the analogue of (ii), with Y moreover regular;

(ii bis) the analogue of (iii), with U regular.

If, in addition, X is affine, then in (ii), respectively in (ii bis), one may suppose Y closed in an open subset of the form $D(f)$, with $f \in \Gamma(X, \mathcal{O}_X)$.

Proof. The equivalence of (i)–(iii) was proved in SGA A, Exposé IX, 2.5 and 5.8. The remaining assertions follow by noetherian induction. ☐

3.1.3. Cohomological dimension. Let X be a locally noetherian prescheme. We denote by $(\mathrm{cdloc})_n$ the following condition. For every sub-prescheme Y of X, every geometric point $\overline{y}$ of Y, and every

$$f \in \Gamma(\widetilde{Y}(\overline{y}), \mathcal{O}_{\widetilde{Y},\overline{y}}),$$

where $\widetilde{Y}(\overline{y})$ denotes the strict localization of Y at $\overline{y}$ (SGA A, Exposé VIII, 7.1), one has

$$\mathrm{cd}_{\mathbf{n}}(\widetilde{Y}(\overline{y}) - V(f)) \leq \dim \mathcal{O}_{Y,\overline{y}} \quad (= \dim \mathcal{O}_{Y,y} \text{ by EGA IV, 6.1.3}),$$

where $\mathbf{n}$ is the set of prime divisors of n.

The condition $(\mathrm{cdloc})_n$ is satisfied if X is locally of finite type over a field (SGA A, Exposé XIV, 3.5), or is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 6.2).[4]

3.1.4. Absolute cohomological purity. Let X be a regular prescheme, and let Y be a regular closed sub-prescheme of X, of codimension d at every point. One says that the pair (X, Y) satisfies the absolute cohomological purity theorem, relative to n, if

$$\underline{\mathrm{H}}^i_Y(A_X) = 0 \qquad (i \neq 2d),$$

and if the "local fundamental class" homomorphism (SGA $4^{1/2}$, Cycle 2.2)

$$A_Y \longrightarrow \underline{\mathrm{H}}^{2d}_Y((\mu_n)^{\otimes d}_X)$$

is an isomorphism.

One says that the absolute cohomological purity theorem is true on X if every pair (U, Y) as above satisfies it, where U is an open subset of X.

Absolute purity is true on X if X is smooth over a perfect field of characteristic prime to n (SGA A, Exposé XVI, 3.9), or if X is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 3.2), or if X is regular of dimension 1 (cf. 5.1).[5]

3.1.5. Hironaka-style resolution of singularities.

a) Let X be a regular prescheme, and let $D \geq 0$ be a divisor on X. One says that D has *strict normal crossings* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma^*(X, \mathcal{O}_X)$ such that

$$D = \sum_{i \in I} \mathrm{Div}(f_i)$$

(EGA IV, 21), and such that, for every $x \in \mathrm{Supp}(D)$, the germs $(f_i)_x$ which lie in $\mathfrak{m}_x$ form part of a regular system of parameters. One says that D has *normal crossings* if, locally for the étale topology, D has strict normal crossings.

The standard example of a normal crossings divisor is furnished by a locally finite union of coordinate hyperplanes in affine space.

b) Let X be a locally noetherian prescheme. We shall say that X is *strongly desingularizable* if the following condition is satisfied: for every finite morphism $f : Y \to X$, where Y is integral and its generic point is separable over its image, and every regular open subset U of Y, there exists a proper birational morphism $h : Y' \to Y$, with Y' regular, such that h induces an isomorphism from $U' = h^{-1}(U)$ onto U, and $Y' - U'$ is a normal crossings divisor.

Here are cases where X is strongly desingularizable:

a) X is excellent of characteristic zero;

b) X is locally noetherian, regular, and of dimension ≤ 1;

[4] Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.
[5] Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

c) X is of finite type over $\mathbb{Z}$ or over a perfect field, and of dimension ≤ 2.[6]

3.1.6. Cohomological finiteness. Let $f : X \to Y$ be a morphism of preschemes, and let $F \in \operatorname{ob} \mathrm{D}_c^+(X)$. One has $\mathrm{R}f_*(F) \in \operatorname{ob} \mathrm{D}_c^+(Y)$ in the following cases:

a) f is proper and of finite presentation (SGA A, Exposé XIV, 1.1);

b) "when resolution of singularities and the purity theorem are available", in particular:

(i) if X and Y are locally of finite type over a field k of characteristic zero and f is a k-morphism of finite type (SGA A, Exposé XVI, 5.1);

(ii) if Y is excellent of characteristic zero and f is of finite type (SGA A, Exposé XIX, 5.1);

(iii) if X is locally of finite type over a field and $\dim(X) \leq 2$ (SGA A, Exposé XIX, 5.1).

3.2. The quasi-injective dimension of A_X

Proposition 3.2.1. Let X be a regular prescheme whose residual characteristics are prime to n. Suppose that X satisfies (reg) (3.1.1) and $(\mathrm{cdloc})_n$ (3.1.3), and that absolute purity is true on X (3.1.4). Then, for every constructible A_X-module F and every $x \in X$, one has

$$\underline{\mathrm{Ext}}^i(F, A_X)_x = 0 \qquad \text{for } i > 2\dim \mathcal{O}_{X,x}. \tag{*}$$

Proof. The question being local, we may suppose X affine. By dévissage (3.1.2), it is enough to verify $(*)$ for $F = i_!(G)$, where $i : Y \to X$ is the immersion of a regular integral sub-prescheme, closed in an open subset $D(f)$, $f \in A(X)$, and G is a locally constant constructible sheaf on Y. We must compute

$$\underline{\mathrm{Ext}}^q(F, A_X) = \mathrm{H}^q(\mathrm{R}\,\underline{\mathrm{Hom}}(F, A_X)).$$

The projection formula for i (SGA A, Exposé XVIII) gives

$$\mathrm{R}\,\underline{\mathrm{Hom}}(i_!G, A_X) \xrightarrow{\sim} \mathrm{R}i_*\mathrm{R}\,\underline{\mathrm{Hom}}(G, \mathrm{R}^!i(A_X)).$$

By purity,

$$\mathrm{R}^!i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes d}[-2d],$$

where $d = \operatorname{codim}(Y, X)$. Put $T = (\mu_n)_Y^{\otimes d}$. Since G is locally constant constructible, for every geometric point $\overline{y}$ of Y and every p one has

$$\underline{\mathrm{Ext}}^p(G, T)_{\overline{y}} \xrightarrow{\sim} \mathrm{Ext}^p(G_{\overline{y}}, T_{\overline{y}}).$$

But $T_{\overline{y}}$ is injective, being isomorphic to $A = \mathbb{Z}/n\mathbb{Z}$. Hence $\underline{\mathrm{Ext}}^p(G, T) = 0$ for $p \neq 0$, and

$$\mathrm{R}\,\underline{\mathrm{Hom}}(G, T) \xrightarrow{\sim} \underline{\mathrm{Hom}}(G, T).$$

[6] Added in 1977: more generally, if X is excellent noetherian of dimension ≤ 2, by H. Hironaka, *Desingularization of excellent surfaces*, notes by B. Bennett, Bowdoin College, 1967, to be supplemented if necessary by H. Hironaka, *Certain numerical characters of singularities*, J. Math. Kyoto Univ. 10 (1970), 151–187.

Thus

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F, A_X) \xrightarrow{\sim} \mathrm{R}i_*(\mathcal{H})[-2d], \qquad \mathcal{H} = \underline{\mathrm{Hom}}(G, T).$$

Let Z be the schematic closure of Y in X (EGA I, 9.5.10). Then $Y = Z \cap D(f)$ (EGA 0_{I}, 2.1.6). Denote by $j : Y \to Z$ and $k : Z \to X$ the canonical immersions. We have $\mathrm{R}i_* = k_* \mathrm{R}j_*$, so we are reduced to computing $\mathrm{R}j_*(\mathcal{H})$. If $\overline{x}$ is a geometric point of Z, let $\widetilde{Z} = \mathrm{Spec}(\mathcal{O}_{Z,\overline{x}})$ be the strict localization, let $\widetilde{f}$ be the image of f in $\mathcal{O}_{Z,\overline{x}}$, let $\widetilde{Y} = \widetilde{Z} \times_Z Y = \widetilde{Z} - V(\widetilde{f})$, and let $\widetilde{\mathcal{H}}$ be the inverse image of $\mathcal{H}$ on $\widetilde{Y}$. By SGA A, Exposé VIII, 5.2,

$$\mathrm{R}^q j_*(\mathcal{H})_{\overline{x}} \xrightarrow{\sim} \mathrm{H}^q(\widetilde{Y}, \widetilde{\mathcal{H}}),$$

hence

$$\underline{\mathrm{Ext}}^q(F, A_X)_{\overline{x}} \xrightarrow{\sim} \mathrm{H}^{q-2d}(\widetilde{Y}, \widetilde{\mathcal{H}}).$$

The condition $(\mathrm{cdloc})_n$ implies that this group is zero for

$$q - 2d > \dim \mathcal{O}_{Z,\overline{x}}.$$

Using

$$d = \mathrm{codim}_{\overline{x}}(Z, X) \leq \dim \mathcal{O}_{X,\overline{x}}$$

(EGA IV, 5.1.3) and

$$\dim \mathcal{O}_{Z,\overline{x}} = \dim \mathcal{O}_{X,\overline{x}} - \mathrm{codim}_{\overline{x}}(Z, X)$$

(EGA 0_{IV}, 16.5.12 and EGA IV, 5.1.9), one obtains

$$\underline{\mathrm{Ext}}^q(F, A_X)_x = 0 \qquad (q > 2 \dim \mathcal{O}_{X,x}),$$

as required. □

Remark 3.2.2. The bound obtained in 3.2.1 is the best possible. Indeed, let x be a point of X, and let Y be the closed integral sub-prescheme of X whose generic point is x. Since X satisfies (reg), there exists a regular open subset U of Y containing x. By purity,

$$\underline{\mathrm{Ext}}^{2d}(A_U, A_X)|U = (\mu_n)_U^{\otimes d},$$

where $d = \dim \mathcal{O}_{X,x} = \mathrm{codim}(Y, X)$.

Corollary 3.2.3. Let X be a regular prescheme of dimension d. If X is smooth over a field k of characteristic prime to n, or if X is excellent of characteristic zero, then X satisfies the conclusion of 3.2.1; consequently

$$\dim . q. inj(A_X) \leq 2d.$$

Proof. If X is excellent of characteristic zero, or if k is perfect of characteristic prime to n and X is smooth over k, then the hypotheses of 3.2.1 are satisfied by 3.1.1, 3.1.3, and 3.1.4. The conclusion is still valid when X is smooth over an arbitrary field k, as follows immediately after the base change $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$, where k' is a perfect closure of k (SGA A, Exposé VIII, 1.1). □

Lemma 3.2.4. Let X be a prescheme satisfying the conclusion of 3.2.1 and of dimension $\leq d$. If there exists an integer N such that X has cohomological dimension $\leq N$, then

$$\dim . inj(A_X) \leq 2d + N.$$

If, moreover, for every closed subset Y of X, the implication

$$\operatorname{codim}(Y, X) \geq p \quad \Rightarrow \quad \operatorname{cd}_{\mathbf{n}}(Y) \leq N - 2p$$

holds, then

$$\dim . inj(A_X) \leq N.$$

Proof. The first assertion follows immediately from 1.5. For the second, one argues as in 1.5. We are reduced to proving, for F constructible on X, the relation

$$\operatorname{Ext}^i(X; F, A_X) = 0 \qquad (i > N).$$

For this, inspect the spectral sequence

$$E_2^{pq} = \mathrm{H}^p(X; \underline{\operatorname{Ext}}^q(F, A_X)) \Rightarrow \operatorname{Ext}^*(X; F, A_X).$$

By 3.2.1, the sheaves $\underline{\operatorname{Ext}}^{2i}(F, A_X)$ and $\underline{\operatorname{Ext}}^{2i-1}(F, A_X)$ are concentrated in codimension $\geq i$. Therefore $E_2^{p,2i} = E_2^{p,2i-1} = 0$ for $p > N - 2i$, by the hypothesis. Hence $E_2^{pq} = 0$ for $p + q > N$, and the assertion follows. □

Corollary 3.2.5. If X is a smooth prescheme of dimension d over a separably closed field k of characteristic prime to n, then

$$\dim . inj(A_X) = 2d.$$

Proof. By 3.2.3 and SGA A, Exposé X, 4.2, we are in the situation of 3.2.4. The hypotheses of SGA A, Exposé X, 4.2 are satisfied by SGA A, Exposé X, 2.1 and 3.2; hence $\dim . inj(A_X) \leq 2d$. The equality follows from purity: if x is a closed point of codimension d in X, then

$$\operatorname{Ext}^{2d}(X, A_x, A_X) = \mathrm{H}_x^{2d}(A_X) \xrightarrow{\sim} (\mu_n)^{\otimes d}(k) \neq 0.$$

□

3.3. Constructibility of the sheaves $\underline{\operatorname{Ext}}^i(F, A_X)$

Proposition 3.3.1. Let X be a prescheme.

a) The following conditions are equivalent:

(i) for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^+(X)$, one has

$$\mathrm{R}\,\underline{\operatorname{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}_c^+(X);$$

(i bis) for every pair (F, G) of constructible A_X-modules, one has $\mathrm{R}\,\underline{\operatorname{Hom}}(F, G) \in \operatorname{ob} \mathrm{D}_c^+(X)$, i.e. the sheaves $\underline{\operatorname{Ext}}^i(F, G)$ are constructible for all i;

(ii) for every open immersion $i : U \to X$ and every constructible A_U-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(ii bis) for every immersion $i : Y \to X$ and every constructible A_Y-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(ii ter) for every immersion $i : Y \to X$ and every locally constant constructible A_Y-module F, the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p;

(iii) for every finite morphism, respectively every closed immersion, $f : Y \to X$, and every $F \in \operatorname{ob} \mathrm{D}_c^+(X)$, one has $\mathrm{R}^! f(F) \in \operatorname{ob} \mathrm{D}_c^+(Y)$.

b) These conditions are satisfied in each of the following two cases:

(i) $\dim(X) \leq 1$;

(ii) resolution of singularities and purity are available on X in the sense of 3.1.6(b).

Proof. The equivalence of (i) and (i bis) follows from the standard spectral sequence passing from sheaf-Ext to hyper-Ext.

(i bis) $\Rightarrow$ (ii). Let $\overline{F}$ be a constructible A_X-module extending F, for example $\overline{F} = i_! F$. By duality (SGA A, Exposé XVIII, 3.1.10),

$$\mathrm{R}i_* F = \mathrm{R}i_* \mathrm{R}\,\underline{\mathrm{Hom}}(A_U, i^! \overline{F}) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(i_! A_U, \overline{F}),$$

and this proves the claim.

(ii) $\Rightarrow$ (ii bis). Factor i as a closed immersion followed by an open immersion.

(ii bis) $\Rightarrow$ (i bis). The question is local, so we may suppose X affine. By the usual dévissage (SGA A, Exposé IX, 2.5), it is enough to verify (ii) for $F = i_! M$, where $i : Y \to X$ is the immersion of a sub-prescheme and M is a locally constant constructible A_Y-module. By the projection formula (SGA A, Exposé XVIII),

$$\mathrm{R}\,\underline{\mathrm{Hom}}(i_! M, G) \xrightarrow{\sim} \mathrm{R}i_* \mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G),$$

and by a spectral sequence we are reduced to proving that

$$\mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G) \in \operatorname{ob} \mathrm{D}_c^+(Y).$$

First prove $\mathrm{R}i^! G \in \operatorname{ob} \mathrm{D}_c^+(Y)$. After factoring i, we may suppose that Y is closed. Let $j : U = X - Y$ be the complementary open immersion. Applying $\mathrm{R}\,\underline{\mathrm{Hom}}(\ , G)$ to

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0$$

gives a distinguished triangle. Since $\mathrm{R}\,\underline{\mathrm{Hom}}(A_X, G) = G$, and

$$\mathrm{R}\,\underline{\mathrm{Hom}}(A_{U,X}, G) \xrightarrow{\sim} \mathrm{R}j_* j^! G$$

belongs to $\mathrm{D}_c^+(X)$ by hypothesis, it follows that

$$\mathrm{R}\,\underline{\mathrm{Hom}}(A_Y, G) \xrightarrow{\sim} i_* \mathrm{R}i^! G$$

belongs to $\mathrm{D}_c^+(X)$. Resolving M locally on Y, for the étale topology, by free A_Y-modules, another dévissage gives

$$\mathrm{R}\,\underline{\mathrm{Hom}}(M, \mathrm{R}i^! G) \in \operatorname{ob} \mathrm{D}_c^+(Y).$$

This proves the implication.

(ii bis) $\Rightarrow$ (ii ter) is trivial.

(ii ter) $\Rightarrow$ (ii). Let $i : U \to X$ be an open immersion and let F be a constructible A_U-module. We must prove $\mathrm{R}i_*F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$. The question being local, suppose X noetherian and argue by noetherian induction on X. We may suppose $U \neq \varnothing$. Then there exists a non-empty open subset $j : V \to U$ such that j^*F is locally constant (SGA A, Exposé IX, 2.4). Let M be the mapping cylinder of

$$F \longrightarrow \mathrm{R}j_*j^*F.$$

Applying $\mathrm{R}i_*$ gives a distinguished triangle

$$\mathrm{R}i_*F \longrightarrow \mathrm{R}i_*\mathrm{R}j_*j^*F \longrightarrow \mathrm{R}i_*M \xrightarrow{[1]} .$$

Condition (ii ter) implies

$$\mathrm{R}i_*\mathrm{R}j_*j^*F \xrightarrow{\sim} \mathrm{R}(ij)_*j^*F \in \mathrm{ob}\,\mathrm{D}_c^+(X).$$

Consequently,

$$\mathrm{R}j_*j^*F \xrightarrow{\sim} i^*\mathrm{R}(ij)_*j^*F \in \mathrm{ob}\,\mathrm{D}_c^+(U).$$

Thus $M \in \mathrm{ob}\,\mathrm{D}_c^+(U)$, since $F \in \mathrm{ob}\,\mathrm{D}_c^+(U)$. Moreover, the cohomology of M is supported on a closed subset Z of U distinct from U. Applying the induction hypothesis to $U \cap \overline{Z} \to \overline{Z}$, we deduce $\mathrm{R}i_*M \in \mathrm{ob}\,\mathrm{D}_c^+(X)$, and hence $\mathrm{R}i_*F \in \mathrm{ob}\,\mathrm{D}_c^+(X)$.

The equivalence of (i)–(ii ter) with condition (iii) is left to the reader; use 1.14.

For b)(i), normalization reduces one to the case where X is regular, and one uses purity, which will be proved in 5.1. For b)(ii), see 3.1.6(b). ☐

3.4. The biduality theorem on regular preschemes

Theorem 3.4.1, cf. [H], V, 2.2. Let X be a regular locally noetherian prescheme of finite dimension. Suppose that X is strongly desingularizable (3.1.5), satisfies (reg) (3.1.1) and $(\mathrm{cdloc})_n$ (3.1.3), and that absolute purity (3.1.4) is true on every prescheme smooth over X. Then A_X is dualizing.

Proof. We must prove that A_X satisfies conditions (i)–(iii) of 1.6. Condition (i) follows from 3.2.1. Condition (ii) follows from 3.3.1(b). It remains to prove condition (iii). Thus, by 1.7, for every constructible A_X-module F we must prove that the canonical homomorphism

$$F \longrightarrow \underline{D}_X\underline{D}_XF, \qquad \underline{D}_X = \mathrm{R}\,\underline{\mathrm{Hom}}(\ ,A_X),$$

is an isomorphism. The question being local, suppose X noetherian. By dévissage (3.1.2, iii bis), it is enough to treat sheaves

$$F = f_*i_!(A'_U),$$

where $i : U \to Y$ is the inclusion of a regular open subset of an integral prescheme, $f : Y \to X$ is finite and the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$ for a prime divisor ℓ of n.

Because X is strongly desingularizable, there is a proper birational morphism $h : Y' \to Y$, with Y' regular, inducing an isomorphism from $U' = h^{-1}U$ to U, and such that $Y' - U'$ is a normal crossings divisor. By base change (SGA A, Exposé XVII),

$$i_!(A'_U) \xrightarrow{\sim} \mathrm{R}h_*i'_!(A'_{U'}),$$

where $i' : U' \to Y'$ is the inclusion. Hence, with $g = fh$,

$$f_* i_!(A'_U) \xrightarrow{\sim} f_* \mathrm{R}h_* i'_!(A'_{U'}) \xrightarrow{\sim} \mathrm{R}g_* i'_!(A'_{U'}).$$

By 1.11, it suffices to prove that the pair

$$(i'_!(A'_{U'}), \mathrm{R}^! g(A_X))$$

is bidualizing. Locally immersing Y' in a prescheme X' smooth over X, and applying purity on X', one sees that $\mathrm{R}^! g(A_X)$ is locally isomorphic to $A_{Y'}$, up to a translation of degrees. We are therefore reduced to the following special case.

Lemma 3.4.2. Let X be a locally noetherian regular prescheme on which absolute purity is true. Let $Y = Y_1 \cup \cdots \cup Y_p$ be a strict normal crossings divisor on X, and put $U = X - Y$. Then the pair $(A'_{U,X}, A_X)$ is bidualizing.

Proof. By the exact sequence

$$0 \longrightarrow A'_{U,X} \longrightarrow A'_X \longrightarrow A'_Y \longrightarrow 0,$$

it is equivalent to verify that (A'_Y, A_X) is bidualizing. We argue by induction on p.

If $p = 1$, let $i : Y \to X$ be the immersion. By 1.11, it is enough to verify that

$$(A'_Y, \mathrm{R}^! i(A_X))$$

is bidualizing. By purity,

$$\mathrm{R}^! i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes 1}[-2].$$

Thus the canonical homomorphism $A'_Y \to \underline{D}_Y \underline{D}_Y(A'_Y)$ is an isomorphism, as is checked trivially fibre by fibre, using Pontryagin duality for A-modules.

Suppose $p \geq 2$, and put $Y' = Y_1 \cup \cdots \cup Y_{p-1}$. We have an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_Y \longrightarrow A'_{Y'} \longrightarrow 0.$$

By the induction hypothesis, $(A'_{Y'}, A_X)$ is bidualizing. By the five lemma, we are reduced to biduality for $(A'_{(Y-Y'),X}, A_X)$. But again there is an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_{Y_p} \longrightarrow A'_{Y_p \cap Y'} \longrightarrow 0.$$

Since (A'_{Y_p}, A_X) is bidualizing, it remains to prove the same for $(A'_{Y_p \cap Y'}, A_X)$. Applying 1.11 again and using purity, we are finally reduced to biduality for this pair on Y_p, where the induction hypothesis applies. $\square$

This completes the proof of Theorem 3.4.1.

Corollary 3.4.3. Let S be an excellent noetherian regular prescheme of characteristic zero. Then A_S is dualizing. Moreover, if $f : X \to S$ is a morphism of finite type, then $\mathrm{R}^! f(A_S)$ is dualizing.

Proof. The first assertion follows because S satisfies the hypotheses of 3.4.1, as recalled in 3.1. For the second, one must first verify that $\mathrm{R}^! f(A_S)$ has finite quasi-injective dimension. Cover S by finitely many affine open subsets S_i such that $X_i = f^{-1}(S_i)$ is a finite union of affine open subsets

X_{ij} immersed as closed subschemes in preschemes X'_{ij} smooth of finite dimension over S_i. From 3.2.3 and 1.6 it follows that

$$\dim . q. inj(\mathrm{R}^! f(A_S)) \leq 2 \sup \dim X'_{ij}.$$

It remains to verify conditions (ii) and (iii) of 1.7. Since these are local on X for the étale topology, we may suppose that f factors as $f = f'i$, where $f' : X' \to S$ is smooth of relative dimension d, and $i : X \to X'$ is a closed immersion. Then

$$\mathrm{R}^! f(A_S) \simeq \mathrm{R}^! i \, \mathrm{R}^! f'(A_S).$$

But $\mathrm{R}^! f'(A_S)$ is locally isomorphic to $A_{X'}$, up to the translation by $2d$, hence is dualizing by the first assertion. By 1.15, $\mathrm{R}^! f(A_S)$ is dualizing, and in particular satisfies conditions (ii) and (iii) of 1.7. ☐

Corollary 3.4.4. Let k be a field, and let $f : X \to S = \mathrm{Spec}(k)$ be a morphism locally of finite type, with $\dim(X) \leq 2$. Then $\mathrm{R}^! f(A_S)$ is dualizing.

Proof. Replacing k by its perfect closure, which is harmless for the étale topology (SGA A, Exposé VIII, 1.1), we may suppose k perfect. Cover X by affine open subsets X_i of finite type over k. Since $\dim(X_i) \leq 2$, the normalization lemma makes X_i finite over a scheme X'_i smooth over S and of dimension ≤ 2. By 1.15 this reduces us to the case where f is smooth, hence locally

$$f^!(A_S) \simeq A_X[2d].$$

Thus it remains to show that A_X is dualizing in this case; this is exactly 3.4.1, since the hypotheses are satisfied (see 3.1). ☐

Remark 3.4.5. We shall see below that if X is regular of dimension ≤ 1, then A_X is dualizing.

4. Local duality

The object of this section is to present the notion of "biduality" developed in the preceding sections in a form closer to geometric intuition.

4.1

Let X be a prescheme, let $i : x \to X$ be a closed point whose residue field is separably closed, and let K_X be a dualizing complex on X (1.7). We know (1.15) that $K_{\{x\}} = \mathrm{R}^! i(K_X)$ is a dualizing complex on x, hence, by (2.1) and (3.4.4), isomorphic to $A[r]$ for some $r \in \mathbb{Z}$ (if A is local). We shall say that K_X is *normalized at* x if $r = 0$ and if an isomorphism $K_{\{x\}} \xrightarrow{\sim} A$ has been given.

For example, if X is smooth of relative dimension d over the spectrum of a separably closed field, the complex $(\mu_n)_X^{\otimes d}[2d]$ is, by the purity theorem (SGA XVI 3, SGA $4^{1/2}$ Cycle 2.3), canonically normalized at every closed point of X.

4.2

Let X be a prescheme, $i : Y \to X$ a closed sub-prescheme, and K_X a dualizing complex on X. Then (1.15) $K_Y = \mathrm{R}^! i(K_X)$ is a dualizing complex on Y, and one has the *complementary induction formula* (1.12 b) (ii)), for $F \in \operatorname{ob} \mathrm{D}_c^b(X)$,

$$(4.2.1) \qquad i^* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y \mathrm{R}^! i(F),$$

where $\underline{D}_X = \mathrm{R}\,\underline{\mathrm{Hom}}(\ ,K_X)$ and $\underline{D}_Y = \mathrm{R}\,\underline{\mathrm{Hom}}(\ ,K_Y)$.

Apply (4.2.1) in particular in the case where $Y = \{x\}$ is a closed point of X with separably closed residue field, K_X being normalized at x (4.1). Taking into account that A_x is injective, (4.2.1) is then written

$$(4.2.2) \qquad \underline{D}_X(F)_x \xrightarrow{\sim} \mathrm{Hom}^*(\mathrm{R}^! i(F), A).$$

Noting that $\mathrm{R}^! i(F)$ is also written, by definition, $\mathrm{R}\Gamma_x(F)$, one therefore has a perfect pairing in $\mathrm{D}(A)$:

$$(4.2.2)' \qquad \underline{D}_X(F)_x \times \mathrm{R}\Gamma_x(F) \longrightarrow A,$$

giving rise to perfect pairings between finite groups:

$$(4.2.2)'' \qquad \underline{\mathrm{H}}^i(\underline{D}_X(F))_x \times \underline{\mathrm{H}}_x^{-i}(F) \longrightarrow A.$$

For example, let X be a prescheme smooth of dimension d over a separably closed field, and let $x \in X$ be a closed point. Then, for $F \in \operatorname{ob} \mathrm{D}_c^b(X)$, one has a canonical perfect duality between finite groups:

$$(4.2.2)''' \qquad \mathrm{Ext}^{2d-i}(F, \mu_n^{\otimes d})_x \times \underline{\mathrm{H}}_x^i(F) \longrightarrow A.$$

Exercise 4.2.3

Let X be a strictly local scheme (SGA VIII 4.2), with closed point x, and let $i : x \to X$ be the canonical morphism. Let K be a dualizing complex on X, normalized at x. Denote by

$$\overset{\mathbf{L}}{\otimes} i_? : \mathrm{D}^b(A) \longrightarrow \mathrm{D}^-(X)$$

the functor defined by

$$\overset{\mathbf{L}}{\otimes} i_?(M) = M \overset{\mathbf{L}}{\otimes}_A K.$$

(If K has finite tor-dimension over A, $\overset{\mathbf{L}}{\otimes} i_?$ extends to a functor from $\mathrm{D}^+(A)$ to $\mathrm{D}(X)$.)

Show that there exists, for $L \in \mathrm{D}^-(X)$ and $M \in \mathrm{D}^b(A)$, a canonical functorial isomorphism

$$(*) \qquad \mathrm{R}\,\mathrm{Hom}(L, \overset{\mathbf{L}}{\otimes} i_?(M)) \xrightarrow{\sim} \mathrm{R}\,\mathrm{Hom}(\mathrm{R}^! i(L), M),$$

(assuming of course that $\mathrm{R}^! i : \mathrm{D}^-(X) \to \mathrm{D}^-(A)$ is defined ...).

Hint: First define a "trace" morphism $\mathrm{R}^! i \overset{\mathbf{L}}{\otimes} i_?(M) \to M$; then, to check that $(*)$ is an isomorphism, reduce by dévissage to the case where $L \in \operatorname{ob} \mathrm{D}_c^b(X)$ and $M = A$, and apply (4.2.2). Note that, for $M = A$, $(*)$ reduces to the perfect pairing ("local duality", cf. [H] V 6.2)

$$\mathrm{R}\Gamma_x(L) \times \mathrm{R}\,\mathrm{Hom}(L, K) \longrightarrow A,$$

or

$$\mathrm{H}_x^i(L) \times \mathrm{Ext}^{-i}(L, K) \longrightarrow A.$$

4.3

We shall now generalize (4.2.2) to the case of a geometric point $\overline{x}$ of X localized at a point x which is not necessarily closed. For this we need a few preliminaries.

Lemma 4.3. Let X be a prescheme, X' the strict localization of X at a geometric point $\overline{x}$, and $f : X' \to X$ the canonical map. Then, for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$, the canonical functorial homomorphism (SGA VI I 3.7)

$$f^*\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is an isomorphism.

Proof. As usual, one reduces to F and G concentrated in degree 0. Since the question is local in a neighborhood of x, one may suppose X affine. Then F admits (SGA IX 2.7) a left resolution by sheaves of the form $i_!(A_U)$, where $i : U \to X$ is étale of finite presentation; hence, by dévissage, one is reduced to $F = i_!(A_U)$. Form the cartesian square

$$\begin{tikzcd} U & U' \arrow[l, "g"'] \\ X \arrow[u, "i"] & X' \arrow[l, "f"] \arrow[u, "i'"'] \end{tikzcd}$$

One has canonical isomorphisms, trivial special cases of duality (SGA XVIII 3.1.10):

$$\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \xleftarrow{\sim} \mathrm{R}i_*i^*G, \qquad \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G) \xleftarrow{\sim} \mathrm{R}i'_*g^*i^*G.$$

Under these isomorphisms, the canonical arrow

$$f^*\mathrm{R}\,\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\,\underline{\mathrm{Hom}}(f^*F, f^*G)$$

is identified with the base-change arrow (SGA XVII)

$$f^*\mathrm{R}i_*i^*(G) \longrightarrow \mathrm{R}i'_*g^*i^*(G).$$

But this is an isomorphism, as one sees at once by passing to the limit (SGA VII 5.11).

Lemma 4.4. Let X be a prescheme, and let $K \in \operatorname{ob} \mathrm{D}_c^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K be dualizing, it is necessary and sufficient that, for every strict localization $f : \overline{X}(\overline{x}) \to X$, f^*K be dualizing.

Proof. Use the fact that every constructible sheaf on a strict localization $\overline{X}(\overline{x})$ is induced by a constructible sheaf on an étale $\overline{x}$-pointed scheme over X (SGA IX 2.7.4), and apply (4.3).

4.5

Let X be a prescheme, $\overline{x}$ a geometric point of X, and $f : X' = \overline{X}(\overline{x}) \to X$ the corresponding strict localization.

Let Y be the integral closed sub-prescheme which is the closure of x in X; denote by $g : \overline{x} \to Y$ and $i : Y \to X$ the canonical maps, and form the cartesian square

$$\begin{tikzcd} Y \arrow[r, "i"] & X \\ \overline{x} \arrow[u, "g"] \arrow[r, "i'"'] & X' \arrow[u, "f"'] \end{tikzcd}$$

($\overline{x}$ is both the closed point of X' and the strict localization of Y at $\overline{x}$).

We shall denote by

$$\mathrm{R}\Gamma_{\overline{x}} : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(\overline{x})$$

the functor defined by

$$(4.5.1) \qquad \mathrm{R}\Gamma_{\overline{x}} = g^*\mathrm{R}^! i.$$

Using the well-known relation between the functors $\mathrm{R}^! i$ and $\mathrm{R}j_*j^*$, where $j : U \to X$ is the open complement of Y, and applying passage to the limit (SGA VII 5.11), one obtains a canonical isomorphism

$$(4.5.2) \qquad \mathrm{R}\Gamma_{\overline{x}} \xrightarrow{\sim} \mathrm{R}^! i' f^*.$$

When x is a closed point with separably closed residue field, the functor $\mathrm{R}\Gamma_{\overline{x}}$ coincides with the usual functor $\mathrm{R}\Gamma_x$ (SGA V 4.3), which justifies the notation.

Let $K_X \in \operatorname{ob}\mathrm{D}^+(X)$, and put $K_Y = \mathrm{R}^! j(K_X)$, $K_{X'} = f^*K_X$, and $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ (warning: $K_{\overline{x}}$ is not the fiber of K_X at $\overline{x}$!). Suppose that K_X is dualizing. Then, by (1.15) and (4.4), the same is true of $K_{X'}$, K_Y, and $K_{\overline{x}}$. Hence (2.1) one has $K_{\overline{x}} \simeq A[d]$ for some $d \in \mathbb{Z}$.

We shall say that K_X is *normalized at* $\overline{x}$ if $d = 0$ and if an isomorphism $K_{\overline{x}} \simeq A$ has been chosen. In the case where x is closed with separably closed residue field, this recovers the notion introduced in (4.1).

Let K_X be a dualizing complex, and put $\underline{D}_X = \mathrm{R}\underline{\mathrm{Hom}}(\ , K_X)$, $\underline{D}_Y = \mathrm{R}\underline{\mathrm{Hom}}(\ , K_Y)$, etc. Let $F \in \operatorname{ob}\mathrm{D}^b_c(X)$; let us compute $(\underline{D}_X F)_{\overline{x}}$. One has:

$$\begin{aligned} (\underline{D}_X F)_{\overline{x}} &\xrightarrow{\sim} g^* i^* \underline{D}_X F \\ &\xrightarrow{\sim} g^* \underline{D}_Y \mathrm{R}^! i F \qquad \text{(by the complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} g^* \mathrm{R}^! i(F) \qquad \text{(by (4.3)), that is,} \end{aligned}$$

$$(4.5.3) \qquad (\underline{D}_X F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}}(F).$$

Of course, one could have made an analogous computation by passing through X' instead of Y; one would then have obtained a canonical isomorphism

$$(\underline{D}_X F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}^! i' f^*(F),$$

compatible with (4.5.3) by means of the identification (4.5.2).

If moreover K_X is normalized at $\overline{x}$, one obtains perfect pairings generalizing (4.2.2)' and (4.2.2)":

$$(4.5.3)' \qquad \mathrm{H}^i_Y(F)_{\overline{x}} \times \mathrm{R}^{-i}\Gamma_{\overline{x}}(F) \longrightarrow A,$$

$$(4.5.3)'' \qquad \mathrm{H}^i(\underline{D}_X(F))_{\overline{x}} \times \mathrm{H}^{-i}_{\overline{x}}(F) \longrightarrow A,$$

where $\mathrm{H}^i_{\overline{x}} = \mathrm{H}^i \mathrm{R}\Gamma_{\overline{x}}$.

Conversely, let $K_X \in \mathrm{ob}\, \mathrm{D}^+(X)$ satisfy conditions (i) and (ii) of (1.7), and suppose that $K_{\overline{x}} = \mathrm{R}\Gamma_{\overline{x}}(K_X)$ is dualizing for every geometric point $\overline{x}$ of X. Then, for every $\overline{x}$, one can define (cf. 1.12 b) (ii)) a canonical arrow

$$(4.5.3)^{\mathrm{bis}} \qquad \underline{D}_X(F)_{\overline{x}} \longrightarrow \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}}(F) \qquad (F \in \mathrm{ob}\, \mathrm{D}^b_c(X)).$$

If this is an isomorphism for every $\overline{x}$ and every F, then K_X is dualizing. Indeed, it remains to verify that the canonical arrow

$$F \longrightarrow \underline{D}_X \underline{D}_X(F)$$

is an isomorphism for every $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$. At an arbitrary geometric point $\overline{x}$, one has

$$\begin{aligned} F_{\overline{x}} &\xrightarrow{\sim} \underline{D}_{\overline{x}} \underline{D}_{\overline{x}}(F_{\overline{x}}) \qquad \text{(since } K_{\overline{x}} \text{ is dualizing)} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}} \underline{D}_X(F) \qquad \text{(by (4.3) and the induction formula 1.12 b) (i))} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} \underline{D}_{\overline{x}}(F)_{\overline{x}} \qquad \text{(by hypothesis),} \end{aligned}$$

and therefore, modulo a compatibility check, (F, K_X) is bidualizing, which proves the assertion. In summary:

Proposition 4.5.4. Let X be a prescheme, and let $K_X \in \mathrm{ob}\, \mathrm{D}^+(X)$ be a complex satisfying conditions (i) and (ii) of (1.7). In order that K_X be dualizing, it is necessary and sufficient that, for every geometric point $\overline{x}$ of X, the complex $K_{\overline{x}}$ be dualizing and the canonical arrow $(4.5.3)^{\mathrm{bis}}$ be an isomorphism for every $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$.

Remark 4.5.5. The existence of dualizing complexes should be regarded, in a sense, as an important complement to the global duality theorem. Indeed, the latter expresses that, for a separated morphism of finite type $f : X \to Y$ (Y quasi-compact), one has a canonical isomorphism

$$\underline{D}_Y \mathrm{R}_! f(F) \xrightarrow{\sim} \mathrm{R} f_* \underline{D}_X(F)$$

(with the notation of 1.12). Its use for computing $\mathrm{R}_! f(F)$ (when $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$) requires information about $\underline{D}_X(F)$. When K_X is dualizing, this information is supplied by the isomorphism (4.5.3), which makes possible, at least in theory, a computation of the fibers of $\underline{D}_X(F)$ as invariants of "purely spatial" cohomology.

Exercise 4.5.6. Let X be a prescheme endowed with a dualizing complex (1.7). The family of functors (cf. (4.5.3)")

$$\mathrm{H}^i_{\overline{x}} : \mathrm{D}^b_c(X) \longrightarrow (A\text{-modules})$$

(i running through $\mathbb{Z}$ and $\overline{x}$ through the set of geometric points of X) is "conservative", i.e. a morphism $F \to G$ in $\mathrm{D}^b_c(X)$ such that the maps $\mathrm{H}^i_{\overline{x}}(F) \to \mathrm{H}^i_{\overline{x}}(G)$ are isomorphisms, is an isomorphism.

4.6

Let X be a prescheme, $j : Y \to X$ a closed sub-prescheme, and $i : \overline{x} \to Y$ a geometric point of Y. Let, on the other hand, K_X be a dualizing complex on X, normalized at $\overline{x}$ (cf. (4.5)). We saw in (4.5) that, if Y is the closure of x, one has, for $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$, a perfect pairing

$$\mathrm{R}^! j(F)_{\overline{x}} \times \underline{D}_X(F)_{\overline{x}} \longrightarrow A.$$

It is not difficult to define an analogous pairing in the general case. Introduce the factorization of i as

$$\overline{x} \xrightarrow{i'} \{\overline{x}\} \xrightarrow{i''} Y.$$

For $F \in \operatorname{ob} \mathrm{D}_c^b(X)$, one has canonical isomorphisms:

$$\begin{aligned} \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F) &\xrightarrow{\sim} \mathrm{R}\Gamma_{\overline{x}} \underline{D}_Y \mathrm{R}^! j(F) && \text{(complementary induction formula (4.2.1))} \\ &\xrightarrow{\sim} i'^* \mathrm{R}^! i'' \underline{D}_Y \mathrm{R}^! j(F) && \text{(definition of } \mathrm{R}\Gamma_{\overline{x}} \text{ (4.5.1))} \\ &\xrightarrow{\sim} i'^* \underline{D}_{\overline{x}} i''^* \mathrm{R}^! j(F) && \text{(ordinary induction (1.12 b) (i))} \\ &\xrightarrow{\sim} \underline{D}_{\overline{x}} i^* \mathrm{R}^! j(F) && \text{(by 4.3)), i.e. finally a} \end{aligned}$$

canonical isomorphism

$$(4.6.1) \qquad \mathrm{R}^! j(F)_{\overline{x}} \xrightarrow{\sim} \underline{D}_{\overline{x}} \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F),$$

and hence perfect pairings

$$(4.6.1)' \qquad \mathrm{R}^! j(F)_{\overline{x}} \times \mathrm{R}\Gamma_{\overline{x}} j^* \underline{D}_X(F) \longrightarrow A,$$

$$(4.6.1)'' \qquad \mathrm{H}_Y^i(F)_{\overline{x}} \times \mathrm{H}_{\overline{x}}^{-i}(\underline{D}_X(F)|Y) \longrightarrow A.$$

Example 4.6.2. Let X be a regular prescheme, let $\overline{x}$ be a geometric point of X, and put $d = \dim(\mathcal{O}_{X,\overline{x}})$. Set $K_X = (\mu_n)_X^{\otimes d}[2d]$. If X satisfies condition (reg) (3.1.1) and the purity theorem (3.1.4), then K_X is canonically normalized at $\overline{x}$. Thus, if K_X is dualizing, one has perfect pairings

$$\mathrm{H}_Y^i(F)_{\overline{x}} \times \mathrm{H}_{\overline{x}}^{2d-i}(F'|Y) \longrightarrow A,$$

for every closed subset Y containing x and every $F \in \operatorname{ob} \mathrm{D}_c^b(X)$, with $F' = \mathrm{R}\operatorname{Hom}(F, (\mu_n)_X^{\otimes d})$.

To finish, let us give a strictly local variant of (4.6.1).

4.7

Let X be a noetherian strictly local scheme (SGA VIII 4.2), with closed point x. Let Y be a non-empty closed subset of X, and consider the canonical immersions

$$U = X - Y \xrightarrow{i} V = X - \{x\} \xrightarrow{j} X.$$

Finally, let K_X be a dualizing complex normalized at x (4.1), giving dualizing complexes "corresponding" on U, V, Y, x (and by hypothesis $\underline{D}_x = \mathrm{R}\operatorname{Hom}(\ , A)$):

$$K_V = j^* K_X, \qquad K_U = i^* K_V, \qquad K_Y = \mathrm{R}\Gamma_Y(K_X).$$

Let $G \in \operatorname{ob} \mathrm{D}_c^b(U)$, and put $G' = \underline{D}_U(G)$. We propose to define a natural pairing

$$(4.7.1) \qquad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A,$$

i.e. a functorial homomorphism

$$(4.7.1)' \qquad \mathrm{R}\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A.$$

(Since the ring $A = \mathbb{Z}/\ell\mathbb{Z}$ has infinite cohomological dimension for $\ell \geqslant 2$, one knows how to define only the tensor products

$$\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \longrightarrow \mathrm{D}^-(A)$$

and

$$\overset{\mathbf{L}}{\otimes} : \mathrm{D}(A) \times \mathrm{D}^b(A)_{\mathrm{torf}} \longrightarrow \mathrm{D}(A) \quad (*).$$

We therefore leave it to the reader to make suitable hypotheses ensuring that the two factors of $\overset{\mathbf{L}}{\otimes}$ appearing in (4.7.1)' are in $\mathrm{D}^b(A)$, or that one of them has finite tor-dimension.)

First note that, by the complementary duality formula (1.12 a) (ii), one has a canonical isomorphism

$$(4.7.2) \qquad i_!(G') \xrightarrow{\sim} \underline{D}_V(\mathrm{R}i_*G).$$

On the other hand, one has the ordinary duality isomorphism

$$(4.7.3) \qquad \mathrm{R}\operatorname{Hom}(A_U, G) \xrightarrow{\sim} \mathrm{R}\operatorname{Hom}(A_V, \mathrm{R}i_*G).$$

By the definition of $\underline{D}_V$ (and independently of the fact that K_V is dualizing), one has a canonical arrow

$$(4.7.4) \qquad \mathrm{R}i_*G \overset{\mathbf{L}}{\otimes} \underline{D}_V(\mathrm{R}i_*G) \longrightarrow K_V.$$

From (4.7.2), (4.7.3), and (4.7.4) one obtains a (canonical) arrow

$$(4.7.5) \qquad \mathrm{R}\Gamma(U, G) \overset{\mathbf{L}}{\otimes}_A \mathrm{R}\Gamma(V, i_!G') \longrightarrow \mathrm{R}\operatorname{Hom}(A_V, K_V).$$

But $K_V = j^*K_X$, and one has the duality isomorphism (SGA 3.1.10):

$$(4.7.6) \qquad \mathrm{R}\operatorname{Hom}(A_V, K_V) \xrightarrow{\sim} \mathrm{R}\operatorname{Hom}(A_{V,X}, K_X).$$

Finally, the exact sequence

$$(4.7.7) \qquad 0 \longrightarrow A_{V,X} \longrightarrow A_X \longrightarrow A_x \longrightarrow 0$$

gives a homomorphism of degree $+1$:

$$(4.7.8) \qquad \mathrm{R}\operatorname{Hom}(A_{V,X}, K_X) \longrightarrow \mathrm{R}\operatorname{Hom}(A_x, K_X) \xrightarrow{\sim} A.$$

Composing (4.7.5), (4.7.6), and (4.7.8), one obtains the announced arrow (4.7.1)'.

(*) (added in 1977) Inexact: in fact it is easy to extend $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}^-(A) \to \mathrm{D}^-(A)$ *to* $\overset{\mathbf{L}}{\otimes} : \mathrm{D}^-(A) \times \mathrm{D}(A) \to \mathrm{D}(A)$.

We shall now see that the pairing (4.7.1) is "perfect", i.e. that the arrow (4.7.1)' identifies $\mathrm{R}\Gamma(V, i_!G')[-1]$ with $\underline{D}_x\mathrm{R}\Gamma(U, G)$. Indeed suppose, as is allowed, that

$$G = (ji)^*F,$$

with $F \in \operatorname{ob}\mathrm{D}^b_c(X)$. Put $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!G'$. One has the exact sequence

$$(4.7.9) \qquad 0 \longrightarrow F'_{U,X} \longrightarrow F' \longrightarrow F'|Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{tikzcd}[column sep=small] & \mathrm{R}\Gamma_x(F'_{U,X}) \arrow[dl] & \\ \mathrm{R}\Gamma_x(F'|Y) \arrow[rr] & & \mathrm{R}\Gamma_x(F') \arrow[ul, "+1"'] \end{tikzcd} \qquad (b)$$

By definition of $\mathrm{R}\Gamma_x$, one has

$$\mathrm{R}\Gamma_x(F'_{U,X}) = \mathrm{R}\operatorname{Hom}(A_x, F'_{U,X});$$

hence, using (4.7.7) and the fact that $(F'_{U,X})_x = \mathrm{R}\Gamma(X, F'_{U,X}) = 0$, one has a canonical isomorphism

$$\mathrm{R}\operatorname{Hom}(A_{V,X}, F'_{U,X})[1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

But one has

$$\begin{aligned} \mathrm{R}\operatorname{Hom}(A_{V,X}, F'_{U,X}) &= \mathrm{R}\operatorname{Hom}(j_! A_V, j_! i_! G') \\ &\xrightarrow{\sim} \mathrm{R}\operatorname{Hom}(A_V, i_! G') \qquad \text{by duality (SGA 3.1.10),} \end{aligned}$$

and hence finally a canonical isomorphism

$$\text{(4.7.10)} \qquad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

On the other hand one has the exact sequence

$$\text{(4.7.11)} \qquad 0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0,$$

which gives rise to a distinguished triangle

$$\begin{tikzcd}[column sep=small] & \mathrm{R}\Gamma(U, F|U) \arrow[dl] & \\ \mathrm{R}\Gamma_Y(F) \arrow[rr] & & \mathrm{R}\Gamma_X(F) \arrow[ul, "+1"'] \end{tikzcd} \qquad (a)$$

By virtue of (4.7.1) and (4.7.10), one has a natural pairing

$$\text{(4.7.12)} \qquad \mathrm{R}\Gamma(U, F|U) \times \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow A.$$

By virtue of (4.6.1), one has a perfect pairing

$$\text{(4.7.13)} \qquad \mathrm{R}\Gamma_Y(F) \times \mathrm{R}\Gamma_x(F'|Y) \longrightarrow A.$$

Finally, by the ordinary induction formula (1.12 b) (i)), one has a perfect pairing

$$\text{(4.7.14)} \qquad \mathrm{R}\Gamma(X, F) \times \mathrm{R}\Gamma_x(F') \longrightarrow A.$$

The reader who has the patience to orient himself in this maze of "canonical arrows" will check that the pairings (4.7.12), (4.7.13), and (4.7.14) are "compatible" with the arrows of the triangles (a) and (b), and consequently that the pairings (4.7.12) and (4.7.1) are perfect.

The pairing of the triangles (a) and (b) gives rise to two exact sequences "transpose to one another":

$$\text{(4.7.15)} \qquad \cdots \longrightarrow \mathrm{H}^i_Y(X, F) \longrightarrow \mathrm{H}^i(X, F) \longrightarrow \mathrm{H}^i(U, F) \longrightarrow \cdots$$

$$\cdots \longleftarrow \mathrm{H}_x^{-i}(Y, F'|Y) \longleftarrow \mathrm{H}_x^{-i}(X, F') \longleftarrow \mathrm{H}_x^{-i-1}(V, F'_{U,V}) \longleftarrow \cdots .$$

In summary, we may state:

Proposition 4.7.16. Let X be a noetherian strictly local scheme, with closed point x. Let Y be a non-empty closed subset of X, let $U = X - Y$, $V = X - x$, and let $i : U \to V$ and $j : V \to X$ be the canonical inclusions. Let K_X be a dualizing complex on X normalized at x (hence giving associated dualizing complexes on V, U, Y). For $G \in \mathrm{ob}\, \mathrm{D}_c^b(U)$, one has a perfect pairing:

$$(*) \qquad \mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_! G')[-1] \longrightarrow A$$

(where $G' = \underline{D}_U(G)$). If $G = F|U$, where $F \in \mathrm{ob}\, \mathrm{D}_c^b(X)$, one has a canonical isomorphism

$$(**) \qquad \mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X})$$

(where $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!(F'|U) = (ji)_!(G')$), and a perfect pairing between the distinguished triangles

$$\mathrm{R}\Gamma_Y(F) \longrightarrow \mathrm{R}\Gamma(X, F) \longrightarrow \mathrm{R}\Gamma(U, F|U) \xrightarrow{+1},$$
$$\mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow \mathrm{R}\Gamma_x(F') \longrightarrow \mathrm{R}\Gamma_x(F'|Y) \xrightarrow{+1} .$$

compatible with the pairing $(*)$ and the isomorphism $(**)$, and giving rise to two exact sequences transpose to one another (4.7.15).

Remark 4.7.17. In the particular case where $Y = \{x\}$, the pairing (4.7.1) is defined under the sole hypothesis that K_X is normalized at x (and not necessarily dualizing), i.e. such that $\mathrm{R}\Gamma_x(K_X) \simeq A$ (the isomorphism being given): the isomorphism (4.7.2) (which used biduality) then reduces to the identity. Since the pairing (4.7.14) is in any case perfect, one sees therefore that local duality on X (in the form (4.2.2)') is equivalent to the "global" duality on $U = X - x$, i.e. to the assertion that the pairing (4.7.1)

$$\mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(U, G')[-1] \longrightarrow A$$

is perfect, or equivalently that the pairings

$$\mathrm{H}^i(U, G) \times \mathrm{H}^{-i-1}(U, G') \longrightarrow A$$

are perfect.

Suppose that one has on X a dualizing complex K_X normalized at x. If X is excellent and U is regular of dimension d (x may therefore be an isolated singularity), one should be able to check that $K_U = K_X|U$ is canonically isomorphic to $(\mu_n)_U^{\otimes d}[2d]$ (this is easily checked, at any rate, if X is the strict localization of a prescheme locally of finite type over a field of characteristic 0). Then local duality on X implies that, for every locally constant constructible A_U-module G, one has perfect pairings:

$$\mathrm{H}^i(U, G) \times \mathrm{H}^{2d-1-i}(U, G^\circ) \longrightarrow A,$$

where $G^\circ = \mathrm{Hom}(G, (\mu_n)_U^{\otimes d})$, corresponding formally to Poincaré duality on a scheme of dimension $2d - 1$.

5. Local duality on curves

Theorem 5.1. Let X be a noetherian regular prescheme of dimension 1. Suppose that n is prime to the residual characteristics of X. Then the purity theorem is true on X (3.1.4), and the complex A_X is dualizing (1.7).

Proof. Suppose purity has been proved. Since X plainly satisfies condition (reg) (3.1.1), and since condition $(\mathrm{cdloc})_n$ is verified by (SGA X 2.2), it follows from (3.2.1) that $\dim .q.inj(A_X) = 2$. On the other hand, condition (ii) of (1.7) is verified by (3.3.1 b) (i)) (which was proved assuming purity). Hence, by (4.4), we are reduced to proving that A_X is dualizing when X is strictly local. But, to prove purity, which is a local question in a neighborhood of a closed point, one may equally well suppose X strictly local. Finally, we are reduced to proving the theorem when X is strictly local.

Thus suppose that X is strictly local. Let x be its closed point, let $(\pi = \mathrm{car}(k(x)))$, let η be its generic point, and let $U = X - x = \mathrm{Spec}(k(\eta))$. Let $G = \mathrm{Gal}(k(\overline{\eta})/k(\eta))$ be the fundamental group of U. It is known [CL, Chap. IV § 2 Exer. 2] that one has a canonical exact sequence

$$(5.1.1) \qquad 1 \longrightarrow P \longrightarrow G \longrightarrow H \longrightarrow 1,$$

where $H = \prod_{\ell \neq \pi} \mathbb{Z}_\ell(1)$, $\mathbb{Z}_\ell(1) = \varprojlim \mu_{\ell^\nu}$, and P is a pro-π-group. (Of course, H is isomorphic, but not canonically, to $\prod_{\ell\neq\pi} \mathbb{Z}_\ell$.)

It is also known (SGA VIII 2) that the étale cohomology of U is interpreted as the Galois cohomology of G, sheaves (respectively constructible sheaves) of A_U-modules corresponding to A-modules (respectively finite type A-modules) on which G acts continuously.

This being so, the $\mathrm{H}^i_x(A_X)$ are determined by the exact sequence

$$(5.1.2) \qquad 0 \longrightarrow \mathrm{H}^0_x(A_X) \longrightarrow \mathrm{H}^0(X, A_X) \longrightarrow \mathrm{H}^0(U, A_U) \longrightarrow \mathrm{H}^1_x(A_X) \longrightarrow 0,$$

and the isomorphisms

$$(5.1.3) \qquad \mathrm{H}^i_x(A_X) \xrightarrow{\sim} \mathrm{H}^{i-1}(U, A_U) \qquad \text{for } i \geqslant 2.$$

But trivially $\mathrm{H}^0(X, A_X) = \mathrm{H}^0(U, A_U) = A$, so $\mathrm{H}^0_x(A_X) = \mathrm{H}^1_x(A_X) = 0$.

On the other hand, for a Galois G-A-module M, one has the Hochschild–Serre spectral sequence

$$E_2^{pq} = \mathrm{H}^p(H, \mathrm{H}^q(P, M)) \Longrightarrow \mathrm{H}^{p+q}(G, M).$$

Since n is prime to π and P is a pro-π-group, one has

$$\mathrm{H}^q(P, M) = 0 \quad \text{for } q \neq 0,$$

hence $E_2^{pq} = 0$ for $q \neq 0$, giving a canonical isomorphism

$$(5.1.4) \qquad \mathrm{H}^p(G, M) \xrightarrow{\sim} \mathrm{H}^p(H, M^P).$$

By an analogous argument, one sees moreover that $\mathrm{H}^p(H, N)$ is identified with $\mathrm{H}^p(\widehat{\mathbb{Z}}(1), N)$ when the factor $\widehat{\mathbb{Z}}(1)$ is made to act trivially on N. Thus, finally, with the preceding convention, one has a canonical isomorphism

$$(5.1.5) \qquad \mathrm{H}^p(G, M) \xrightarrow{\sim} \mathrm{H}^p(\widehat{\mathbb{Z}}(1), M^P).$$

Now the Galois cohomology of $\widehat{\mathbb{Z}}$ is well known (CG chap. I p. 31): for a G-A-module N, one has

$$(5.1.6) \qquad \mathrm{H}^i(\widehat{\mathbb{Z}}, N) = 0 \quad \text{for } i \geqslant 2, \qquad \mathrm{H}^0(\widehat{\mathbb{Z}}, N) = N^{\widehat{\mathbb{Z}}}, \qquad \mathrm{H}^1(\widehat{\mathbb{Z}}, N) = N_{\widehat{\mathbb{Z}}}.$$

From (5.1.3), (5.1.5), and (5.1.6) one immediately deduces that

$$\mathrm{H}^i(U, A_U) = 0 \quad \text{for } i \geqslant 2,$$

and

$$\mathrm{H}^1(U, A_U) \xrightarrow{\sim} A.$$

This last isomorphism is not canonical, since it depends on the choice of an identification of $\widehat{\mathbb{Z}}(1)$ with $\widehat{\mathbb{Z}}$. But, using the Kummer exact sequence, one finds (SGA XIX 1.3) a canonical isomorphism (given by the "local fundamental class")

$$(5.1.7) \qquad \mathrm{H}^1(U, \mu_n) \xrightarrow{\sim} A.$$

Purity is therefore proved.

Let us now prove that A_X is dualizing. By (4.7.17), it is a matter of showing that, for every locally constant constructible A_U-module M, the pairing

$$(5.1.8) \qquad \mathrm{H}^i(U, M) \times \mathrm{H}^{1-i}(U, \mathrm{Hom}(M, A_U)) \longrightarrow \mathrm{H}^1(U, A_U) \xrightarrow{\sim} (\mu_n)_x^{\otimes -1}$$

is a perfect duality. We shall interpret this pairing in terms of Galois cohomology. We shall use the following elementary lemma.

Lemma 5.1.9. Let $D = \mathrm{Hom}(\ , A)$ denote the Pontrjagin dualizing functor on the category of finite type A-modules. Let M be a finite type A-module on which a group G acts A-linearly.

(i) One has a canonical isomorphism

$$D(M^G) \xrightarrow{\sim} (DM)_G.$$

(ii) If G is a profinite group acting continuously on M, and if G has pro-order prime to n, then the canonical arrow

$$M^G \longrightarrow M_G$$

is an isomorphism.

Proof. By definition one has

$$M^G = \varprojlim_{s \in G} \mathrm{Ker}(M \xrightarrow{s-1} M),$$

hence

$$D(M^G) \xrightarrow{\sim} \varinjlim_{s \in G} \mathrm{Coker}(DM \xrightarrow{s-1} DM) \xrightarrow{\sim} (DM)_G,$$

which proves (i).

Let us prove (ii). Let M' be the A-module defined by the exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M_G \longrightarrow 0.$$

One has $\mathrm{H}^1(G, M') = 0$ because G has pro-order prime to n. The exact cohomology sequence therefore implies that the arrow $M^G \to M_G$ is an epimorphism. But, applying the functor D and taking (i) into account, one deduces that it is a monomorphism, which completes the proof.

Applying (5.1.9) to M and P, one finds that $\mathrm{Hom}(M, A)^P$ is canonically identified with $\mathrm{Hom}(M^P, A)$. If one chooses an isomorphism from $\widehat{\mathbb{Z}}(1)$ to $\widehat{\mathbb{Z}}$, the pairing (5.1.8) is written, taking (5.1.5) into account, as

$$(5.1.8)' \qquad \mathrm{H}^i(\widehat{\mathbb{Z}}, M^P) \times \mathrm{H}^{1-i}(\widehat{\mathbb{Z}}, D(M^P)) \longrightarrow \mathrm{H}^1(\widehat{\mathbb{Z}}, A) \xrightarrow{\sim} A.$$

It follows immediately from (5.1.6) and (5.1.9(i)) that (5.1.8)' is a perfect pairing. This completes the proof of (5.1).

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Appendix

by L. Illusie

The preceding theory extends easily to base Rings more general than the constant Ring $\mathbb{Z}/n\mathbb{Z}$. We shall briefly indicate how the theory is written for locally constant base Rings, killed by $n > 0$, and whose geometric fibres are noetherian rings. This is, moreover, only a provisional framework[7]: sooner or later one will have to globalize, by taking as "coefficients" on a prescheme X relative preschemes over the site $(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z})$.

1. Constructible sheaves

1.1. Let X be a prescheme, and let A_X be a sheaf of rings on X (for the étale topology), locally constant, whose geometric fibres are left noetherian rings. We denote by A^X the category of left

[7] According to Grothendieck!

A_X-Modules, and by $\mathrm{D}_A(X)$ (or $\mathrm{D}(X)$ if there is no risk of confusion) the corresponding derived category. The notion of a constructible left A_X-Module is defined as in (SGAA IX 2.3), by replacing the expression "locally constant with value of finite presentation" by "of finite presentation as an A_U-Module".

We denote by $\mathrm{D}_c(X)$ the full subcategory of $\mathrm{D}(X)$ consisting of complexes E such that $\mathrm{H}^i(E)$ is a constructible A_X-Module for every i. It is a triangulated subcategory of $\mathrm{D}(X)$.

1.2. Let $E \in \mathrm{ob}\,\mathrm{D}(X)$. We say (cf. Exp. II 3.11) that E is pseudo-coherent if, for every i, $\mathrm{H}^i(E)$ is an A_X-Module of finite presentation (or locally constant constructible, if one prefers). We denote by $\mathrm{D}(X)_{\mathrm{coh}}$ (or $\mathrm{D}_{\mathrm{coh}}(X)$) the full subcategory of $\mathrm{D}(X)$ consisting of pseudo-coherent objects. It follows from (SGAA IX 2.1) that this is a triangulated subcategory of $\mathrm{D}_c(X)$.

Proposition 1.3 (dévissage lemma). Let X be a quasi-compact and quasi-separated prescheme (respectively a noetherian prescheme), and let C be a strictly full subcategory of the category $\mathrm{Cons}(X)$ of constructible left A_X-Modules. Suppose that C is stable under extensions. The following conditions are equivalent:

(i) $C = \mathrm{Cons}(X)$;

(ii) C contains the sheaves of the form $i_!(F)$, where $i : Y \to X$ is a sub-prescheme of finite presentation (respectively an integral sub-prescheme), and F is locally constant constructible on Y.

If A_X is constant with value A, a left noetherian ring killed by an integer $n > 0$, the preceding conditions are equivalent to:

(iii) C is stable under direct factors and contains the sheaves of the form $f_* i_!(M_U)$, where $i : U \to Y$ is an open immersion (respectively an open subset of an integral prescheme), $f : Y \to X$ is a finite morphism (respectively finite and such that the generic point of Y is separable over its image), and M_U is a left A_U-Module constant with value a monogenic A-module killed by a prime divisor of n.

Finally, if X is noetherian and satisfies condition (reg) (I 3.1.1), condition (ii) (respectively (iii)) is equivalent to (i) if one further assumes Y (respectively U) regular.

Proof. The equivalence of conditions (i) and (ii) is proved as in the case where A_X is constant; see (SGAA IX 2.4 and 2.5).

Suppose that A_X is constant with value A and killed by n, and prove that (iii) implies (ii). Let $E \in \mathrm{ob}\,\mathrm{Cons}(X)$; we show that $E \in \mathrm{ob}\,C$. By the equivalence (i)$\Leftrightarrow$(ii), we may suppose that $E = j_! F$, where $j : Z \to X$ is the immersion of a sub-prescheme of finite presentation (respectively an integral one), and F is a constructible locally constant A_Z-Module, which we may even suppose, after a little more dévissage, to be trivialized by an étale covering $g : Z' \to Z$ with group G. The sheaf F is therefore defined by a finite type A-module M on which G acts. If ℓ is a prime divisor of n, the subgroup of ℓ-torsion in M is a sub-G-A-module M_ℓ of M, and

$$M = \bigoplus_\ell M_\ell,$$

where ℓ runs over the set of prime divisors of n. Since C is stable under direct factors, we are reduced to the case where M is ℓ-torsion. Let H be an ℓ-Sylow subgroup of G, and let $h : U = Z'/H \to Z$

be the corresponding intermediate covering. Since the degree of h is prime to ℓ, the "trace method" (SGAA IX 5.1) shows that the adjunction map

$$F \longrightarrow h_* h^*(F)$$

is an isomorphism onto a direct factor; it is therefore enough to prove that $j_! h_* h^*(F) \in \operatorname{ob} C$. By the "Main theorem" (EGA IV 8.12.6), there exists a commutative diagram

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[d, "h"'] & Y \arrow[d, "f"] \\ Z \arrow[r, "j"'] & X, \end{tikzcd}$$

where i is an open immersion and f is finite. We have

$$j_! h_* h^*(F) \simeq f_* i_! h^*(F).$$

But $h^*(F)$ is defined by M considered as an H-A-module. Thus (iii)⇒(ii) follows from the next lemma.

Lemma 1.3.1. Let A be a left noetherian ring, ℓ a prime number, H a finite ℓ-group, and M a left H-A-module of finite type killed by ℓ^ν ($\nu \geq 1$). There exists a finite filtration of M by sub-H-A-modules such that the successive quotients are monogenic A-modules killed by ℓ, and on which H acts trivially.

Proof. The exact sequence of H-A-modules

$$0 \longrightarrow \ell M \longrightarrow M \longrightarrow M/\ell M \longrightarrow 0$$

reduces, by induction on ν, to the case where M is killed by ℓ. We show that then M has a finite filtration by sub-H-A-modules on whose successive quotients H acts trivially. Arguing by noetherian induction on the set of sub-H-A-modules of M having the stated property, we are reduced to proving that $M \neq 0$ implies $M^H \neq 0$. But the kernel of the augmentation $\mathbb{F}_\ell[H] \to \mathbb{F}_\ell$ is plainly nilpotent, and Nakayama's lemma implies that every nonzero H-$\mathbb{F}_\ell$-module has a nonzero invariant vector. This proves the assertion. Thus M has a finite filtration whose successive quotients are finite type A-modules killed by ℓ and on which H acts trivially. By noetherian induction one may even arrange for the successive quotients to be monogenic A-modules, completing the proof of (1.3.1).

The last assertion of (1.3) is immediate and left to the reader.

2. Quasi-injective dimension

With the notation of (1.1), one defines, as in (I 1.1 and 1.4), the notions of a quasi-injective A_X-Module and of quasi-injective dimension of an object of $\mathrm{D}(X)$. Propositions (I 1.3, 1.5 and 1.6) extend without change.

It is sometimes convenient to introduce a notion of pointwise quasi-injective dimension. Let $x \in X$ and let $K \in \operatorname{ob} \mathrm{D}^+(X)$. We say that K has quasi-injective dimension $\leq N$ at $\bar{x}$ if, for every constructible A_X-Module F, one has

$$\underline{\mathrm{Ext}}^i(F, K)_{\bar{x}} = 0 \qquad \text{for } i > N.$$

The quasi-injective dimension of K at $\bar{x}$, denoted $\dim.\mathrm{q.inj}_{\bar{x}}(K)$, is the greatest lower bound of the N satisfying the preceding condition. Of course,

$$\dim.\mathrm{q.inj}(K) = \sup_{\bar{x}} \dim.\mathrm{q.inj}_{\bar{x}}(K),$$

where $\bar{x}$ runs through the set of geometric points of X.

It is interesting to compare this with the notion of strict topological dimension introduced by Verdier in [1]. We say that $K \in \mathrm{ob}\,\mathrm{D}^+(X)$ has strict topological dimension $\leq N$ at a geometric point $\bar{x}$ of X if, for every sub-prescheme Y of X,

$$\underline{\mathrm{Ext}}^i(A_Y, K)_{\bar{x}} = \underline{\mathrm{H}}^i_Y(K)_{\bar{x}} = 0 \qquad \text{for } i > N.$$

The strict topological dimension of K at $\bar{x}$, denoted $\dim.\mathrm{top}_{\bar{x}}(K)$, is the greatest lower bound of the N satisfying the preceding condition. Whereas quasi-injective dimension depends a priori on the base Ring, strict topological dimension depends only on the underlying complex of abelian sheaves.

One plainly has

$$\dim.\mathrm{top}_{\bar{x}}(K) \leq \dim.\mathrm{q.inj}_{\bar{x}}(K).$$

We shall see below examples where equality holds.

3. Dualizing complexes: definition and formal properties

Keep the preceding notation, but suppose A_X is commutative. The notions of bidualizing pair and dualizing complex are defined as in (I 1.7). The statements (I 1.9, 1.11, 1.14) remain valid without modification. The statements (I 1.12, 1.13, 1.15) are also valid, but on condition that A be killed by an integer $n > 0$ prime to the residual characteristics of X. Finally, the uniqueness theorem (I 2.1) is valid provided one assumes $\mathrm{Spec}(A_{\bar{x}})$ connected for every geometric point $\bar{x}$ of X.

4. Absolute purity

Return to the situation of absolute cohomological purity (I 3.1.4). Let X be a locally noetherian regular prescheme, let $n > 0$ be an integer prime to the residual characteristics of X, and let $i : Y \to X$ be a regular sub-prescheme of codimension $d > 0$ at every point. To say that the pair (Y, X) satisfies absolute cohomological purity means, recall, that the "local fundamental class" homomorphism (SGA $4^{1/2}$ Cycle 2.2)

$$(\mu_n^{\otimes d})_Y[-2d] \longrightarrow \mathrm{R}i^!(\mathbb{Z}/n\mathbb{Z})_X \tag{4.1}$$

is an isomorphism.

Now put on X a sheaf of rings A_X killed by n. For the moment set $G = (\mathbb{Z}/n\mathbb{Z})_X$. Then, for every $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$, one can define a canonical arrow in $\mathrm{D}^+_A(X)$:

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G) \longrightarrow \mathrm{R}i^!(F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G). \tag{4.2}$$

Indeed, by duality (SGAA XVIII 3.1.6), this is equivalent to defining an arrow

$$i_!(i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G)) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G;$$

we take this to be the composite of the Künneth arrow (SGAA XVII 5.4.2.2)

$$i_!(i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} \mathrm{R}i^!(G)) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} i_!\mathrm{R}i^!(G),$$

and the arrow deduced from the trace homomorphism (SGAA XVIII 3.1.6)

$$F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} i_!\mathrm{R}i^!(G) \longrightarrow F \otimes^L_{\mathbb{Z}/n\mathbb{Z}} G.$$

Notice that in the definition of (4.2) only the fact that G and $\mathrm{R}i^!(G)$ have finite Tor-dimension over $\mathbb{Z}/n\mathbb{Z}$ was used. Suppose on the other hand that the functor $i^!$ has finite cohomological dimension on the category of $(\mathbb{Z}/n\mathbb{Z})_X$-Modules: this is the case, for example, if X is of finite type over a field (SGAA XIV 3.1), or more generally if X is excellent of equal characteristic (SGAA XIX 6.1) when resolution of singularities is available, or again if X is of finite type over $\mathrm{Spec}(\mathbb{Z})$ (SGAA X 6). Then the arrow (4.2) is defined for all $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$.

Composing (4.1) and (4.2), one obtains, for $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$, a canonical arrow

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n^{\otimes d})_Y[-2d] \longrightarrow \mathrm{R}i^!(F). \tag{4.3}$$

Assume that A_X is locally constant, and let $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$ be such that every $\mathrm{H}^i(F)$ is a locally constant A_X-Module. Then, if $i^!$ has finite cohomological dimension, the arrow (4.2), and hence the arrow (4.3) if (Y, X) satisfies purity, is an isomorphism.

Indeed, by dévissage, one first reduces to the case where F is concentrated in degree zero. Since the question is local, one may suppose X affine and A_X and F constant. Resolving F on the left by free A_X-Modules and applying the hypothesis that $i^!$ has finite cohomological dimension, one reduces to the case where F is free. By an immediate passage to the limit, one reduces to F free of finite type, and finally to $F = A_X$. It is therefore enough to prove the assertion when $A_X = (\mathbb{Z}/n\mathbb{Z})_X$. But the preceding dévissage reduces to $F = (\mathbb{Z}/n\mathbb{Z})_X$, and in that case (4.2) is the identity.

Definition 4.4. Given the Ring A_X (locally constant and killed by n), we shall say that the pair (Y, X) satisfies cohomological purity in the strong sense if, for every $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$ such that the $\mathrm{H}^i(F)$ are locally constant, the canonical arrow (4.3) is an isomorphism.

We shall say that the purity theorem is true in the strong sense on X if, locally for the étale topology, every pair (Y, X) as above satisfies purity in the strong sense. From what has been shown, for (Y, X) to satisfy purity in the strong sense it is enough that (Y, X) satisfy ordinary purity (I 3.1.) and that the functor $i^!$ have finite cohomological dimension.

5. Bound for quasi-injective dimension

Proposition 5.1. Let X be a locally noetherian regular prescheme, equipped with a locally constant sheaf of rings A_X, killed by an integer $n > 0$ prime to the residual characteristics of X, whose geometric fibres are left noetherian rings. Suppose that X satisfies conditions (reg) (I 3.1.1) and $(\mathrm{cdloc})_n$ (I 3.1.3), and that the purity theorem in the strong sense is true on X (4.4) (conditions all satisfied if X is smooth over a perfect field, or if X is excellent of characteristic zero).

Let $F \in \mathrm{ob}\,\mathrm{D}^+_A(X)$ be such that the $\mathrm{H}^i(F)$ are locally constant, and let x be a point of X such that $F_{\bar{x}} \neq 0$ and $\dim.\mathrm{inj}_{A_{\bar{x}}}(F_{\bar{x}}) < +\infty$. Then

$$\dim.\mathrm{q.inj}_{\bar{x}}(F) = 2\dim(\mathcal{O}_{X,x}) + \dim.\mathrm{inj}(F_{\bar{x}}),$$

with the notation of no. 2.

Proof. One merely repeats step by step the proof of (I 3.2.1). First show the inequality

$$\dim .\mathrm{q.inj}_{\bar{x}}(F) \leq 2\dim(\mathcal{O}_{X,x}) + \dim .\mathrm{inj}(F_{\bar{x}}) = N,$$

that is, the relation

$$\underline{\mathrm{Ext}}^i(E,F)_{\bar{x}} = 0 \quad \text{for } i > N \text{ and every constructible } A_X\text{-Module } E. \tag{*}$$

Since the question is local, by dévissage (1.3) one may restrict to checking $(*)$ for $E = i_!(M)$, where $i : Y \to X$ is an integral regular sub-prescheme closed in an open $D(f)$ (with $f \in \Gamma(X, \mathcal{O}_X)$), of codimension d at every point of $D(f)$, and M is a locally constant constructible A_Y-Module ($A_Y = i^* A_X$).

Now there is the duality isomorphism (SGAA XVIII)

$$\mathrm{R}\underline{\mathrm{Hom}}(i_!(M), F) \simeq \mathrm{R}i_*\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F).$$

By the purity theorem, there is also the isomorphism (4.3)

$$i^*(F) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n^{\otimes d})_Y[-2d] \xrightarrow{\sim} \mathrm{R}i^!(F).$$

Let $\bar{y}$ be a geometric point of Y. Since M is locally constant constructible, one has canonically

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)_{\bar{y}} \simeq \mathrm{R}\,\mathrm{Hom}(M_{\bar{y}}, (\mathrm{R}i^! F)_{\bar{y}}),$$

hence, by (4.3),

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)_{\bar{y}} \simeq \mathrm{R}\,\mathrm{Hom}(M_{\bar{y}}, F_{\bar{y}})[-2d].$$

After replacing X by a connected neighbourhood of x, one has $F_{\bar{x}} \simeq F_{\bar{y}}$ and $A_{\bar{x}} \simeq A_{\bar{y}}$, since A_X and the $\mathrm{H}^i(F)$ are locally constant. Thus

$$\mathrm{Ext}^i(M_{\bar{y}}, F_{\bar{y}}) = 0 \quad \text{for } i > \dim .\mathrm{inj}(F_{\bar{x}}),$$

and hence

$$\underline{\mathrm{Ext}}^q(M, \mathrm{R}i^! F) = 0 \quad \text{for } q > 2d + \dim .\mathrm{inj}(F_{\bar{x}}).$$

Set $G = \mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^! F)$ and compute $\mathrm{R}i_*(G)_{\bar{x}}$. Introduce, as in (I 3.2.1), the schematic closure Z of Y in X, giving a cartesian diagram of immersions

$$\begin{tikzcd} Y \arrow[r] \arrow[d, "j"'] & D(f) \arrow[d] \\ Z \arrow[r, "k"'] & X. \end{tikzcd}$$

Then $\mathrm{R}i_*(G) \simeq k_*\mathrm{R}j_*(G)$. Assume of course that $x \in Z$, since otherwise $\mathrm{R}i_*(G)_{\bar{x}} = 0$. Let $\bar{Z} = \mathrm{Spec}(\mathcal{O}_{Z,\bar{x}})$ be the strict localization of Z at $\bar{x}$, let $\bar{f}$ be the image of f in $\mathcal{O}_{Z,\bar{x}}$, let $\bar{Y} = \bar{Z} \times_Z Y = \bar{Z} \cap D(f)$, and let $\bar{G}$ be the inverse image of G on $\bar{Y}$. By (SGAA VIII 5.2),

$$\mathrm{R}i_*(G)_{\bar{x}} = \mathrm{R}j_*(G)_{\bar{x}} \simeq \mathrm{R}\Gamma(\bar{Y}, \bar{G}).$$

Condition $(\mathrm{cdloc})_n$ therefore implies, by a spectral sequence argument, that

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > 2d + \dim .\mathrm{inj}(F_{\bar{x}}) + \dim(\mathcal{O}_{Z,x}).$$

But (cf. the proof of I 3.2.1)

$$\dim(\mathcal{O}_{Z,x}) = \dim(\mathcal{O}_{X,x}) - d \geq 0,$$

so

$$R^p i_*(G)_{\bar{x}} = 0 \quad \text{for } p > d + \dim(\mathcal{O}_{X,x}) + \dim .\mathrm{inj}(F_{\bar{x}}),$$

and a fortiori for $p > 2\dim(\mathcal{O}_{X,x}) + \dim .\mathrm{inj}(F_{\bar{x}})$. This proves $(*)$.

It remains to prove that the bound $(*)$ is the best possible. Put $f = \dim .\mathrm{inj}(F_{\bar{x}})$, and choose a finite type $A_{\bar{x}}$-Module M' such that

$$\mathrm{Ext}^f(M', F_{\bar{x}}) \neq 0.$$

Let Z be the integral closed sub-prescheme of X which is the closure of x, and let $Y = U \cap Z$ be a regular open subset of Z containing x and of codimension $d = \dim(\mathcal{O}_{X,x})$ at every point. After possibly shrinking Y, there exists a locally constant constructible A_Y-Module M such that $M_{\bar{x}} = M'$. Applying the purity theorem, one finds

$$\underline{\mathrm{Ext}}^{2d+f}(M, F|U)_{\bar{x}} \simeq \mathrm{Ext}^f(M'_{\bar{x}}, F_{\bar{x}}) \neq 0,$$

which completes the proof.

If F is an A_X-Module such that $F_{\bar{x}}$ is injective, then in the last part of the proof one may take $M = A_Y$; hence:

Corollary 5.1.1. Under the hypotheses of (5.1), suppose that F is a locally constant A_X-Module such that $F_{\bar{x}}$ is injective. Then

$$\dim .\mathrm{q.inj}_{\bar{x}}(F) = \dim .\mathrm{top}_{\bar{x}}(F) = 2\dim(\mathcal{O}_{X,x}),$$

with the notation of no. 2.

Remarks 5.1.2. a) With the notation of (5.1), if $F_{\bar{x}} = 0$, then $F = 0$ in a neighbourhood of x, hence $\mathrm{R}\underline{\mathrm{Hom}}(E, F)_{\bar{x}} = 0$ for all $E \in \mathrm{ob}\, \mathrm{D}^-(X)$.

b) If $\dim .\mathrm{inj}(F_{\bar{x}}) = +\infty$, then $\dim .\mathrm{q.inj}_{\bar{x}}(F) = +\infty$; indeed, for every p there exists a finite type $A_{\bar{x}}$-module M' such that $\mathrm{Ext}^p(M', F_{\bar{x}}) \neq 0$, hence a constructible A_X-Module M, locally constant in a neighbourhood of x, such that $\underline{\mathrm{Ext}}^p(M, F)_{\bar{x}} \neq 0$.

6. Constructibility of $\mathrm{R}\underline{\mathrm{Hom}}$

The equivalence of conditions a) of (I 3.5.1) remains valid when the Ring $(\mathbb{Z}/n\mathbb{Z})_X$ is replaced by a commutative Ring, locally constant, whose geometric fibres are noetherian rings: nothing has to be changed in the proof.

The preceding conditions are verified when one has resolution of singularities and purity in the strong sense (4.4), in particular if X has dimension ≤ 1, or if X is excellent of characteristic zero, or locally of finite type over a field and of dimension ≤ 2.

7. Existence of dualizing complexes

In this number, X denotes a prescheme equipped with a commutative sheaf of rings A_X, locally constant, whose geometric fibres are noetherian rings.

Lemma 7.1. If $F \in \operatorname{ob} \mathrm{D}^-_{\mathrm{coh}}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$, then $\mathrm{R}\underline{\mathrm{Hom}}(F,G) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ (notation of (1.2)).

Proof. This is standard. One reduces, by a spectral sequence, to F and G concentrated in degree zero, and then argues as in (EGA 0_{III} 12.3.3).

Lemma 7.2. Let $g : X' \to X$ be a morphism of preschemes, and endow X' with the sheaf of rings $A_{X'} = g^*(A_X)$. Then, for $F \in \operatorname{ob} \mathrm{D}^-_{\mathrm{coh}}(X)$ and $G \in \operatorname{ob} \mathrm{D}^+(X)$, the canonical map

$$g^*\mathrm{R}\underline{\mathrm{Hom}}(F,G) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(g^*F, g^*G)$$

is an isomorphism.

Proof. This too is standard: one reduces by dévissage to $F = A_X$ (cf. EGA 0_{III} 12.3.5).

Remark 7.2.1. The preceding lemmas are special cases of more general statements appearing in (SGA 6 I).

Lemma 7.3. Suppose X is quasi-compact, and let $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ be such that, for every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ is a dualizing complex in the sense of [H] V 2.1. Then, for every $F \in \operatorname{ob} \mathrm{D}^b_{\mathrm{coh}}(X)$, one has $\mathrm{R}\underline{\mathrm{Hom}}(F,K) \in \operatorname{ob} \mathrm{D}^b_{\mathrm{coh}}(X)$, and F is reflexive relative to K.

Proof. It suffices, of course, to check this for F concentrated in degree zero. By (7.1), $\mathrm{R}\underline{\mathrm{Hom}}(F,K) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$. To prove boundedness, let $\bar{x}$ be a geometric point of X. By (7.2),

$$\mathrm{R}\underline{\mathrm{Hom}}(F,K)_{\bar{x}} \simeq \mathrm{R}\operatorname{Hom}(F_{\bar{x}}, K_{\bar{x}}).$$

Putting $\dim.\mathrm{inj}(K_{\bar{x}}) = N(\bar{x})$, one obtains

$$\mathrm{Ext}^i(F,K)_{\bar{x}} = 0 \quad \text{for } i > N(\bar{x}).$$

Since the sheaves $\underline{\mathrm{Ext}}^i(F,K)$ are locally constant, there is an open neighbourhood U of x such that $\underline{\mathrm{Ext}}^i(F,K)|U = 0$ for $i > N(\bar{x})$, and quasi-compactness of X gives the assertion. It remains to check that the canonical arrow

$$F \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F,K),K)$$

is an isomorphism. It is enough to evaluate at a geometric point $\bar{x}$ of X. In the commutative diagram

$$\begin{tikzcd}
F_{\bar{x}} \arrow[r] \arrow[d, "\mathrm{Id}"'] & \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(F,K),K)_{\bar{x}} \arrow[d] \\
F_{\bar{x}} \arrow[r] & \mathrm{R}\operatorname{Hom}(\mathrm{R}\operatorname{Hom}(F_{\bar{x}},K_{\bar{x}}),K_{\bar{x}}),
\end{tikzcd}$$

the right vertical arrow is an isomorphism by (7.2), and the bottom horizontal arrow is an isomorphism because $K_{\bar{x}}$ is dualizing; hence the top horizontal arrow is an isomorphism, q.e.d.

Lemma 7.4. Suppose X is a noetherian regular prescheme, A_X is killed by an integer $n > 0$ prime to the residual characteristics of X, and the purity theorem in the strong sense (4.4) is true on X.

Let D be a divisor with normal crossings on X (I 3.1.5 a)), and denote by $j : Y = V(D) \to X$ and $i : X - Y = U \to X$ the canonical inclusions. Let $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ be such that, for every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ is a dualizing complex for the ring $A_{\bar{x}}$ (in the sense of [H] V 2.1). Then, for every $E \in \operatorname{ob} \mathrm{D}^b_{\mathrm{coh}}(X)$, one has $\mathrm{R}\underline{\mathrm{Hom}}(i_!i^*(E),K) \in \operatorname{ob} \mathrm{D}^b_c(X)$, and the pair $(i_!i^*(E),K)$ is bidualizing.

Proof. We may suppose E is concentrated in degree zero, i.e. a locally constant constructible A_X-Module. Since the question is local on X for the étale topology, we may suppose that (a) A_X is constant with value A, and E is constant with value E; and (b) D has strict normal crossings, say $D = \sum_{i \in I} \operatorname{div}(s_i)$ with $s_i \in \Gamma(X, \mathcal{O}_X)$.

Argue by induction on $p = \operatorname{card}(I)$. For $p = 0$, the assertions follow from (7.3). Suppose $I = \{1, \ldots, p\}$, put $Y_i = V(s_i)$, $Y' = V(\sum_{1 \le i \le p-1} \operatorname{div}(s_i))$, and $U' = X - Y'$. We get a cartesian diagram of immersions

$$\begin{tikzcd}
\varnothing \arrow[r] \arrow[d] & U' \cap Y_p \arrow[r, "i'_p"] \arrow[d, "j'_p"'] & Y_p \arrow[d, "j_p"] \\
U \arrow[r] & U' \arrow[r, "i'"'] & X.
\end{tikzcd}$$

Consider the exact sequence

$$0 \longrightarrow E_{U,X} \longrightarrow E_{U',X} \longrightarrow E_{U' \cap Y_p, X} \longrightarrow 0.$$

By the induction hypothesis the theorem is true for $(E_{U',X}, K)$. For fixed K, the F such that $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \operatorname{ob} \mathrm{D}^b_c(X)$ and (F, K) is bidualizing form a triangulated subcategory of $\mathrm{D}_c(X)$; hence we are reduced to the case $F = E_{U' \cap Y_p, X}$. But

$$E_{U' \cap Y_p, X} = j_{p*} i'_{p!}(E_{U' \cap Y_p}),$$

and, by (I 1.13), it is equivalent to prove the theorem for $(i'_{p!}(E_{U' \cap Y_p}), \mathrm{R}j^!_p(K))$ on Y_p. By purity, however,

$$\mathrm{R}j^!_p(K) \simeq j^*_p(K) \otimes^L_{\mathbb{Z}/n\mathbb{Z}} (\mu_n)^{\otimes -1}_{Y_p}[-2].$$

Thus $\mathrm{R}j^!_p(K) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(Y_p)$ and, for every geometric point $\bar{y}$ of Y_p, $(\mathrm{R}j^!_p(K))_{\bar{y}}$ is dualizing. The induction hypothesis then applies.

Theorem 7.5. Assume the following: X is a noetherian regular prescheme; A_X is killed by an integer $n > 0$ prime to the residual characteristics of X; X is strongly desingularizable (I 3.1.5), satisfies conditions (reg) (I 3.1.1) and $(\mathrm{cdloc})_n$ (I 3.1.3); and the purity theorem in the strong sense (4.4) is true on every prescheme smooth over X (conditions satisfied if X is excellent of characteristic zero, or of dimension ≤ 2 and smooth over a perfect field).

Then, if $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$ is such that $K_{\bar{x}}$ is dualizing (in the sense of [H] V 2.1) for every geometric point $\bar{x}$ of X, the complex K is dualizing.

Proof. We must show that K satisfies conditions (i) to (iii) of (I 1.7). For every geometric point $\bar{x}$ of X, $K_{\bar{x}}$ has finite injective dimension, and since $K \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(X)$, the function $x \mapsto \dim.\mathrm{inj}(K_{\bar{x}})$ is locally constant; it is therefore bounded because X is noetherian. It follows from (5.1) that K has finite quasi-injective dimension.

By what was seen in no. 6, condition (ii) of loc. cit. is also satisfied.

It remains to prove condition (iii), i.e. that every $F \in \operatorname{ob} \mathrm{D}^b_c(X)$ is reflexive relative to K. By dévissage we may suppose F is concentrated in degree zero. Since the question is local on X for the étale topology, we may suppose A_X constant with value A. Dévissage (1.3) then reduces us to $F = f_* i_!(E_U)$, where $f : Y \to X$ is finite, Y is integral with generic point separable over its image, $i : U \to Y$ is a nonempty regular open, and E_U is an A_U-Module constant with value a finite type A-module E. Since X is strongly desingularizable, there exists (I 3.1.5) a projective birational morphism $h : Y' \to Y$ such that Y' is regular, h induces an isomorphism $h^{-1}(U) \to U$, and $h^{-1}(U)$

is the complement of a normal crossings divisor in Y'. Identifying $h^{-1}(U)$ with U via h, we have a commutative diagram

$$\begin{tikzcd} & Y' \arrow[dl, "i'"'] \arrow[d, "h"] \arrow[dr, "g"] & \\ U \arrow[r, "i"'] & Y \arrow[r, "f"'] & X. \end{tikzcd}$$

Consequently,

$$f_* i_!(E_U) \simeq f_* \mathrm{R}h_* i'_!(E_U) \simeq \mathrm{R}g_* i'_!(E_U) \simeq \mathrm{R}g_* i'_! i'^*(E_{Y'}).$$

By (I 1.13), it is enough to prove biduality for the pair $(i'_! i'^*(E_{Y'}), \mathrm{R}g^!(K))$. But, locally embedding Y' into a prescheme X' smooth over X and applying strong purity on X' (4.4), one finds that $\mathrm{R}g^!(K)$ is locally isomorphic, up to degree shift, to $g^*(K)$. Hence $\mathrm{R}g^!(K) \in \operatorname{ob} \mathrm{D}^+_{\mathrm{coh}}(Y')$ and $\mathrm{R}g^!(K)_{\bar{x}}$ is dualizing for every geometric point $\bar{x}$ of Y'. It remains only to apply (7.4).

The preceding proof also shows that $\mathrm{R}\underline{\mathrm{Hom}}(F, K) \in \operatorname{ob} \mathrm{D}^b_c(X)$ (using the finiteness theorem for a proper morphism (SGAA XIV)), which will reassure the reader who did not trust the spiel of no. 6.

This completes the proof.

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Bibliography

[1] Verdier, J.-L., *Dimension des espaces localement compacts*, Note aux C.R. Ac. Sc. Paris, 261, pp. 5293–5296 (1965). See also: Verdier, J.-L., *Dualité dans la cohomologie des espaces localement compacts*, Séminaire Bourbaki, no. 300, 1965–66.

SGA A = SGA 4.

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SGA 5 — Exposé III
LEFSCHETZ FORMULA

by A. Grothendieck, written up by L. Illusie

In this exposé we prove the Lefschetz formula in étale cohomology, for correspondences between complexes of sheaves, in Verdier's formulation (cf. [18]). In an earlier version, distributed by the IHES about ten years ago, the formula was stated, under resolution hypotheses, but the required compatibilities were not proved. Deligne's finiteness theorems (SGA $4^{1/2}$, Finiteness) made it possible to dispense with these hypotheses, provided one works with schemes separated and of finite type over a field. As for the compatibilities, the editor has verified them (painfully).

Let us say at once that the Lefschetz formula in question here, just like the trace formula for constant coefficients (SGA $4^{1/2}$, Cycle), which it generalizes, is essentially trivial: it only translates certain compatibilities in the formalism of duality. Roughly speaking, it says that, under certain conditions, the trace of an endomorphism of a hypercohomology complex $R\Gamma(X, L)$, where L is a complex of sheaves on X, can be calculated as an "integral" of certain cohomology classes on $X \times X$. In practice, the endomorphisms one considers are associated with correspondences $C \subset X \times X$, and the classes recovered on the product have support in $C \cap \Delta$, where Δ is the diagonal; these classes are "local in nature" in a neighbourhood of the fixed points, which in principle facilitates their calculation. See 4.2 and 4.7 for precise statements. In fact, the calculation of these "local" classes in terms of more comprehensible invariants, without which the present formula would have no interest, has so far been undertaken only in very special situations (see 4.10, 4.11, and the following exposé, where in particular the case of the Frobenius correspondence on a curve is treated).

For lack of a good derived category of ℓ-adic sheaves (Jouanolou's thesis unfortunately not having been published), we work systematically with torsion coefficients (prime to the residual characteristics). Despite minor technical difficulties in the definition of traces, this point of view in fact leads to finer formulae, also more manageable insofar as, for the calculation of local terms, one wants to trivialize the sheaves involved by coverings (see notably (SGA $4^{1/2}$, Report), III B, and XII).

In no. 1, we fix the notation and recall various Künneth formulae from (SGA $4^{1/2}$, Finiteness), which play an essential role in what follows. We also introduce a certain number of fibred and cofibred categories, whose consideration later facilitates certain verifications of compatibilities. No. 2, which is very technical, may be skipped on first reading. Its main result (2.5) expresses a compatibility between external tensor product of RHom, direct image, and base change. In no. 3, we define cohomological correspondences between complexes, and show how they act on hypercohomology; the result quoted above makes it possible to describe this action in several equivalent ways. The core of the exposé is no. 4, where Verdier's pairing between cohomological correspondences is defined and the Lefschetz formula is proved, in a relative situation appreciably more general than the one usually considered, with a view to later applications to computations of local terms. We give the most evident corollaries for correspondences with constant coefficients. In no. 5, we indicate rapidly how to generalize to correspondences between complexes certain usual constructions on correspondences with constant coefficients (cf. [12]), such as passage to the dual, composition, and external tensor product. Most of the verifications, analogous to those in the preceding numbers, have been omitted. Finally, in the appendix, we paraphrase the preceding theory in the context of coherent sheaves, using the formalism of duality of [8]. Here one may work over any noetherian base S, provided that, when necessary, one imposes hypotheses of properness, finite tor-dimension, and tor-independence on the S-schemes considered. We show that the very general Lefschetz formula thereby obtained contains as a special case the Woods Hole formula [1], and we point out some applications.

Contents

2 Notation and recalls of Künneth formulae

1.1

We denote by S the spectrum of a field. Unless otherwise stated, the S-schemes considered in what follows will be S-schemes separated and of finite type.

We denote by A a commutative noetherian ring, annihilated by an integer invertible on S. If X is an S-scheme, we denote by $D(X) = D(X, A)$ the derived category of complexes of A-modules on X (for the étale topology), and by $D_{\mathrm{ctf}}(X)$ the full subcategory consisting of complexes of finite tor-dimension, with constructible cohomology. The category $D_{\mathrm{ctf}}(X)$ is a triangulated subcategory of $D(X)$. If $E, F \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, it follows readily from the definitions that $E \otimes^L F \in \operatorname{ob} D_{\mathrm{ctf}}(X)$. According to (SGA $4^{1/2}$, Finiteness Theorem 1.6, 1.7), one also has $\mathrm{RHom}(E, F) \in \operatorname{ob} D_{\mathrm{ctf}}(X)$.

1.2

Let $f : X \to Y$ be an S-morphism. One has trivially

$$f^* D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

and, by (loc. cit.),

$$Rf_* D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \qquad Rf_! D_{\mathrm{ctf}}(X) \subset D_{\mathrm{ctf}}(Y), \qquad Rf^! D_{\mathrm{ctf}}(Y) \subset D_{\mathrm{ctf}}(X),$$

(the inclusion relative to $Rf_!$ uses only the finiteness theorem for proper morphisms (SGA 4 XIV) and the sorites (SGA 4 XVII 5.2.11), and hence holds more generally for every compactifiable morphism whose fibres have bounded dimension). In what follows, we shall sometimes write f_*, $f_!$, $f^!$ instead of Rf_*, $Rf_!$, $Rf^!$; the hypotheses of 1.1 imply that these functors are defined on the whole derived category.

1.3

For $f : X \to S$, we shall put

$$K_X = f^! A,$$

and shall denote by D_X (or D) the functor $\mathrm{RHom}(-, K_X)$. According to (SGA $4^{1/2}$, Finiteness Theorem 4.3), the complex K_X is dualizing: for $E \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, one has canonically

$$E \xrightarrow{\sim} DDE.$$

1.4

Let $f : X \to Y$ be an S-morphism, $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)$. We put

$$\operatorname{Hom}^{(1)}_f(L, M) = \operatorname{Hom}(f^*M, L) \quad (\simeq \operatorname{Hom}(M, f_*L)),$$

$$\operatorname{Hom}^{(2)}_f(L, M) = \operatorname{Hom}(L, f^*M),$$

$$\operatorname{Hom}^{(3)}_f(L, M) = \operatorname{Hom}(f_!L, M) \quad (\simeq \operatorname{Hom}(L, f^!M)),$$

$$\operatorname{Hom}^{(4)}_f(L, M) = \operatorname{Hom}(f^!M, L).$$

The elements of $\operatorname{Hom}^{(i)}_f(L, M)$ will be called *arrows of type i from L to M over f* (or f-arrows of type i from L to M). The transitivity isomorphisms for the functors f^*, f_*, $f_!$, $f^!$ allow one to compose arrows of type i for fixed i, and one thus obtains certain fibred or cofibred categories over the category of S-schemes: for example, for $i = 1$ (resp. $i = 3$) one obtains the fibred and cofibred category considered in (SGA 4 XVII 4.1.3) (resp. (SGA 4 XVIII 3.1.13)), whose fibre over X is the category opposite to $D(X)$ (resp. $D(X)$).

1.5

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, and $L_i \in \operatorname{ob} D^-(X_i)$. We put

$$L_1 \otimes^L_S L_2 = \operatorname{pr}_1^* L_1 \otimes^L \operatorname{pr}_2^* L_2,$$

where $\operatorname{pr}_i : X \to X_i$ is the canonical projection.

1.6

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \to Y = Y_1 \times_S Y_2,$$

$L_i \in \operatorname{ob} D^-(X_i)$, $M_i \in \operatorname{ob} D^-(Y_i)$. The evident canonical isomorphism

$$f_1^*M_1 \otimes^L_S f_2^*M_2 \xrightarrow{\sim} f^*(M_1 \otimes^L_S M_2) \tag{1.6.1}$$

allows one to define, for $u_i \in \operatorname{Hom}^{(1)}_{f_i}(L_i, M_i)$ (resp. $\operatorname{Hom}^{(2)}_{f_i}(L_i, M_i)$),

$$u_1 \otimes^L_S u_2 \in \operatorname{Hom}^{(1)}_f(L, M) \quad (\text{resp. } \operatorname{Hom}^{(2)}_f(L, M)), \tag{1.6.2}$$

where $L = L_1 \otimes^L_S L_2$, $M = M_1 \otimes^L_S M_2$. If u_i, v_i are composable, one has

$$(v_1u_1) \otimes^L_S (v_2u_2) = (v_1 \otimes^L_S v_2)(u_1 \otimes^L_S u_2). \tag{1.6.3}$$

Taking $M_i = f_{i*}L_i$, and u_i of type 1, given by the identity of $f_{i*}L_i$, 1.6.2 gives a *Künneth arrow* (cf. SGA 4 XVII 5.4.1.4)

$$f_{1*}L_1 \otimes^L_S f_{2*}L_2 \longrightarrow f_*(L_1 \otimes^L_S L_2). \tag{1.6.4}$$

According to (SGA $4^{1/2}$, Finiteness Theorem 1.9), 1.6.4 is an isomorphism: by 1.6.3 one is indeed reduced to the case where $X_2 = Y_2$, $f_2 = 1$, and one applies (loc. cit. App.). When f_1 and f_2 are proper, it is not necessary to invoke (loc. cit.): 1.6.4 is a special case of the Künneth isomorphism (SGA 4 XVII 5.4.3).

1.7

Choose a compactification $f_i = p_i j_i$, where j_i is an open immersion and p_i is a proper morphism. Composing the evident canonical isomorphism (where $j = j_1 \times_S j_2$)

$$j_{1!}L_1 \otimes_S^L j_{2!}L_2 \xrightarrow{\sim} j_!(L_1 \otimes_S^L L_2)$$

with the isomorphism 1.6.4 relative to $(p_i, j_{i!}L_i)$, one obtains a Künneth isomorphism (SGA 4 XVII 5.4.3)

$$f_{1!}L_1 \otimes_S^L f_{2!}L_2 \xrightarrow{\sim} f_!(L_1 \otimes_S^L L_2). \tag{1.7.1}$$

One checks easily (loc. cit.) that 1.7.1 does not depend on the compactifications chosen, and is compatible with the transitivity isomorphisms for composites $g_i f_i$ (cf. (loc. cit. 5.2.4)). Thanks to 1.7.1, one can define, for $u_i \in \mathrm{Hom}_{f_i}^{(3)}(L_i, M_i)$,

$$u_1 \otimes_S^L u_2 \in \mathrm{Hom}_f^{(3)}(L, M) \tag{1.7.2}$$

with L and M as in 1.6.2, and one has a transitivity formula analogous to 1.6.3.

For $L_i = f_i^! M_i$, and u_i given by the identity of L_i, 1.7.2 gives a Künneth arrow

$$f_1^! M_1 \otimes_S^L f_2^! M_2 \longrightarrow f^!(M_1 \otimes_S^L M_2). \tag{1.7.3}$$

Proposition 1.7.4

The arrow 1.7.3 is an isomorphism.

By the transitivity formula, it suffices to treat the case where $X_2 = Y_2$, $f_2 = 1$. By dévissage, one reduces to $M_2 = g_* j_! A_V$, where $j : V \to U$ is an open immersion and $g : U \to Y_2$ a finite morphism, and then, by (SGA 4 XVIII 3.1.12.3), to the case where $M_2 = A_{Y_2}$. By localization on X_2, one may suppose moreover that $f_1 = g_1 j_1$, where j_1 is a closed immersion and g_1 a smooth morphism purely of relative dimension n_1. By transitivity, it suffices to treat the cases

$$\text{a) } f_1 = g_1, \qquad \text{b) } f_1 = j_1.$$

In case a), the trace morphism supplies by adjunction (SGA 4 XVIII 3.2.5) an isomorphism

$$f_1^* M_1(n_1)[2n_1] \xrightarrow{\sim} f_1^! M_1;$$

this is compatible with the base change $Y_2 \to S$, because the trace morphism is so (SGA 4 XVIII 2.9), whence 1.7.4 in this case. In case b), if i_1 denotes the inclusion of the complementary open, one has an exact sequence

$$0 \longrightarrow j_{1*} R j_1^! M_1 \longrightarrow M_1 \longrightarrow R i_{1*} i_1^* M_1 \longrightarrow 0,$$

and the compatibility of $R i_{1*}$ with base change, a special case of 1.6.4, gives the conclusion.

Thanks to 1.7.3, one can define, for $u_i \in \mathrm{Hom}_{f_i}^{(4)}(L_i, M_i)$,

$$u_1 \otimes_S^L u_2 \in \mathrm{Hom}_f^{(4)}(L, M) \tag{1.7.5}$$

(L and M as in 1.6.2), with a transitivity formula analogous to 1.6.3.

On the other hand, with the notation of 1.3, 1.7.3 gives an isomorphism

$$K_{X_1} \otimes_S^L K_{X_2} \xrightarrow{\sim} K_X. \tag{1.7.6}$$

1.8

Let $f : X \to Y$ be an S-morphism, $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)$, S' an S-scheme, $P \in \operatorname{ob} D(S')$, and $f' : X' \to Y'$ the morphism deduced from f by the base change $S' \to S$. The identity arrow of P may be regarded indifferently as an arrow of type i ($1 \le i \le 4$) over the identity arrow of S'; for $u \in \operatorname{Hom}_f^{(i)}(L, M)$ ($1 \le i \le 4$), one may therefore, by 1.6.2, 1.7.2, 1.7.5, consider

$$u \otimes_S^L \operatorname{Id}_P \in \operatorname{Hom}_{f'}^{(i)}(L \otimes_S^L P, M \otimes_S^L P). \tag{1.8.1}$$

We shall sometimes write $u \otimes_S^L P$ instead of $u \otimes_S^L \operatorname{Id}_P$.

3 Functoriality of RHom and external tensor products

2.1

Let $f : X \to Y$ be an S-morphism, $P \in \operatorname{ob} D(Y)$, $Q \in \operatorname{ob} D^+(Y)$. The evident canonical arrow at the level of complexes

$$f^*\mathcal{H}om^\bullet(P, Q) \longrightarrow \mathcal{H}om^\bullet(f^*P, f^*Q)$$

gives by derivation a canonical arrow

$$f^*\mathrm{RHom}(P, Q) \longrightarrow \mathrm{RHom}(f^*P, f^*Q). \tag{2.1.1}$$

Let $L \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D^+(X)$. Put

$$\operatorname{Hom}_f^{(i,j)}((L, M), (P, Q)) = \operatorname{Hom}_f^{(i)}(L, P) \times \operatorname{Hom}_f^{(j)}(M, Q), \tag{2.1.2}$$

with the notation of 1.4. For $(u, v) \in \operatorname{Hom}_f^{(2,1)}((L, M), (P, Q))$ one defines

$$\mathrm{RHom}(u, v) \in \operatorname{Hom}_f^{(1)}(\mathrm{RHom}(L, M), \mathrm{RHom}(P, Q)) \tag{2.1.3}$$

by composing 2.1.1 with the arrow $\mathrm{RHom}(f^*P, f^*Q) \to \mathrm{RHom}(L, M)$ defined by u and v. For $L = f^*P$, $Q = f_*M$, u (resp. v) given by the identity of L (resp. Q), $\mathrm{RHom}(u, v)$ is the "trivial duality" isomorphism

$$\mathrm{RHom}(P, f_*M) \xrightarrow{\sim} f_*\mathrm{RHom}(f^*P, M). \tag{2.1.4}$$

For $(u, v) \in \operatorname{Hom}_f^{(3,4)}((L, M), (P, Q))$, one defines

$$\mathrm{RHom}(u, v) \in \operatorname{Hom}_f^{(1)}(\mathrm{RHom}(L, M), \mathrm{RHom}(P, Q)) \tag{2.1.5}$$

as the composite of the arrows

$$\mathrm{RHom}(P, Q) \xrightarrow{a} \mathrm{RHom}(f_!L, Q) \xrightarrow{b} f_*\mathrm{RHom}(L, f^!Q) \xrightarrow{c} f_*\mathrm{RHom}(L, M),$$

where a (resp. c) is defined by u (resp. v), and b is the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6).

It follows easily from the definitions that the formation of $\mathrm{RHom}(u, v)$ is compatible with composition: for composable (u, v), (u', v') of type $(2, 1)$ (resp. $(3, 4)$), one has

$$\mathrm{RHom}(u'u, v'v) = \mathrm{RHom}(u', v')\,\mathrm{RHom}(u, v). \tag{2.1.6}$$

2.2

Let X be an S-scheme. Recall that for $L \in \operatorname{ob} D^-(X)$, $M \in \operatorname{ob} D(X)$, $N \in \operatorname{ob} D^+(X)$, one has the adjunction isomorphism ("dear to Cartan," as Grothendieck says)

$$\operatorname{Hom}(L \otimes^L M, N) = \operatorname{Hom}(L, \operatorname{RHom}(M, N)). \tag{2.2.1}$$

For $E \in \operatorname{ob} D^-(X)$, $F \in \operatorname{ob} D^+(X)$, the identity arrow of $\operatorname{RHom}(E, F)$ defines via 2.2.1 an arrow

$$E \otimes^L \operatorname{RHom}(E, F) \longrightarrow F, \tag{2.2.2}$$

called the *evaluation arrow*: it is derived from the arrow at the level of complexes

$$\mathcal{H}om^\bullet(E, F) \otimes E \longrightarrow F, \qquad f \otimes x \longmapsto f(x).$$

Let, for $i = 1, 2$, $E_i, F_i \in \operatorname{ob} D_{\mathrm{ctf}}(X)$. By adjunction (2.2.1), the arrow

$$E_1 \otimes^L E_2 \otimes \operatorname{RHom}(E_1, F_1) \otimes \operatorname{RHom}(E_2, F_2) \longrightarrow F_1 \otimes^L F_2,$$

composed of a symmetry isomorphism and of the tensor product of the evaluation arrows relative to (E_1, F_1) and (E_2, F_2), supplies an arrow

$$\operatorname{RHom}(E_1, F_1) \otimes \operatorname{RHom}(E_2, F_2) \longrightarrow \operatorname{RHom}(E_1 \otimes^L E_2, F_1 \otimes^L F_2). \tag{2.2.3}$$

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the canonical projection, $E_i, F_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, $E = E_1 \otimes^L_S E_2$, $F = F_1 \otimes^L_S F_2$. One has a canonical arrow

$$\operatorname{RHom}(E_1, F_1) \otimes^L_S \operatorname{RHom}(E_2, F_2) \longrightarrow \operatorname{RHom}(E, F), \tag{2.2.4}$$

obtained by composing the tensor product of the arrows $p_i^* \operatorname{RHom}(E_i, F_i) \to \operatorname{RHom}(p_i^* E_i, p_i^* F_i)$ and 2.2.3 for $(p_i^* E_i, p_i^* F_i)$.

Proposition 2.3

The arrow 2.2.4 is an isomorphism.

By localization and dévissage, one reduces to $E_i = f_{i!} A_{U_i}$ for $f_i : U_i \to X_i$ étale. One has canonical isomorphisms (duality)

$$\operatorname{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} f_{i*} f_i^* F_i, \qquad \operatorname{RHom}(E, F) \xrightarrow{\sim} f_* f^* F \quad (f = f_1 \times_S f_2). \tag{1}$$

Taking into account the Künneth formula 1.6.4, it suffices, to conclude, to verify that these isomorphisms make the square

$$\begin{tikzcd} \operatorname{RHom}(f_{1!} A_{U_1}, F_1) \otimes^L_S \operatorname{RHom}(f_{2!} A_{U_2}, F_2) \arrow[r, "2.2.3"] \arrow[d, "\sim"'] & \operatorname{RHom}(f_! A_U, F) \arrow[d, "\sim"] \\ f_{1*} f_1^* F_1 \otimes^L_S f_{2*} f_2^* F_2 \arrow[r, "1.6.4"'] & f_* f^* F \end{tikzcd}$$

commute, where $U = U_1 \times_S U_2$. Since f_i is étale, the isomorphism (1) relative to f_i (resp. f) is adjoint to the composite arrow

$$f_i^* \operatorname{RHom}(f_{i!} A_{U_i}, F_i) \xrightarrow{\sim} \operatorname{RHom}(f_i^* f_{i!} A_{U_i}, f_i^* F_i) \longrightarrow f_i^* F_i, \tag{3}$$

where the second arrow is deduced from the adjunction $A_{U_i} \to f_i^* f_{i!} A_{U_i}$ (resp. the analogous composite arrow with f). The commutativity of (2) is equivalent to that of the square

$$\begin{tikzcd}
f_1^* \mathrm{RHom}(f_{1!}A_{U_1}, F_1) \otimes_S^L f_2^* \mathrm{RHom}(f_{2!}A_{U_2}, F_2) \arrow[r] \arrow[d] & f^* \mathrm{RHom}(f_! A_U, F) \arrow[d] \\
f_1^* F_1 \otimes_S^L f_2^* F_2 \arrow[r, "\sim"'] & f^* F,
\end{tikzcd}$$

where the vertical arrows are given by (3). But the commutativity of this square follows from the fact that 2.2.3 is functorial in E_i and reduces to the identity for $E_i = A$, whence the proposition.

2.4

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X \to Y,$$

$E_i, F_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$, $P_i, Q_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y_i)$,

$$E = E_1 \otimes_S^L E_2, \quad F = F_1 \otimes_S^L F_2, \quad P = P_1 \otimes_S^L P_2, \quad Q = Q_1 \otimes_S^L Q_2.$$

Consider the diagram

$$\begin{tikzcd}
(E_1, F_1) \arrow[d, "f_1"'] & X_1 & X_1 \times_S Y_2 \arrow[l] \arrow[d] & X \arrow[l] \arrow[d] \arrow[ddl] & (E, F) \\
(P_1, Q_1) & Y_1 & Y \arrow[l] \arrow[d] & Y_1 \times_S X_2 \arrow[l, "f"] \arrow[d] & \\
S & & Y_2 \arrow[ll] & X_2 \arrow[l, "f_2"] & (E_2, F_2).
\end{tikzcd} \tag{2.4.0}$$

Give ourselves $(u_1, v_1) \in \mathrm{Hom}_{f_1}^{(3,4)}((E_1, F_1), (P_1, Q_1))$, $(u_2, v_2) \in \mathrm{Hom}_{f_2}^{(2,1)}((E_2, F_2), (P_2, Q_2))$ (notation of 2.1.2 and 1.4). By 1.8, one deduces from (u_i, v_i) a square

$$\begin{tikzcd}
(E, F) \arrow[d, "(u_1 \otimes_S^L E_2,\, v_1 \otimes_S^L F_2)"] & (E_1 \otimes_S^L P_2, F_1 \otimes_S^L Q_2) \arrow[l, "(E_1 \otimes_S^L u_2,\, F_1 \otimes_S^L v_2)"'] \arrow[d, "(u_1 \otimes_S^L P_2,\, v_1 \otimes_S^L Q_2)"] \\
(P_1 \otimes_S^L E_2, Q_1 \otimes_S^L F_2) \arrow[r, "(P_1 \otimes_S^L u_2,\, Q_1 \otimes_S^L v_2)"'] & (P, Q),
\end{tikzcd} \tag{2.4.1}$$

where the vertical arrows (resp. horizontal arrows) are of type $(3, 4)$ (resp. $(2, 1)$).

We may now state the main compatibility of this number, which will play an essential role in the verification of the Lefschetz–Verdier formula.

Proposition 2.5

The diagram 2.5.1 below, where the square on the right is deduced from 2.4.1 by applying RHom in the sense of 2.1.3 and 2.1.5, is commutative:

$$
\begin{tikzcd}
\mathrm{RHom}(E_1, F_1) \otimes^L_S \mathrm{RHom}(E_2, F_2) \arrow[r, "2.2.4"] \arrow[d] & \mathrm{RHom}(E, F) \arrow[dr] & \\
\mathrm{RHom}(u_1, v_1) \otimes^L_S \mathrm{RHom}(u_2, v_2) \arrow[d] & & \mathrm{RHom}(E_1 \otimes^L_S P_2, F_1 \otimes^L_S Q_2) \arrow[d] \\
\mathrm{RHom}(P_1, Q_1) \otimes^L_S \mathrm{RHom}(P_2, Q_2) & \mathrm{RHom}(P, Q) \arrow[l, "2.2.4"'] \arrow[r, dash] & \mathrm{RHom}(P_1 \otimes^L_S E_2, Q_1 \otimes^L_S F_2)
\end{tikzcd}
\tag{2.5.1}
$$

2.6

The proof of 2.5 will occupy the rest of no. 2. It follows from 1.6.3 that, if $w_i : L_i \to M_i$ is an arrow of type 1 over f_i, the arrow $w = w_1 \otimes^L_S w_2$ makes the diagram

$$
\begin{tikzcd}
& L_1 \otimes^L_S L_2 \arrow[dl, "L_1 \otimes^L_S w_2"'] \arrow[dr, "w_1 \otimes^L_S L_2"] & \\
L_1 \otimes^L_S M_2 \arrow[dr, "w_1 \otimes^L_S M_2"'] & & M_1 \otimes^L_S L_2 \arrow[dl, "M_1 \otimes^L_S w_2"] \\
& M_1 \otimes^L_S M_2 &
\end{tikzcd}
$$

commute. Applying this remark to $w_i = \mathrm{RHom}(u_i, v_i)$, one sees that it suffices to prove the following special cases of 2.5.

Corollary 2.6.1

Let $f : X \to Y$ be an S-morphism, S' an S-scheme, $f' : X' \to Y'$ the morphism deduced from f by the base change $S' \to S$, $E, F \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, $P, Q \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, $P', Q' \in \mathrm{ob}\, D_{\mathrm{ctf}}(S')$, and $(u, v) \in \mathrm{Hom}^{(2,1)}_f((E, F), (P, Q))$ (resp. $\mathrm{Hom}^{(3,4)}_f((E, F), (P, Q))$). Then one has a commutative square, whose vertical arrows are of type 1 over f':

$$
\begin{tikzcd}
\mathrm{RHom}(E, F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "2.2.4"] \arrow[d, "\mathrm{RHom}(u,v) \otimes^L_S \mathrm{RHom}(P',Q')"'] & \mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \arrow[d, "\mathrm{RHom}(u \otimes^L_S P', v \otimes^L_S Q')"] \\
\mathrm{RHom}(P, Q) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "2.2.4"'] & \mathrm{RHom}(P \otimes^L_S P', Q \otimes^L_S Q').
\end{tikzcd}
\tag{2.6.1.1}
$$

First examine the case where (u, v) is of type $(2, 1)$. By functoriality, one may suppose that $E = f^*P$, $F = f^*Q$, (u, v) being given by the identity of (f^*P, f^*Q). The commutativity of 2.6.1.1 in the case

$P' = Q' = A_{S'}$ is evident; to establish it in the general case, it therefore suffices to verify that, for $L, M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$, the square

$$\begin{tikzcd}
f^*\mathrm{RHom}(E,F) \otimes^L f^*\mathrm{RHom}(L,M) \arrow[r, "f^*(2.2.3)"] \arrow[d, "2.1.1\otimes 2.1.1"'] & f^*\mathrm{RHom}(E \otimes^L L, F \otimes^L M) \arrow[d, "2.1.1"] \\
\mathrm{RHom}(f^*E, f^*F) \otimes^L \mathrm{RHom}(f^*L, f^*M) \arrow[r, "2.2.3"'] & \mathrm{RHom}(f^*(E \otimes^L L), f^*(F \otimes^L M))
\end{tikzcd}$$

is commutative. But one sees immediately that the two composites of this square are adjoint to the tensor product of the evaluation arrows

$$f^*E \otimes^L f^*\mathrm{RHom}(E,F) \longrightarrow f^*F \quad \text{and} \quad f^*L \otimes^L f^*\mathrm{RHom}(L,M) \longrightarrow f^*M,$$

which proves the assertion.

Now pass to the case where (u,v) is of type $(3,4)$. By functoriality, one may suppose that $P = f_!E$, $F = f^!Q$, (u,v) being defined by $(\mathrm{Id}_P, \mathrm{Id}_F)$. One must prove the commutativity of the square

$$\begin{tikzcd}
f_*\mathrm{RHom}(E, f^!Q) \otimes_S^L \mathrm{RHom}(P',Q') \arrow[r, "b"] \arrow[d, "a\otimes\mathrm{RHom}(P',Q')"'] & f'_*\mathrm{RHom}(E \otimes_S^L P', f^!Q \otimes_S^L Q') \arrow[d, "a'"] \\
\mathrm{RHom}(f_!E, Q) \otimes_S^L \mathrm{RHom}(P',Q') \arrow[r, "2.2.4"'] & \mathrm{RHom}(f_!E \otimes_S^L P', Q \otimes_S^L Q'),
\end{tikzcd}$$

where b is the composite of $f'_*(2.2.4)$ and a base-change isomorphism, a is the duality isomorphism (SGA 4 XVIII 3.1.9.6), and a' is the composite of the duality isomorphism (relative to $E \otimes_S^L P'$ and $Q \otimes_S^L Q'$) and the base-change isomorphisms

$$f'_!(E \otimes_S^L P') \xrightarrow{\sim} f_!E \otimes_S^L P', \qquad f^!Q \otimes_S^L Q' \xrightarrow{\sim} f'^!(Q \otimes_S^L Q').$$

Recall that a is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_*\mathrm{RHom}(E, f^!Q) \longrightarrow \mathrm{RHom}(f_!E, f_!f^!Q)$$

and of the arrow defined by the adjunction arrow $f_!f^!Q \to Q$. There is an analogous description of a', as the composite of the quoted arrow

$$f'_*\mathrm{RHom}(E \otimes_S^L P', f'^!(Q \otimes_S^L Q')) \longrightarrow \mathrm{RHom}(f'_!(E \otimes_S^L P'), f'_!f'^!(Q \otimes_S^L Q')),$$

of the arrows defined by c and d, and of the arrow defined by the adjunction arrow

$$f'_!f'^!(Q \otimes_S^L Q') \longrightarrow Q \otimes_S^L Q'.$$

By definition, d is adjoint to the arrow

$$f'_!(f^!Q \otimes_S^L Q') \longrightarrow Q \otimes_S^L Q'$$

defined by the base-change isomorphism $f'_!(f^!Q \otimes_S^L Q') \simeq f_!f^!Q \otimes_S^L Q'$ and the adjunction arrow $f_!f^!Q \to Q$; in other words, the square

$$\begin{tikzcd}
f'_!(f^!Q \otimes_S^L Q') \arrow[r, "f'_!d"] \arrow[d, "\text{base ch. } \sim"'] & f'_!f'^!(Q \otimes_S^L Q') \arrow[d, "\mathrm{adj}"] \\
f_!f^!Q \otimes_S^L Q' \arrow[r, "\mathrm{adj}\otimes 1"'] & Q \otimes_S^L Q'
\end{tikzcd}$$

is commutative. It follows that one may also describe a' as the composite of the arrow above, of the arrow defined by c, and of those defined by the base-change isomorphism

$$f'_!(f^!Q \otimes^L_S Q') \xrightarrow{\sim} f_!f^!Q \otimes^L_S Q'$$

and the adjunction arrow $f_!f^!Q \to Q$. This decomposition of a' and that of a given above, together with the commutativity of the square

$$\begin{tikzcd}
\mathrm{RHom}(f_!E, f_!f^!Q) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r] \arrow[d] & \mathrm{RHom}(f_!E \otimes^L_S P', f_!f^!Q \otimes^L_S Q') \arrow[d] \\
\mathrm{RHom}(f_!E, Q) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r] & \mathrm{RHom}(f_!E \otimes^L_S P', Q \otimes^L_S Q'),
\end{tikzcd}$$

defined by 2.2.4 and the adjunction arrow $f_!f^!Q \to Q$, show that, to establish the commutativity of 2.6.2, it suffices to prove the following compatibility.

Lemma 2.6.3

Let f, f', E, F, P', Q' be as in 2.6.1. Then one has a commutative square

$$\begin{tikzcd}
f_*\mathrm{RHom}(E, F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(1)"] \arrow[d, "(2)"'] & f'_*\mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \arrow[d, "(4)"] \\
\mathrm{RHom}(f_!E, f_!F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(3)"'] & \mathrm{RHom}(f_!E \otimes^L_S P', f_!F \otimes^L_S Q'),
\end{tikzcd}$$

where (1) is the composite of the base-change isomorphism

$$f_*\mathrm{RHom}(E, F) \otimes^L_S - \xrightarrow{\sim} f'_*(\mathrm{RHom}(E, F) \otimes^L_S -)$$

and $f'_*(2.2.4)$, (2) is deduced from the canonical arrow

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_!E, f_!F)$$

(SGA 4 XVIII 3.1.9.5) by applying $\otimes^L_S\mathrm{RHom}(P', Q')$, (3) is given by 2.2.4, and finally (4) is the composite of the canonical arrow

$$f'_*\mathrm{RHom}(E \otimes^L_S P', F \otimes^L_S Q') \longrightarrow \mathrm{RHom}(f'_!(E \otimes^L_S P'), f'_!(F \otimes^L_S Q'))$$

(loc. cit.) and the arrows defined by the base-change isomorphisms

$$f'_!(E \otimes^L_S P') \xrightarrow{\sim} f_!E \otimes^L_S P', \qquad f'_!(F \otimes^L_S Q') \xrightarrow{\sim} f_!F \otimes^L_S Q'.$$

Returning to the definition of the arrow (SGA 4 XVIII 3.1.9.5), choose a compactification of f[8]; one is then reduced to proving the lemma separately in the case where f is an open immersion and in the case where f is a proper morphism.

[8]The arrow quoted is defined with the aid of such a compactification, but it does not depend on it: this point, tacitly admitted in (loc. cit.), is easily verified.

a) Case where f is an open immersion. By adjunction (2.2.1), it suffices to see that, if one tensors the square 2.6.3 by $f_!E \otimes^L_S P'$ and composes with the evaluation arrow 2.2.2

$$(f_!E \otimes^L_S P') \otimes^L \mathrm{RHom}(f_!E \otimes^L_S P', f_!F \otimes^L_S Q') \longrightarrow f_!F \otimes^L_S Q',$$

the square obtained is commutative. But the sums of this square are zero outside X', so that it suffices to verify that the square induced on X' commutes, which is trivial.

b) Case where f is a proper morphism. Then $f_! = f_*$, $f'_! = f'_*$. One sees easily that the composite of

$$f_*\mathrm{RHom}(E, F) \longrightarrow \mathrm{RHom}(f_*E, f_*F)$$

with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E, f_*F) \xrightarrow{\sim} f_*\mathrm{RHom}(f^*f_*E, F)$$

is none other than the arrow deduced from the adjunction arrow $f^*f_*E \to E$ by applying $f_*\mathrm{RHom}(-, F)$. Similarly, the composite of (4) with the trivial-duality isomorphism

$$\mathrm{RHom}(f_*E \otimes^L_S P', f_*F \otimes^L_S Q') \xrightarrow{\sim} f'_*\mathrm{RHom}(f^*f_*E \otimes^L_S P', F \otimes^L_S Q')$$

($f_*F \otimes^L_S Q'$ being identified with $f'_*(F \otimes^L_S Q')$ by the canonical isomorphism) is none other than the arrow deduced from the adjunction arrow $f^*f_*E \to E$ by applying $f'_*\mathrm{RHom}(- \otimes^L_S P', F \otimes^L_S Q')$. Thus one is reduced to verifying the commutativity of the square

$$\begin{tikzcd}
\mathrm{RHom}(f_*E, f_*F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(3)"] \arrow[d] & \mathrm{RHom}(f_*E \otimes^L_S P', f_*F \otimes^L_S Q') \arrow[d] \\
f_*\mathrm{RHom}(f^*f_*E, F) \otimes^L_S \mathrm{RHom}(P', Q') \arrow[r, "(1')"'] & f'_*\mathrm{RHom}(f^*f_*E \otimes^L_S P', F \otimes^L_S Q'),
\end{tikzcd}$$

where the vertical arrows are the trivial-duality isomorphisms and $(1')$ is (1) with E replaced by f^*f_*E. But this is a special case of 2.6.1.1 in the case, already treated, where (u, v) is of type $(2, 1)$ (take $u : f^*f_*E \to f_*E$ given by the identity of f^*f_*E, and $v : F \to f_*F$ given by the identity of f_*F). We have therefore proved 2.6.3. As we have seen, the commutativity of 2.6.2 follows, which completes the proof of 2.6.1 and, finally, of 2.5.

4 Cohomological Correspondences

3.1

Let, for $i = 1, 2$, X_i be an S-scheme and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, and let

$$X = X_1 \times_S X_2, \qquad p_i : X \longrightarrow X_i, \qquad q_i : X_i \longrightarrow S$$

be the canonical projections:

$$\begin{tikzcd} & X \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 \arrow[dr, "q_1"'] & & X_2 \arrow[dl, "q_2"] \\ & S & \end{tikzcd}$$

With the notation of 1.3, one has a canonical isomorphism (2.2.4)

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2).$$

Composing with the Künneth isomorphism

$$q_1^! A_S \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} p_2^! L_2 \qquad (1.7.3),$$

one obtains an isomorphism

$$DL_1 \overset{\mathbf{L}}{\otimes}_S L_2 \xrightarrow{\sim} \mathrm{RHom}(p_1^* L_1, p_2^! L_2). \tag{3.1.1}$$

This isomorphism will play a fundamental role throughout. For $X_i = S$, 3.1.1 reduces to

$$L_1^\vee \overset{\mathbf{L}}{\otimes} L_2 \xrightarrow{\sim} \mathrm{RHom}(L_1, L_2),$$

a special case of an isomorphism valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.7). We set

$$\mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}_S(L_1, L_2), \qquad \mathrm{Hom}(p_1^* L_1, p_2^! L_2) = \mathrm{Hom}_S(L_1, L_2). \tag{3.1.2}$$

The elements of $\mathrm{Hom}_S(L_1, L_2)$ will be called *cohomological correspondences from L_1 to L_2*. Instead of "cohomological correspondence from A_{X_1} to A_{X_2}", we shall simply say *cohomological correspondence from X_1 to X_2* (understood: with coefficients in A).

If X_1 is smooth, purely of relative dimension r_1, then the same is true of p_2, and by the duality theorem (SGA 4 XVIII 3.2.5) one has

$$p_2^! L_2 \xrightarrow{\sim} p_2^* L_2(r_1)[2r_1],$$

therefore

$$\mathrm{RHom}_S(L_1, L_2) = \mathrm{RHom}(p_1^* L_1, p_2^* L_2)(r_1)[2r_1], \qquad \mathrm{Hom}_S(L_1, L_2) = \mathrm{H}^{2r_1}(X, \mathrm{RHom}(p_1^* L_1, p_2^* L_2)(r_1)). \tag{3.1.3}$$

In particular, the group of cohomological correspondences from X_1 to X_2 is identified in this case with $\mathrm{H}^{2r_1}(X, A(r_1))$.

3.2

The cohomological correspondences met in practice are associated with S-morphisms $X_2 \to X_1$, or more generally with cycles on X, or with schemes over X. Let

$$c : C \longrightarrow X$$

be a scheme over X (one sometimes says that c is a correspondence between X_1 and X_2), and put $c_i = p_i c : C \to X_i$ for the projections. We shall call *cohomological correspondences from L_1 to L_2 with support in c* (or in C, if there is no risk of confusion) the elements of the group

$$\mathrm{H}^0(C, \mathrm{RHom}(c_1^* L_1, c_2^! L_2)) \quad (= \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \text{ if } c \text{ is proper}).$$

By the induction formula (SGA 4 XVIII 3.1.12.2), one has a canonical isomorphism

$$c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}(c_1^* L_1, c_2^! L_2), \tag{3.2.1}$$

whence, applying $c_!$ and composing with the adjunction arrow $c_! c^! \to \mathrm{Id}$, canonical arrows

$$c_! \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{RHom}(p_1^* L_1, p_2^! L_2), \tag{3.2.2}$$

$$\mathrm{H}^0(C, \mathrm{RHom}(c_1^* L_1, c_2^! L_2)) \longrightarrow \mathrm{Hom}_S(L_1, L_2). \tag{3.2.3}$$

Thus every correspondence from L_1 to L_2 with support in C canonically defines, by 3.2.3, a correspondence (without support) from L_1 to L_2.

Examples. a) Suppose that c is proper and that c_2 has finite tor-dimension and relative dimension $\leq N$. The trace morphism (SGA $4^{1/2}$ Cycle 2.3.3)

$$\mathrm{Tr}_{c_2} : c_{2!} A_C \longrightarrow A_{X_2}(-N)[-2N]$$

defines by adjunction a cohomological correspondence from A_{X_1} to $A_{X_2}(-N)[2N]$ with support in C, which will be called the *class of* (c, N), and will be denoted

$$\mathrm{cl}(c, N) \in \mathrm{H}^{-2N}(C, c_2^! A_{X_2}(-N)). \tag{3.2.4}$$

The most important case is that in which $N = 0$, i.e. c_2 is finite (and of finite tor-dimension); in that case $\mathrm{cl}(c)$ is a cohomological correspondence from X_1 to X_2 with support in C. If X_1 is smooth, purely of relative dimension r_1, then, by 3.1.3 and 3.2.1,

$$\mathrm{H}^{-2N}(C, c_2^! A_{X_2}(-N)) = \mathrm{H}^{2(r_1 - N)}(C, c^! A_X(r_1 - N)). \tag{3.2.5}$$

If moreover c is a closed immersion, the second member rewrites as

$$\mathrm{H}_C^{2(r_1 - N)}(X, A_X(r_1 - N)),$$

and the class of (c, N), considered as an element of this group, is precisely the cohomology class associated with c in the sense of (SGA $4^{1/2}$ Cycle 2.3.1), as follows immediately from the definitions.

b) Let $F : X_2 \to X_1$ be an S-morphism, and denote by $c(F) : C(F) \to X$ its graph, i.e. the image of the section of p_2 defined by F ("the set of points (Fx, x)"): c_2 is an isomorphism and $c_1 c_2^{-1} = F$. One therefore has canonical isomorphisms

$$\mathrm{RHom}(c_1(F)^* L_1, c_2(F)^! L_2) = \mathrm{RHom}(F^* L_1, L_2), \qquad \mathrm{Hom}(c_1(F)^* L_1, c_2(F)^! L_2) = \mathrm{Hom}(F^* L_1, L_2). \tag{3.2.6}$$

In other words, the cohomological correspondences from L_1 to L_2 with support in the graph of F are the "lifts" of F to a homomorphism $F^*L_1 \to L_2$. Here are two important examples of such correspondences:

(i) $X_1 = X_2$, $L_1 = L_2$, $F = 1$ is the identity of X_1, whose graph Δ is the diagonal. The correspondence with support in Δ defined by the identity of L_1 is called the *diagonal correspondence*. It is the element

$$1 \in \mathrm{Hom}(L_1, L_1) = \mathrm{H}^0(X, \Delta_X^! \mathrm{RHom}_S(L_1, L_1))$$

(a group that is identified with $\mathrm{H}^{2r}_{\Delta}(X, \mathrm{RHom}_S(L_1, L_1)(r))$ when X_1 is smooth, purely of relative dimension r).

(ii) Suppose that S is the spectrum of a finite field $\mathbb{F}_q$, that $X_1 = X_2$, $L_1 = L_2$, and take for $F : X_1 \to X_1$ the relative Frobenius endomorphism over $\mathbb{F}_q$, the identity on the underlying space and raising to the q-th power on $\mathcal{O}_{X_1}$ (cf. XIV), and take as lift of F to L_1 the canonical isomorphism

$$F^*L_1 \xrightarrow{\sim} L_1$$

(loc. cit. §2, Prop. 1) (more precisely, the a-th iterate of the isomorphism of (loc. cit.) if $q = p^a$). The cohomological correspondence from L_1 to L_1 with support in the graph of F so defined is called the *Frobenius correspondence* (relative to $\mathbb{F}_q$).

c) Certain cohomological correspondences between sheaves on curves, related to Hecke operators, have been studied by Langlands in ([13] §7).

d) Cohomological correspondences between constant sheaves (the most frequent ones in nature) furnish in a natural way, by "integration", cohomological correspondences between complexes, as we shall now see.

3.3

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism, and put

$$f = f_1 \times_S f_2 : X \to Y, \qquad f_1' = f_1 \times_S Y_2, \qquad f_2' = Y_1 \times_S f_2,$$

$$p_1' = X_1 \times_S f_2, \qquad p_2' = f_1 \times_S X_2.$$

$$\begin{tikzcd}
X_1 \times_S Y_2 & X \arrow[l, "p_1'"'] \arrow[dl, "f"'] \arrow[d, "p_2'"] \\
Y \arrow[d, equal] & Y_1 \times_S X_2 \arrow[l, "f_2'"] \\
Y_1 \times_S Y_2 &
\end{tikzcd} \tag{3.3.0}$$

Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. There exists a unique isomorphism

$$f_* \mathrm{RHom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{RHom}_S(f'_{1!} L_1, f'_{2*} L_2) \tag{3.3.1}$$

whose inverse makes the square

$$\begin{tikzcd}
D(f'_{1!}L_1) \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2 \arrow[r, "3.1.1"] & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \arrow[d] \\
f_{1!}DL_1 \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2 \arrow[r, "(2)"'] & f_*\mathrm{RHom}_S(L_1, L_2),
\end{tikzcd}$$

commutative, where (1) is deduced from the inverse of the duality isomorphism (SGA 4 XVIII 3.1.9.6)

$$Df'_{1!}L_1 \xrightarrow{\sim} f_{1*}DL_1$$

by applying $- \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2$, and (2) is the composite of the Künneth isomorphism

$$f_{1*}DL_1 \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2 \xrightarrow{\sim} f_*(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2)$$

and $f_*(3.1.1)$. Applying $R\Gamma(Y, -)$ to 3.3.1, one obtains an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f'_{1!}L_1, f'_{2*}L_2). \tag{3.3.2}$$

For $u \in \mathrm{Hom}_S(L_1, L_2)$, the correspondence f_*u will be called the *direct image* of u by f. When $Y_1 = Y_2 = S$, we shall write u_* rather than f_*u. If $Y_1 = Y_2 = S$ is the spectrum of a separably closed field, 3.3.2 reads

$$\mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)). \tag{3.3.3}$$

Thus cohomological correspondences "operate" on cohomology. The following statement will allow us to interpret this operation in several ways, and in particular to relate it to the usual definitions in situations such as a) or b) above.

Proposition 3.4. With the notation of 3.3.0, one has a commutative diagram of isomorphisms in $D_{\mathrm{ctf}}(Y)$:

$$\begin{tikzcd}
& \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \arrow[dl, "(1)"'] \arrow[d] \arrow[dr, "(2)"] & \\
f'_{1*}\mathrm{RHom}_S(L_1, f'_{2*}L_2) \arrow[dr, "f'_{1*}(3)"'] & f_*\mathrm{RHom}_S(L_1, L_2) & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2) \arrow[dl, "f'_{2*}(4)"] \\
& f_*\mathrm{RHom}_S(L_1, L_2) &
\end{tikzcd} \tag{3.4.1}$$

where (5) is the inverse of 3.3.1 and the arrows (1) to (4) are defined by the commutative diagrams below; in these diagrams "base ch." denotes an arrow deduced from a base-change or Künneth isomorphism of one of the types considered in no. 1, "dual." an arrow deduced from a duality isomorphism of type 2.1.4 or (SGA 4 XVIII 3.1.9.6), and "triv." a trivial isomorphism:

$$\begin{tikzcd}
\mathrm{RHom}(f'_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2) \arrow[r, "\text{base ch.}"] \arrow[d, "\text{base ch.}"'] & \mathrm{RHom}_S(f'_{1!}L_1, f'_{2*}L_2) \\
\mathrm{RHom}(f'_{1!}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2) \arrow[d, "\text{dual.}"'] & \\
f'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, f'^{!}_1(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S f'_{2*}L_2)) \arrow[d, "\text{base ch.}"'] & \\
f'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) \arrow[r, "\text{base ch.}"'] & f'_{1*}\mathrm{RHom}_S(L_1, f'_{2*}L_2),
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[r, "\text{base ch.}"] \arrow[d, "\text{base ch.}"'] & \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, f'_{2*}(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2)) \arrow[d, "\text{dual.}"'] & \\
f'_{2*}\mathrm{RHom}(f'^{*}_2(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[d, "\text{triv.}"'] & \\
f'_{2*}\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[r, "\text{base ch.}"'] & f'_{2*}\mathrm{RHom}_S(f_{1!}L_1, L_2),
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S f_{2*}L_2) \arrow[r, "\text{base ch.}"] \arrow[d, "\text{base ch.}"'] & \mathrm{RHom}_S(L_1, f_{2*}L_2) \\
\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}, p'_{1*}(K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2)) \arrow[d, "\text{dual.}"'] & \\
p'_{1*}\mathrm{RHom}(p'^{*}_1(L_1 \overset{\mathbf{L}}{\otimes}_S A_{Y_2}), K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[d, "\text{triv.}"'] & \\
p'_{1*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) \arrow[r, "\text{base ch.}"'] & p'_{1*}\mathrm{RHom}_S(L_1, L_2),
\end{tikzcd}$$

$$\begin{array}{ccc}
\mathrm{RHom}(f_{1!}L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow{\text{base ch.}} & \mathrm{RHom}_S(f_{1!}L_1, L_2) \\
\downarrow{\scriptstyle \text{base ch.}} & & \\
\mathrm{RHom}(p'_{2!}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}), K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2) & & \\
\downarrow{\scriptstyle \text{dual.}} & & \\
p'_{2*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, p'^!_2(K_{Y_1} \overset{\mathbf{L}}{\otimes}_S L_2)) & & \\
\downarrow{\scriptstyle \text{base ch.}} & & \\
p'_{2*}\mathrm{RHom}(L_1 \overset{\mathbf{L}}{\otimes}_S A_{X_2}, K_{X_1} \overset{\mathbf{L}}{\otimes}_S L_2) & \xrightarrow[\text{base ch.}]{} & p'_{2*}\mathrm{RHom}_S(L_1, L_2).
\end{array}$$

This is only an explication of 2.5 in the particular case where $(u_1, v_1) : (L_1, K_{X_1}) \to (f_{1!}L_1, K_{Y_1})$ is given by the identity of $f_{1!}L_1$ and the canonical isomorphism $f_1^! K_{Y_1} \xrightarrow{\sim} K_{X_1}$, and where $(u_2, v_2) : (A_{X_2}, L_2) \to (A_{Y_2}, f_{2*}L_2)$ is given by the canonical isomorphism $A_{X_2} \xrightarrow{\sim} f_2^! A_{Y_2}$ and the identity of $f_{2*}L_2$.

Corollary 3.5. Assume that $Y_1 = Y_2 = S$. Let $u \in \mathrm{Hom}_S(L_1, L_2)$. The direct image $f_*u \in \mathrm{Hom}(f_{1!}L_1, f_{2*}L_2)$ (3.3.2) can be obtained in the following two equivalent ways:

a) Identify u, by trivial duality, with an arrow

$$L_1 \longrightarrow p_{1*}p_2^! L_2;$$

compose with the base-change isomorphism

$$p_{1*}p_2^! L_2 \xrightarrow{\sim} f_1^! f_{2*}L_2;$$

f_*u is the adjoint of this composite arrow.

b) Identify u, by adjunction, with an arrow

$$p_{2!}p_1^* L_1 \longrightarrow L_2;$$

compose with the base-change isomorphism

$$f_2^* f_{1!}L_1 \xrightarrow{\sim} p_{2!}p_1^* L_1;$$

f_*u is the adjoint of this composite arrow.

Description a) (respectively b)) indeed corresponds to the path $(f_{1*}(3) \circ (1))^{-1}$ (respectively $f_{2*}(4) \circ (2)$) in 3.4.1.

3.6

In practice, it is mainly description 3.5 b) that is used. Let us spell it out at the level of cohomology classes, assuming, in order to lighten the notation, that S is the spectrum of a separably closed field. Let

$$u \in \mathrm{Hom}_S(L_1, L_2) = \mathrm{Hom}(p_1^* L_1, p_2^! L_2).$$

For $x \in \mathrm{H}^i_c(X_1, L_1)$, $u_*(x) \in \mathrm{H}^i(X_2, L_2)$ is defined as follows: consider the element $p_1^* x \in \mathrm{H}^i(X_2, p_{2!} p_1^* L_1)$ deduced from x by inverse image, apply u to it, i.e. form the product

$$u \cdot p_1^* x \in \mathrm{H}^i(X_2, p_{2!} p_2^! L_2),$$

then apply the arrow

$$a : \mathrm{H}^i(X_2, p_{2!} p_2^! L_2) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

defined by the adjunction arrow $p_{2!} p_2^! \to \mathrm{Id}$; in short:

$$u_*(x) = a(u \cdot p_1^* x). \tag{3.6.1}$$

Let $c : C \to X$ be a proper morphism, and let $u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2)$. Denote by

$$u_* \in \mathrm{Hom}\big(R\Gamma_c(X_1, L_1), R\Gamma(X_2, L_2)\big)$$

the image of u under the composite of 3.2.3 and 3.3.3. It follows from 3.6.1 that, for $x \in \mathrm{H}^i_c(X_1, L_1)$, one has

$$u_*(x) = a(u \cdot c_1^* x), \tag{3.6.2}$$

where $c_1^* x$ is the inverse image of x in $\mathrm{H}^i(X_2, c_{2!} c_1^* L_1)$ and

$$a : \mathrm{H}^i(X_2, c_{2!} c_2^! L_2) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

is the arrow deduced from the adjunction $c_{2!} c_2^! L_2 \to \mathrm{Id}$. In the situation of Example 3.2 a), one thus finds

$$\mathrm{cl}(c, N)_*(x) = \mathrm{Tr}_{c_2}(c_1^* x), \tag{3.6.3}$$

where

$$\mathrm{Tr}_{c_2} : \mathrm{H}^i(X_2, Rc_{2!} A_C) \longrightarrow \mathrm{H}^{i-2N}(X_2, A(-N))$$

is the arrow induced by the trace morphism relative to c_2. For $N = 0$, this formula agrees with (SGA $4^{1/2}$ Cycle 3.6): $\mathrm{cl}(c)_*$ is the homomorphism denoted γ^*_{X,X_1} there. If X_1 is smooth, purely of relative dimension r_1, and c is a closed immersion, it follows from 3.6.1 and from the remark made at the end of 3.2 a) that one also has

$$\mathrm{cl}(c, N)_*(x) = \mathrm{Tr}_{p_2}(\mathrm{cl}(c)\, p_1^* x), \tag{3.6.4}$$

where $\mathrm{cl}(c) \in \mathrm{H}_C^{2(r_1 - N)}(X, A(r_1 - N))$ is the cohomology class of C and

$$\mathrm{Tr}_{p_2} : \mathrm{H}^*(X, Rp_{2!} A_X) \longrightarrow \mathrm{H}^{*-2r_1}(X_2, A(-r_1))$$

is the arrow defined by the trace morphism relative to p_2. This recovers the usual definition of the homomorphism induced on cohomology by a correspondence ([12] 1.3), (SGA $4^{1/2}$ Cycle 3.2).

3.6.5

In the situation of 3.2 b), it follows from 3.6.2 that, for $u \in \mathrm{Hom}(F^* L_1, L_2)$,

$$u_* : \mathrm{H}^i_c(X_1, L_1) \longrightarrow \mathrm{H}^i(X_2, L_2)$$

is the composite of the "inverse image" homomorphism

$$\mathrm{H}^i_c(X_1, L_1) \longrightarrow \mathrm{H}^i(X_2, F^* L_1)$$

and the homomorphism induced by u (more generally, $f_*u : Rf_{1!}L_1 \to Rf_{2*}L_2$ is the composite of the canonical homomorphism

$$Rf_{1!}L_1 \longrightarrow Rf_{2*}F^*L_1$$

and of $Rf_{2*}u$). In particular, the diagonal correspondence induces the canonical arrow

$$Rf_{1!}L_1 \longrightarrow Rf_{1*}L_1$$

(the identity if X_1 is proper over S).

3.7

Return to the situation of the beginning of 3.3, assuming in addition that a commutative square

$$\begin{tikzcd} X & C \arrow[l, "c"'] \arrow[d, "g"] \\ Y \arrow[u, "f"] & D \arrow[l, "d"] \end{tikzcd} \tag{3.7.1}$$

is given, with c proper. Put $c_i = p_i c$, $d_i = q_i d$, where $p_i : X \to X_i$, $q_i : Y \to Y_i$ are the projections. From 3.7.1 one deduces a commutative diagram whose inner square is cartesian:

$$\begin{tikzcd} & C \arrow[dl, "c"'] \arrow[d, "i"] \arrow[dr, "g"] & \\ X & C' \arrow[l, "c'"'] \arrow[d, "g'"] & D \arrow[l] \\ Y & D \arrow[l, "d"] & \end{tikzcd} \tag{3.7.2}$$

For $K \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, there is a canonical arrow

$$f_*c_!c^!K \longrightarrow d_*d^!f_*K, \tag{3.7.3}$$

which is the composite of the arrow

$$f_*c_!c^!K = f_*c'_*(i_*i^!)c'^!K \longrightarrow f_*c'_!c'^!K = d_*g'_!c'^!K$$

defined by the adjunction arrow $i_*i^! \to \mathrm{Id}$, and the isomorphism

$$d_*g'_!c'^!K \xrightarrow{\sim} d_*d^!f_*K$$

defined by the canonical isomorphism $g'_!d'^! \xrightarrow{\sim} d^!f_*$ (SGA 4 XVIII 3.1.12.3). Taking $K = \mathrm{RHom}_S(L_1, L_2)$, and composing 3.7.3 with the isomorphism deduced from 3.3.1, one obtains a canonical arrow

$$f_*c_!c^!\mathrm{RHom}_S(L_1, L_2) \longrightarrow d_*d^!\mathrm{RHom}_S(f_{1!}L_1, f_{2*}L_2), \tag{3.7.4}$$

which, by 3.2.1, may be rewritten as

$$f_*c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \longrightarrow d_*\mathrm{RHom}(d_1^*(f_{1!}L_1), d_2^!(f_{2*}L_2)). \tag{3.7.5}$$

One deduces

$$f_* : \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \longrightarrow \mathrm{Hom}(d_1^*(f_{1!}L_1), d_2^!(f_{2*}L_2)). \tag{3.7.6}$$

The arrow 3.7.5 (respectively 3.7.6) generalizes 3.3.1 (respectively 3.3.2), with which it coincides when c and d are the identity. Also note that, for c proper, 3.2.2 (respectively 3.2.3) is nothing other than 3.7.5 (respectively 3.7.6) when f_1, f_2, and d are the identity arrows. When $Y_1 = Y_2 = S = D$, we shall abbreviate f_*u as u_*, as above.

5 Pairings of Correspondences. Lefschetz Formula

4.1

Let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the projections, $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$, and K_{X_i} the dualizing complex (1.3). The canonical arrows

$$p_i^* \mathrm{RHom}(L_i, K_{X_i}) \overset{\mathbf{L}}{\otimes} p_i^* L_i \longrightarrow p_i^* K_{X_i}, \tag{4.1.1}$$

composed from 2.1.1 and the evaluation arrow 2.2.2, give, by tensor product on X and composition with the Künneth isomorphism 1.7.6, a pairing

$$(DL_1 \overset{\mathbf{L}}{\otimes}_S L_2) \overset{\mathbf{L}}{\otimes} (L_1 \overset{\mathbf{L}}{\otimes}_S DL_2) \longrightarrow K_X, \tag{4.1.2}$$

which may be rewritten, by 3.1.1 and 3.1.2, as

$$\mathrm{RHom}_S(L_1, L_2) \overset{\mathbf{L}}{\otimes} \mathrm{RHom}_S(L_2, L_1) \longrightarrow K_X. \tag{4.1.3}$$

One deduces a pairing, called cup product,

$$\langle\ ,\ \rangle : \mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_1) \longrightarrow \mathrm{H}^0(X, K_X). \tag{4.1.4}$$

One may also regard 4.1.2 as the external tensor product of the evaluation arrows

$$DL_i \overset{\mathbf{L}}{\otimes} L_i \longrightarrow K_{X_i}.$$

In view of the Künneth isomorphism 2.2.4, one thus sees that 4.1.2 is a “perfect duality”, i.e. identifies by the adjunction isomorphism 2.2.1 each of the complexes $DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$ and $L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$ with the dual of the other (with values in K_X).

When $X_1 = X_2 = S$, 4.1.3 is a special case of a pairing valid for perfect complexes on an arbitrary ringed topos (SGA 6 I 7.8), and, with the notation of (SGA 6 I 8.3), one has

$$\langle u, v \rangle = \mathrm{Tr}(vu) = \mathrm{Tr}(uv). \tag{4.1.5}$$

4.2

Let $c : C \to X$, $d : D \to X$, and $e = c \times_X d : E = C \times_X D \to X$. For $P, Q \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, one has a canonical arrow (which is not in general an isomorphism)

$$c^! P \overset{\mathbf{L}}{\otimes}_X d^! Q \longrightarrow e^!(P \overset{\mathbf{L}}{\otimes} Q), \tag{4.2.1}$$

defined, in the same way as 1.7.3, as the adjoint of the composite arrow

$$e_!(c^! P \overset{\mathbf{L}}{\otimes}_X d^! Q) \xrightarrow[(1)]{\sim} c_! c^! P \overset{\mathbf{L}}{\otimes} d_! d^! Q \xrightarrow{(2)} P \overset{\mathbf{L}}{\otimes} Q,$$

where (1) is the Künneth isomorphism (SGA 4 XVII 5.4.2.2) and (2) is the tensor product of the adjunction arrows. Applying $e_!$ to 4.2.1 and composing with the Künneth isomorphism just cited, one obtains a canonical arrow

$$c_! c^! P \overset{\mathbf{L}}{\otimes} d_! d^! Q \longrightarrow e_! e^!(P \overset{\mathbf{L}}{\otimes} Q). \tag{4.2.2}$$

Similarly, composing the Künneth arrow

$$c_*c^!P \overset{\mathbf{L}}{\otimes} d_*d^!Q \longrightarrow e_*(c^!P \overset{\mathbf{L}}{\otimes}_X d^!Q)$$

with the arrow deduced from 4.2.1 by applying e_*, one obtains a canonical arrow

$$c_*c^!P \overset{\mathbf{L}}{\otimes} d_*d^!Q \longrightarrow e_*e^!(P \overset{\mathbf{L}}{\otimes} Q). \tag{4.2.3}$$

For $P = DL_1 \overset{\mathbf{L}}{\otimes}_S L_2$, $Q = L_1 \overset{\mathbf{L}}{\otimes}_S DL_2$, one deduces from 4.2.2 and 4.2.3, by composition with 4.1.2 and the isomorphism $e^!K_X = K_E$, and with the identifications 3.1.1 and 3.2.1, pairings

$$c_!\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \overset{\mathbf{L}}{\otimes} d_!\mathrm{RHom}(d_2^*L_2, d_1^!L_1) \longrightarrow e_!K_E, \tag{4.2.4}$$

$$c_*\mathrm{RHom}(c_1^*L_1, c_2^!L_2) \overset{\mathbf{L}}{\otimes} d_*\mathrm{RHom}(d_2^*L_2, d_1^!L_1) \longrightarrow e_*K_E, \tag{4.2.4'}$$

$$\langle\ ,\ \rangle : \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \otimes \mathrm{Hom}(d_2^*L_2, d_1^!L_1) \longrightarrow \mathrm{H}^0(E, K_E). \tag{4.2.5}$$

These pairings are compatible with étale localization: given a commutative square (not necessarily cartesian)

$$\begin{tikzcd} & E' \arrow[dl] \arrow[dr] & \\ C' \arrow[dr] & & D' \arrow[dl] \\ & X' & \end{tikzcd} \quad \text{étale over} \quad \begin{tikzcd} & E \arrow[dl] \arrow[dr] & \\ C \arrow[dr] & & D \arrow[dl] \\ & X & \end{tikzcd}$$

if c', d', e' denote the composites $C' \to C \to X$, $D' \to D \to X$, $E' \to E \to X$, one has a commutative square.

$$\begin{tikzcd} \mathrm{Hom}(c_1^*L_1, c_2^!L_2) \otimes \mathrm{Hom}(d_2^*L_2, d_1^!L_1) \arrow[r, "4.2.5"] \arrow[d] & \mathrm{H}^0(E, K_E) \arrow[d] \\ \mathrm{Hom}(c_1'^*L_1, c_2'^!L_2) \otimes \mathrm{Hom}(d_2'^*L_2, d_1'^!L_1) \arrow[r] & \mathrm{H}^0(E', K_{E'}) \end{tikzcd} \tag{4.2.6}$$

where the vertical arrows are the restrictions, and the lower horizontal arrow is the composite of the arrow analogous to 4.2.5 and the restriction

$$\mathrm{H}^0(E'', K_{E''}) \longrightarrow \mathrm{H}^0(E', K_{E'}),$$

with $E'' = C' \times_{X'} D'$ (there are analogous commutative squares with arrows of type 4.2.4, 4.2.4', which we leave to the reader to write down). If Z is étale over E, we shall denote, for

$$u \in \mathrm{Hom}(c_1^*L_1, c_2^!L_2), \qquad v \in \mathrm{Hom}(d_2^*L_2, d_1^!L_1),$$

$$\langle u, v\rangle_Z \in \mathrm{H}^0(Z, K_Z) \tag{4.2.7}$$

the image of $\langle u, v\rangle$ in $\mathrm{H}^0(Z, K_Z)$ by restriction (in practice, Z will be a connected component of E). If z is an isolated closed point of E, and $\bar{z}$ a geometric point above z, it follows from 4.2.6 that

$$\langle u, v\rangle_z \in \mathrm{H}^0(z, K_z) = A$$

depends only on the situation deduced by strict localization at the images of $\bar{z}$ in C, D, X.

Example 4.3. Assume that X_i is smooth, purely of relative dimension r_i, and that c and d are closed immersions, with c_2 (respectively d_1) finite. For $L_1 = A_{X_1}$, $L_2 = A_{X_2}$, we have, by 3.1.3,

$$DL_1 \otimes^{\mathbf{L}}_S L_2 = A_X(r_1)[2r_1], \qquad L_1 \otimes^{\mathbf{L}}_S DL_2 = A_X(r_2)[2r_2],$$

and 4.1.2 is identified with the product

$$A_X(r_1)[2r_1] \otimes^{\mathbf{L}}_X A_X(r_2)[2r_2] \longrightarrow A_X(r)[2r],$$

where $r = r_1 + r_2$, and hence, taking 3.2.1 into account, 4.2.5 is identified with the usual cup product

$$\mathrm{H}^{2r_1}_C(X, A(r_1)) \otimes \mathrm{H}^{2r_2}_D(X, A(r_2)) \longrightarrow \mathrm{H}^{2r}_{C\cap D}(X, A(r)).$$

Take as cohomological correspondence from X_1 to X_2 (respectively from X_2 to X_1) the class of c (respectively d) (3.2 a)). It follows from (SGA $4^{1/2}$ Cycle 2.3.8) that, at an isolated point z of $C \cap D$, $\langle \mathrm{cl}(c), \mathrm{cl}(d)\rangle_z$ is nothing but the image in A of the intersection multiplicity of C and D at z.

We now give the main result of the exposé.

Theorem 4.4. Let, for $i = 1, 2$, $f_i : X_i \longrightarrow Y_i$ be a proper S-morphism,

$$f = f_1 \times_S f_2 : X \longrightarrow Y, \qquad p_i : X \longrightarrow X_i, \qquad q_i : Y \longrightarrow Y_i$$

the projections. Let moreover

$$\begin{tikzcd}
 & C' \arrow[rr] \arrow[dl, "c'"'] \arrow[dd, "c''" near start] & & C \arrow[dl, "c"] \arrow[dd, "g"] \\
X \arrow[dd, "f"'] & & C'' \arrow[ll, "f'" near end, crossing over] & \\
 & D' \arrow[rr, "f''" near end] \arrow[dl, "d'"'] & & D \arrow[dl, "d"] \\
Y & & D'' \arrow[ll, "d''"] \arrow[from=uu, crossing over] &
\end{tikzcd} \tag{4.4.0}$$

be a commutative cube with cartesian horizontal faces, with c', c'', d', d'' proper. Put

$$c'_i = p_i c', \qquad c''_i = p_i c'', \qquad d'_i = q_i d', \qquad d''_i = q_i d''.$$

Finally, let $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(X_i)$. Then there is a commutative square

$$\begin{tikzcd}
f_* c'_* \mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes^{\mathbf{L}} f_* c''_* \mathrm{RHom}(c_2''^* L_2, c_1''^! L_1) \arrow[r, "(1)"] \arrow[d, "(2)"'] & f_* c_* K_C \arrow[d, "(4)"] \\
d'_* \mathrm{RHom}(d_1'^! M_1, d_2'^! M_2) \otimes^{\mathbf{L}} d''_* \mathrm{RHom}(d_2''^! M_2, d_1''^! M_1) \arrow[r, "(3)"'] & d_* K_D,
\end{tikzcd} \tag{4.4.1}$$

where $M_i = f_{i*} L_i$, (3) is given by 4.2.4, (1) is deduced from 4.2.4 by applying f_* and composing with the canonical arrow

$$f_* c'_*(-) \otimes^{\mathbf{L}} f_* c''_*(-) \longrightarrow f_*(c'_*(-) \otimes^{\mathbf{L}} c''_*(-)),$$

(2) is the tensor product of the arrows 3.7.5 relative to the faces containing f, and (4) is defined by the adjunction arrow $g_*K_C = g_*g^!K_D \longrightarrow K_D$.

Proof. Put

$$DL_1 \otimes_S^{\mathbf{L}} L_2 = P, \qquad L_1 \otimes_S^{\mathbf{L}} DL_2 = Q, \qquad DM_1 \otimes_S^{\mathbf{L}} M_2 = E, \qquad M_1 \otimes_S^{\mathbf{L}} DM_2 = F.$$

Consider the diagram

$$\begin{tikzcd}
f_*c'_*c'^!P \otimes^{\mathbf{L}} f_*c''_*c''^!Q \arrow[rr, "(1)"] \arrow[d, "(8)"'] & & f_*c'_*(P \otimes^{\mathbf{L}} Q) \arrow[r, "(2)"] \arrow[d, "(9)"'] & f_*c'_*K_X \arrow[d, "(10)"] \\
d'_*d'^!f_*P \otimes^{\mathbf{L}} d''_*d''^!f_*Q \arrow[r, "(3)"'] \arrow[d, "(11)"'] & d_*d'^!(f_*P \otimes^{\mathbf{L}} f_*Q) \arrow[r, "(4)"'] \arrow[d, "(12)"'] & d_*d'^!(P \otimes^{\mathbf{L}} Q) \arrow[r, "(5)"'] & d_*d'^!K_X \arrow[d, "(13)"] \\
d'_*d'^!E \otimes^{\mathbf{L}} d''_*d''^!F \arrow[r, "(6)"'] & d_*d'^!(E \otimes^{\mathbf{L}} F) \arrow[rr, "(7)"'] & & d_*d^!K_Y.
\end{tikzcd} \tag{4.4.2}$$

The arrows are as follows: (1) is deduced from 4.2.2 by applying f_*; (2) is defined by 4.1.2; (3) is given by 4.2.2; (4) is given by the canonical arrow $f_*P \otimes^{\mathbf{L}} f_*Q \longrightarrow f_*(P \otimes^{\mathbf{L}} Q)$; (5) by 4.1.2; (6) by 4.2.2; (7) by 4.1.2; (8) is the tensor product of arrows 3.7.3; (9) and (10) are defined by 3.7.3; (11) and (12) by the isomorphisms $f_*P \longrightarrow E$, $f_*Q \longrightarrow F$ (3.3.1); and (13) by the adjunction arrow $f_*K_X \longrightarrow K_Y$. By definition, 4.4.1 is identified, via 3.1.1 and 3.2.1, with the outer boundary of 4.4.2. It remains only to prove that the squares (A), (B), (C), (D) are commutative. The commutativity of (B) and (C) is clear by functoriality. It remains to prove that of (A) and (D).

a) Commutativity of (D). It is enough to prove that the square

$$\begin{tikzcd}
f_*P \otimes^{\mathbf{L}} f_*Q \arrow[r] \arrow[d] & f_*K_X \arrow[d] \\
E \otimes^{\mathbf{L}} F \arrow[r] & K_Y
\end{tikzcd} \tag{4.4.3}$$

(corresponding to (D) for $d = \mathrm{Id}$) commutes. Recall that the pairing

$$P \otimes^{\mathbf{L}} Q \longrightarrow K_X \quad (\text{respectively } E \otimes^{\mathbf{L}} F \longrightarrow K_Y)$$

is the external tensor product of the evaluation arrows

$$DL_i \otimes^{\mathbf{L}} L_i \longrightarrow K_{X_i} \quad (\text{respectively } DM_i \otimes^{\mathbf{L}} M_i \longrightarrow K_{Y_i}).$$

By the Künneth isomorphisms

$$f_{1*}DL_1 \otimes_S^{\mathbf{L}} f_{2*}L_2 \xrightarrow{\sim} f_*P, \qquad f_{1*}L_1 \otimes_S^{\mathbf{L}} f_{2*}DL_2 \xrightarrow{\sim} f_*Q,$$

4.4.3 is therefore identified with the external tensor product of the squares

$$\begin{tikzcd}
f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i \arrow[r, "(1)"] \arrow[d, "(3)"'] & f_{i*}K_{X_i} \arrow[d, "(4)"] \\
DM_i \otimes^{\mathbf{L}} M_i \arrow[r, "(2)"'] & K_{Y_i},
\end{tikzcd} \tag{4.4.4}$$

where (2) is the evaluation arrow, (1) the composite of the canonical arrow

$$f_{i*}DL_i \otimes^{\mathbf{L}} f_{i*}L_i \longrightarrow f_{i*}(DL_i \otimes^{\mathbf{L}} L_i)$$

and the arrow defined by the evaluation arrow, (3) is the tensor product of the duality isomorphism $f_{i*}DL_i \longrightarrow DM_i$ and the identity arrow of M_i, and (4) is the adjunction arrow. It is therefore enough to verify the commutativity of the squares 4.4.4. Since the above duality isomorphism is the composite of the canonical arrow (SGA 4 XVIII 3.1.9.5)

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i})$$

and the arrow deduced from the adjunction arrow

$$f_{i*}K_{X_i} \longrightarrow K_{Y_i},$$

it is enough to see that the following square commutes:

$$\begin{tikzcd}
f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i \arrow[r, "(1)"] \arrow[d, "(3)"'] & f_{i*}K_{X_i} \arrow[d, "\mathrm{Id}"] \\
\mathrm{RHom}(f_{i*}L_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} f_{i*}L_i \arrow[r, "(2)"'] & f_{i*}K_{X_i},
\end{tikzcd} \tag{4.4.5}$$

where (1) is the arrow (1) of 4.4.4, (2) the evaluation arrow, and (3) the tensor product by $f_{i*}L_i$ of the canonical arrow just mentioned. More generally, for any arrow $N_i \longrightarrow f_{i*}L_i$ in $D_{\mathrm{ctf}}(X_i)$, i.e. any arrow $L_i \xrightarrow{u} N_i$ of type 1 over f_i, the triangle

$$\begin{tikzcd}
f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \otimes^{\mathbf{L}} N_i \arrow[d] \arrow[dr] & \\
\mathrm{RHom}(N_i, f_{i*}K_{X_i}) \otimes^{\mathbf{L}} N_i \arrow[r] & f_{i*}K_{X_i}
\end{tikzcd}$$

deduced from 4.4.5 commutes: indeed, by functoriality, one may assume that $L_i = f_i^*N_i$, with u given by the identity of L_i; the arrow

$$f_{i*}\mathrm{RHom}(L_i, K_{X_i}) \longrightarrow \mathrm{RHom}(N_i, f_{i*}K_{X_i})$$

is then the inverse of the trivial duality isomorphism, and the commutativity of the triangle is verified immediately at the level of complexes, resolving N_i flatly and K_{X_i} injectively.

b) Commutativity of (A). Referring to the definition of 3.7.3, one sees that it is enough to verify this commutativity in each of the following cases:

(i) $X_i = Y_i$, $\quad f_i =$ the identity arrow of X_i; $\qquad$ (ii) the lateral faces of 4.4.0 are cartesian.

Place ourselves in case (i). The square (A) is the outer boundary of the diagram

$$\begin{tikzcd}
c'_*(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) \arrow[r] \arrow[d] & c_*(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) \arrow[r] \arrow[d] & c_*c'^!(P \otimes Q) \arrow[d] \\
d'_*d'^!P \otimes^{\mathbf{L}}_X d''_*d''^!Q \arrow[r] & d_*(d'^!P \otimes^{\mathbf{L}}_X d''^!Q) \arrow[r] & d_*d'^!(P \otimes Q),
\end{tikzcd}$$

in which the horizontal isomorphisms of A_1 are Künneth isomorphisms, the horizontal arrows of A_2 are given by 4.2.1, and the middle vertical arrow is deduced by applying d_* to the arrow

$$g_*(c'^!P \otimes^{\mathbf{L}}_X c''^!Q) \longrightarrow d'^!P \otimes^{\mathbf{L}}_X d''^!Q$$

defined, via Künneth, by the external tensor product of the adjunction arrows

$$f'_*c'^!P \longrightarrow d'^!P, \qquad f''_*c''^!Q \longrightarrow d''^!Q.$$

These adjunction arrows and the identity arrows of $d'^!d'^!P$, $d''^!d''^!Q$ define commutative triangles of arrows of type 3 (1.4)

$$\begin{tikzcd}
& c'^!P \arrow[dl] \arrow[d] \\
d'^!d'^!P & d'^!P \arrow[l]
\end{tikzcd}
\qquad
\begin{tikzcd}
& c''^!Q \arrow[dl] \arrow[d] \\
d''^!d''^!Q & d''^!Q \arrow[l]
\end{tikzcd}$$

over

$$\begin{tikzcd}
& C' \arrow[dl, "c'"'] \arrow[d, "f'"] \\
X & D' \arrow[l, "d'"]
\end{tikzcd}
\qquad
\begin{tikzcd}
& C'' \arrow[dl, "c''"'] \arrow[d, "f''"] \\
X & D'' \arrow[l, "d''"]
\end{tikzcd} .$$

By external tensor product (over X), these triangles give a commutative triangle

$$\begin{tikzcd}
& c'^!P \otimes^{\mathbf{L}}_X c''^!Q \arrow[dl] \arrow[d] \\
d'^!d'^!P \otimes^{\mathbf{L}} d''^!d''^!Q & d'^!P \otimes^{\mathbf{L}}_X d''^!Q \arrow[l]
\end{tikzcd}
\quad \text{over} \quad
\begin{tikzcd}
& C \arrow[dl, "c"'] \arrow[d, "g"] \\
X & D \arrow[l, "d"]
\end{tikzcd} .$$

The commutativity of this triangle is equivalent to that of the square deduced from A_1 by inverting the horizontal arrows. On the other hand, there are commutative triangles of arrows of type 3

$$\begin{tikzcd}
& c'^!P \arrow[dl] \arrow[d] \\
P & d'^!P \arrow[l]
\end{tikzcd}
\qquad
\begin{tikzcd}
& c''^!Q \arrow[dl] \arrow[d] \\
Q & d''^!Q \arrow[l]
\end{tikzcd}$$

defined by the identity arrows of $d'^!P$, $c'^!P$, $d''^!Q$, $c''^!Q$. By external tensor product, one obtains a commutative triangle

$$\begin{tikzcd}
& c'^!P \otimes^{\mathbf{L}}_X c''^!Q \arrow[dl] \arrow[d] \\
P \otimes Q & d'^!P \otimes^{\mathbf{L}}_X d''^!Q \arrow[l]
\end{tikzcd}$$

which can be interpreted as a morphism of arrows of type 3 over $g : C \longrightarrow D$:

$$\begin{tikzcd} c'^! P \otimes^{\mathbf{L}}_X c''^! Q \arrow[r] \arrow[d] & c'^!(P \otimes^{\mathbf{L}} Q) \arrow[d] \\ d'^! P \otimes^{\mathbf{L}}_X d''^! Q \arrow[r] & d'^!(P \otimes Q), \end{tikzcd}$$

or again as a commutative square over D:

$$\begin{tikzcd} g_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q) \arrow[r] \arrow[d] & g_* c'^!(P \otimes^{\mathbf{L}} Q) \arrow[d] \\ d'^! P \otimes^{\mathbf{L}}_X d''^! Q \arrow[r] & d^!(P \otimes Q). \end{tikzcd}$$

But the square deduced from the latter by applying d_* is none other than A_2, whence the commutativity of (A) in case (i).

It remains to treat case (ii). Since the faces of 4.4.0 containing f are cartesian, the arrows

$$f'_* c'^! P \longrightarrow d'^! f_* P, \qquad f''_* c''^! Q \longrightarrow d''^! f_* Q$$

(3.7.3) are isomorphisms. Their inverses define arrows of type 1

$$c'^! P \longrightarrow d'^! f_* P, \qquad c''^! Q \longrightarrow d''^! f_* Q$$

over f', f'', whence, by direct image, commutative squares of arrows of type 1

$$\begin{tikzcd} c'_* c'^! P & c'^! P \arrow[l] \arrow[d] \\ d'_* d'^! P & d'^! f_* P \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} c''_* c''^! Q & c''^! Q \arrow[l] \arrow[d] \\ d''_* d''^! Q & d''^! f_* Q \arrow[l] \end{tikzcd}$$

over the faces of 4.4.0 containing f. Taking the tensor product, one obtains a commutative square in $D_{\mathrm{ctf}}(Y)$

$$\begin{tikzcd} d'_* d'^! f_* P \otimes^{\mathbf{L}}_Y d''_* d''^! f_* Q \arrow[r] \arrow[d] & d_*(d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q) \arrow[d] \\ f_* c'_* c'^! P \otimes^{\mathbf{L}} f_* c''_* c''^! Q \arrow[r] & d_* g_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q), \end{tikzcd} \tag{4.4.6}$$

where the left vertical arrow is the inverse of (8), the horizontal arrows are given by Künneth, and the right vertical arrow by the external tensor product over Y of the arrows

$$f'_* c'^! P \longrightarrow d'^! f_* P, \qquad f''_* c''^! Q \longrightarrow d''^! f_* Q.$$

The same tensor product defines the left vertical arrow of the square

$$\begin{tikzcd} d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q \arrow[r] \arrow[d] & d'^! f_*(P \otimes^{\mathbf{L}} Q) \arrow[d] \\ g_*(c'^! P \otimes^{\mathbf{L}}_X c''^! Q) \arrow[r] & g_* c'^!(P \otimes Q), \end{tikzcd} \tag{4.4.7}$$

where the right vertical arrow is 3.7.3, and the horizontal arrows are defined by 4.2.1. The outer boundary of the diagram composed of 4.4.6 and $d_*(4.4.7)$ is the square deduced from (A) by inverting the vertical arrows. It is therefore enough to prove that 4.4.7 commutes, in other words that the arrows 4.2.1

$$c'^! P \otimes^{\mathbf{L}}_X c''^! Q \longrightarrow c'^!(P \otimes^{\mathbf{L}} Q), \qquad d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q \longrightarrow d'^! f_*(P \otimes^{\mathbf{L}} Q)$$

define a morphism from

$$c'^! P \otimes^{\mathbf{L}}_X c''^! Q \longrightarrow d'^! f_* P \otimes^{\mathbf{L}}_Y d''^! f_* Q$$

to

$$c'^!(P \otimes Q) \longrightarrow d'^! f_*(P \otimes Q),$$

viewed as arrows of type 1 over g. By adjunction, we are reduced to verifying that the tensor products of adjunction arrows

$$c'_* c'^! P \otimes^{\mathbf{L}}_X c''_* c''^! Q \longrightarrow P \otimes^{\mathbf{L}} Q, \qquad d'_* d'^! f_* P \otimes^{\mathbf{L}}_Y d''_* d''^! f_* Q \longrightarrow f_* P \otimes^{\mathbf{L}} f_* Q$$

define a morphism, between arrows of type 1 over f, from the tensor product of the arrows

$$c'_* c'^! P \longrightarrow d'_* d'^! f_* P, \qquad c''_* c''^! Q \longrightarrow d''_* d''^! f_* Q$$

to the tensor product of the arrows

$$P \longrightarrow f_* P, \qquad Q \longrightarrow f_* Q.$$

For this it is enough to verify that the adjunction arrows

$$c'_* c'^! P \longrightarrow P, \qquad d'_* d'^! f_* P \longrightarrow f_* P$$

define a morphism from

$$c'_* c'^! P \longrightarrow d'_* d'^! f_* P$$

to

$$P \longrightarrow f_* P,$$

and likewise for the pair (c'', d''). It is enough to consider the case of (c', d'), the statement being the same in the two cases. We have to see that the triangle

$$\begin{tikzcd}
 & f_* c'_* c'^! P \arrow[dl] \arrow[d, "\sim"] \\
f_* P & d'_* d'^! f_* P \arrow[l]
\end{tikzcd}$$

where the horizontal and oblique arrows are given by adjunction and the vertical arrow by the isomorphism 3.7.3, is commutative. Taking into account the definition of 3.7.3 (SGA 4 XVIII 3.1.12.3) as the transpose of a base-change isomorphism, we are reduced, by applying the functor $\mathrm{Hom}(R, -)$ $(R \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y))$, to verifying the commutativity of the square

$$\begin{tikzcd}
c'_* c'^* f^* R \arrow[r, "\mathrm{Id}"] & c'_* f'^* d'^* R \\
f^* R \arrow[u] \arrow[r] & f^* d'_* d'^* R \arrow[u, "\sim"']
\end{tikzcd}$$

where the left vertical arrow (respectively the lower horizontal arrow) is defined by the adjunction arrow $1 \longrightarrow c'_* c'^*$ (respectively $1 \longrightarrow d'_* d'^*$), and the right vertical arrow by the base-change isomorphism

$$f^* d'_* \xrightarrow{\sim} c'_* f'^*.$$

But this square commutes by the very definition of the base-change arrow. The commutativity of (A) is therefore established, and this completes the proof of the theorem.

Corollary 4.5. Under the hypotheses of 4.4, there is a commutative square

$$\begin{tikzcd}
\mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1) \arrow[r, "4.2.5"] \arrow[d, "3.7.6 \otimes 3.7.6"'] & \mathrm{H}^0(C, K_C) \arrow[d, "g_*"] \\
\mathrm{Hom}(d_1'^* f_{1*} L_1, d_2'^! f_{2*} L_2) \otimes \mathrm{Hom}(d_2''^* f_{2*} L_2, d_1''^! f_{1*} L_1) \arrow[r, "4.2.5"'] & \mathrm{H}^0(D, K_D),
\end{tikzcd}$$

where g_* is defined by the adjunction arrow

$$g_* K_C = g_* g^! K_D \longrightarrow K_D.$$

In other words, for

$$u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2), \qquad v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1),$$

one has

$$\langle f_* u, f_* v \rangle = g_* \langle u, v \rangle. \tag{4.5.1}$$

For every proper S-scheme $t : T \longrightarrow S$, denote by

$$\int_T : \mathrm{H}^0(T, K_T) \longrightarrow \mathrm{H}^0(S, K_S) = \mathrm{H}^0(S, A) \tag{4.6}$$

the arrow defined by the adjunction arrow

$$t_! K_T \longrightarrow A.$$

Corollary 4.7 – Lefschetz–Verdier formula. Under the hypotheses of 4.4, suppose that $Y_1 = Y_2 = S = D' = D''$. For

$$u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2), \qquad v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1),$$

one has

$$\langle u_*, v_* \rangle = \int_C \langle u, v \rangle, \tag{4.7.1}$$

where

$$u_* \in \mathrm{Hom}(f_{1*} L_1, f_{2*} L_2), \qquad v_* \in \mathrm{Hom}(f_{2*} L_2, f_{1*} L_1)$$

are the homomorphisms deduced from u, v by direct image over S (3.7.6), and

$$\langle u_*, v_* \rangle = \mathrm{Tr}(u_* v_*) = \mathrm{Tr}(v_* u_*)$$

is the cup product, in the sense of (SGA 6 I 8.3), of the homomorphisms of perfect complexes u_*, v_* (4.1.5).

Taking 3.6.5 into account, in particular one has:

Corollary 4.8. Let $t : T \longrightarrow S$ be a proper S-scheme, $\Delta \subset T \times_S T$ the diagonal, $F : T \longrightarrow T$ an S-endomorphism of T, $T^F =$ (graph of F) $\times_{(T\times_S T)} \Delta$ the scheme of fixed points of F, $L \in \operatorname{ob} D_{\mathrm{ctf}}(T)$, $u \in \operatorname{Hom}(F^*L, L)$ a correspondence from L to L with support in the graph of F, and $1 \in \operatorname{Hom}(L, L)$ the identity correspondence, with support in Δ (3.2 b)). Then

$$\operatorname{Tr}(u_*) = \int_{T^F} \langle u, 1\rangle,$$

where u_* is the endomorphism of t_*L obtained by composing $t_*L \longrightarrow t_*F^*L$ (inverse image) with $t_*u : t_*F^*L \longrightarrow t_*L$.

From the remarks of 3.2 a) and 3.6 one also deduces:

Corollary 4.9. Under the hypotheses of 4.7, assume that X_i is smooth over S, purely of relative dimension r_i, and that C' (respectively C'') is a closed subscheme of X finite and of finite tor-dimension over X_2 (respectively X_1). Let

$$\operatorname{cl}(C') \in \mathrm{H}^{2r_1}_{C'}(X, A(r_1)), \qquad \operatorname{cl}(C'') \in \mathrm{H}^{2r_2}_{C''}(X, A(r_2))$$

be the cohomology classes of C' and C'', and let

$$\operatorname{cl}(C')_* \in \operatorname{Hom}(\mathrm{R}f_{1*}A, \mathrm{R}f_{2*}A), \qquad \operatorname{cl}(C'')_* \in \operatorname{Hom}(\mathrm{R}f_{2*}A, \mathrm{R}f_{1*}A)$$

be the corresponding homomorphisms (3.2 a), 3.7.6). Then

$$\langle \operatorname{cl}(C')_*, \operatorname{cl}(C'')_*\rangle = \int_{C'\cap C''} \operatorname{cl}(C)\operatorname{cl}(D),$$

where

$$\operatorname{cl}(C)\operatorname{cl}(D) \in \mathrm{H}^{2(r_1+r_2)}_{C'\cap C''}(X, A(r_1+r_2)) = \mathrm{H}^0(C'\cap C'', K_{C'\cap C''})$$

is the usual cup product.

Taking 4.3 into account, in particular one has:

Corollary 4.10. Under the hypotheses of 4.9, suppose that $C'\cap C''$ is finite. For $z \in C'\cap C''$, denote by $(C'\cdot C'')_z$ the image in A of the intersection multiplicity of C' and C'' at z. Then

$$\langle \operatorname{cl}(C')_*, \operatorname{cl}(C'')_*\rangle = \sum_{z\in C'\cap C''} (C'\cdot C'')_z.$$

4.11. Under the hypotheses of 4.9, one has in general

$$\langle \operatorname{cl}(C')_*, \operatorname{cl}(C'')_*\rangle = \sum_{Z\in\pi_0(C'\cap C'')} (\operatorname{cl}(C')\operatorname{cl}(C''))_Z, \tag{4.11.1}$$

where $\pi_0(C'\cap C'')$ denotes the set of connected components of $C'\cap C''$, and $(\operatorname{cl}(C')\operatorname{cl}(C''))_Z$ the restriction of $\operatorname{cl}(C')\operatorname{cl}(C'')$ to $\mathrm{H}^0(Z, K_Z)$. When Z is a component of dimension ≥ 1, the calculation of $(\operatorname{cl}(C')\operatorname{cl}(C''))_Z$ presents problems, because one is in an "excess" situation. If Z is smooth, purely of dimension d, if C' and C'' are smooth in a neighbourhood of Z, and if X is projective, one can show that

$$\int_Z (\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z = \int_Z c_d(N_Z), \tag{4.11.2}$$

where N_Z is the locally free sheaf of rank d defined by the exact sequence of normal bundles

$$0 \longrightarrow N_{Z/C'} \oplus N_{Z/C''} \longrightarrow N_{Z/X} \longrightarrow N_Z \longrightarrow 0,$$

and $c_d(N_Z) \in \mathrm{H}^{2d}(Z, A(d))$ denotes its d-th Chern class (SGA 5 VII) (use (SGA 6 XIV 6.3) to reduce to a calculation of intersection number in K-theory, and apply (SGA 6 VII 2.5), cf. ([10] 4.1); the projectivity hypothesis is no doubt unnecessary).

Let T be a smooth projective S-scheme; take as correspondence C' (respectively C'') the graph of an S-endomorphism F (respectively the diagonal), and suppose that the scheme of fixed points $T^F = C' \cap C''$ is smooth. The sheaf $N_{(T^F)}$ above is nothing other than the tangent sheaf to T^F, so that 4.11.2 implies, by the Gauss–Bonnet formula (SGA 5 VII 4.9), the analogue of a formula well known to topologists:

$$\mathrm{Tr}(F, \mathrm{R}\Gamma(T, A)) = \mathrm{EP}(T_s^F), \tag{4.11.3}$$

where $\mathrm{EP}(T_s^F)$ denotes the Euler–Poincaré characteristic of T_s^F, for s a geometric point above S (loc. cit.).

The calculation of $(\mathrm{cl}(C')\,\mathrm{cl}(C''))_Z$ for a non-smooth component Z of dimension ≥ 1 has not been treated in general, at least to the knowledge of the editor.

4.12. In the situation of 4.7, let z be an isolated point of C, with projection z_i on X_i, a' (respectively a'') on C' (respectively C''), and let $\bar{z}$ be a geometric point above z, with image $\bar{z}_i$ above z_i. Assume that c_2' (respectively c_1'') is quasi-finite and flat at a' (respectively a''), that L_i is perfect in a neighbourhood of $\bar{z}_i$, and that u (respectively v) is given in a neighbourhood of a' (respectively a'') by

$$\mathrm{Tr}_{c_2'} : c'_{2!}c_2'^{*}L_2 \longrightarrow L_2 \quad \text{and} \quad s \in \mathrm{Hom}(c_1'^{*}L_1, c_2'^{*}L_2)$$

(respectively

$$\mathrm{Tr}_{c_1''} : c''_{1!}c_1''^{*}L_1 \longrightarrow L_1 \quad \text{and} \quad t \in \mathrm{Hom}(c_2''^{*}L_2, c_1''^{*}L_1)).$$

Let $s_{\bar{z}} \in \mathrm{Hom}(L_{1\bar{z}_1}, L_{2\bar{z}_2})$ (respectively $t_{\bar{z}} \in \mathrm{Hom}(L_{2\bar{z}_2}, L_{1\bar{z}_1})$) be the homomorphism deduced from s (respectively t) by passage to the fibres. Let again, as in 3.2 a), $\mathrm{cl}(c')$ (respectively $\mathrm{cl}(c'')$) denote the correspondence from X_1 to X_2 (respectively from X_2 to X_1) defined, in a neighbourhood of a' (respectively a''), by c_2' (respectively c_1''), and let $(C' \cdot C'')_z$ be the corresponding local term $\langle \mathrm{cl}(c'), \mathrm{cl}(c'')\rangle_z$ (equal to the image in A of the intersection multiplicity of C' and C'' at z when C' and C'' are closed subschemes of X, with X_1 and X_2 smooth). Then:

$$\langle u, v\rangle_z = (C' \cdot C'')_z \langle s_{\bar{z}}, t_{\bar{z}}\rangle. \tag{4.12.1}$$

This follows easily from 4.2.6.

4.13. If T is a scheme over S, denote by $DF(T)$ the filtered derived category of complexes of A_T-modules with finite filtration ([9] V 1), and by $DF_{\mathrm{ctf}}(T)$ the full subcategory formed by the L such that $\mathrm{gr}(L) \in \mathrm{ob}\, D_{\mathrm{ctf}}(T)$. The kind of compatibility of filtered derived functors with passage to associated graded objects implies that DF_{ctf} is stable under the six operations f^*, f_*, $f_!$, $f^!$, $\otimes^{\mathbf{L}}$, RHom. The definitions and results of the preceding numbers are valid, mutatis mutandis, in DF_{ctf}. The pairing 4.1.2 is compatible with passage to associated graded objects (K_X being regarded as endowed with the trivial filtration), and it follows that, with the notation of 4.2.5, for $L_i \in \mathrm{ob}\, DF_{\mathrm{ctf}}(X_i)$,

$$u \in \mathrm{Hom}_{DF}(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}_{DF}(d_2^* L_2, d_1^! L_1),$$

one has the following equality in $\mathrm{H}^0(E, K_E)$:

$$\langle u, v\rangle = \sum_i \langle \mathrm{gr}^i\, u, \mathrm{gr}^i\, v\rangle. \tag{4.13.1}$$

As Deligne observes, this remark applies in particular to the situation considered by Langlands in ([13] §7). The correspondences that he considers in (loc. cit. 7.11) indeed preserve, near the fixed points, finite filtrations on the sheaves with associated graded objects inducing, on the strict localization at the image of a fixed point, either a constant sheaf or a constant sheaf on the complement of the closed point extended by zero. The calculation of the local terms is easily reduced, by 4.13, to 4.12.1 and to the case of sheaves concentrated on the closed points that are the images of the fixed points, a case treated directly. Applying 4.7, one obtains the Lefschetz formula required in ([13] 7.11). We point out that the general statement (loc. cit. 7.12) is true; the proof, more delicate, will be given in III B.

5. Complements

5.1. Dual of a correspondence

Let, for $i = 1, 2$, X_i be an S-scheme, let $X = X_1 \times_S X_2$, and let $p_i : X \to X_i$ be the projections. Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. Every cohomological correspondence u from L_1 to L_2 gives, by applying D_X and identifying $Dp_2^!$ with $p_2^* D$ and Dp_1^* with $p_1^! D$, a correspondence Du from DL_2 to DL_1, called the *dual of u*. In other words, D defines a homomorphism

$$D : \mathrm{Hom}(p_1^* L_1, p_2^! L_2) \longrightarrow \mathrm{Hom}(p_2^* DL_2, p_1^! DL_1). \tag{5.1.1}$$

One checks that 5.1.1 is obtained, by applying $H^0(X, -)$, from the isomorphism

$$D : \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}(p_2^* DL_2, p_1^! DL_1) \tag{5.1.2}$$

which makes the square

$$\begin{tikzcd} p_1^* DL_1 \otimes^{\mathbf{L}} p_2^* L_2 \arrow[r, "3.1.1"] \arrow[d] & \mathrm{RHom}(p_1^* L_1, p_2^! L_2) \arrow[d] \\ p_2^* DDL_2 \otimes^{\mathbf{L}} p_1^* DL_1 \arrow[r, "3.1.1"] & \mathrm{RHom}(p_2^* DL_2, p_1^! DL_1), \end{tikzcd} \tag{5.1.3}$$

commutative. Here the left vertical arrow is the composite of the symmetry isomorphism and the one defined by the biduality isomorphism $L_2 \to DDL_2$. It follows easily from the definitions that the following square is commutative:

$$\begin{tikzcd} \mathrm{RHom}_S(L_1, L_2) \otimes^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_1) \arrow[r, "4.1.3"] \arrow[d, "D\otimes D"'] & K_X \arrow[d, "\mathrm{Id}"] \\ \mathrm{RHom}_S(DL_2, DL_1) \otimes^{\mathbf{L}} \mathrm{RHom}_S(DL_1, DL_2) \arrow[r, "4.1.3"] & K_X. \end{tikzcd} \tag{5.1.4}$$

Let $c : C \to X$ be an S-morphism and put $c_i = p_i c$. Applying $c_* c^!$ to 5.1.2, and using (3.2.1), one deduces an isomorphism

$$D : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \xrightarrow{\sim} \mathrm{Hom}(c_2^* DL_2, c_1^! DL_1), \tag{5.1.5}$$

which is checked to identify with the one obtained by applying the functor D_X and the identifications of Dc_1^* with $c_1^! D$ and $Dc_2^!$ with $c_2^* D$. If $c' : C' \to X$ and $c'' : C'' \to X$ are S-morphisms, it follows from 5.1.4 that, for

$$u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2), \qquad v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1),$$

one has

$$\langle u, v \rangle = \langle Du, Dv \rangle, \tag{5.1.6}$$

with the notation of 4.2.5.

On the other hand, one checks that, in the situation of 3.3, the following square is commutative:

$$\begin{tikzcd} f_* \mathrm{RHom}_S(L_1, L_2) \arrow[r, "3.3.1"] \arrow[d, "f_* D"'] & \mathrm{RHom}_S(f_1^! L_1, f_2^* L_2) \arrow[d, "D"] \\ f_* \mathrm{RHom}_S(DL_2, DL_1) \arrow[r, "3.3.1"] & \mathrm{RHom}_S(f_2^! DL_2, f_1^* DL_1), \end{tikzcd} \tag{5.1.7}$$

and consequently, in the situation of 3.7.6, one has

$$f_* Du = D f_* u. \tag{5.1.8}$$

5.2. Composition of correspondences

Let, for $i = 1, 2, 3$, X_i be an S-scheme, let $X_{ij} = X_i \times_S X_j$, $X = X_1 \times_S X_2 \times_S X_3$, and let $p_i : X \to X_i$, $p_{ij} : X \to X_{ij}$ be the projections. Let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. One has $X = X_{12} \times_{X_2} X_{23}$, and

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_{X_2}^{\mathbf{L}} (DL_2 \otimes_S^{\mathbf{L}} L_3) = p_1^* DL_1 \otimes^{\mathbf{L}} p_2^* L_2 \otimes^{\mathbf{L}} p_2^* DL_2 \otimes^{\mathbf{L}} p_3^* L_3.$$

The evaluation arrow $L_2 \otimes^{\mathbf{L}} DL_2 \to K_{X_2}$ therefore defines a pairing

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_{X_2}^{\mathbf{L}} (DL_2 \otimes_S^{\mathbf{L}} L_3) \longrightarrow p_{13}^! (DL_1 \otimes_S^{\mathbf{L}} L_3), \tag{5.2.1}$$

the second member being identified with $(DL_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} K_{X_2}$ by Künneth (1.7.3). Taking 3.1.1 and 3.1.2 into account, one may rewrite 5.2.1 in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_3) \longrightarrow p_{13}^! \mathrm{RHom}_S(L_1, L_3) = \mathrm{RHom}(p_1^* L_1, p_3^! L_3), \tag{5.2.2}$$

the identification with the second member being given by the induction isomorphism (SGA 4 XVIII 3.1.12.2). Hence one obtains a pairing, called *composition of correspondences*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_2, L_3) \longrightarrow \mathrm{Hom}(p_1^* L_1, p_3^! L_3), \tag{5.2.3}$$

which will be denoted $(u, v) \mapsto vu$. One checks that, if $p_{ij,i} : X_{ij} \to X_i$ and $p_{ij,j} : X_{ij} \to X_j$ denote the projections, 5.2.2 can also be obtained by composing the arrow

$$\mathrm{RHom}_S(L_1, L_2) \otimes_{X_2}^{\mathbf{L}} \mathrm{RHom}_S(L_2, L_3) \longrightarrow \mathrm{RHom}(p_{12}^* p_{12,1}^* L_1, p_{12}^! p_{12,2}^* L_2) \otimes \mathrm{RHom}(p_{23}^! p_{23,2}^* L_2, p_{23}^! p_{23,3}^! L_3)$$

(the tensor product of the canonical arrows $p_{12}^*\mathrm{RHom}_S(L_1, L_2) \to \mathrm{RHom}(p_{12}^*L_1, p_{12}^!L_2)$ and $p_{23}^*\mathrm{RHom}_S(L_2, L_3) \to \mathrm{RHom}(p_{23}^!L_2, p_{23}^!L_3)$), the arrow defined by the base-change arrow $p_{12}^! p_{12,2}^* L_2 \to p_{23}^* p_{23,2}^* L_2$, and a usual composition arrow on the RHom's.

Let $c' : C' \to X_{12}$ and $c'' : C'' \to X_{23}$ be correspondences, and let $c : C = C' \times_{X_2} C'' \to X$ be the "composite" correspondence. From 5.2.2 one deduces, or else defines directly by a composition analogous to the one just described, a pairing

$$\mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes^{\mathbf{L}}_{X_2} \mathrm{RHom}(c_2''^* L_2, c_3''^! L_3) \longrightarrow \mathrm{RHom}(c_1^* L_1, c_3^! L_3), \tag{5.2.4}$$

hence a "composition" pairing $(u, v) \mapsto vu$:

$$\mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_3''^! L_3) \longrightarrow \mathrm{Hom}(c_1^* L_1, c_3^! L_3). \tag{5.2.5}$$

One checks that composition of correspondences is compatible with direct images in the following sense. Let, for $1 \le i \le 3$, $f_i : X_i \to Y_i$ be an S-morphism, with f_2 proper, and let c', c'', c be morphisms as above, with c' and c'' proper. Let $d' : D' \to Y_{12}$ and $d'' : D'' \to Y_{23}$ be proper morphisms, $d : D = D' \times_{Y_2} D'' \to Y$, and suppose given the commutative squares

$$\begin{tikzcd} X_{12} & C' \arrow[l] \arrow[d] \\ Y_{12} \arrow[u, "f_{12}"] & D' \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} X_{23} & C'' \arrow[l] \arrow[d] \\ Y_{23} \arrow[u, "f_{23}"] & D'' \arrow[l] \end{tikzcd}$$

Then one has a commutative square

$$\begin{tikzcd} \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_2''^* L_2, c_3''^! L_3) \arrow[r, "5.2.5"] \arrow[d, "f_{12*} \otimes f_{23*}"'] & \mathrm{Hom}(c_1^* L_1, c_3^! L_3) \arrow[d, "f_{13*}"] \\ \mathrm{Hom}(d_1'^* M_1, d_2'^! M_2) \otimes \mathrm{Hom}(d_2''^* M_2, d_3''^! M_3) \arrow[r, "5.2.5"] & \mathrm{Hom}(d_1^* M_1, d_3^! M_3), \end{tikzcd} \tag{5.2.6}$$

where $M_i = f_{i!} L_i$.

Finally, suppose that $X_1 = X_3$ and $L_1 = L_3$. The fundamental pairing 4.1.2 may be interpreted as the composite of the pairing

$$q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* L_2 \otimes^{\mathbf{L}} q_2^* DL_2 \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \tag{5.2.7}$$

defined by $L_2 \otimes^{\mathbf{L}} DL_2 \to K_{X_2}$, or, equivalently, induced by 5.2.1 on X_{12} via the diagonal $X_{12} \to X_1 \times_S X_2 \times_S X_1$ ($q_i : X_{12} \to X_i$ denoting the projection), and the pairing

$$q_1^* DL_1 \otimes^{\mathbf{L}} q_2^* K_{X_2} \otimes^{\mathbf{L}} q_1^* L_1 \longrightarrow K_{X_{12}} \tag{5.2.8}$$

defined by $DL_1 \otimes^{\mathbf{L}} L_1 \to K_{X_1}$. It follows that, for $u \in \mathrm{Hom}_S(L_1, L_2)$ and $v \in \mathrm{Hom}_S(L_2, L_1)$, one has, in $H^0(X_1 \times_S X_2, K_{X_1 \times_S X_2})$,

$$\langle u, v \rangle = \langle vu, 1 \rangle, \tag{5.2.9}$$

where 1 denotes the identity correspondence from L_1 to L_1 on $X_1 \times_S X_1$, supported on the diagonal (3.2 b)), and the cup product in the second member is taken in the sense of 4.2.5, taking account of the fact that the fibred product $X \times_{(X_1 \times_S X_1)} X_1$ (via $p_{13} : X \to X_{13} = X_1 \times_S X_1$ and the diagonal $X_1 \to X_1 \times_S X_1$) identifies with $X_1 \times_S X_2$. Likewise, more generally, if $c' : C' \to X_{12}$ and $c'' : C'' \to X_{21}$ are correspondences, if $c : C = C' \times_{X_{12}} C'' \to X_1 \times_S X_2 \times_S X_1$ is the composite correspondence, and if $d : C' \times_{X_{12}} C'' = D \to X_{12}$ is the intersection correspondence, then, for $u \in \mathrm{Hom}(c_1'^* L_1, c_2'^! L_2)$ and $v \in \mathrm{Hom}(c_2''^* L_2, c_1''^! L_1)$, one has the following equality in $H^0(D, K_D)$:

$$\langle u, v \rangle = \langle vu, 1 \rangle, \tag{5.2.10}$$

where 1 denotes, as above, the diagonal correspondence, and D is identified with the fibred product of $C \to X_1 \times_S X_1$ and the diagonal $X_1 \to X_1 \times_S X_1$. Formulas 5.2.9 and 5.2.10 generalize a well-known formula in the case of constant coefficients ([12] 1.3.6 (ii) c)).

5.3. External tensor product of correspondences

Let, for $1 \leq i \leq 4$, X_i be an S-scheme, let $X = X_1 \times_S X_2 \times_S X_3 \times_S X_4$, $X_{ij} = X_i \times_S X_j$, and let $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}(X_i)$. The Künneth isomorphism (1.7.6, 2.2.4)

$$DL_1 \otimes_S^{\mathbf{L}} DL_3 \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3)$$

defines an isomorphism on X

$$(DL_1 \otimes_S^{\mathbf{L}} L_2) \otimes_S^{\mathbf{L}} (DL_3 \otimes_S^{\mathbf{L}} L_4) \xrightarrow{\sim} D(L_1 \otimes_S^{\mathbf{L}} L_3) \otimes_S^{\mathbf{L}} (L_2 \otimes_S^{\mathbf{L}} L_4), \tag{5.3.1}$$

which, by 3.1.1, may be rewritten in the form

$$\mathrm{RHom}_S(L_1, L_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}_S(L_3, L_4) \xrightarrow{\sim} \mathrm{RHom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4). \tag{5.3.2}$$

Hence a pairing, called the *external tensor product*,

$$\mathrm{Hom}_S(L_1, L_2) \otimes \mathrm{Hom}_S(L_3, L_4) \longrightarrow \mathrm{Hom}_S(L_1 \otimes_S^{\mathbf{L}} L_3, L_2 \otimes_S^{\mathbf{L}} L_4), \tag{5.3.3}$$

which will be denoted $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$. The pairing 5.3.2 may also be interpreted as the composite of the external tensor product

$$\mathrm{RHom}(E_1, E_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}(E_3, E_4) \xrightarrow{\sim} \mathrm{RHom}(E_1 \otimes_S^{\mathbf{L}} E_3, E_2 \otimes_S^{\mathbf{L}} E_4)$$

(where $E_1 = p_{12}^* p_{12,1}^* L_1$, etc., with the evident notation) and of the isomorphism defined by Künneth

$$E_2 \otimes_S^{\mathbf{L}} E_4 \xrightarrow{\sim} p_{24}^!(L_2 \otimes_S^{\mathbf{L}} L_4) \quad (1.7.3).$$

Let $c' : C' \to X_{12}$ and $c'' : C'' \to X_{34}$ be correspondences, and let $c : C = C' \times_S C'' \to X = X_{13} \times_S X_{24}$ be the product correspondence. From 5.3.2 one deduces, or else defines analogously, a pairing

$$\mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes_S^{\mathbf{L}} \mathrm{RHom}(c_3''^* L_3, c_4''^! L_4) \longrightarrow \mathrm{RHom}(c_{13}^*(L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^!(L_2 \otimes_S^{\mathbf{L}} L_4)), \tag{5.3.4}$$

hence an external tensor product pairing, $(u, v) \mapsto u \otimes_S^{\mathbf{L}} v$,

$$\mathrm{Hom}(c_1'^* L_1, c_2'^! L_2) \otimes \mathrm{Hom}(c_3''^* L_3, c_4''^! L_4) \longrightarrow \mathrm{Hom}(c_{13}^*(L_1 \otimes_S^{\mathbf{L}} L_3), c_{24}^!(L_2 \otimes_S^{\mathbf{L}} L_4)). \tag{5.3.5}$$

One checks that these pairings are compatible with direct images, in the following sense. Let, for $1 \leq i \leq 4$, $f_i : X_i \to Y_i$ be an S-morphism, let c', c'', c be as above, with c' and c'' proper, let

$d' : D' \to Y_{12}$ and $d'' : D'' \to Y_{34}$ be proper morphisms, let $d : D = D' \times_S D'' \to Y = Y_{12} \times_S Y_{34}$, and suppose given the commutative squares

$$\begin{tikzcd} X_{12} & C' \arrow[l] \arrow[d] \\ Y_{12} \arrow[u, "f_{12}"] & D' \arrow[l] \end{tikzcd} \qquad \begin{tikzcd} X_{34} & C'' \arrow[l] \arrow[d] \\ Y_{34} \arrow[u, "f_{34}"] & D'' \arrow[l] \end{tikzcd}$$

Then one has a commutative square

$$\begin{tikzcd} \mathrm{Hom}(c_1'^*L_1, c_2'^!L_2) \otimes \mathrm{Hom}(c_3''^*L_3, c_4''^!L_4) \arrow[r, "5.3.5"] \arrow[d, "f_{12*} \otimes f_{34*}"'] & \mathrm{Hom}(c_{13}^*L_{13}, c_{24}^!L_{24}) \arrow[d, "f_{13,24*}"] \\ \mathrm{Hom}(d_1'^*M_1, d_2'^!M_2) \otimes \mathrm{Hom}(d_3''^*M_3, d_4''^!M_4) \arrow[r, "5.3.5"] & \mathrm{Hom}(d_{13}^*M_{13}, d_{24}^!M_{24}), \end{tikzcd} \tag{5.3.6}$$

where $M_i = f_{i!}L_i$ if $i = 1, 3$ and $f_{i*}L_i$ otherwise, $L_{ij} = L_i \otimes_S^{\mathbf{L}} L_j$, and $M_{ij} = M_i \otimes_S^{\mathbf{L}} M_j$ (there is also an analogous square with the RHom's).

Suppose that $X_3 = X_2$, $X_4 = X_1$, $L_3 = L_2$, and $L_4 = L_1$. A reasoning analogous to that of the last paragraph of 5.2 shows that, for $u \in \mathrm{Hom}_S(L_1, L_2)$ and $v \in \mathrm{Hom}_S(L_2, L_1)$, one has

$$\langle u, v \rangle = \langle u \otimes_S^{\mathbf{L}} v, 1 \rangle, \tag{5.3.7}$$

where 1 denotes the diagonal correspondence from L_{12} to L_{12} on X_{12}, and the cup product in the second member is taken in the sense of 4.2.5. More generally, if $c' : C' \to X_{12}$ and $c'' : C'' \to X_{21}$ are correspondences, their "intersection" $C = C' \times_{X_{12}} C''$ identifies with the fibred product of $C' \times_S C'' \to X = X_{12} \times_S X_{21}$ and the diagonal $X_{12} \to X$, and, for $u \in \mathrm{Hom}(c_1'^*L_1, c_2'^!L_2)$ and $v \in \mathrm{Hom}(c_2''^*L_2, c_1''^!L_1)$, one has the following equality in $H^0(C, K_C)$:

$$\langle u, v \rangle = \langle u \otimes_S^{\mathbf{L}} v, 1 \rangle, \tag{5.3.8}$$

where 1 denotes, as above, the diagonal correspondence. Thus there are two "diagonal" procedures (5.2.10, 5.3.8) for the calculation of the "Verdier local term" $\langle u, v \rangle$.

6. Appendix. Lefschetz formula for coherent sheaves

In this number, S denotes a noetherian scheme. The S-schemes considered will be assumed separated and of finite type. Recall that, by Nagata, every S-morphism between such schemes is compactifiable, i.e. factors as an open immersion followed by a proper morphism. For every S-scheme X, we shall write $D(X)$ for the derived category of the category of $\mathcal{O}_X$-modules.

6.1

Recall ([8] App.) that for every S-morphism $f : X \to Y$ one has a functor

$$f^! : D^+(Y) \longrightarrow D^+(X)$$

(and transitivity isomorphisms $(gf)^! = f^! g^!$ for a composite gf, with a cocycle condition). If f is proper, $f^!$ is right "adjoint" to the functor $f_* : D^+(X)_{\mathrm{coh}} \to D^+(Y)_{\mathrm{coh}}$ (where the subscript "coh" denotes the full subcategory formed by complexes with coherent cohomology), this adjunction being made precise by an isomorphism

$$f_* \mathrm{RHom}(E, f^! F) \xrightarrow{\sim} \mathrm{RHom}(f_* E, F) \tag{6.1.1}$$

for $E \in \mathrm{ob}\, D(X)_{\mathrm{coh}}$, $F \in \mathrm{ob}\, D^+(Y)$, defined analogously to the isomorphism (SGA 4 XVIII 3.1.9.6). The considerations of ([8] App.5), together with the duality theorem for projective space, show that, if f is smooth, purely of dimension d, there is a canonical isomorphism

$$f^! L \xleftarrow{\sim} f^* L \otimes \Omega^d_{X/Y}[d]; \tag{6.1.2}$$

and, if moreover f is proper, the arrow

$$\mathrm{Tr}_f : f_*(f^* L \otimes \Omega^d_{X/Y}[d]) \longrightarrow L \tag{6.1.3}$$

deduced, by 6.1.2, from the adjunction arrow $f_* f^! \to I$, coincides with the trace morphism ([8] VI).

6.2

Contrary to what happens in the case of discrete coefficients, there is no reasonable subcategory of D here that is stable under the "six operations" $(f^*, f_*, f^!, f_!, \otimes^{\mathbf{L}}, \mathrm{RHom})$. The most natural finiteness notion seems to be relative perfection (SGA 6 III 4). If X is an S-scheme and $L \in \mathrm{ob}\, D(X)$, recall that L is said to be perfect relative to S if L has finite tor-dimension relative to S and coherent cohomology. We shall denote by $D(X)_{\mathrm{parf}/S}$ the full subcategory of $D(X)$ formed by the complexes perfect relative to S. It is not stable in general either under $\otimes^{\mathbf{L}}$ or under RHom. However, if E is perfect in the absolute sense and F is perfect relative to S, then $\mathrm{RHom}(E, F)$ is perfect relative to S (SGA 6 III 4.9). On the other hand, let, for $i = 1, 2$, X_i be an S-scheme and $L_i \in \mathrm{ob}\, D^-(X_i)$. We shall put, as in 1.5,

$$L_1 \otimes^{\mathbf{L}}_S L_2 = Lp_1^* L_1 \otimes^{\mathbf{L}} Lp_2^* L_2,$$

where $p_i : X_1 \times_S X_2 \to X_i$ is the projection. Considering this object is reasonable only if X_1 and X_2 are tor-independent over S, i.e. (SGA 6 III 1.5) if $\mathrm{Tor}_i^S(\mathcal{O}_X, \mathcal{O}_Y) = 0$ for $i \neq 0$. Suppose this is the case: then, if L_1 and L_2 are perfect relative to S, so is $L_1 \otimes^{\mathbf{L}}_S L_2$ (SGA 6 III 4.7).

Let $f : X \to Y$ be an S-morphism. If f is proper, then $f_* D(X)_{\mathrm{parf}/S} \subset D(Y)_{\mathrm{parf}/S}$ (SGA 6 III 4.8). If f has finite tor-dimension, then $f^* D(Y)_{\mathrm{parf}/S} \subset D(X)_{\mathrm{parf}/S}$ (SGA 6 III 4.5), and $f^! D(Y)_{\mathrm{parf}/S} \subset D(X)_{\mathrm{parf}/S}$ (SGA 6 III 4.9).

For every S-scheme $a : X \to S$, we put

$$K_X = a^! \mathcal{O}_S, \tag{6.2.1}$$

and write D_X (or simply D) for the functor $\mathrm{RHom}(-, K_X)$. If X has finite tor-dimension over S, then K_X is perfect relative to S and, for every $L \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$, one has $DL \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$, and the canonical arrow $L \to DDL$ is an isomorphism (SGA 6 III 4.9).

Let $f : X \to Y$ be an S-morphism, $L \in \mathrm{ob}\, D(X)$, and $M \in \mathrm{ob}\, D(Y)$. If $f^* M$ is defined (for example if $M \in \mathrm{ob}\, D^-(Y)$, or if f has finite tor-dimension), put

$$\mathrm{Hom}_f^{(1)}(L, M) = \mathrm{Hom}(f^* M, L), \qquad \mathrm{Hom}_f^{(2)}(L, M) = \mathrm{Hom}(L, f^* M).$$

If $f^!M$ is defined (for example if $M \in \operatorname{ob} D^+(Y)$, or if f is smooth, or finite of finite tor-dimension, or a composite of such morphisms, since in these cases $f^!$ extends, by 6.1.1 and 6.1.2, to a functor defined on the whole category $D(Y)$), put

$$\operatorname{Hom}_f^{(3)}(L, M) = \operatorname{Hom}(L, f^!M), \qquad \operatorname{Hom}_f^{(4)}(L, M) = \operatorname{Hom}(f^!M, L).$$

The elements of $\operatorname{Hom}_f^{(i)}(L, M)$ will be called, as in 1.4, *arrows of type i from L to M over f* (or f-arrows of type i from L to M). We leave to the reader the task of making explicit the composition of arrows of type i, for fixed i, and the corresponding fibred and cofibred categories. We recall only the partial adjunction formulas

$$\operatorname{Hom}_f^{(1)}(L, M) = \operatorname{Hom}(M, f_*L) \quad (M \in \operatorname{ob} D^-(Y),\ L \in \operatorname{ob} D^+(X)),$$

$$\operatorname{Hom}_f^{(3)}(L, M) = \operatorname{Hom}(f_*L, M) \quad (f \text{ proper},\ L \in \operatorname{ob} D(X)_{\mathrm{coh}},\ M \in \operatorname{ob} D^+(Y)).$$

6.3

In the situation of 1.6, suppose that X_1 and X_2 are tor-independent over S, and likewise Y_1 and Y_2. Then, for $L_i \in \operatorname{ob} D(X_i)_{\mathrm{parf}/S}$, the Künneth arrow 1.6.4 is defined (use the fact that finite relative tor-dimension is preserved by direct image (SGA 6 III 3.7.1)) and is an isomorphism (to verify this, one may assume S and the Y_i affine, and, for a Čech calculation, reduce to also assuming X_1 and X_2 affine; the conclusion is then a trivial consequence of the tor-independence hypotheses). On the other hand, for $M_i \in \operatorname{ob} D(Y_i)_{\mathrm{parf}/S}$, with $f_1^!M_1$ or $f_2^!M_2$ in D^b, the Künneth arrow 1.7.3 is defined[9], and is an isomorphism when f_1 and f_2 have finite tor-dimension, as is easily checked with the help of the Künneth isomorphism of type 1.6.4 just indicated, conjugated with 6.1.2, and with 6.1.1 in the case where f_i is a closed immersion. In particular, if X_1 and X_2 have finite tor-dimension over S and are tor-independent, 1.7.6 is an isomorphism.

One defines $\operatorname{RHom}(u, v)$ as in 2.1 for (u, v) of type $(2, 1)$ (respectively $(3, 4)$, with f proper in order to avoid the pro-objects caused by an $f_!$). Under suitable degree hypotheses (see for example ([9] V 2.3)), the evaluation arrows (2.2.2) and tensor-product arrows (2.2.3, 2.2.4) are defined, and the arrows 2.2.3 and 2.2.4 (provided they are defined) are isomorphisms as soon as E_1 and E_2 are perfect in the absolute sense.

In the situation of 2.4, suppose that the pairs (X_1, X_2), (X_1, Y_2), (Y_1, X_2), (Y_1, Y_2) are tor-independent over S, that f_1 is proper, that E_i, F_i, P_i, Q_i, $\operatorname{RHom}(E_i, F_i)$, $\operatorname{RHom}(P_i, Q_i)$ are perfect relative to S, and that the Künneth arrows

$$f_1^!Q_1 \otimes_S^{\mathbf{L}} Q_2 \to (f_1 \times_S Y_2)^!(Q_1 \otimes_S^{\mathbf{L}} Q_2), \qquad f_1^!Q_1 \otimes_S^{\mathbf{L}} F_2 \to (f_1 \times_S Y_2)^!(Q_1 \otimes_S^{\mathbf{L}} F_2)$$

are isomorphisms. Then 2.4.1 and the diagram of 2.5 are defined, and the latter is commutative (however, the horizontal arrows are not necessarily isomorphisms). The verification is the same as for 2.5.

6.4

Let, for $i = 1, 2$, X_i be an S-scheme, let $X = X_1 \times_S X_2$, and let $p_i : X \to X_i$, $q_i : X_i \to S$ be the canonical projections. Suppose that X_1 and X_2 are tor-independent and that q_1 has finite

[9] In the proper or lissifiable case, the definition of the arrow is immediate; the general case can be reduced to the lissifiable case by cohomological descent.

tor-dimension. Let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. By the analogue of 2.2.4, one has a canonical arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \mathrm{RHom}(L_1 \otimes^{\mathbf{L}} \mathcal{O}_{X_2}, q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2), \tag{6.4.1}$$

and hence, by composing with the Künneth isomorphism (6.3) $q_1^! \mathcal{O}_S \otimes_S^{\mathbf{L}} L_2 \xrightarrow{\sim} p_2^! L_2$, an arrow

$$DL_1 \otimes_S^{\mathbf{L}} L_2 \longrightarrow \mathrm{RHom}(p_1^* L_1, p_2^! L_2). \tag{6.4.2}$$

The arrow 6.4.1, and therefore also 6.4.2, is an isomorphism, as one sees by reducing to the case where q_1 is a closed immersion.

As in 3.1.2, we shall put

$$\mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}_S(L_1, L_2), \qquad \mathrm{Hom}(p_1^* L_1, p_2^! L_2) = \mathrm{Hom}_S(L_1, L_2), \tag{6.4.3}$$

and call the elements of $\mathrm{Hom}_S(L_1, L_2)$ *cohomological correspondences from L_1 to L_2*. If $c : C \to X$ is a correspondence (proper in practice), then by the induction formula ([8] III 8.8)[10]

$$c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2) = \mathrm{RHom}(c_1^* L_1, c_2^! L_2), \tag{6.4.4}$$

and the elements of

$$\mathrm{Hom}(c_1^* L_1, c_2^! L_2) = H^0(X, c_* c^! \mathrm{RHom}(p_1^* L_1, p_2^! L_2))$$

will be called *cohomological correspondences from L_1 to L_2 with support in C*. We shall see below examples analogous to those of 3.2.

6.5

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be a proper S-morphism, and let $f = f_1 \times_S f_2 : X \to Y$. Suppose that the X_i and Y_i have finite tor-dimension over S, and that the squares of 2.4.0 are tor-independent. Let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. As in 3.3.1, one defines an isomorphism

$$f_* \mathrm{RHom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{RHom}_S(f_{1*} L_1, f_{2*} L_2), \tag{6.5.1}$$

hence, by applying $R\Gamma(Y, -)$, an isomorphism

$$f_* : \mathrm{Hom}_S(L_1, L_2) \xrightarrow{\sim} \mathrm{Hom}_S(f_{1*} L_1, f_{2*} L_2), \tag{6.5.2}$$

which we shall simply denote by $u \mapsto u_*$ when $Y_1 = Y_2 = S$. There is a commutative diagram analogous to 3.4.1, as follows from the analogue of 2.5 indicated in 6.3. One deduces from it a corollary analogous to 6.5 and, in the case where $Y_1 = Y_2 = S$ is the spectrum of a field, a description, analogous to that of 3.6, of the operations u_* on cohomology.

Given a commutative square 3.7.1, with c proper, one has, for $K \in \mathrm{ob}\, D^+(C)$, a canonical arrow

$$f_* c_* c^! K \longrightarrow d_* d^! f_* K, \tag{6.5.3}$$

defined in the same way as 3.7.3 (the isomorphism $g'_* c^! \xrightarrow{\sim} d^! f_*$ is also valid in the present context: definition of the arrow by transposition of a base-change arrow, and verification that it is an isomorphism by reduction to the cases where d is smooth and where d is a closed immersion). Taking $K = \mathrm{RHom}_S(L_1, L_2)$, one deduces from 6.5.3 and 6.5.1 canonical arrows

$$f_* c_* \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \longrightarrow d_* \mathrm{RHom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \tag{6.5.4}$$

$$f_* : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \longrightarrow \mathrm{Hom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \tag{6.5.5}$$

analogous to 3.7.5 and 3.7.6.

[10]The "lissifiability" hypothesis of (loc. cit.) is unnecessary.

6.6

Let, for $i = 1, 2$, X_i be S-schemes of finite tor-dimension and tor-independent over S, and let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. The external tensor product of the evaluation arrows $DL_i \otimes^{\mathbf{L}} L_i \to K_{X_i}$ defines, as in 4.1, "cup-product" pairings analogous to 4.1.2, 4.1.3, and 4.1.4.

Let $c : C \to X$ and $d : D \to X$ be proper morphisms, let $e = c \times_X d : E \to X$, and let $P, Q \in \mathrm{ob}\, D(X)_{\mathrm{parf}/S}$. By 6.1.1, one has

$$c_* c^! P = \mathrm{RHom}(Rc_* \mathcal{O}_C, P)$$

(and an analogous isomorphism with d). Assuming c and d of finite tor-dimension at the points above $e(E)$, one deduces, by tensor product, a canonical arrow

$$c_* c^! P \otimes^{\mathbf{L}} d_* d^! Q \longrightarrow \mathrm{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, P \otimes^{\mathbf{L}} Q). \tag{6.6.1}$$

When c and d are tor-independent, the second member is rewritten, by Künneth and 6.1.1, as $e_* e^! (P \otimes^{\mathbf{L}} Q)$, and 6.6.1 takes the form 4.2.3. For $P = DL_1 \otimes^{\mathbf{L}}_S L_2$, $Q = L_1 \otimes^{\mathbf{L}}_S DL_2$, one deduces from 6.6.1 and the pairings indicated above pairings analogous to 4.2.4 and 4.2.5:

$$c_* \mathrm{RHom}(c_1^* L_1, c_2^! L_2) \otimes^{\mathbf{L}} d_* \mathrm{RHom}(d_2^* L_2, d_1^! L_1) \longrightarrow \mathrm{RHom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X), \tag{6.6.2}$$

$$\langle\ ,\ \rangle : \mathrm{Hom}(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}(d_2^* L_2, d_1^! L_1) \longrightarrow \mathrm{Hom}(Rc_* \mathcal{O}_C \otimes^{\mathbf{L}} Rd_* \mathcal{O}_D, K_X). \tag{6.6.3}$$

Just like the pairings of 4.2, these are compatible with étale localization. When c and d are tor-independent, the second member of 6.6.2 (respectively 6.6.3) identifies canonically with $e_* K_E$ (respectively $H^0(E, K_E)$).

Theorem 4.4 has the following analogue:

Theorem 6.7. Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be a proper S-morphism, let $f = f_1 \times_S f_2 : X \to Y$, and let $p_i : X \to X_i$, $q_i : Y \to Y_i$ be the projections. Suppose given, on the other hand, a cube 4.4.0 with cartesian horizontal faces, with c', c'', d', d'' proper, and with c', c'' (respectively d', d'') of finite tor-dimension at the points above $c(C)$ (respectively $d(D)$). Put $c_i' = p_i c'$, $c_i'' = p_i c''$, $d_i' = q_i d'$, and $d_i'' = q_i d''$. Finally, let $L_i \in \mathrm{ob}\, D(X_i)_{\mathrm{parf}/S}$. Assume that the pairs (X_1, X_2), (X_1, Y_2), (Y_1, X_2), (Y_1, Y_2) are tor-independent over S, and that X_i and Y_i have finite tor-dimension over S. Then one has a commutative square

$$\begin{CD}
f_* c'_* \mathrm{RHom}(c_1'^* L_1, c_2'^! L_2) \otimes f_* c''_* \mathrm{RHom}(c_2''^* L_2, c_1''^! L_1) @>{(1)}>> f_* \mathrm{RHom}(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''}, K_X) \\
@V{(2)}VV @VV{(4)}V \\
d'_* \mathrm{RHom}(d_1'^* M_1, d_2'^! M_2) \otimes d''_* \mathrm{RHom}(d_2''^* M_2, d_1''^! M_1) @>{(3)}>> \mathrm{RHom}(Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''}, K_Y),
\end{CD} \tag{6.7.1}$$

where $M_i = f_{i!} L_i$, (3) is given by 6.6.2, (1) is deduced from 6.6.2 by application of f_* and composition with the canonical arrow $f_* c'_* \otimes f_* c''_* \to f_*(c'_* \otimes c''_*)$, (2) is the tensor product of the arrows 6.5.4 relative to the faces containing f, and (4) is the composite of the duality isomorphism 6.1.1 and of the arrow defined by the canonical arrow

$$Rd'_* \mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_* \mathcal{O}_{D''} \longrightarrow f_*(Rc'_* \mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_* \mathcal{O}_{C''})$$

which is adjoint to the tensor product of the base-change arrows $f^* Rd'_* \mathcal{O}_{D'} \to Rc'_* \mathcal{O}_{C'}$, $f^* Rd''_* \mathcal{O}_{D''} \to Rc''_* \mathcal{O}_{C''}$.

The proof is essentially the same as that of 4.4; we leave to the reader the small modifications to be made.

When c' and c'' are tor-independent, and likewise d' and d'', the arrow (4) of 6.7.1 is rewritten in the form $f_*c_*K_C \to d_*K_D$; it is simply defined by the adjunction arrow $g_*K_C = g_*g^!K_D \to K_D$ (as is arrow (4) of 4.4.1). However, it would not be natural to consider only this case, since that would exclude correspondences with excess intersection.

From 6.7 one deduces formulas analogous to 4.5.1 and 4.7.1: under the hypotheses of 6.7, one has, for $u \in \mathrm{Hom}(c_1'^*L_1, c_2'^!L_2)$ and $v \in \mathrm{Hom}(c_2''^*L_2, c_1''^!L_1)$,

$$\langle f_*u, f_*v\rangle = f_*\langle u, v\rangle, \tag{6.7.2}$$

where f_*u (respectively f_*v) denotes the direct image correspondence of u (respectively v) in the sense of 6.5.5, and f_* in the second member is the homomorphism

$$\mathrm{Hom}(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''}, K_X) \longrightarrow \mathrm{Hom}(Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''}, K_Y) \tag{6.7.3}$$

which is the composite of the duality isomorphism 6.1.1 and the arrow induced by the canonical arrow

$$Rd'_*\mathcal{O}_{D'} \otimes^{\mathbf{L}} Rd''_*\mathcal{O}_{D''} \longrightarrow f_*(Rc'_*\mathcal{O}_{C'} \otimes^{\mathbf{L}} Rc''_*\mathcal{O}_{C''})$$

cited in 6.7. When $Y_1 = Y_2 = S = D' = D''$, 6.7.2 is the *Lefschetz-Verdier formula.*

As an illustration, we shall show that this formula implies the Woods Hole formula [1].

6.8

For this purpose we shall need a few reminders on the "Hodge" cohomology class associated with a regularly embedded subscheme of a smooth scheme and on Grothendieck residues ([8] III 9). Let $f : X \to S$ be a smooth morphism purely of relative dimension N, and let $i : Y \hookrightarrow X$ be a regular immersion of codimension d, i.e. ((SGA IV 16.9.2), (SGA 6 VII 1)) a closed subscheme defined locally by a regular sequence of d equations, or, in equivalent terminology ((8 III 1), (SGA 6 VIII 1)), a closed immersion, locally a complete intersection, of codimension d. Put $g = fi$. Then one associates with Y an element

$$\mathrm{cl}^d_{X/S}(Y) \in \mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}), \tag{6.8.1}$$

called the *cohomology class associated with* Y, defined as follows. By the "purity" theorem ([8] III 7.2), one has

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}) = \begin{cases} 0, & i \neq d, \\ \mathrm{Hom}(\bigwedge^d N_{Y/X}, \Omega^d_{X/S} \otimes \mathcal{O}_Y), & i = d, \end{cases} \tag{6.8.2}$$

where $N_{Y/X} = I/I^2$ is the conormal sheaf ($I = \mathrm{Ker}\ \mathcal{O}_X \to \mathcal{O}_Y$). Therefore

$$\mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S}) = H^0(Y, \underline{\mathrm{Hom}}(\bigwedge^d N_{Y/X}, \Omega^d_{X/S} \otimes \mathcal{O}_Y)). \tag{6.8.3}$$

The differential $d_{X/S} : \mathcal{O}_X \to \Omega^1_{X/S}$ induces an $\mathcal{O}_Y$-linear map

$$d_{X/S} \otimes 1 : N_{Y/X} \longrightarrow \Omega^1_{X/S} \otimes \mathcal{O}_Y.$$

The d-th exterior power of this map is a global section, on Y, of $\Lambda^d N^\vee_{Y/X} \otimes \Omega^d_{X/S}$; its image in $\mathrm{Ext}^d_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^d_{X/S})$ under the isomorphism 6.8.3 is, by definition, the class 6.8.1.

Assume g finite and flat (hence $d = N$). Let $\omega \in H^0(Y, \Lambda^N N^\vee_{Y/X} \otimes \Omega^N_{X/S})$. By 6.8.2 and 6.1, one may view ω as a homomorphism

$$\mathcal{O}_Y \longrightarrow g^! \mathcal{O}_S (= \Lambda^N N^\vee_{Y/X} \otimes \Omega^N_{X/S}),$$

and therefore, by adjunction, as a homomorphism $g_* \mathcal{O}_Y \to \mathcal{O}_S$. The residue (relative to g) of ω is defined as the section of $\mathcal{O}_S$ which is the image of 1 under this homomorphism, and is denoted $\mathrm{Res}_{Y/S}(\omega)$ (cf. ([8] III 9)). Let, on the other hand, $i_*\omega \in H^N(X, \Omega^N_{X/S})$ denote the image of ω by the composite of the isomorphism 6.8.3 and the canonical arrow

$$\mathrm{Ext}^N_{\mathcal{O}_X}(\mathcal{O}_Y, \Omega^N_{X/S}) \longrightarrow H^N(X, \Omega^N_{X/S}).$$

When f is proper, it follows from transitivity for trace morphisms (or adjunction arrows) $i_* i^! \to 1$, $f_* f^! \to 1$, that

$$\mathrm{Res}_{Y/S}(\omega) = \mathrm{Tr}_f(i_*\omega), \tag{6.8.4}$$

where $\mathrm{Tr}_f : H^N(X, \Omega^N_{X/S}) \to H^0(S, \mathcal{O}_S)$ is defined by the adjunction arrow $f_* f^! \mathcal{O}_S = Rf_* \Omega^N_{X/S}[N] \to \mathcal{O}_S$ (6.1). On the other hand, without any properness hypothesis, one has the important compatibility ([8] III 9 R6)

$$\mathrm{Res}_{Y/S}(y \,\mathrm{cl}^N_{X/S}(Y)) = \mathrm{Tr}_{Y/S}(y), \tag{6.8.5}$$

for $y \in H^0(Y, \mathcal{O}_Y)$, where $\mathrm{Tr}_{Y/S} : g_* \mathcal{O}_Y \to \mathcal{O}_S$ denotes the trace in the classical sense, i.e. $\mathrm{Tr}_{Y/S}(y)$ is the trace of the endomorphism of $g_* \mathcal{O}_Y$ defined by multiplication by y (this compatibility is not proved in loc. cit.; the case $N = 1$ is treated in [Raynaud, Anneaux locaux henséliens, VII 1, Lecture Notes in Math. n° 169, Springer Verlag, 1970]). In other words, $\mathrm{Tr}_{Y/S}$ corresponds, by adjunction, to the homomorphism $\mathcal{O}_Y \to g^! \mathcal{O}_S$ defined by multiplication by $\mathrm{cl}^N_{X/S}(Y)$. It follows that, for $F \in \mathrm{ob}\, D(S)_{\mathrm{coh}}$, the arrow

$$g^* F \longrightarrow g^! F = g^* F \otimes^{\mathbf{L}} g^! \mathcal{O}_S \tag{6.8.6}$$

given by multiplication by $\mathrm{cl}^N_{X/S}(Y)$ corresponds by adjunction to the arrow

$$g_* g^* F = F \otimes^{\mathbf{L}} g_* \mathcal{O}_Y \longrightarrow F \tag{6.8.7}$$

defined by $\mathrm{Tr}_{Y/S} : g_* \mathcal{O}_Y \to \mathcal{O}_S$.

6.9

Let X (respectively Y) be a proper smooth S-scheme of relative dimension m (respectively n), $Z = X \times_S Y$, and let $a : A \hookrightarrow Z$, $b : B \hookrightarrow Z$ be regular immersions such that $a_2 : A \to Y$, $b_1 : B \to X$ are finite and flat (write $p_1 : Z \to X$, $p_2 : Z \to Y$ for the projections, and $a_i = p_i a$, $b_i = p_i b$). Let $c : C = A \times_Z B \to Z$, $L \in \mathrm{ob}\, D(X)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D(Y)_{\mathrm{parf}}$ (the subscript parf means "perfect in the absolute sense", which is also equivalent to being perfect rel. to S, since X and Y are smooth). Assume that C is finite and flat over S. The hypotheses made therefore imply that c is a regular immersion and that a and b are tor-independent.

One has $p_1^* \Omega^1_{X/S} = \Omega^1_{Z/Y}$, $p_2^* \Omega^1_{Y/S} = \Omega^1_{Z/X}$, and a decomposition

$$\Omega^1_{Z/S} = \Omega^1_{Z/Y} \oplus \Omega^1_{Z/X},$$

whence, for every integer r, a decomposition

$$\Omega^r_{Z/S} = \bigoplus_{i+j=r} \Omega^i_{Z/Y} \otimes \Omega^j_{Z/X}. \tag{6.9.0}$$

On the other hand, by 6.1.2, one has $p_2^! M = p_2^* M \otimes \Omega^m_{Z/Y}[m]$, hence

$$H^0(Z, a_* a^! \mathrm{RHom}(p_1^* L, p_2^! M)) = \mathrm{Ext}^m_{\mathcal{O}_Z}(a_1^* L, a_2^* M \otimes \Omega^m_{Z/Y}),$$

whence, by 6.4.4 and 6.8.2, an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) = H^0(A, \mathrm{RHom}(a_1^* L, a_2^* M) \otimes \Lambda^m N^\vee_{A/Z} \otimes \Omega^m_{Z/Y}). \tag{6.9.1}$$

Similarly one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) = H^0(B, \mathrm{RHom}(b_2^* M, b_1^* L) \otimes \Lambda^n N^\vee_{B/Z} \otimes \Omega^n_{Z/X}). \tag{6.9.2}$$

Let $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$, and write

$$\langle u, v \rangle \in H^0(C, \mathcal{O}_C) \tag{6.9.3}$$

for the section $\mathrm{Tr}(u_C v_C) = \mathrm{Tr}(v_C u_C)$, where $u_C \in \mathrm{Hom}(c_1^* L, c_2^* M)$ and $v_C \in \mathrm{Hom}(c_2^* M, c_1^* L)$ are induced by u and v. Consider the cohomological correspondence $u^! \in \mathrm{Hom}(a_1^* L, a_2^! M)$ (respectively $v^! \in \mathrm{Hom}(b_2^* M, b_1^! L)$) deduced from u (respectively v) by composition with the trace homomorphism 6.8.7 $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$), or equivalently (6.8.6) by product with $\mathrm{cl}_{Z/Y}(A)$ (respectively $\mathrm{cl}_{Z/X}(B)$). Note in passing that $\mathrm{cl}_{Z/Y}(A)$ (respectively $\mathrm{cl}_{Z/X}(B)$) is the component in $H^0(A, \Lambda^m N^\vee_{A/Z} \otimes \Omega^m_{Z/Y})$ (respectively $H^0(B, \Lambda^n N^\vee_{B/Z} \otimes \Omega^n_{Z/X})$) of $\mathrm{cl}_{Z/S}(A)$ (respectively $\mathrm{cl}_{Z/S}(B)$) defined by the projection $\Omega^m_{Z/S} \to \Omega^m_{Z/Y}$ (respectively $\Omega^n_{Z/S} \to \Omega^n_{Z/X}$) of 6.9.0. One checks easily that the cup-product

$$\langle u^!, v^! \rangle \in \mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) = H^0(C, \Lambda^{m+n} N^\vee_{C/Z} \otimes \Omega^{m+n}_{Z/S})$$

defined in 6.6.3 is given by

$$\langle u^!, v^! \rangle = \langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B). \tag{6.9.4}$$

On the other hand one checks that the homomorphism 6.7.3

$$\mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) \longrightarrow H^0(S, \mathcal{O}_S)$$

is the composite of the canonical arrow $\mathrm{Ext}^{m+n}_{\mathcal{O}_Z}(\mathcal{O}_C, \Omega^{m+n}_{Z/S}) \to H^{m+n}(Z, \Omega^{m+n}_{Z/S})$ and the trace morphism $\mathrm{Tr}_{Z/S} : H^{m+n}(Z, \Omega^{m+n}_{Z/S}) \to H^0(S, \mathcal{O}_S)$. The Lefschetz-Verdier formula 6.7.2 therefore gives the relation

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Tr}_{Z/S}(\langle u, v \rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B)). \tag{6.9.5}$$

Thanks to 6.8.4 one may rewrite the second member as a residue; thus, in summary, one has obtained:

Theorem 6.10. Under the hypotheses of the first paragraph of 6.9, let $u \in \mathrm{Hom}(a_1^* L, a_2^* M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^* L)$, and consider the cohomological correspondence $u^!$ (respectively $v^!$) from L to M (respectively from M to L), with support in A (respectively B), deduced from u (respectively v) by composition with the trace $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$) (6.8.7), and the homomorphism $(u^!)_* : f_* L \to g_* M$ (respectively $(v^!)_* : g_* M \to f_* L$) deduced from $u^!$ (respectively $v^!$) by direct image (6.5.5) ($f : X \to S$, $g : Y \to S$ denoting the projections). Then, with the notation of 6.8, 6.9.3, one has

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Res}_{C/S}(\langle u, v\rangle \, \mathrm{cl}_{Z/Y}(A) \, \mathrm{cl}_{Z/X}(B)). \tag{6.10.1}$$

Remarks 6.11. a) As in 3.6, using the description 3.5 b) of the direct image of a correspondence, one checks that, if $h : Z \to S$ is the projection, $(u^!)_*$ (respectively $(v^!)_*$) is the composite of the arrows

$$f_*L \xrightarrow{(1)} (ha)_* a_1^* L \xrightarrow{(ha)_* u} (ha)_* a_2^* M = g_* a_{2*} a_2^* M \xrightarrow{(2)} g_* M,$$

$$(\text{respectively } g_* M \xrightarrow{(1)} (hb)_* b_2^* M \xrightarrow{(hb)_* v} (hb)_* b_1^* L = f_* b_{1*} b_1^* L \xrightarrow{(2)} f_* L),$$

where (1) is the usual functoriality arrow (defined by the adjunction arrow $1 \to a_{1*} a_1^*$ (respectively $1 \to b_{2*} b_2^*$)) and (2) is given by the trace $a_{2*} a_2^* M \to M$ (respectively $b_{1*} b_1^* L \to L$) (6.8.7).

b) Suppose that S is the spectrum of a field. The intersection $C = A \times_{(X \times_S Y)} B$ is then formed of isolated points. At each point z of C, choose, near the images of z, regular sequences $(s_1, \ldots, s_m)$, $(t_1, \ldots, t_n)$ of equations of A, B. Then 6.10.1 is written, with the notation of the residue symbols of ([8] III 9), as

$$\langle (u^!)_*, (v^!)_* \rangle = \sum_{z \in C} \mathrm{Res}_z \left(\langle u, v \rangle \prod_{1 \le i \le m} \frac{d_{Z/Y} s_i}{s_i} \prod_{1 \le j \le n} \frac{d_{Z/X} t_j}{t_j} \right), \tag{6.11.1}$$

where $d_{Z/Y} s_i$ (respectively $d_{Z/X} t_j$) denotes the component of type $(1, 0)$ (respectively $(0, 1)$) of $d_{Z/S} s_i$ (respectively $d_{Z/S} t_j$) in the decomposition 6.9.0.

Corollary 6.12. (Woods-Hole formula [1]). Assume that S is the spectrum of a field. Let X be a proper smooth S-scheme, F an S-endomorphism of X, L an object of $D(X)_{\mathrm{parf}}$ (i.e. a perfect complex on X), and $u \in \mathrm{Hom}(F^* L, L)$. Assume that the scheme of fixed points X^F is finite and consists of transverse points (i.e. points where the graph of F meets the diagonal transversely). Then

$$\sum_i (-1)^i \, \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \frac{\mathrm{Tr}(u_x)}{\det(1 - dF_x)},$$

where (F, u) is the endomorphism of $H^i(X, L)$ which is the composite of $H^i(X, L) \to H^i(X, F^* L)$ (inverse image) and $H^i(X, u)$, $\mathrm{Tr}(u_x)$ denotes the value at x of the endomorphism induced by u on the fibre of L at x, and $\det(1 - dF_x)$ is the value at x of the determinant of the map $(1 -$ map induced by F on $\Omega^1_{X,x})$.

Proof. Apply 6.10, taking account of 6.11 a), b). Put $Z = X \times_S X$. Let $x \in X^F$. Near x, choose a system of local coordinates $(x_1, \ldots, x_n)$ (defining an étale morphism into $\mathbb{A}^n_S$), put $F_i = x_i F$, and write $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ for the corresponding local coordinates on Z near $z = (x, x)$ ($x_i = x_i \otimes 1$, $y_i = 1 \otimes x_i$). Then, near z, $(x_1 - y_1, \ldots, x_n - y_n)$ (respectively $(y_1 - F_1, \ldots, y_n - F_n)$) is a regular system of equations for the diagonal A (respectively for the graph B of F). The contribution at z of the second member of 6.11.1 is

$$\langle u, 1 \rangle_z = \mathrm{Res}_z \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n \, dy_1 \cdots dy_n}{(x_1 - y_1) \cdots (x_n - y_n)(y_1 - F_1) \cdots (y_n - F_n)} \right).$$

Observe that in the numerator one may replace dy_i by

$$dy_i - dF_i = dy_i - \sum_{1 \le j \le n} (dF_i / dx_j) dx_j.$$

Applying then formula ([8] III 9 R3) to the situation $(B \hookrightarrow Z \to S)$, $(s_1, \ldots, s_p) = (y_1 - F_1, \ldots, y_n - F_n)$, $(t_1, \ldots, t_n) = (x_1 - y_1, \ldots, x_n - y_n)$, $\omega = (\mathrm{Tr}(u) dx_1 \cdots dx_n)$, one obtains

$$\langle u, 1 \rangle_z = \mathrm{Res}_x \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right). \tag{*}$$

Since x is transverse, the $x_i - F_i$ form a regular system of parameters of $\mathcal{O}_{X,x}$, and hence one may write, near x,

$$x_i = \sum_j (x_j - F_j) c_{ij},$$

where (c_{ij}) is a matrix of sections of $\mathcal{O}_X$ invertible at x, whose determinant has value at x equal to $\det(1 - dF_x)^{-1}$. Thanks to ([8] III 9 R1), one therefore deduces from (*) that

$$\langle u, 1 \rangle_z = \mathrm{Res}_x \left(\mathrm{Tr}(u) \det(c_{ij}) \frac{dx_1 \cdots dx_n}{x_1 \cdots x_n} \right),$$

whence, by ([8] III 9 R6),

$$\langle u, 1 \rangle_z = \mathrm{Tr}(u(x)) \det(c_{ij}(x)) = \mathrm{Tr}(u(x)) \det(1 - dF_x)^{-1}.$$

The corollary therefore follows from 6.11.1.

Remark 6.12.1. If X^F is assumed only finite, the preceding calculation shows that one has

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \mathrm{Res}_x \left(\mathrm{Tr}(u) \frac{dx_1 \cdots dx_n}{(x_1 - F_1) \cdots (x_n - F_n)} \right),$$

where (x_i) is a system of local coordinates at x, and the F_i are the coordinates of F (near x) in this system.

The reader will convince himself, moreover, that the preceding formula is still valid if X is proper and is assumed only to be smooth at the points of X^F.

6.13

Assume that S is the spectrum of an algebraic closure of the finite field $\mathbb{F}_q$ and that X/S is obtained by extension of scalars from a proper smooth scheme $X_0/\mathbb{F}_q$. Take for F the Frobenius endomorphism of X/S "defined by raising coordinates to the q-th power". One has $X^F = X_0(\mathbb{F}_q)$, and X^F consists of transverse points since $dF = 0$. If L is a locally free sheaf of finite type on X and $u \in \mathrm{Hom}(F^*L, L)$, the Woods-Hole formula is therefore written as

$$\sum_i (-1)^i \mathrm{Tr}((F, u), H^i(X, L)) = \sum_{x \in X_0(\mathbb{F}_q)} \mathrm{Tr}(u_x, L_x). \tag{6.13.1}$$

In particular, for $L = \mathcal{O}_X$, and u the canonical isomorphism $F^*\mathcal{O}_X \xrightarrow{\sim} \mathcal{O}_X$, one finds:

$$\sum_i (-1)^i \mathrm{Tr}(F, H^i(X, \mathcal{O}_X)) = \mathrm{Card}(X_0(\mathbb{F}_q)) \pmod p. \tag{6.13.2}$$

Artin-Schreier theory makes it possible to rewrite the first member in the form $\sum_i (-1)^i \mathrm{Tr}(F, H^i(X, \mathbb{F}_p))$, where $H^*(X, \mathbb{F}_p)$ denotes the étale cohomology of X with values in the constant sheaf $\mathbb{F}_p$. By a refinement of this argument, Deligne, in (SGA $4^{1/2}$ Fonctions L), deduced from 6.13.1 the following generalization of 6.13.2:

Theorem 6.13.3 (Deligne). Let X_0 be a separated scheme of finite type over $\mathbb{F}_q$, let X be the scheme deduced from X_0 by extension of scalars to an algebraic closure k of $\mathbb{F}_q$, and let F be the Frobenius endomorphism of X/k. Then, for every $L_0 \in \operatorname{ob} D^b_{\mathrm{ctf}}(X_0, \mathbb{F}_p)$, if L denotes the inverse image of L_0 on X, one has

$$\sum_i (-1)^i \operatorname{Tr}(F, H^i_c(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(F, L_x).$$

Let us point out that the particular case of 6.13.3 corresponding to $L_0 = \mathbb{F}_p$ was established by Katz in (SGA 7 XXII) by a completely different method, based on Dwork's calculation of the zeta function of a hypersurface. Moreover, 6.13.2 in the proper smooth case is also an immediate consequence of the Lefschetz formula in crystalline cohomology ([5] VII 3.1.9). In fact, crystalline cohomology gives the more precise information that each factor $\det(1 - Ft, H^i(X_0/W)) \in W[t]$ of the zeta function is congruent mod p to $\det(1 - Ft, H^i(X_0, \mathcal{O}_{X_0}))$ (at least if $H^*(X_0/W)$ is torsion-free). Katz showed [11] that, using estimates of B. Mazur [14], one could obtain higher congruences (involving the $H^i(X_0, \Omega^j_{X_0})$ and the Cartier operator) for smooth hypersurfaces (or complete intersections) for which the Hodge numbers (in the interesting dimension) are assumed to vanish up to a certain level.

On the other hand, if X_0 is a proper scheme over $\mathbb{F}_q$, geometrically integral, and such that $H^i(X_0, \mathcal{O}_{X_0}) = 0$ for $i > 0$, the Lefschetz-Verdier formula 6.7.2, applied to $X = X_0 \otimes k$ and the Frobenius endomorphism of X, immediately implies that X_0 has at least one rational point over $\mathbb{F}_q$ (since $\operatorname{Tr}(F, R\Gamma(X, \mathcal{O}_X)) = 1$). Serre asks whether this conclusion is still true when one replaces $\mathbb{F}_q$ by a field satisfying (C_1) ([17] II 3), for example $\mathbb{C}(t)$ (Tsen), or $\mathbb{C}((t))$ (Lang) (see loc. cit. for other examples); indeed, he observes that if X_0 is a complete intersection of multidegree $(m_1, \ldots, m_r)$ in $\mathbb{P}^n$, the condition of triviality of cohomology is equivalent ([16] no. 78) to the condition $m_1 + \cdots + m_r < n + 1$, which is also the condition that occurs when one writes condition (C_1).

6.14

The Woods Hole formula has many other applications, for which we refer to [2] and to Beauville's exposé [4].

6.15

The editor has not examined the case of components of fixed points of dimension ≥ 1, but it is not impossible that one could deduce from 6.7.2 (probably conjugated with Riemann-Roch) Lefschetz-Riemann-Roch formulae in the style of those of Donovan [7], Iversen [10], Nielsen [15] (at least with values in the base field). Nor is it impossible, on the other hand, that the formula 6.7.2, applied over the dual numbers, might provide residue formulae for vector fields analogous to those of Bott [6], [3].

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Exposé III B
Calculations of Local Terms

by Luc Illusie

Introduction

This exposé, written in January 1977, does not correspond to any oral exposé of the seminar. It comprises two independent parts. In the first, we compute the Lefschetz–Verdier local terms for cohomological correspondences between smooth curves over an algebraically closed field, supported on correspondences satisfying certain general-position hypotheses. The formula that we obtain gives as corollaries Verdier's formula [6] for transversal endomorphisms (which applies in particular to the case of the Frobenius endomorphism), and the formula stated by Langlands in ([4] 7.12). The method of proof is parallel to that followed by Artin–Verdier in [6]. One reduces to proving the vanishing of the local term when the given complexes of sheaves have zero fibre at the images of the fixed point under consideration, and in fact one proves a stronger property, namely that a certain restriction map is zero (2.5). To establish this property, one first reduces to the tamely ramified case by covers of the same degree of the two given curves (so as not to destroy the general-position hypotheses), and then treats this case directly by means of Abhyankar's lemma and the relative purity theorem. The second part of this exposé, of a much more technical nature, is inspired by the method used by Grothendieck to establish the Lefschetz formula for certain cohomological correspondences on curves (see XII and (SGA $4^{1/2}$ Rapport)). In no. 5, for complexes of modules on a topos, we develop a theory of non-commutative traces which generalizes the commutative theory of (SGA 6 I) and, in the case of a punctual topos, recovers the non-commutative theory developed by Grothendieck in the oral seminar (this was written up in Exposé XI, unfortunately lost). We apply the techniques of no. 5 to define, in no. 6, Lefschetz–Verdier local terms for cohomological correspondences between complexes of modules over rings that are not necessarily commutative. These local terms behave like non-commutative traces and make it possible, in particular when one takes as base ring the algebra of a group, to divide canonically the ordinary local terms attached to equivariant correspondences. As an application, we prove Grothendieck's divisibility conjecture of (XII 4.5), in a more general form. We also show that the local terms defined by Grothendieck in the Lefschetz formula of XII are indeed the Lefschetz–Verdier local terms, and we generalize the formula of (loc. cit.).

I thank P. Deligne for the useful suggestions he made to me during many discussions; I am particularly grateful to him for having pointed out, and helped me correct, an error in an early proof of Theorem 1.2.

Summary

I. Correspondences in General Position between Curves

In this part, S denotes the spectrum of an algebraically closed field k of characteristic p, and Λ a noetherian commutative ring killed by an integer prime to p. For every scheme X over S we denote by $D(X)$ the derived category of sheaves of Λ-modules on X, and by $D_{\mathrm{ctf}}(X)$ the full subcategory of $D(X)$ formed by complexes of finite tor-dimension and constructible cohomology.

1. Statement of the theorem and corollaries

1.1

Let X (resp. Y) be the strict localization, at a closed point, of a smooth separated S-curve, let A (resp. B) be a strictly local S-scheme, and let

$$a : A \longrightarrow X \times_S Y, \qquad b : B \longrightarrow X \times_S Y$$

be S-morphisms. We denote by $p_1 : X \times_S Y \to X$, $p_2 : X \times_S Y \to Y$ the projections, put $a_i = p_i a$, $b_i = p_i b$, and denote by x (resp. y) the closed point of X (resp. Y). We assume that a_2 and b_1 are finite and surjective, and that $A \times_{(X \times_S Y)} B$ is reduced to the closed point z whose image in $X \times_S Y$ is (x, y).

Let $L \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, $M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$, $u \in \operatorname{Hom}(a_1^* L, a_2^! M)$, and $v \in \operatorname{Hom}(b_2^* M, b_1^! L)$. Denote by

$$u_z \in \operatorname{Hom}(L_x, M_y) \qquad (\text{resp. } v_z \in \operatorname{Hom}(M_y, L_x))$$

the composite homomorphism

$$L_x = R\Gamma(X, L) \xrightarrow{a_1^*} R\Gamma(A, a_1^* L) \xrightarrow{R\Gamma(A,u)} R\Gamma(A, a_2^! M) \xrightarrow{a_{2*}} R\Gamma(Y, M) = M_y,$$

where a_{2*} is defined by the adjunction map $a_{2*} a_2^! M \to M$ (resp. the analogous composite homomorphism defined by means of (b, v)). Since L_x and M_y are perfect complexes of Λ-modules, one can form the cup-product (SGA 6 I 8.3)

$$\langle u_z, v_z \rangle = \operatorname{Tr}(u_z v_z) = \operatorname{Tr}(v_z u_z) \in \Lambda. \tag{1.1.1}$$

Put $\Lambda_X(1)[2] = K_X$, $\Lambda_Y(1)[2] = K_Y$. By (I 5) (or (SGA $4^{1/2}$ Duality 1, or Finitude 4.1)) the functor $D_X = \operatorname{RHom}(-, K_X)$ (resp. $D_Y = \operatorname{RHom}(-, K_Y)$) on $D_{\mathrm{ctf}}(X)$ (resp. $D_{\mathrm{ctf}}(Y)$) is dualizing, i.e. for every $E \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$ (resp. $E \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$), one has $E \xrightarrow{\sim} D_X D_X E$ (resp. $E \xrightarrow{\sim} D_Y D_Y E$). We shall sometimes write D instead of D_X, D_Y.

On the other hand, denote by $p_i^!$ the functor $p_i^*(-)(1)[2]$. We have trivially

$$K_X \otimes_S^{\mathbf{L}} M \xrightarrow{\sim} p_2^! M, \qquad L \otimes_S^{\mathbf{L}} K_Y \xrightarrow{\sim} p_1^! L,$$

and it follows from (III 2.3), by passage to the limit, that the Künneth maps analogous to (III 3.1.1)

$$DL \otimes_S^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(p_1^* L, p_2^! M), \qquad L \otimes_S^{\mathbf{L}} DM \longrightarrow \mathrm{RHom}(p_2^* M, p_1^! L)$$

are isomorphisms. Since a_2 is finite, a is finite, and hence the functor $a^!$ is defined; one verifies easily that one has the transitivity isomorphism $a^! p_2^! = a_2^!$. By the induction formula for a (SGA 4 XVIII 3.1.12.2), one therefore has a canonical isomorphism analogous to (III 3.2.1)

$$a^! \mathrm{RHom}(p_1^* L, p_2^! M) \xrightarrow{\sim} \mathrm{RHom}(a_1^* L, a_2^! M),$$

whence an isomorphism

$$\mathrm{Hom}(a_1^* L, a_2^! M) \xrightarrow{\sim} H^0(X \times_S Y, a_* a^! (DL \otimes_S^{\mathbf{L}} M)). \tag{*}$$

Likewise one has an isomorphism

$$\mathrm{Hom}(b_2^* M, b_1^! L) \xrightarrow{\sim} H^0(X \times_S Y, b_* b^! (L \otimes_S^{\mathbf{L}} DM)). \tag{**}$$

As in (III 4.1.2), one defines a natural pairing

$$(DL \otimes_S^{\mathbf{L}} M) \otimes^{\mathbf{L}} (L \otimes_S^{\mathbf{L}} DM) \longrightarrow K_X \otimes_S^{\mathbf{L}} K_Y = \Lambda_{X \times_S Y}(2)[4] \overset{\mathrm{dfn}}{=} K_{X \times_S Y}.$$

Composing with the pairing analogous to (III 4.2.3)

$$a_* a^! P \otimes^{\mathbf{L}} b_* b^! Q \longrightarrow c_* c^! (P \otimes^{\mathbf{L}} Q),$$

where $P = DL \otimes_S^{\mathbf{L}} M$, $Q = L \otimes_S^{\mathbf{L}} DM$, and $c : Z = A \times_{(X \times_S Y)} B \to X \times_S Y$ is the canonical projection, and taking into account the isomorphisms (*), (**), one deduces a pairing analogous to (III 4.2.5)

$$\langle\ ,\ \rangle : \mathrm{Hom}(a_1^* L, a_2^! M) \otimes \mathrm{Hom}(b_2^* M, b_1^! L) \longrightarrow H^0(X \times_S Y, c_* c^! K_{X \times_S Y}).$$

One verifies easily, by passage to the limit, that $c^! K_{X \times_S Y} = \Lambda_z$, so that $H^0(X \times_S Y, c_* c^! K_{X \times_S Y}) = \Lambda$. Thus one has a "cup-product"

$$\langle u, v \rangle \in \Lambda. \tag{1.1.2}$$

1.1.3

Let X' (resp. Y') be a smooth S-curve, A' (resp. B') a separated S-curve,

$$a' : A' \longrightarrow X' \times_S Y', \qquad b' : B' \longrightarrow X' \times_S Y'$$

finite S-morphisms, x' (resp. y') a closed point of X' (resp. Y'), z' an isolated closed point of $A' \times_{(X' \times_S Y')} B'$, with image s' (resp. t') in A' (resp. B'), and such that $a_2'(= p_2 a')$ (resp. $b_1'(= p_1 b')$) is quasi-finite at s' (resp. t'),

$$L' \in \mathrm{ob}\, D_{\mathrm{ctf}}(X'), \quad M' \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y'), \quad u' \in \mathrm{Hom}(a_1'^* L', a_2'^! M'), \quad v' \in \mathrm{Hom}(b_2'^* M', b_1'^! L').$$

One can then consider the local term (III 4.2.7) $\langle u', v'\rangle_{z'} \in \Lambda$. On the other hand, if X, Y, A, B are the strict localizations of X', Y', A', B' at x', y', s', t', if $a : A \to X \times_S Y$, $b : B \to X \times_S Y$ are the morphisms deduced from a', b', if $L = L'|X$, $M = M'|Y$, and if $u \in \operatorname{Hom}(a_1^* L, a_2^! M)$, $v \in \operatorname{Hom}(b_2^* M, b_1^! L)$ are the maps deduced from u', v', then the data (X, Y, a, b, u, v) satisfy the hypotheses at the beginning of 1.1, and hence a cup-product 1.1.2 $\langle u, v\rangle \in \Lambda$ is defined. It follows from the compatibility of the formation of local terms (III 4.2.7) with localization (III 4.2.6) that $\langle u', v'\rangle_{z'} = \langle u, v\rangle$. It is also easy to see that any data (X, Y, a, b, u, v) satisfying the hypotheses at the beginning of 1.1 are induced by data $(X', Y', a', b', x', y', z', s', t', u', v')$ as above, where one may also assume, if desired, that $A' \times_{(X' \times_S Y')} B'$ is reduced to z'.

In general the cup-products 1.1.1 and 1.1.2 are unequal. Nevertheless one has the following theorem, which is the main result of this first part:

Theorem 1.2. Under the hypotheses at the beginning of 1.1, assume moreover that A and B are in general position, i.e. that the intersection of the tangent cones at (x, y) to the images of A and B in $X \times_S Y$ is reduced to (x, y), and that likewise A and $X \times \{y\}$ are in general position, as are $\{x\} \times Y$ and B. Then

$$\langle u_z, v_z\rangle = \langle u, v\rangle. \tag{1.2.1}$$

By the Lefschetz–Verdier formula (III 4.7), taking 1.1.3 into account, one deduces:

Corollary 1.3. Let F be an endomorphism of a proper smooth curve X/S such that the fixed-point scheme X^F is finite and formed of points where the graph of F is transverse to the diagonal. Let $L \in \operatorname{ob} D_{\mathrm{ctf}}(X)$, and $u \in \operatorname{Hom}(F^* L, L)$. Then

$$\operatorname{Tr}((F, u), R\Gamma(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(u_x, L_x),$$

where the trace on both sides is taken in the sense of (SGA 6 I 8).

For L concentrated in degree 0, 1.3 is the result of Artin–Verdier ([6] 4.1). By a passage to the limit explained in XV, one deduces from 1.3 an analogous result in ℓ-adic cohomology ([6] 1.1), (XV §2 no. 3 prop. 2):

Corollary 1.3.1. With X and F as in 1.3, let L be a constructible $\mathbb{Q}_\ell$-sheaf on X (ℓ prime to p), and let $u \in \operatorname{Hom}(F^* L, L)$. Then:

$$\sum_i (-1)^i \operatorname{Tr}((F, u), H^i(X, L)) = \sum_{x \in X^F} \operatorname{Tr}(u_x, L_x).$$

The hypotheses of 1.3 and 1.3.1 apply in particular to the case where (F, u) is a Frobenius correspondence (XV 2), whence Grothendieck's trace formula ([2] 5.1), (SGA $4^{1/2}$ Rapport 3.2).

1.4

With the notation of 1.1, assume that b_2 and a_1 are finite and surjective (instead of a_2 and b_1), and consider the dual correspondences (III 5.1)

$$Du \in \operatorname{Hom}(a_2^* DM, a_1^! DL), \qquad Dv \in \operatorname{Hom}(b_1^* DL, b_2^! DM),$$

and the homomorphisms $(Du)_z \in \mathrm{Hom}((DM)_y, (DM)_x)$, $(Dv)_z \in \mathrm{Hom}((DL)_x, (DM)_y)$. Assume that $(A, \{x\} \times Y)$ is in general position, as are $(X \times \{y\}, B)$ and (A, B). Applying 1.2 to Du, Dv, one obtains

$$\langle Du, Dv\rangle = \langle (Du)_z, (Dv)_z\rangle,$$

whence, by (III 5.1.6) and (III 4.2.6),

$$\langle u, v\rangle = \langle (Du)_z, (Dv)_z\rangle. \tag{1.4.1}$$

From this remark and from 1.2 one deduces, by the Lefschetz–Verdier formula (III 4.7):

Corollary 1.5. Let X, Y be proper smooth curves over S,

$$a : A \longrightarrow X \times_S Y, \qquad b : B \longrightarrow X \times_S Y$$

correspondences where A and B are proper curves over S, let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(X)$, $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, and $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$. Assume that $C = A \times_{(X \times_S Y)} B$ is finite over S and of the form $C = C_1 \amalg C_2$, with the following properties: for $z \in C_1$, the morphisms deduced from a_2, b_1 by strict localization at the images of z are finite, and, if (x, y) denotes the image of z in $X \times_S Y$, each of the pairs (A, B), $(A, X \times \{y\})$, $(\{x\} \times Y, B)$ is in general position at (x, y); for $z \in C_2$, the morphisms deduced from a_1, b_2 by strict localization at the images of z are finite, and each of the pairs (A, B), $(A, \{x\} \times Y)$, $(B, X \times \{y\})$ is in general position at (x, y) ("general position" meaning that the intersection of the tangent cones at (x, y) to the images of the curves of the pair under consideration is reduced to (x, y)). Then, if $u_* \in \mathrm{Hom}(R\Gamma(X, L), R\Gamma(Y, M))$, $v_* \in \mathrm{Hom}(R\Gamma(Y, M), R\Gamma(X, L))$ are the morphisms defined by u, v (III 3.7.6), one has, with the notation of 1.1.1, (III 5.1.5),

$$\langle u_*, v_*\rangle = \sum_{z \in C_1} \langle u_z, v_z\rangle + \sum_{z \in C_2} \langle (Du)_z, (Dv)_z\rangle. \tag{1.5.1}$$

1.6

Assuming the formalism of ℓ-adic cohomology, one deduces from 1.5, by the passage-to-the-limit technique of XIV, that, under the hypotheses of 1.5 on X, Y, a, b, for $L \in \mathrm{ob}\, D^b(X, \mathbb{Q}_\ell)$ (ℓ prime $\neq p$), $M \in \mathrm{ob}\, D^b(Y, \mathbb{Q}_\ell)$, $u \in \mathrm{Hom}(a_1^* L, a_2^! M)$, $v \in \mathrm{Hom}(b_2^* M, b_1^! L)$, one has the analogue of 1.5.1 in $\mathbb{Q}_\ell$. This formula contains, as a special case, Proposition 7.12 of Langlands [4]: apply the formula with $X = Y$, M the sheaf F considered by Langlands; $L = \alpha M$ ($\alpha \in \mathbb{Q}_\ell^*$), A the diagonal of $X \times X$, b the correspondence Φ of (loc. cit.), u the correspondence supported on A given by $\alpha^{-1} : L \to M$, v the correspondence Φ of (loc. cit.); with the notation of (loc. cit.), C_1 (resp. C_2) is the set of fixed points where $a < d$ (resp. $a > d$).

1.7

Under the hypotheses of 1.2, take for u (resp. v) the correspondence $\mathrm{cl}(A)$ (resp. $\mathrm{cl}(B)$) defined by $\mathrm{Tr}_{a_2} : a_{2*}\Lambda \to \Lambda$ (resp. $\mathrm{Tr}_{b_1} : b_{1*}\Lambda \to \Lambda$). Formula 1.2.1 becomes

$$\deg(a_2)\deg(b_1) = \langle \mathrm{cl}(A), \mathrm{cl}(B)\rangle, \tag{1.7.1}$$

where $\deg(a_2)$ (resp. $\deg(b_1)$) denotes the generic degree of a_2 (resp. b_1). Taking into account the hypotheses of general position, the left hand side of 1.7.1 is none other than the intersection multiplicity at (x, y) of the cycles a_*A, b_*B, the direct images of A and B. It is not difficult to deduce 1.7.1 directly from the compatibility of the formation of the cohomology class associated with a cycle with intersections (SGA $4^{1/2}$ Cycle 2.3.8).

Exposé III B: Calculations of Local Terms

2. Reduction to a vanishing theorem

2.1

Under the hypotheses of 1.1, let s (resp. t) denote the image of z in A (resp. B), and let $i : U = X - \{x\} \hookrightarrow X$ and $j : V = Y - \{y\} \hookrightarrow Y$ be the canonical inclusions. Since a_2 and b_1 are finite, one has $a_2^{-1}(y) = s$, $b_1^{-1}(x) = t$, hence

$$a_1^{-1}(U) \subset a_2^{-1}(V), \qquad b_2^{-1}(V) \subset b_1^{-1}(U),$$

and there are commutative diagrams

$$\begin{tikzcd} U & a_1^{-1}(U) \arrow[l, "a_1'"'] \arrow[r, "a_2'"] \arrow[d] & V \arrow[d, "j"] \\ X & A \arrow[l, "a_1"] \arrow[r, "a_2"'] & Y \end{tikzcd} \qquad \begin{tikzcd} U & b_2^{-1}(V) \arrow[l, "b_1'"'] \arrow[r, "b_2'"] \arrow[d] & V \arrow[d, "j"] \\ X & B \arrow[l, "b_1"] \arrow[r, "b_2"'] & Y, \end{tikzcd}$$

where the squares corresponding to $U \leftarrow a_1^{-1}(U) \to A \to X$ and to $b_2^{-1}(V) \to V \to Y$ are cartesian.

Lemma 2.2. To prove 1.2, it is enough to consider the following two cases:

$$\text{a) } L_x = 0, \quad M_y = 0; \qquad \text{b) } L|U = 0, \quad M|V = 0.$$

Consider the decreasing filtrations on L and M defined by the subcomplexes

$$F^nL = \begin{cases} L, & n \leq 0, \\ i_!i^*L, & n = 1, \\ 0, & n > 1, \end{cases} \qquad F^nM = \begin{cases} M, & n \leq 0, \\ j_!j^*M, & n = 1, \\ 0, & n > 1. \end{cases}$$

The complex $a_{2*}a_1^*L$ (resp. $b_{1*}b_2^*M$) is then filtered by

$$F^n a_{2*}a_1^*L = a_{2*}a_1^*F^nL.$$

We have $F^n a_{2*}a_1^*L = a_{2*}a_1^*L$ for $n \leq 0$, and 0 for $n > 1$; moreover, since a_2 is finite and the square is cartesian,

$$F^1 a_{2*}a_1^*L = j_!(a_2')_!(a_1')^*i^*L.$$

Similarly,

$$F^n b_{1*}b_2^*M = \begin{cases} b_{1*}b_2^*M, & n \leq 0, \\ i_!(b_1')_!(b_2')^*j^*M, & n = 1, \\ 0, & n > 1. \end{cases}$$

We may suppose that u (resp. v) is defined by a homomorphism of complexes

$$u : a_{2*}a_1^*L \longrightarrow M \quad (\text{resp. } v : b_{1*}b_2^*M \longrightarrow L).$$

The preceding formulas show that the composite

$$F^1 a_{2*}a_1^*L \longrightarrow a_{2*}a_1^*L \xrightarrow{u} M$$

(resp. $F^1 b_{1*}b_2^*M \to b_{1*}b_2^*M \xrightarrow{v} L$) factors uniquely through F^1M (resp. F^1L); in other words, u (resp. v) is a homomorphism of filtered complexes. Thus u (resp. v) underlies a filtered correspondence, in the sense of (III 4.13), which we again denote by

$$u \in \operatorname{Hom}_{DF}(a_1^*L, a_2^!M), \qquad v \in \operatorname{Hom}_{DF}(b_2^*M, b_1^!L).$$

By additivity of cup-products ([3] V 3.8.7), (III 4.13.1), one has

$$\langle u_z, v_z\rangle = \langle \operatorname{gr}^0 u_z, \operatorname{gr}^0 v_z\rangle + \langle \operatorname{gr}^1 u_z, \operatorname{gr}^1 v_z\rangle,$$

and

$$\langle u, v\rangle = \langle \operatorname{gr}^0 u, \operatorname{gr}^0 v\rangle + \langle \operatorname{gr}^1 u, \operatorname{gr}^1 v\rangle.$$

The lemma follows, since $\operatorname{gr}^1 L = i_!i^*L$ (resp. $\operatorname{gr}^1 M = j_!j^*M$) has zero fibre at x (resp. y), whereas $\operatorname{gr}^0 L$ (resp. $\operatorname{gr}^0 M$) is concentrated at x (resp. y).

Lemma 2.3. Under the hypotheses of 1.1, suppose that $L|U = 0$ and $M|V = 0$. Then

$$\langle u_z, v_z\rangle = \langle u, v\rangle.$$

The idea of the proof is that then u and v are direct images of correspondences between complexes concentrated at the closed points, and that the formula to be proved merely expresses the Lefschetz formula for the projections $x \to S$, $y \to S$. In view of 1.1.3, the data of 1.1 may be replaced by the following data: X (resp. Y) is a smooth S-curve endowed with a closed point x (resp. y), A (resp. B) is a separated S-curve, $a : A \to X \times_S Y$, $b : B \to X \times_S Y$ are finite S-morphisms such that $A \times_{X\times_S Y} B$ is reduced to a closed point z above (x, y),

$$u \in \operatorname{Hom}(a_1^*L, a_2^!M), \qquad v \in \operatorname{Hom}(b_2^*M, b_1^!L),$$

with

$$L = h_{1*}L', \quad M = h_{2*}M', \quad L' \in \operatorname{ob} D_{\mathrm{ctf}}(\{x\}), \quad M' \in \operatorname{ob} D_{\mathrm{ctf}}(\{y\}),$$

where $h_1 : \{x\} \to X$ and $h_2 : \{y\} \to Y$ are the canonical inclusions. We must show that

$$\langle u_z, v_z\rangle = \langle u, v\rangle_z,$$

where

$$u_z \in \operatorname{Hom}(\mathrm{R}\Gamma(X, L), \mathrm{R}\Gamma(Y, M)), \qquad v_z \in \operatorname{Hom}(\mathrm{R}\Gamma(Y, M), \mathrm{R}\Gamma(X, L))$$

are defined as in 1.1, and $\langle u, v\rangle_z$ is the value at z of the cup-product (III 4.2.7). We have $\mathrm{R}\Gamma(X, L) = \mathrm{R}\Gamma_c(X, L) = L_x$, $\mathrm{R}\Gamma(Y, M) = \mathrm{R}\Gamma_c(Y, M) = M_y$, and u_z (resp. v_z) is nothing other than the direct image u_* (resp. v_*) of u (resp. v) on S in the sense of (III 3.7.6), as is shown by the calculation procedure (III 3.5 b)).

On the other hand, if $h = h_1 \times_S h_2 : (x,y) \to X \times_S Y$ is the inclusion, and if $a' : A' \to (x,y)$, $b' : B' \to (x,y)$ denote the fibres of a, b at (x,y), each reduced to a point, then there are canonical isomorphisms

$$h_* a'_* a'^! (DL_x \overset{\mathrm{L}}{\otimes}_\Lambda M_y) \xrightarrow{\sim} a_* a^! (DL \overset{\mathrm{L}}{\otimes}_\Lambda M),$$

$$h_* b'_* b'^! (L_x \overset{\mathrm{L}}{\otimes}_\Lambda DM_y) \xrightarrow{\sim} b_* b^! (L \overset{\mathrm{L}}{\otimes}_\Lambda DM),$$

and the arrows

$$h_* : \operatorname{Hom}(a_1'^* L_x, a_2'^! M_y) \to \operatorname{Hom}(a_1^* L, a_2^! M),$$

$$h_* : \operatorname{Hom}(b_2'^* M_y, b_1'^! L_x) \to \operatorname{Hom}(b_2^* M, b_1^! L)$$

(III 3.7.6) are isomorphisms. The correspondences u and v are therefore written uniquely as

$$u = h_* u', \qquad v = h_* v',$$

with

$$u' \in \operatorname{Hom}(a_1'^* L_x, a_2'^! M_y), \qquad v' \in \operatorname{Hom}(b_2'^* M_y, b_1'^! L_x).$$

By the Lefschetz formula (III 4.5.1) for $(h_1, h_2, A' \to A, B' \to B)$, one has

$$\langle u, v \rangle_z = \langle u', v' \rangle_z.$$

On the other hand, by the Lefschetz formula for $\{x\} \to S$, $\{y\} \to S$, one has

$$\langle u_*, v_* \rangle = \langle u', v' \rangle_z.$$

In view of the preceding remark, the conclusion follows by comparing these two equalities.

Lemma 2.4. To prove 1.2, one may suppose that a and b are closed immersions, and it is enough to prove that, if $L_x = 0$ and $M_y = 0$, the restriction map

$$H^0_A(X \times_S Y, DL \overset{\mathrm{L}}{\otimes}_\Lambda M) \longrightarrow H^0_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)) \tag{2.4.1}$$

is zero.

Put $X \times_S Y = Z$. The closed subschemes $a' : A' \to Z$, $b' : B' \to Z$, images of a and b, satisfy the same hypotheses as a and b. Let u' (resp. v') be the correspondence supported on A' (resp. B') which is the direct image of u (resp. v) by the arrow deduced from the adjunction arrow $f_* f^! \to 1$ (resp. $g_* g^! \to 1$), where $f : A \to A'$ (resp. $g : B \to B'$) is the projection. It is immediate that $u_z = u'_z$ and $v_z = v'_z$. On the other hand, it follows easily from the definition of the cup-product 1.1.2 that

$$\langle u, v \rangle = \langle u', v' \rangle,$$

hence the first assertion.

By 2.2 and 2.3, one may suppose that $L_x = 0$ and $M_y = 0$, and the conclusion of 1.2 then becomes $\langle u, v \rangle = 0$. Let $P, Q \in \operatorname{ob} D^b_{\mathrm{ctf}}(Z)$. The cup-product

$$H^0_A(Z, P) \otimes H^0_B(Z, Q) \longrightarrow H^0_z(Z, P \overset{\mathrm{L}}{\otimes}_\Lambda Q)$$

decomposes as

$$H^0_A(Z, P) \otimes H^0_B(Z, Q) \xrightarrow{b^* \otimes 1} H^0_z(B, b^* P) \otimes H^0_B(Z, Q) \longrightarrow H^0_z(Z, P \overset{\mathrm{L}}{\otimes}_\Lambda Q),$$

where b^* is the restriction and the second arrow, a variant of the product considered in (SGA $4^{1/2}$ Cycle I 2.1), is derived from the evident product at the level of sections of sheaves. This assertion is easily checked, for example by means of the flasque resolutions of (SGA 4 XVII 4.2). If $a' : \{z\} \to B$, $b' : \{z\} \to A$, $c : \{z\} \to Z$ are the inclusions, one may also invoke the standard decomposition of the Künneth arrow (III 4.2.1)

$$a'^! P \overset{\mathrm{L}}{\otimes}_\Lambda b'^! Q \longrightarrow c^! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)$$

into a base-change arrow

$$a'^! P \overset{\mathrm{L}}{\otimes}_\Lambda b'^! Q \longrightarrow a'^! b^* P \overset{\mathrm{L}}{\otimes}_\Lambda a^* b'^! Q,$$

followed by projection arrows

$$a'^! b^* P \overset{\mathrm{L}}{\otimes}_\Lambda a^* b'^! Q \longrightarrow a'^! (b^* P \overset{\mathrm{L}}{\otimes}_\Lambda b'^! Q) \longrightarrow a'^! b'^! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q).$$

Taking $P = DL \overset{\mathrm{L}}{\otimes}_\Lambda M'$, $Q = L \overset{\mathrm{L}}{\otimes}_\Lambda DM$, and returning to the definition of $\langle u, v \rangle$ (III 4.2.5), one sees that it is enough to prove that $b^* u = 0$, whence the conclusion.

In view of 2.4, 1.2 will therefore follow from the following more precise result.

Proposition 2.5. Under the hypotheses of 1.2, suppose that a and b are closed immersions and that $L_x = 0$, $M_y = 0$. Let T be the strict localization of $X \times_S Y$ at $z = (x, y)$, and identify A, B with closed subschemes of T. Then the restriction arrows

$$b^* : \mathrm{R}\Gamma_A(T, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T) \longrightarrow \mathrm{R}\Gamma_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)),$$

$$a^* : \mathrm{R}\Gamma_B(T, (L \overset{\mathrm{L}}{\otimes}_\Lambda DM)|T) \longrightarrow \mathrm{R}\Gamma_z(A, a^*(L \overset{\mathrm{L}}{\otimes}_\Lambda DM))$$

are zero.

2.6

The proof of 2.5 will occupy the rest of number 2 and numbers 3 and 4. By symmetry of the statement, it is enough to prove the assertion concerning b^*. We shall rewrite it in a form which will be more convenient. First note that

$$\mathrm{R}\Gamma(T, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T) = (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)_z = (DL)_x \overset{\mathrm{L}}{\otimes}_\Lambda M_y = 0,$$

because $M_y = 0$, and similarly

$$\mathrm{R}\Gamma(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)) = (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)_z = 0.$$

Consequently, the relative cohomology triangles give isomorphisms

$$\mathrm{R}\Gamma(T - A, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T - A) \xrightarrow{\sim} \mathrm{R}\Gamma_A(T, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T)[1],$$

$$\mathrm{R}\Gamma(B - z, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|B - z) \xrightarrow{\sim} \mathrm{R}\Gamma_z(B, b^*(DL \overset{\mathrm{L}}{\otimes}_\Lambda M))[1],$$

under which the arrow b^* of 2.5, shifted by 1, is identified with the restriction

$$b^* : \mathrm{R}\Gamma(T - A, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T - A) \longrightarrow \mathrm{R}\Gamma(B - z, (DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|B - z). \tag{2.6.1}$$

Since $L_x = 0$, $M_y = 0$, one may write

$$L = i_! P, \qquad M = j_! Q, \tag{2.6.2}$$

where $i : X - x \to X$, $j : Y - y \to Y$ are the inclusions, and $P \in \operatorname{ob} D_{\mathrm{ctf}}(X - x)$, $Q \in \operatorname{ob} D_{\mathrm{ctf}}(Y - y)$. By the duality isomorphism (SGA 4 XVIII 3.19.6), together with passage to the limit, one has

$$DL = Ri_* DP = Ri_* P,$$

hence

$$DL \overset{\mathrm{L}}{\otimes}_\Lambda M = Ri_* P \overset{\mathrm{L}}{\otimes}_\Lambda j_! Q.$$

Put $X \times_S Y = Z$, and identify $X \times \{y\}$ and $\{x\} \times Y$ with closed subschemes of Z (resp. T), which we shall simply denote by X and Y. Let

$$f' : Z - Y \hookrightarrow Z, \qquad g' : Z - (X \cup Y) \hookrightarrow Z - Y,$$

$$f : T - Z \hookrightarrow T, \qquad g : T - (X \cup Y) \hookrightarrow T - Y$$

be the inclusions. The Künneth formula (III 1.6.4) gives, by passage to the limit,

$$DL \overset{\mathrm{L}}{\otimes}_\Lambda M = Rf'_* g'_! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q),$$

and hence

$$(DL \overset{\mathrm{L}}{\otimes}_\Lambda M)|T = Rf_* g_! (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T - (X \cup Y). \tag{2.6.3}$$

With this identification, 2.6.1 becomes

$$b^* : \mathrm{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) \longrightarrow \mathrm{R}\Gamma(B - z, g_!(E)|B - z), \tag{2.6.4}$$

where

$$E = (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T - (X \cup Y).$$

In the next two numbers, we shall show that the arrow 2.6.4 is zero: first we shall reduce to the case where the cohomology sheaves of E have tame ramification, more precisely where the inertia action occurs through an ℓ-group, for a prime number $\ell \neq p$, and then we shall treat that case directly by means of a purity theorem.

3. Reduction to the tame case

3.1

Recall that the base ring Λ is annihilated by an integer prime to p, hence is a direct product of noetherian rings Λ_i, each Λ_i being annihilated by a power of a prime number $\ell_i \neq p$. For every scheme S' over S, the derived category $D(S', \Lambda)$ is the product of the categories $D(S', \Lambda_i)$, so that, in order to prove the vanishing of 2.6.4, we may suppose that Λ is annihilated by a power of a prime number $\ell \neq p$. We may also suppose that P (resp. Q) is bounded and has locally free components of finite type.

3.2

Put $X - x = U$, $Y - y = V$. Let U''/U be a finite Galois covering with group G which trivializes the components of P, let H be an ℓ-Sylow subgroup of G, let $U' = U''/H$ be the quotient covering, and let X' be the normalized strictly local trait of X in U'. Then X'/X is a finite connected covering of degree d_1 prime to ℓ, and the representation of $\pi_1(U')$ defined by the components of $P' = P|U'$ factors through a finite ℓ-group. Proceeding similarly with Q, one finds a finite connected covering Y'/Y of degree d_2 prime to ℓ, inducing on V an étale covering V' such that the representation of $\pi_1(V')$ defined by the components of $Q' = Q|V'$ factors through a finite ℓ-group. Let

$$d = d_1 e_1 = d_2 e_2 = \mathrm{lcm}(d_1, d_2).$$

After adjoining an e_1-th root (resp. an e_2-th root) of a uniformizer of X' (resp. Y'), we may suppose that X' (resp. Y') has the same properties as above, and moreover that $d_1 = d_2 = d$.

Let x' (resp. y') be the closed point of X' (resp. Y'), T' the strict localization of $X' \times_S Y'$ at (x', y'), z' the closed point of T', A' (resp. B') the inverse image of A (resp. B) in T', and

$$E' = (P \overset{\mathrm{L}}{\otimes}_\Lambda Q)|T' - (X' \cup Y') = (P' \overset{\mathrm{L}}{\otimes}_\Lambda Q')|T' - (X' \cup Y').$$

We then have a commutative square of restriction arrows

$$\begin{array}{ccc}
\mathrm{R}\Gamma(T - (Y \cup A), g_!(E)|T - (Y \cup A)) & \xrightarrow{b^*} & \mathrm{R}\Gamma(B - z, g_!(E)|B - z) \\
\downarrow & & \downarrow{\scriptstyle (1)} \\
\mathrm{R}\Gamma(T' - (Y' \cup A'), g'_!(E')|T' - (Y' \cup A')) & \xrightarrow{b'^*} & \mathrm{R}\Gamma(B' - z', g'_!(E')|B' - z'),
\end{array}$$

where $b' : B' \to T'$, $g' : T' - (X' \cup Y') \to T' - Y'$ are the inclusions. The covering T'/T is finite, of degree d^2, and étale outside $X \cup Y$, hence induces an étale covering of degree d^2 from $B' - z'$ onto $B - z$. The composite of (1) and the trace morphism

$$\mathrm{R}\Gamma(B' - z', g'_!(E')|B' - z') \longrightarrow \mathrm{R}\Gamma(B - z, g_!(E)|B - z)$$

is therefore multiplication by d^2, which is an isomorphism, since ℓ does not divide d. To prove that the arrow 2.6.4 is zero, it is therefore enough to prove that $b'^* = 0$. We are thus reduced to the case where the representation of $\pi_1(U)$ (resp. $\pi_1(V)$) defined by the components of P (resp. Q) factors through a finite ℓ-group, provided one ensures that A', B' satisfy the general-position hypotheses of 1.2. This will follow from the following lemma.

Lemma 3.3. Under the hypotheses of 1.2, a and b being closed immersions, let X'/X, Y'/Y be finite coverings of strictly local S-traits, of the same degree d. Let A' (resp. B') denote the inverse image of A (resp. B) in $X' \times_S Y'$. Then the pairs (A', B'), (A', X'), (Y', B') are each in general position in the sense of 1.2.

The proof of 3.3 will be given after a few preliminaries.

Lemma 3.4. Let $X \xleftarrow{s_1} W \xrightarrow{s_2} Y$ be finite morphisms of S-traits, let $s = (s_1, s_2) : W \to X \times_S Y$, and let u, v, w be uniformizers of X, Y, W respectively, so that

$$s_1^* u = a w^m \pmod{w^{m+1}}, \qquad s_2^* v = b w^n \pmod{w^{n+1}},$$

with $a, b \in k^*$. Then: (i) if $m < n$, $s(W)$ is tangent to X at $z = (x, y)$; (ii) if $m = n$, $s(W)$ is tangent to the subscheme of $X \times_S Y$ with equation

$$ap_2^*(v) - bp_1^*(u) = 0.$$

Let $Z = X \times_S Y$, and let W' be the reduced closed subscheme which is the image of s. In the tangent space to Z at z, the slope, with respect to the tangent to X, of the tangent to W' is the value at the origin of the function induced on W' by $p_2^*(v)/p_1^*(u)$; this is also the value at the origin of the function induced by $p_2^*(v)/p_1^*(u)$ on W, and is therefore 0 (resp. b/a) in case (i) (resp. (ii)).

In the situation of 3.4, we shall define the slope of W with respect to $((X, u), (Y, v))$ as the element of $k \cup \{\infty\}$ equal to 0 if $m < n$, to ∞ if $m > n$, and to b/a if $m = n$. This element is independent of the choice of w.

Lemma 3.5. Let $f : X' \to X$, $g : Y' \to Y$ be finite morphisms of S-traits with the same ramification index d, and let u, v, u', v' be uniformizers of X, Y, X', Y' respectively. Then there exists $C \in k^*$ such that, for every commutative diagram of finite morphisms of S-traits

$$\begin{tikzcd} X' & W' \arrow[l, "s_1'"'] \arrow[r, "s_2'"] \arrow[d, "h"] & Y' \arrow[d, "g"] \\ X & W \arrow[l, "s_1"] \arrow[r, "s_2"'] & Y, \end{tikzcd}$$

one has $\lambda = C{\lambda'}^d$, where λ (resp. λ') is the slope of W (resp. W') with respect to $((X, u), (Y, v))$ (resp. $((X', u'), (Y', v'))$).

By hypothesis,

$$f^*u = \alpha {u'}^d \pmod{{u'}^{d+1}}, \qquad g^*v = \beta {v'}^d \pmod{{v'}^{d+1}},$$

with $\alpha, \beta \in k^*$. We show that $C = \beta/\alpha$ has the required properties. Let w (resp. w') be a uniformizer of W (resp. W'). We have

$$s_1^*u = aw^m \pmod{w^{m+1}}, \qquad s_2^*v = bw^n \pmod{w^{n+1}},$$

$${s_1'}^*u' = a'{w'}^{m'} \pmod{{w'}^{m'+1}}, \qquad {s_2'}^*v' = b'{w'}^{n'} \pmod{{w'}^{n'+1}},$$

$$h^*w = \gamma {w'}^r \pmod{{w'}^{r+1}},$$

with $a, b, a', b', \gamma \in k^*$. Commutativity of the squares in the diagram implies

$$m'd = rm, \qquad n'd = rn,$$

hence $m'/n' = m/n$. Suppose first that $m = n$. Then $\lambda = b/a$, $\lambda' = b'/a'$. If $N = m'd = rm = n'd = rn$, the preceding congruences give

$$(fs_1')^*u = \alpha {a'}^d {w'}^N \pmod{{w'}^{N+1}}, \qquad (s_1h)^*u = a\gamma^m {w'}^N \pmod{{w'}^{N+1}},$$

whence $\alpha {a'}^d = a\gamma^m$. Similarly $\beta {b'}^d = b\gamma^m$, whence $\lambda = C{\lambda'}^d$, for $\lambda \neq 0, \infty$. But, since $m'/n' = m/n$, $\lambda = 0$ (resp. ∞) is equivalent to $\lambda' = 0$ (resp. ∞), which completes the proof.

We now prove 3.3. Let $Z = X \times_S Y$, $Z' = X' \times_S Y'$, let $\widetilde{A}$ (resp. $\widetilde{B}$) be the normalization of A (resp. B), and let $\widetilde{A}'$ (resp. $\widetilde{B}'$) be the normalization of A' (resp. B'). Since $\widetilde{A}'$ (resp. $\widetilde{B}'$) is also the normalization of $\widetilde{A} \times_Z Z'$ (resp. $\widetilde{B} \times_Z Z'$), we have commutative squares

$$\begin{tikzcd}
\widetilde{A}' \arrow[r] \arrow[d] & Z' & \widetilde{B}' \arrow[l] \arrow[d] \\
\widetilde{A} \arrow[r] & Z & \widetilde{B}. \arrow[l]
\end{tikzcd}$$

Let $(x_i)_{1\le i\le m}$ (resp. $(y_j)_{1\le j\le n}$) be the points of $\widetilde{A}$ (resp. $\widetilde{B}$) above x (resp. y), and let $(x'_i)_{1\le i\le m'}$ (resp. $(y'_j)_{1\le j\le n'}$) be the points of $\widetilde{A}'$ (resp. $\widetilde{B}'$) above x' (resp. y'). Since A and B are strictly local, we have decompositions

$$\widetilde{A} = \coprod_{1\le i\le m} A_i, \qquad \widetilde{B} = \coprod_{1\le j\le n} B_j, \qquad \widetilde{A}' = \coprod_{1\le i\le m'} A'_i, \qquad \widetilde{B}' = \coprod_{1\le j\le n'} B'_j,$$

where A_i, B_j, A'_i, B'_j are strictly local traits with respective closed points x_i, y_j, x'_i, y'_j. Choose uniformizers u, v, u', v' of X, Y, X', Y' respectively; let λ_i (resp. μ_j) be the slope of A_i (resp. B_j) with respect to $((X,u),(Y,v))$, and let λ'_i (resp. μ'_j) be that of A'_i (resp. B'_j) with respect to $((X',u'),(Y',v'))$. The general-position hypotheses on A and B are expressed by

$$\lambda_i \neq 0 \text{ for all } i, \qquad \mu_j \neq \infty \text{ for all } j, \qquad \lambda_i \neq \mu_j \text{ for all } i,j, \tag{1}$$

and we have to show that

$$\lambda'_i \neq 0 \text{ for all } i, \qquad \mu'_j \neq \infty \text{ for all } j, \qquad \lambda'_i \neq \mu'_j \text{ for all } i,j. \tag{2}$$

Let $i' \in [1, m']$, $j' \in [1, n']$, and let x_i (resp. y_j) be the image of $x'_{i'}$ (resp. $y'_{j'}$). The preceding diagram provides commutative squares of finite morphisms of S-traits

$$\begin{tikzcd}
X' & A'_{i'} \arrow[l] \arrow[r] \arrow[d] & Y' \arrow[d] & X' & B'_{j'} \arrow[l] \arrow[r] \arrow[d] & Y' \arrow[d] \\
X & A_i \arrow[l] \arrow[r] & Y & X & B_j \arrow[l] \arrow[r] & Y.
\end{tikzcd}$$

By 3.5, there exists $C \in k^*$ such that

$$\lambda_i = C{\lambda'_{i'}}^d, \qquad \mu_j = C{\mu'_{j'}}^d.$$

Thus (1) implies (2), which completes the proof of 3.3.

4. End of the proof of 1.2

4.1

To complete the proof of 2.5, hence of 1.2, it remains to establish the vanishing of 2.6.4 under the following additional hypotheses: Λ is annihilated by a power of a prime number $\ell \neq p$, and

P and Q are bounded, with locally free components of finite type, such that the corresponding representations of the fundamental groups factor through finite ℓ-groups. We have

$$g_!(E)|T-(Y\cup A)=g_{1!}(E_1),$$

where $g_1: T-(X\cup Y\cup A)\hookrightarrow T-(Y\cup A)$ is the inclusion, and

$$E_1=(P\overset{\mathrm{L}}{\otimes}_\Lambda Q)|T-(X\cup Y\cup A).$$

Now E_1 is a complex with bounded, constructible, locally constant cohomology such that the corresponding representation of $\pi_1(T-(X\cup Y\cup A))$ factors through an ℓ-group. The assertion to be proved will therefore follow from the following more general statement.

Proposition 4.2. Let T be the strict localization at a closed point of a smooth S-scheme of dimension 2, let t be the closed point of T, and let A, B, X be closed subschemes of dimension 1 of T. Let $i: T-(A\cup X)\hookrightarrow T-A$ be the inclusion. Suppose that X is regular, and that A is in general position with respect to B and X; that is, the intersection of the tangent cones at t to A and $B\cup X$ is reduced to t. Let $E\in \operatorname{ob} D^b(T-(A\cup X))$ be such that $H^*(E)$ is constructible and locally constant, and that the corresponding representation of $\pi_1(T-(A\cup X))$ factors through an ℓ-group. Then the restriction arrow

$$\mathrm{R}\Gamma(T-A, i_!(E))\longrightarrow \mathrm{R}\Gamma(B-t, i_!(E)|B-t)$$

is zero.

The proof will rely on the following lemma.

Lemma 4.3. Let Z be a connected regular noetherian scheme of dimension 2, with residual characteristics prime to ℓ, and let X, Y be regular divisors meeting transversely at a point z, giving a diagram of inclusions

$$\begin{tikzcd} Z-(X\cup Y) \arrow[r, "i'"] \arrow[d, "j"] & Z-Y \arrow[d, "j"] \\ Z-X \arrow[r, "i"'] \arrow[d, "m"] & Z \arrow[d, "m"] \\ Y-z \arrow[r, "i_1"'] & Y. \end{tikzcd}$$

Assume the purity hypothesis

$$(P)\qquad R^q j_*(\mathbb{Z}/\ell)=0\quad\text{for } q>1.$$

Let $L\in \operatorname{ob} D^b(Z-(X\cup Y),\Lambda)$ have constructible locally constant cohomology, with the corresponding representation of $\pi_1(Z-(X\cup Y))$ factoring through an ℓ-group. Then:

a) There is a canonical isomorphism

$$i_{1!}m'^*Rj_*i'_!L\xrightarrow{\sim} m^*Rj_*(i'_!L),$$

and the complex $m'^*Rj_*i'_!L$ has constructible, locally constant cohomology.

b) Suppose that Y is connected, and let $\bar{s}$ be a geometric point of Y localized at a closed point $s \neq z$. Then the restriction arrow

$$\mathrm{R}\Gamma(Y, m^* Rj_*(i'_! L)) \longrightarrow (Rj_*(i'_! L))_{\bar{s}} \tag{4.3.1}$$

is zero.

c) Under the hypotheses of b), let C be a closed subscheme of dimension 1 of Z such that $Y \cap C = s$, and let $\widetilde{C}$ be the strict localization of C at $\bar{s}$. Then the restriction arrow

$$\mathrm{R}\Gamma(Z - Y, i'_! L) \longrightarrow \mathrm{R}\Gamma(\widetilde{C} - \bar{s}, i'_!(L)|\widetilde{C} - \bar{s}) \tag{4.3.2}$$

is zero.

The trivial isomorphism

$$i^* Rj_* i'_! L \xrightarrow{\sim} Rj_*(i'^* i'_! L) = Rj_* L$$

gives an isomorphism

$$m'^* Rj_* i'_! L \xrightarrow{\sim} i^* m^* Rj_* i'_! L,$$

hence, by adjunction, an arrow

$$i_{1!} m'^* Rj_* i'_! L \longrightarrow m^* Rj_*(i'_! L),$$

which induces an isomorphism outside z. We show that it is an isomorphism, that is, that if $\bar{z}$ is a geometric point above z, then

$$(Rj_*(i'_! L))_{\bar{z}} = 0.$$

For this, one may replace Λ by $\mathbb{Z}/\ell^\nu$, where ℓ^ν annihilates Λ, and by dévissage and passage to the limit reduce to the case where L is a locally constant $\mathbb{Z}/\ell^\nu$-module of finite type corresponding to a representation of π_1 through an ℓ-group. Since the only finite abelian group of ℓ-torsion which is simple under the action of an ℓ-group is $\mathbb{Z}/\ell$ with trivial action, a further dévissage reduces to the case where L is the constant sheaf $\mathbb{Z}/\ell$.

Let $\widetilde{Z}$ be the strict localization of Z at $\bar{z}$, $\widetilde{X} = X \times_Z \widetilde{Z}$, $\widetilde{Y} = Y \times_Z \widetilde{Z}$, and let $i' : \widetilde{Z} - (\widetilde{X} \cup \widetilde{Y}) \hookrightarrow \widetilde{Z} - \widetilde{Y}$ be the inclusion. One has

$$(Rj_*(i'_!(\mathbb{Z}/\ell)))_{\bar{z}} = \mathrm{R}\Gamma(\widetilde{Z} - \widetilde{Y}, i'_!(\mathbb{Z}/\ell)),$$

so it remains to prove that the restriction arrow

$$H^n(\widetilde{Z} - \widetilde{Y}, \mathbb{Z}/\ell) \longrightarrow H^n(\widetilde{X} - z, \mathbb{Z}/\ell)$$

is an isomorphism for all n. For $n \leq 1$ this is a consequence of Abhyankar's lemma, and for $n > 1$ both sides vanish by the purity hypothesis. The first assertion of a) is therefore proved. We prove the second. By dévissage, we may suppose L concentrated in degree 0. The hypothesis (P) implies, by dévissage, that $R^q j_* L = 0$ for $q > 2$. It remains to prove constructibility and local constancy of $m'^* R^q j_* i'_! L$ for $q \leq 1$, which is standard. For constructibility, resolving L on the right by sheaves of the form $p_* p^* L$, where $p : Z' \to Z$ is a finite covering tamely ramified along Y, reduces to the case where L is constant, of value L_0: then

$$m'^* Rj_* i'_! L = (L_0)_{Y-z}, \qquad m'^* R^1 j_* i'_! L = (L_0)_{Y-z}(-1)$$

by Abhyankar. For local constancy, one has to see that the specialization arrows are isomorphisms. One may, for this, replace Λ by $\mathbb{Z}/\ell^{\nu}$ as above, and then reduce by dévissage to the case where L is the constant sheaf $\mathbb{Z}/\ell$; the conclusion follows from Abhyankar's lemma.

We have therefore proved a). We prove b). Let $\bar{\eta}$ be a geometric generic point of Y. Choose specialization arrows
$$a : \bar{\eta} \to Y(\bar{z}), \qquad b : \bar{\eta} \to Y(\bar{s}),$$
where $Y(\bar{z})$ (resp. $Y(\bar{s})$) is the strict localization of Y at $\bar{z}$ (resp. $\bar{s}$). Put
$$M = m^* Rj_*(i'_! L).$$

The commutative square
$$\begin{tikzcd} Y \arrow[r] & Y(\bar{s}) \\ Y(\bar{z}) \arrow[u] \arrow[r, "a"'] & \bar{\eta} \arrow[u, "b"'] \end{tikzcd}$$
induces a commutative square
$$\begin{tikzcd} \mathrm{R}\Gamma(Y, M) \arrow[r, "4.3.1"] \arrow[d] & M_{\bar{s}} \arrow[d, "b^*"] \\ M_{\bar{z}} \arrow[r, "a^*"'] & M_{\bar{\eta}}. \end{tikzcd}$$

Since $M|Y - z$ has locally constant cohomology, b^* is an isomorphism. On the other hand, since $M = i_{1!} i_1^* M$, one has $M_{\bar{z}} = 0$, hence the arrow 4.3.1 is zero. It remains to prove c). We have $\widetilde{C} = C \times_Z \widetilde{Z}$, where $\widetilde{Z}$ denotes, as above, the strict localization of Z at $\bar{z}$. Put $\widetilde{Y} = Y \times_Z \widetilde{Z}$. Then there is a commutative diagram of restriction arrows
$$\begin{tikzcd} \mathrm{R}\Gamma(Z - Y, i'_! L) = \mathrm{R}\Gamma(Z, Rj_*(i'_! L)) \arrow[r, "4.3.2"] \arrow[d] & \mathrm{R}\Gamma(\widetilde{C} - \bar{s}, i'_!(L)|\widetilde{C} - \bar{s}) \\ \mathrm{R}\Gamma(Y, m^* Rj_*(i'_! L)) \arrow[r, "4.3.1"'] & (Rj_*(i'_! L))_{\bar{s}} = \mathrm{R}\Gamma(\widetilde{Z} - \widetilde{Y}, (i'_! L)|\widetilde{Z} - \widetilde{Y}) \arrow[u] \end{tikzcd}$$

The vanishing of 4.3.2 therefore follows from b), completing the proof of 4.3.

We prove 4.2. Let $f : T' \to T$ be the blowing-up of the maximal ideal of T, let $D = f^{-1}(z) (\simeq \mathbb{P}^1_S)$ be the exceptional divisor, and let A', B', X' be the respective strict transforms of A, B, X. Then T' is regular, connected, of dimension 2, X' is a regular divisor meeting D transversely at one point z', and the general-position hypotheses on A, B, X mean that A' is disjoint from $B' \cup X'$. Let E' be the inverse image of E on $T' - (A' \cup D \cup X')$, and let
$$i' : T' - (A' \cup D \cup X') \hookrightarrow T' - (A' \cup D)$$
be the inclusion. We must prove that the restriction arrow
$$\mathrm{R}\Gamma(T' - (A' \cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B' \cup D, i'_!(E')|B' - (B' \cup D)) \tag{*}$$
is zero. But B' is a disjoint union of strictly local schemes B'_j $(1 \leq j \leq n)$, of dimension 1, such that for every j, $B'_j \cap D$ is reduced to a point b_j. Applying 4.3 c) to $(Z = T' - A', X = X', Y =$

$D-(D\cap A'), L=E', C=B'_j)$, the hypothesis (P) being verified by virtue of the relative purity theorem (SGA 4 XVI 3), since (Z,Y) is a filtered projective limit of smooth pairs of codimension 1 over S, with affine étale transition morphisms, one obtains that the restriction arrow

$$\mathrm{R}\Gamma(T'-(A'\cup D), i'_!(E')) \longrightarrow \mathrm{R}\Gamma(B'_j - b_j, i'_!(E')|B'_j - b_j)$$

is zero. Therefore (*) is zero, which completes the proof of 4.2, hence of 2.5 and 1.2.

5.9. Traces and extension of scalars

Let $A\to B$ be a morphism of flat K-algebras. It induces a morphism of K-algebras $A^e\to B^e$, hence an extension-of-scalars functor

$$-\overset{\mathrm{L}}{\otimes}_{A^e} B^e : D(A^e)\longrightarrow D(B^e).$$

For E,F as in 5.3.5, if $F\overset{\mathrm{L}}{\otimes}_{A^e} B^e\in \mathrm{ob}\, D^b(B^e)$ and $\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\in \mathrm{ob}\, D^b((B^e)^\circ)$, there is a commutative square in $D(K)$

$$\begin{tikzcd}
\mathrm{RHom}_A(E,F)\overset{\mathrm{L}}{\otimes}_{A^e} E \arrow[r, "(1)"] \arrow[d, "(3)"'] & F\overset{\mathrm{L}}{\otimes}_{A^e} A \arrow[d, "(4)"] \\
\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_{B^e}(B\overset{\mathrm{L}}{\otimes}_A E) \arrow[r, "(2)"'] & (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_{B^e} B,
\end{tikzcd} \tag{5.9.1}$$

where the arrows (1) and (2) are defined by 5.3.6, (4) by $A\to B$, and (3) by the canonical arrow $F\to F\overset{\mathrm{L}}{\otimes}_{A^e} B^e$. The verification is immediate. Similarly, for E,F,G as in 5.4.2, if $F\overset{\mathrm{L}}{\otimes}_{A^e} B^e\in \mathrm{ob}\, D^+(B^e)$ and $(F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A G)\in \mathrm{ob}\, D^+(B)$, one has a commutative square in $D(K)$

$$\begin{tikzcd}
\mathrm{RHom}_A(E,F)\overset{\mathrm{L}}{\otimes}_A G \arrow[r, "(1)"] \arrow[d, "(3)"'] & \mathrm{RHom}_A(E,F\overset{\mathrm{L}}{\otimes}_A G) \arrow[d, "(4)"] \\
\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A G) \arrow[r, "(2)"'] & \mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A G)),
\end{tikzcd} \tag{5.9.2}$$

where (1) and (2) are defined by 5.4.2, and (3) and (4) by $F\to F\overset{\mathrm{L}}{\otimes}_{A^e} B^e$. From this one deduces, for E,F as in 5.6.1, with $F\overset{\mathrm{L}}{\otimes}_{A^e} B^e\in \mathrm{ob}\, D^b(B^e)$, a commutative square in $D(K)$, analogous to the preceding ones, in which the vertical arrows are defined by 5.6.1:

$$\begin{tikzcd}
\mathrm{RHom}_A(E,F\overset{\mathrm{L}}{\otimes}_A E) \arrow[r, "\mathrm{Tr}_A"] \arrow[d] & F\overset{\mathrm{L}}{\otimes}_{A^e} A \arrow[d] \\
\mathrm{RHom}_B(B\overset{\mathrm{L}}{\otimes}_A E, (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_B (B\overset{\mathrm{L}}{\otimes}_A E)) \arrow[r, "\mathrm{Tr}_B"'] & (F\overset{\mathrm{L}}{\otimes}_{A^e} B^e)\overset{\mathrm{L}}{\otimes}_{B^e} B.
\end{tikzcd} \tag{5.9.3}$$

Finally, if one is given $F' \in \operatorname{ob} D^b(B^e)$ such that $\mathrm{RHom}_A(E, F') \in \operatorname{ob} D^b((B^\circ))$, and $f : F \overset{\mathrm{L}}{\otimes}_{A^e} B^e \to F'$, then there is a compatibility slightly more general than 5.9.1, expressed by an analogous commutative square, with F' in place of $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$ in the lower row and with the vertical arrows defined by f. One has similarly commutative squares analogous to 5.9.2 and 5.9.3 with F' in place of $F \overset{\mathrm{L}}{\otimes}_{A^e} B^e$. Taking in particular $F = A$ and, for $f : A \overset{\mathrm{L}}{\otimes}_{A^e} B^e \to B$, the arrow defined by $A \to B$, one obtains a commutative square

$$\begin{tikzcd}
\mathrm{RHom}_A(E,E) \arrow[r, "\mathrm{Tr}_A"] \arrow[d] & A \overset{\mathrm{L}}{\otimes}_{A^e} A \arrow[d] \\
\mathrm{RHom}_B(B \overset{\mathrm{L}}{\otimes}_A E, B \overset{\mathrm{L}}{\otimes}_A E) \arrow[r, "\mathrm{Tr}_B"'] & B \overset{\mathrm{L}}{\otimes}_{B^e} B,
\end{tikzcd} \tag{5.9.4}$$

where the left vertical arrow, respectively the right vertical arrow, is given by extension of scalars, respectively is the canonical arrow deduced from $A \to B$.

5.10. Traces and restriction of scalars

Let $A \to B$ be a morphism of flat K-algebras such that B, as a left A-module, is flat and of finite presentation, i.e. locally a direct factor of a free A-module of finite type. This is the case for example if $A \to B$ is the morphism $K[H] \to K[G]$ defined by an injective morphism of discrete groups $H \hookrightarrow G$ of finite index. The scalar-restriction homomorphism

$$\mathrm{Hom}_B(B, B) \longrightarrow \mathrm{Hom}_A(B, B),$$

where B is considered as a left module, identifies, by the canonical isomorphisms

$$B \xrightarrow{\sim} \mathrm{Hom}_B(B, B), \qquad \mathrm{Hom}_A(B, A) \otimes_A B \xrightarrow{\sim} \mathrm{Hom}_A(B, B),$$

with an arrow of B-bimodules

$$B \longrightarrow \mathrm{Hom}_A(B, A) \otimes_A B. \tag{5.10.1}$$

The composite of 5.10.1 with the evaluation arrow $\mathrm{Hom}_A(B, A) \otimes_A B \to A_\natural$ (5.3.3) is the homomorphism $f \mapsto \mathrm{Tr}_A(f)$ ($= \mathrm{Tr}_A(d_f)$, where d_f is right multiplication by f); hence, by the formula $\mathrm{Tr}_A(f_g) = \mathrm{Tr}_A(gf)$ (5.8.5), it factors through $B_\natural$ as an arrow again denoted $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$. Thus there is a commutative square

$$\begin{tikzcd}
B \arrow[r, "\mathrm{Tr}_B"] \arrow[d, "5.10.1"'] & B_\natural \arrow[d, "\mathrm{Tr}_{B/A}"] \\
\mathrm{Hom}_A(B, A) \otimes_A B \arrow[r, "\mathrm{ev}"'] & A_\natural.
\end{tikzcd} \tag{5.10.2}$$

In the situation $({}_B E, {}^B P_A)$, one has a homomorphism of right B-modules

$$\mathrm{Hom}_B(E, P) \otimes_A B \longrightarrow \mathrm{Hom}_B(E, P \otimes_A B), \qquad u \otimes b \longmapsto (x \longmapsto \nu(x) \otimes b), \tag{5.10.3}$$

which is an isomorphism, as is checked immediately: it is enough to see that the underlying K-linear homomorphism is an isomorphism, and this follows from the fact that B is locally a direct factor of a free A-module of finite type. This isomorphism derives to an isomorphism of $D(B)$

$$\mathrm{RHom}_B(E, P) \overset{\mathrm{L}}{\otimes}_A B \xrightarrow{\sim} \mathrm{RHom}_B(E, P \overset{\mathrm{L}}{\otimes}_A B), \tag{5.10.4}$$

for $E \in \mathrm{ob}\, D^-(B)$, $P \in \mathrm{ob}\, D^+(B \otimes_K A^\circ)$: resolve E by modules of the form $B \otimes_A L$, where L is flat, and P by injective $(B \otimes_K A^\circ)$-modules, which are necessarily injective as B-modules, since A is flat over K; this allows one to remove the R on the left, and the resolution chosen for E also allows one to remove it on the right.

For $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^+(K)$, $F \in \mathrm{ob}\, D^-(B)$, define an arrow in $D(K)$

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F \longrightarrow \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F \tag{5.10.5}$$

as the composite of the arrows

$$\begin{aligned}
\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F &\xrightarrow{(1)} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A) \overset{\mathrm{L}}{\otimes}_A (B \overset{\mathrm{L}}{\otimes}_K M)) \overset{\mathrm{L}}{\otimes}_B F \\
&\xrightarrow{\quad(2)\quad} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A) \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A B \overset{\mathrm{L}}{\otimes}_B F \\
&\xrightarrow{\quad(3)\quad} \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, A \overset{\mathrm{L}}{\otimes}_K M)) \overset{\mathrm{L}}{\otimes}_A F \\
&\xrightarrow{\quad(4)\quad} \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F,
\end{aligned}$$

where (1) is defined by 5.10.1, (2) by the inverse isomorphism of 5.10.4 with $P = \mathrm{Hom}_A(B, A) \overset{\mathrm{L}}{\otimes}_K M$, (3) by the canonical isomorphism

$$\mathrm{Hom}_A(B, A) \overset{\mathrm{L}}{\otimes}_K M \xrightarrow{\sim} \mathrm{Hom}_A(B, A \overset{\mathrm{L}}{\otimes}_K M)$$

(5.4.2), and (4) by the adjunction isomorphism 5.2.2.

Proposition 5.10.6. Suppose that $M \overset{\mathrm{L}}{\otimes}_K F \in \mathrm{ob}\, D^+(B)$. Then there is a commutative square in $D(K)$

$$\begin{tikzcd}
\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F \arrow[r, "5.4.2"] \arrow[d, "5.10.5"'] & \mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K F) \arrow[d, "\mathrm{res}"] \\
\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F \arrow[r, "5.4.2"'] & \mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K F),
\end{tikzcd} \tag{5.10.7}$$

where the right vertical arrow is restriction of scalars.

One has to verify that there is a commutative square

$$\begin{tikzcd} E \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B F \arrow[r] \arrow[d] & M \overset{\mathrm{L}}{\otimes}_K F \arrow[d, "\mathrm{res}"] \\ E \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A F \arrow[r] & M \overset{\mathrm{L}}{\otimes}_K F, \end{tikzcd}$$

where the horizontal arrows are given by the evaluations (5.3.6). This square is deduced, by applying $- \overset{\mathrm{L}}{\otimes}_B F$, from the analogous square with $F = B$, so that one may suppose $F = B$. The commutativity of 5.10.7 in this case is equivalent to that of the triangle

$$\begin{tikzcd} \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \arrow[d, "(1)"'] \arrow[dr, "(2)"] & \\ \mathrm{RHom}_B(E, \mathrm{Hom}_A(B, B \overset{\mathrm{L}}{\otimes}_K M)) \arrow[r, "(3)"'] & \mathrm{RHom}_A(E, B \overset{\mathrm{L}}{\otimes}_K M), \end{tikzcd}$$

where (1) is defined by 5.10.1, (2) is restriction of scalars, and (3) is the adjunction isomorphism. But the arrow $B \overset{\mathrm{L}}{\otimes}_K M \to \mathrm{Hom}_A(B, B \overset{\mathrm{L}}{\otimes}_K M)$ given by 5.10.1 is nothing other than the adjunction arrow corresponding to the functors restriction of scalars and $\mathrm{Hom}_A(B, -)$; the commutativity follows at once.

Let $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^b(K)$ be such that $\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(B^\circ)$. Then, by 5.3.5, there is an evaluation arrow

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_{B^e} E \longrightarrow M \overset{\mathrm{L}}{\otimes}_K (B \overset{\mathrm{L}}{\otimes}_{B^e} B),$$

which, by composition with the canonical arrow $B \overset{\mathrm{L}}{\otimes}_{B^e} B \to B_\natural$, gives an arrow

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E \longrightarrow M \overset{\mathrm{L}}{\otimes}_K B_\natural. \tag{5.10.8}$$

Proposition 5.10.9. Suppose that A is of finite presentation and flat as a K-module, and that $A_\natural$, $B_\natural$ are flat over K. Let $E \in \mathrm{ob}\, D^-(B)$, $M \in \mathrm{ob}\, D^b(K)$ be such that $\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(B^\circ)$. Then $\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \in \mathrm{ob}\, D^b(A^\circ)$, and there is a commutative square

$$\begin{tikzcd} \mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E \arrow[r, "5.10.8"] \arrow[d, "5.10.5"'] & M \overset{\mathrm{L}}{\otimes}_K B_\natural \arrow[d, "M \overset{\mathrm{L}}{\otimes} \mathrm{Tr}_{B/A}"] \\ \mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E \arrow[r, "5.10.8"'] & M \overset{\mathrm{L}}{\otimes}_K A_\natural. \end{tikzcd} \tag{5.10.10}$$

The additional hypotheses on A and B (verified, for example, if A and B are K-algebras of finite groups) are no doubt useless, and one should have a commutative square analogous to 5.10.10 with $A_\natural$ (resp. $B_\natural$) replaced by $A \overset{\mathrm{L}}{\otimes}_{A^e} A$ (resp. $B \overset{\mathrm{L}}{\otimes}_{B^e} B$).

The first assertion of 5.10.9 is trivial. We prove the second. We may suppose that M has injective components and that E has components of the form $B \otimes_K L$, with L flat. Then

$$\mathrm{RHom}_B(E, B \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_B E = \mathrm{Hom}^\bullet_B(E, B \otimes_K M) \otimes_B E,$$

$$\mathrm{RHom}_A(E, A \overset{\mathrm{L}}{\otimes}_K M) \overset{\mathrm{L}}{\otimes}_A E = \mathrm{Hom}^\bullet_A(E, A \otimes_K M) \otimes_A E,$$

and the arrows in 5.10.10 are defined at the level of complexes. Thus we are reduced to verifying that, if L, M are K-modules and E a left B-module, there is a commutative square

$$\begin{tikzcd}
\mathrm{Hom}_B(B \otimes_K L, B \otimes_K M) \otimes_B (B \otimes_K L) \arrow[r, "5.3.3"] \arrow[d, "(1)"'] & M \otimes_K B_\natural \arrow[d, "M \otimes_K \mathrm{Tr}_{B/A}"] \\
\mathrm{Hom}_A(B \otimes_K L, A \otimes_K M) \otimes_A (B \otimes_K L) \arrow[r, "5.3.3"'] & M \otimes_K A_\natural,
\end{tikzcd}$$

where the arrow (1) is defined by sending $B \otimes_K M$ into $\mathrm{Hom}_A(B, A) \otimes_A B \otimes_K M$ by 5.10.1 and composing with an isomorphism of the type 5.10.3 and an adjunction isomorphism. Since A is flat and of finite presentation as a K-module, the tensor-product arrows

$$B \otimes_K \mathrm{Hom}_K(L, M) = \mathrm{Hom}_B(B, B) \otimes_K \mathrm{Hom}_K(L, M) \to \mathrm{Hom}_B(B \otimes_K L, B \otimes_K M),$$

$$\mathrm{Hom}_A(B, A) \otimes_K \mathrm{Hom}_K(L, M) \to \mathrm{Hom}_A(B \otimes_K L, A \otimes_K M)$$

are isomorphisms, by which (1) identifies with the tensor product over K of 5.10.1 by $\mathrm{Hom}_K(L, M) \otimes_K L$. On the other hand, the upper horizontal arrow, respectively the lower horizontal arrow, of the square identifies under these isomorphisms with the tensor product over K of $\mathrm{Tr}_B : B \to B_\natural$ (resp. $\mathrm{ev} : \mathrm{Hom}_A(B, A) \otimes_A B \to A_\natural$) by the evaluation arrow $\mathrm{Hom}_K(L, M) \otimes_K L \to M$. The commutativity therefore follows from that of 5.10.2, completing the proof of 5.10.9.

Corollary 5.10.11. Under the hypotheses of 5.10.9, for $E \in \mathrm{ob}\, D(B)_{\mathrm{parf}}$, $M \in \mathrm{ob}\, D^b(K)$, there is a commutative square in $D(K)$

$$\begin{tikzcd}
\mathrm{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\mathrm{Tr}_B"] \arrow[d] & M \overset{\mathrm{L}}{\otimes}_K B_\natural \arrow[d, "M \overset{\mathrm{L}}{\otimes} \mathrm{Tr}_{B/A}"] \\
\mathrm{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\mathrm{Tr}_A"'] & M \overset{\mathrm{L}}{\otimes}_K A_\natural,
\end{tikzcd} \tag{5.10.12}$$

where the left vertical arrow is restriction of scalars.

It is enough indeed to conjugate 5.10.6 and 5.10.9.

5.11. Application to finite groups: divisibility properties

Let G be a finite group, with neutral element denoted e, and put $K[G] = B$. The K-module $B_\natural$ is free with basis the set $G_\natural$ of conjugacy classes of G; every section s of $B_\natural$ is written uniquely

$$s = \sum_{x \in G_\natural} s(x) x.$$

Proposition 5.11.1. Cf. SGA $4^{1/2}$, Rapport 4.5. Let $E \in \operatorname{ob} D(B)_{\mathrm{parf}}$, $M \in \operatorname{ob} D^b(K)$. For $u \in \operatorname{Hom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E)$, one has, in $H^0(T, M)$,

$$\operatorname{Tr}_K(u) = |G| \operatorname{Tr}_B(u)(e). \tag{5.11.1}$$

It is enough to apply 5.10.11 with $A = K$, noting that $\operatorname{Tr}_{B/K} : B_\natural \to K$ is given by

$$\operatorname{Tr}_{B/K}(g) = \begin{cases} |G|, & g = e, \\ 0, & g \neq e. \end{cases} \tag{5.11.2}$$

Proposition 5.11.3. Under the hypotheses of 5.11.1, let $g \in G$, let Z_g be the centralizer of g in G, and put $A = K[Z_g]$. Let $u \in \operatorname{Hom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E)$, and let $gu \in \operatorname{Hom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E)$ be the composite of the image of u in $\operatorname{Hom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E)$ with left multiplication by g. Then, in $H^0(T, M)$,

$$\operatorname{Tr}_A(gu)(e) = \operatorname{Tr}_B(u)(g^{-1}). \tag{5.11.4}$$

We shall show more generally that there is a commutative square

$$\begin{tikzcd}
\operatorname{RHom}_B(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\operatorname{Tr}_B"] \arrow[d] & M \overset{\mathrm{L}}{\otimes}_K B_\natural \arrow[r, "x \mapsto x(g^{-1})"] & M \\
\operatorname{RHom}_A(E, M \overset{\mathrm{L}}{\otimes}_K E) \arrow[r, "\operatorname{Tr}_A"'] & M \overset{\mathrm{L}}{\otimes}_K A_\natural \arrow[ur, "x \mapsto x(e)"'] &
\end{tikzcd}$$

where the left vertical arrow is the composite of restriction of scalars and left multiplication by g. This square is the boundary of the diagram obtained by inserting $\operatorname{Tr}_{B/A}$ and the left multiplication s_g by g; the central square commutes by 5.10.11, and the lower one by the definition of Tr_A. It remains only to prove that, for $h \in G$,

$$\operatorname{Tr}_A(x \mapsto gxh)(e) = \text{the value at } g^{-1} \text{ of the image of } h \text{ in } B_\natural,$$

i.e.

$$\operatorname{Tr}_A(x \mapsto gxh)(e) = \begin{cases} 1, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases} \tag{*}$$

For this one may replace T by the punctual topos and K by $\mathbb{Z}$, and it is then enough to prove (*) after multiplication by $|Z_g|$. In view of 5.11.1, we are therefore reduced to showing that

$$\operatorname{Tr}_{\mathbb{Z}}(x \mapsto gxh) = \begin{cases} |Z_g|, & g^{-1} \text{ is conjugate to } h, \\ 0, & \text{otherwise.} \end{cases}$$

But this follows from the fact that the left hand side is the number of fixed points of the map $x \mapsto gxh$ from G to G, whence the commutativity of 5.11.4 and the conclusion of 5.11.3.

Corollary 5.11.5. Under the hypotheses of 5.11.3, one has

$$\operatorname{Tr}_K(gu) = |Z_g| \operatorname{Tr}_B(u)(g^{-1}) = |Z_g| \operatorname{Tr}_A(gu)(e). \tag{5.11.5}$$

5.12. Traces and external tensor products

Let, for $i = 1, 2$, A_i be a flat K-algebra, and put $A = A_1 \otimes_K A_2$. Let $E_i \in \operatorname{ob} D^-(A_i)$, $F_i \in \operatorname{ob} D^b(A_i^e)$ be such that $\mathrm{RHom}_{A_i}(E_i, F_i) \in \operatorname{ob} D^b(A_i^\circ)$, let $F = F_1 \overset{\mathrm{L}}{\otimes}_K F_2 \in \operatorname{ob} D^b(A^e)$, and suppose $\mathrm{RHom}_A(E, F) \in \operatorname{ob} D^b(A^\circ)$, where $E = E_1 \overset{\mathrm{L}}{\otimes}_K E_2$. Then there is a commutative square

$$
\begin{tikzcd}
(\mathrm{RHom}_{A_1}(E_1,F_1) \overset{\mathrm{L}}{\otimes}_{A_1} E_1) \overset{\mathrm{L}}{\otimes}_K (\mathrm{RHom}_{A_2}(E_2,F_2) \overset{\mathrm{L}}{\otimes}_{A_2} E_2) \arrow[r, "(1)"] \arrow[d, "(2)"'] & (F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \overset{\mathrm{L}}{\otimes}_K (F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \arrow[d, "(3)"] \\
\mathrm{RHom}_A(E,F) \overset{\mathrm{L}}{\otimes}_A E \arrow[r] & F \overset{\mathrm{L}}{\otimes}_{A^e} A,
\end{tikzcd}
\tag{5.12.1}
$$

where (1) is the tensor product of the arrows 5.3.5, (2) is defined by the tensor-product arrow

$$
\mathrm{RHom}_{A_1}(E_1,F_1) \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2,F_2) \longrightarrow \mathrm{RHom}_A(E,F), \tag{5.12.2}
$$

and (3) is the evident canonical isomorphism. The verification is essentially trivial. One checks in the same way that, for $E_i \in \operatorname{ob} D^-(A_i)$, $F_i \in \operatorname{ob} D^+(A_i^e)$, $G_i \in \operatorname{ob} D^-(A_i)$, with $F_i \overset{\mathrm{L}}{\otimes}_{A_i} G_i \in \operatorname{ob} D^b(A_i)$, $\mathrm{RHom}_{A_i}(E_i, F_i) \in \operatorname{ob} D^b(A_i^\circ)$, $\mathrm{RHom}_{A_i}(E_i, F_i \overset{\mathrm{L}}{\otimes}_{A_i} G_i) \in \operatorname{ob} D^b(K)$, $F \overset{\mathrm{L}}{\otimes}_A G \in \operatorname{ob} D^b(A)$, where $G = G_1 \overset{\mathrm{L}}{\otimes}_K G_2$, there is a commutative square

$$
\begin{tikzcd}
(\mathrm{RHom}_{A_1}(E_1,F_1) \overset{\mathrm{L}}{\otimes}_{A_1} G_1) \overset{\mathrm{L}}{\otimes}_K (\mathrm{RHom}_{A_2}(E_2,F_2) \overset{\mathrm{L}}{\otimes}_{A_2} G_2) \arrow[r, "(1)"] \arrow[d, "(2)"'] & \mathrm{RHom}_{A_1}(E_1, F_1 \overset{\mathrm{L}}{\otimes}_{A_1} G_1) \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2, F_2 \arrow[d, "(3)"] \\
\mathrm{RHom}_A(E,F) \overset{\mathrm{L}}{\otimes}_A G \arrow[r, "5.4.2"'] & \mathrm{RHom}_A(E, F \overset{\mathrm{L}}{\otimes}_A G),
\end{tikzcd}
\tag{5.12.3}
$$

where (2) and (3) are tensor-product arrows of the type 5.12.2, and (1) is the tensor product of the arrows 5.4.2. From the commutativity of the squares 5.12.1 and 5.12.3 it follows that, for $E_i \in \operatorname{ob} D(A_i)_{\mathrm{parf}}$, $F_i \in \operatorname{ob} D^b(A_i^e)$ such that $F = F_1 \overset{\mathrm{L}}{\otimes}_K F_2 \in \operatorname{ob} D^b(A^e)$, there is a commutative square

$$
\begin{tikzcd}
\mathrm{RHom}_{A_1}(E_1, F_1 \overset{\mathrm{L}}{\otimes}_{A_1} E_1) \overset{\mathrm{L}}{\otimes}_K \mathrm{RHom}_{A_2}(E_2, F_2 \overset{\mathrm{L}}{\otimes}_{A_2} E_2) \arrow[r, "(1)"] \arrow[d, "(2)"'] & (F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \overset{\mathrm{L}}{\otimes}_K (F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \arrow[d, "(3)"] \\
\mathrm{RHom}_A(E, F \overset{\mathrm{L}}{\otimes}_A E) \arrow[r, "5.6.1"'] & F \overset{\mathrm{L}}{\otimes}_{A^e} A,
\end{tikzcd}
\tag{5.12.4}
$$

where (1) is the tensor product of the arrows Tr_{A_i} (5.6.1), (2) is defined by 5.12.2, and (3) is the canonical isomorphism appearing in 5.12.1. Consequently, for $u_i \in \mathrm{Hom}_{A_i}(E_i, F_i \overset{\mathrm{L}}{\otimes}_{A_i} E_i)$, one has

$$
\mathrm{Tr}_A(u_1 \overset{\mathrm{L}}{\otimes}_K u_2) = \mathrm{Tr}_{A_1}(u_1)\, \mathrm{Tr}_{A_2}(u_2), \tag{5.12.5}
$$

the product on the right being defined by the canonical arrow

$$
H^0(T, F_1 \overset{\mathrm{L}}{\otimes}_{A_1^e} A_1) \otimes H^0(T, F_2 \overset{\mathrm{L}}{\otimes}_{A_2^e} A_2) \longrightarrow H^0(T, F \overset{\mathrm{L}}{\otimes}_{A^e} A).
$$

5.13. Traces of semilinear endomorphisms

Let A be a K-algebra and φ an endomorphism of A. If E, F are left A-modules, recall that an additive homomorphism $u : E \to F$ is called φ-A-linear if $u(ax) = \varphi(a)u(x)$ for all $a \in A$, $x \in E$, i.e. if u is an A-linear homomorphism from E to the A-module ${}_{(\varphi)}F$ deduced from F by restriction of scalars along φ. If F is an A-bimodule and G a left A-module, there is an evident isomorphism

$$ {}_{(\varphi)}F \otimes_A G \xrightarrow{\sim} {}_{(\varphi)}(F \otimes_A G), $$

which derives trivially to

$$ {}_{(\varphi)}F \overset{\mathrm{L}}{\otimes}_A G \xrightarrow{\sim} {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A G). $$

Assuming A flat over K, one deduces from 5.4.2 the arrow

$$ \mathrm{RHom}_A(E, {}_{(\varphi)}F) \overset{\mathrm{L}}{\otimes}_A G \longrightarrow \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A G)) \tag{5.13.1} $$

for $E \in \mathrm{ob}\, D^-(A)$, $F \in \mathrm{ob}\, D^+(A^e)$, $G \in \mathrm{ob}\, D^-(A)$, with $F \overset{\mathrm{L}}{\otimes}_A G \in \mathrm{ob}\, D^+(A)$. For $E \in \mathrm{ob}\, D(A)_{\mathrm{parf}}$, 5.13.1 is an isomorphism, which allows one to define, by 5.3.5, a trace arrow

$$ \mathrm{Tr}_{A,\varphi} : \mathrm{RHom}_A(E, {}_{(\varphi)}(F \overset{\mathrm{L}}{\otimes}_A E)) \longrightarrow {}_{(\varphi)}F \overset{\mathrm{L}}{\otimes}_{A^e} A. \tag{5.13.2} $$

In particular, for $F = A$, one obtains

$$ \mathrm{Tr}_{A,\varphi} : \mathrm{Hom}_A(E, {}_{(\varphi)}E) \longrightarrow H^0(T, A_{\natural,\varphi}), \tag{5.13.3} $$

where $A_{\natural,\varphi} = {}_{(\varphi)}A \otimes_{A^e} A$ is the quotient K-module of A by the submodule locally generated by local sections of the form $ab - \varphi(b)a$. In the case where $A = K[G]$, G is a discrete group, and φ is defined by an endomorphism of G, $A_{\natural,\varphi}$ is the free K-module with basis $G_{\natural,\varphi}$, where $G_{\natural,\varphi}$ is the quotient of G by the action $(g, x) \mapsto \varphi(g)xg^{-1}$; the orbits are sometimes called φ-conjugacy classes of G. The traces 5.13.2 coincide, for T punctual, with those considered by Grothendieck and discussed in Exposé XII.

In the situation of 5.10, if φ, ψ are respective endomorphisms of A, B, compatible with $A \to B$, there is a unique arrow

$$ \mathrm{Tr}_A : B_{\natural,\psi} \longrightarrow A_{\natural,\varphi} $$

making the square, analogous to 5.10.2,

$$ \begin{tikzcd} {}_{(\psi)}B = \mathrm{Hom}_B(B, {}_{(\psi)}B) \arrow[r, "\mathrm{Tr}_{B,\psi}"] \arrow[d] & B_{\natural,\psi} \arrow[d, "\mathrm{Tr}_A"] \\ \mathrm{Hom}_A(B, {}_{(\varphi)}A) \otimes_A B \arrow[r, "\mathrm{ev}"'] & A_{\natural,\varphi}, \end{tikzcd} \tag{5.13.4} $$

commute, where ev is the evaluation arrow, and the left vertical arrow is restriction of scalars. One defines an arrow analogous to 5.10.5, with $A \otimes_K M$ (resp. $B \otimes_K M$) replaced by ${}_{(\varphi)}A \otimes_K M$ (resp. ${}_{(\psi)}B \otimes_K M$), and there are compatibilities analogous to 5.10.6, 5.10.9, 5.10.11, which we leave to the reader to state.

Let G be a finite group and φ an endomorphism of G. For $g \in G$, let $Z_{g,\varphi}$ be the φ-centralizer of g in G, i.e. the set of $x \in G$ such that $\varphi(x)gx^{-1} = g$. Put $A = K[G]$, and let $E \in \mathrm{ob}\, D(A)_{\mathrm{parf}}$,

$M \in \operatorname{ob} D^b(K)$. The preceding compatibilities easily imply that, for $u \in \operatorname{Hom}_A(E, {}_{(\varphi)}(M \overset{\mathrm{L}}{\otimes}_K E))$ and $g \in G$,

$$\operatorname{Tr}_K(gu) = |Z_{g^{-1},\varphi}|\ \operatorname{Tr}_{A,\varphi}(u)(g^{-1}), \tag{5.13.5}$$

the details of the verification being left to the reader.

6. Equivariant Correspondences and Divisibility of Local Terms

6.1. Notation

In this number, S denotes the spectrum of a field of characteristic p, and Λ a noetherian commutative ring annihilated by an integer not divisible by p. The S-schemes considered will be assumed separated and of finite type. The Λ-algebras considered will be associative, unital, and of finite type, but not necessarily commutative. Most often, and in particular for everything concerning "non-commutative Lefschetz–Verdier local terms," they will be assumed flat; in applications, they will be algebras of finite groups.

If X is an S-scheme and A a Λ-algebra, we write $D({}_AX)$ (resp. $D(X_A)$) for the derived category of the category of sheaves of left (resp. right) A-modules on X. For A commutative, we shall also write $D(X, A)$ for $D({}_AX)$, and $D(X) = D(X, \Lambda)$. We denote by $D_c(-)$ (resp. $D_{\mathrm{ctf}}(-)$) the full subcategory formed by complexes with constructible cohomology (resp. finite tor-dimension and constructible cohomology). If A, B are Λ-algebras, we write $D({}_AX_B)$ for the derived category of sheaves of (A, B)-bimodules, left over A and right over B. Finally, we use the notations A^o, A^e, $A_\natural$ of 5.1.

6.2

The formalism of (III 1, 2, 3) extends without difficulty to the "non-commutative" case. We briefly indicate the main points of this generalization.

Let X be an S-scheme and let A, B be Λ-algebras. If $L \in \operatorname{ob} D_{\mathrm{ctf}}(X_A)$ and $M \in \operatorname{ob} D({}_AX_B)$, with M of finite tor-dimension and constructible cohomology as an object of $D(X_B)$, then

$$L \overset{\mathrm{L}}{\otimes}_A M \in \operatorname{ob} D_{\mathrm{ctf}}(X_B)$$

as is immediate. On the other hand, it follows from (SGA $4^{1/2}$ Finiteness 1.6) that, for M as above and $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_AX)$, one has

$$\operatorname{RHom}_A(L, M) \in \operatorname{ob} D_{\mathrm{ctf}}(X_B).$$

It also follows from loc. cit. that, if f is an S-morphism, $D_{\mathrm{ctf}}({}_A-)$ is stable under the operations $(f^*, f_*, f_!, f^!)$, the case of f^* being trivial.

Let, for $i = 1, 2$, $f_i : X_i \to Y_i$ be an S-morphism,

$$f = f_1 \times_S f_2 : X = X_1 \times_S X_2 \longrightarrow Y = Y_1 \times_S Y_2,$$

and let A be a Λ-algebra. For $L_1 \in \operatorname{ob} D^-(X_{1A})$, $L_2 \in \operatorname{ob} D^-(X^o_{2A})$, there is a Künneth arrow in $D^-(Y)$

$$f_{1!}L_1 \overset{\mathrm{L}}{\otimes}_{S,A} f_{2!}L_2 \longrightarrow f_*(L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2), \tag{6.2.1}$$

where

$$L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \overset{\mathrm{dfn}}{=} p_1^* L_1 \overset{\mathrm{L}}{\otimes}_A p_2^* L_2.$$

As in (III 1.6), one deduces that it is an isomorphism. If B, C are Λ-algebras, with B or C assumed flat over Λ in order to define

$$\overset{\mathrm{L}}{\otimes}_A : D^-(X_{BA}) \times D^-(X_{AC}) \longrightarrow D^-(X_{BC}),$$

there is an isomorphism analogous to 6.2.1 in $D^-(Y_{BC})$, for $L_1 \in \operatorname{ob} D^-(X_{B1A})$, $L_2 \in \operatorname{ob} D^-(X_{A2B})$. One checks similarly that, for $M_1 \in \operatorname{ob} D^-(Y_{1A})$, $M_2 \in \operatorname{ob} D^-(Y^o_{2A})$, there is a Künneth isomorphism in $D^-(X)$

$$f^! M_1 \overset{\mathrm{L}}{\otimes}_{S,A} f^! M_2 \longrightarrow f^!(M_1 \overset{\mathrm{L}}{\otimes}_{S,A} M_2), \tag{6.2.2}$$

and an analogous isomorphism in $D^-({}_B X_C)$ for $M_1 \in \operatorname{ob} D^-(Y_{B1A})$, $M_2 \in \operatorname{ob} D^-(Y_{A2C})$.

Finally, let, for $i = 1, 2$, A_i be a Λ-algebra, A_2 being flat, $E_1 \in \operatorname{ob} D_{\mathrm{ctf}}({}_{A_1}X_1)$, $F_2 \in \operatorname{ob} D^+(X_{A_2})$, $F_1 \in \operatorname{ob} D({}_{A_1}X_{1A_2})$, with F_1 of finite tor-dimension and constructible cohomology as an object of $D(X_{1A_2})$. Then

$$\mathrm{RHom}_{A_1}(E_1, F_1) \in \operatorname{ob} D_{\mathrm{ctf}}(X_{1A_2}),$$

as seen above, and there is a canonical arrow in $D^+(X)$

$$\mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{S,A_2} F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1 \overset{\mathrm{L}}{\otimes}_{S,A_2} F_2), \tag{6.2.3}$$

defined by composing the "inverse-image" arrow (III 2.1.1)

$$p_1^* \mathrm{RHom}_{A_1}(E_1, F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2$$

with the arrow analogous to 5.4.2

$$\mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2 \longrightarrow \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1 \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2),$$

adjoint, by 5.2.3, to the arrow

$$p_1^* E_1 \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1) \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2 \longrightarrow p_1^* F_1 \overset{\mathrm{L}}{\otimes}_{A_2} p_2^* F_2$$

defined by the evaluation arrow 5.3.6. One deduces from 6.2.1, as in (III 2.3), that 6.2.3 is an isomorphism. In particular, for $A_2 = \Lambda$, $F_2 = \Lambda_{X_2}$, 6.2.3 gives an isomorphism

$$p_1^* \mathrm{RHom}_{A_1}(E_1, F_1) \xrightarrow{\sim} \mathrm{RHom}_{A_1}(p_1^* E_1, p_1^* F_1). \tag{6.2.4}$$

6.3

If $t : T \to S$ is an S-scheme, and A is a Λ-algebra, we put

$$t^!_A = KA_T \qquad (\in \operatorname{ob} D^+({}_A T_A)). \tag{6.3.1}$$

For $A = \Lambda$, we write K_T instead of $K\Lambda_T$. It follows from 6.2.2 that there is a canonical isomorphism in $D^+({}_A T_A)$

$$K_T \overset{\mathrm{L}}{\otimes}_\Lambda A \xrightarrow{\sim} KA_T. \tag{6.3.2}$$

We denote by D_A the functor

$$\mathrm{RHom}_A(-, KA_T).$$

It sends $D_{\mathrm{ctf}}({}_AT)$ into $D_{\mathrm{ctf}}(T_A)$ and conversely, and the proof of (SGA $4^{1/2}$ Finiteness Theorem 4) shows that, for $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AT)$, the canonical arrow

$$L \longrightarrow D_A D_A L$$

is an isomorphism. For $A = \Lambda$, we write D instead of D_A.

6.4

Let A be a flat Λ-algebra, and let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, $p_i : X \to X_i$ the projections, and $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AX_i)$. By 6.2.3 applied to $A_1 = A_2 = A$, $E_1 = L_1$, $F_2 = L_2$, $E_2 = KA_{X_1}$, there is an isomorphism

$$D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^* L_1, KA_{X_1} \overset{\mathrm{L}}{\otimes}_{S,A} L_2),$$

whence, by composing with the Künneth isomorphism 6.2.2

$$KA_{X_1} \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} p_2^! L_2,$$

an isomorphism

$$D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2 \xrightarrow{\sim} \mathrm{RHom}_A(p_1^* L_1, p_2^! L_2), \tag{6.4.1}$$

which generalizes (III 3.1.1).

Let $c : C \to X$ be an S-morphism, and put $c_i = p_i c$. By the induction formula (SGA 4 XVIII 3.1.12.2), there is a canonical isomorphism, analogous to (III 3.2.1),

$$c^! \mathrm{RHom}_A(p_1^* L_1, p_2^! L_2) \xrightarrow{\sim} \mathrm{RHom}_A(c_1^* L_1, c_2^! L_2), \tag{6.4.2}$$

and hence, by composing with 6.4.1, an isomorphism

$$H^0(X, c_* c^! (D_A L_1 \overset{\mathrm{L}}{\otimes}_{S,A} L_2)) \xrightarrow{\sim} \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2). \tag{6.4.3}$$

The elements of $\mathrm{Hom}_A(c_1^* L_1, c_2^! L_2)$ will be called cohomological correspondences from L_1 to L_2 with support in c. The terminology differs slightly from that of loc. cit.; the two terminologies coincide when c is proper, which is the case in practice.

Given S-morphisms $f_i : X_i \to Y_i$ $(i = 1, 2)$ and a commutative square (III 3.7.1) with c proper, one defines, as in loc. cit., direct-image arrows analogous to (III 3.7.5), (III 3.7.6), for $L_i \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_AX_i)$.

6.5

Let A be a flat Λ-algebra, X an S-scheme, $X^2 = X \times_S X$, $p_1, p_2 : X^2 \to X$ the canonical projections, $\delta : X \to X^2$ the diagonal, and

$$c : C \longrightarrow X^2$$

an S-morphism. Put $c_i = p_i c$, and let $X^c = C \times_{X^2} X$ be the scheme of fixed points of c, giving a cartesian square

$$\begin{tikzcd} X^c \arrow[r, "i^c"] \arrow[d, "\delta^c"'] & X \arrow[d, "\delta"] \\ C \arrow[r, "c"'] & X^2. \end{tikzcd} \tag{6.5.1}$$

Put

$$KA_{X,L_\natural} = K_X \overset{\mathrm{L}}{\otimes}_\Lambda (A \overset{\mathrm{L}}{\otimes}_{A^e} A), \qquad KA_{X,\natural} = K_X \overset{\mathrm{L}}{\otimes}_\Lambda A_\natural. \tag{6.5.2}$$

Let $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_AX)$. We define a "local trace" homomorphism

$$\mathrm{Tr}_A : \delta^* c_* \mathrm{RHom}_A(p_1^* L, p_2^! L) \longrightarrow KA_{X,L_\natural}, \tag{6.5.3}$$

by composing the isomorphism

$$\delta^* \mathrm{RHom}_A(p_1^* L, p_2^! L) \xrightarrow{\sim} \delta^*(D_A L \overset{\mathrm{L}}{\otimes}_{S,A} L) \tag{6.5.4}$$

deduced from the inverse of 6.4.1 by applying δ^*, the trivial isomorphism

$$\delta^*(D_A L \overset{\mathrm{L}}{\otimes}_{S,A} L) = D_A L \overset{\mathrm{L}}{\otimes}_A L, \tag{6.5.5}$$

and the evaluation arrow 5.3.5

$$D_A L \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X,L_\natural}. \tag{6.5.6}$$

It follows at once from the definition that, for $X = S$, 6.5.3 coincides with the arrow

$$\mathrm{Tr}_A : \mathrm{RHom}_A(L, L) \longrightarrow A \overset{\mathrm{L}}{\otimes}_{A^e} A$$

of 5.6.1. Composing the arrow deduced from 6.5.3 by applying $i^c c^!$ with the arrow

$$\delta^* c^c_* \mathrm{RHom}_A(c_1^* L, c_2^! L) \longrightarrow i^c_* i^{c!} \delta^* \mathrm{RHom}_A(p_1^* L, p_2^! L) \tag{6.5.7}$$

deduced from 6.4.2 and the base-change arrow

$$\delta^* c_* c^! \longrightarrow i^c_* i^{c!} \delta^*,$$

one obtains an arrow

$$\mathrm{Tr}_A : \delta^* c_* \mathrm{RHom}_A(c_1^* L, c_2^! L) \longrightarrow i^c_* i^{c!} KA_{X,L_\natural} = i^c_* KA_{X^c,L_\natural}, \tag{6.5.8}$$

and hence the arrow

$$\mathrm{Tr}_A : \mathrm{Hom}_A(c_1^* L, c_2^! L) \longrightarrow H^0(X^c, KA_{X^c,L_\natural}). \tag{6.5.9}$$

We again write Tr_A for the composites of 6.5.3, 6.5.8, 6.5.9 with the arrows deduced from the canonical arrow

$$KA_{X,L_\natural} \longrightarrow KA_{X,\natural}.$$

One checks easily that, for $A = \Lambda$, one has, for $u \in \mathrm{Hom}(c_1^* L, c_2^! L)$,

$$\mathrm{Tr}(u) = \langle u, 1 \rangle, \tag{6.5.10}$$

with the notation of (III 4.2.5), where 1 denotes the diagonal correspondence. More precisely, the arrow

$$c_* \mathrm{RHom}(c_1^* L, c_2^! L) \longrightarrow \delta_* i^c_* K_{X^c}$$

adjoint to 6.5.8 is the one defined by (III 4.2.4') with $X_1 = X_2$, $L_1 = L_2$, $d = \delta$, and the diagonal correspondence $1 \in \mathrm{Hom}(L, L)$.

6.6

Let A be a flat Λ-algebra, and let, for $i = 1, 2$, X_i be an S-scheme, $X = X_1 \times_S X_2$, and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$. We define a pairing

$$\mathrm{RHom}_A(p_1^* L_1, p_2^! L_2) \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(p_2^* L_2, p_1^! L_1) \longrightarrow KA_{X, L_\natural} \tag{6.6.1}$$

as follows. One identifies the left hand side with

$$\mathrm{RHom}_A(p_1^* L_1, p_1^* KA_{X_1}) \overset{\mathrm{L}}{\otimes} p_2^* L_2 \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_A(p_2^* L_2, p_2^* KA_{X_2}) \overset{\mathrm{L}}{\otimes} p_1^* L_1$$

by the isomorphisms 6.4.1, then with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} \mathrm{RHom}_A(\ ) \overset{\mathrm{L}}{\otimes}_\Lambda p_2^* L_2 \overset{\mathrm{L}}{\otimes}_\Lambda \mathrm{RHom}_A(\ ) \overset{\mathrm{L}}{\otimes}_\Lambda p_1^* L_1 \overset{\mathrm{L}}{\otimes}_{A^e} A$$

by 5.3.1. This object is sent to

$$A \overset{\mathrm{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\mathrm{L}}{\otimes}_\Lambda KA_{X_2}) \overset{\mathrm{L}}{\otimes}_{A^e} A = R$$

by the tensor product of the evaluation arrows 5.3.6. Finally R is identified with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} (KA_{X_1} \overset{\mathrm{L}}{\otimes}_\Lambda KA_{X_2})$$

by 5.8.2, and then with $KA_{X, L_\natural}$ by 6.5.2 and (III 1.7.6).

If $c : C \to X$, $d : D \to X$ are correspondences, and

$$e = c \times_X d : E \to X,$$

one deduces from 6.6.1, as in (III 4.2), a pairing

$$c_* \mathrm{RHom}_A(c_1^* L_1, c_2^! L_2) \overset{\mathrm{L}}{\otimes}_\Lambda d_* \mathrm{RHom}_A(d_2^* L_2, d_1^! L_1) \longrightarrow e_* KA_{E, L_\natural}, \tag{6.6.2}$$

and hence a cup-product

$$\langle\ ,\ \rangle_A : \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2) \otimes \mathrm{Hom}_A(d_2^* L_2, d_1^! L_1) \longrightarrow H^0(E, KA_{E, L_\natural}). \tag{6.6.3}$$

It is not difficult to extend to the non-commutative case the notion of composition of correspondences studied in (III 5.2), and one checks that, for

$$u \in \mathrm{Hom}_A(c_1^* L_1, c_2^! L_2), \qquad v \in \mathrm{Hom}_A(d_2^* L_2, d_1^! L_1),$$

one has the formula analogous to (III 5.2.9)

$$\langle u, v \rangle_A = \mathrm{Tr}_A(\bar{v} u) = \mathrm{Tr}_A(u \bar{v}). \tag{6.6.4}$$

The Lefschetz formula (III 4.4) generalizes without difficulty in this context. Given proper S-morphisms $f_i : X_i \to Y_i$ $(i = 1, 2)$ and a commutative cube (III 4.4.0), for $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_A X_i)$ one has a commutative square analogous to (III 4.4.1), with K_C (resp. K_D) replaced by $KA_{C, L_\natural}$ (resp. $KA_{D, L_\natural}$), and a formula analogous to (III 4.5.1), in $H^0(D, KA_{D, L_\natural})$:

$$\langle f_* u, f_* v \rangle_A = g_* \langle u, v \rangle_A. \tag{6.6.5}$$

The arrow

$$f_* c_* KA_{C, L_\natural} \longrightarrow d_* KA_{D, L_\natural}$$

is deduced from arrow (4) of (III 4.4.1) by tensoring with

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A.$$

6.7

Let X and c be as in 6.5, and let $A \to B$ be a morphism of flat Λ-algebras. With respect to extension and restriction of scalars, the arrows Tr_A (6.5.3, 6.5.8, 6.5.9) behave like the arrows Tr_A of 5.6.1, 5.6.2. The point is that, if f is an S-morphism, the functors $(f^*, f_*, f_!, f^!)$ commute with extension and restriction of scalars; consequently, the study of the behavior of Tr_A (6.5.3) with respect to extension and restriction of scalars reduces to the behavior of the "tensor-product" arrow

$$\mathrm{RHom}_A(p_1^*L, p_1^*KA_X) \overset{\mathrm{L}}{\otimes}_A p_2^*L \longrightarrow \mathrm{RHom}_A(p_1^*L, p_1^*KA_X \overset{\mathrm{L}}{\otimes}_A p_2^*L)$$

and of the "evaluation" arrow

$$\mathrm{RHom}_A(L, KA_X) \overset{\mathrm{L}}{\otimes}_A L \longrightarrow KA_{X,L_\natural}.$$

Thus the commutativity of the squares 5.9.1, 5.9.2 implies the commutativity of the square

$$\begin{tikzcd} \delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) \arrow[r, "\mathrm{Tr}_A"] \arrow[d] & i_*^c KA_{X^c,L_\natural} \arrow[d] \\ \delta^* c_* \mathrm{RHom}_B(c_1^*(B \overset{\mathrm{L}}{\otimes}_A L), c_2^!(B \overset{\mathrm{L}}{\otimes}_A L)) \arrow[r, "\mathrm{Tr}_B"'] & i_*^c KB_{X^c,L_\natural}, \end{tikzcd} \tag{6.7.1}$$

where the left vertical arrow is extension of scalars and the right vertical arrow is defined by the canonical arrow

$$A \overset{\mathrm{L}}{\otimes}_{A^e} A \longrightarrow B \overset{\mathrm{L}}{\otimes}_{B^e} B.$$

It follows that, for $u \in \mathrm{Hom}_A(c_1^*L, c_2^!L)$,

$$\mathrm{Tr}_B(B \overset{\mathrm{L}}{\otimes}_A u) = \varphi\, \mathrm{Tr}_A(u), \tag{6.7.2}$$

where

$$\varphi : H^0(X^c, KA_{X^c,L_\natural}) \longrightarrow H^0(X^c, KB_{X^c,L_\natural})$$

is the canonical map.

Assume A is finite type and flat as a Λ-module, B finite type and flat as a left A-module, and $A_\natural$, $B_\natural$ flat over Λ. There is a map $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$; we shall again denote by

$$\mathrm{Tr}_{B/A} : KB_{X,\natural} \longrightarrow KA_{X,\natural} \tag{6.7.3}$$

the arrow deduced from it by applying $K_X \overset{\mathrm{L}}{\otimes} -$ (6.5.2). It follows from 5.10.6 and 5.10.9 that, for $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_BX)$, there is a commutative square

$$\begin{tikzcd} \delta^* c_* \mathrm{RHom}_B(c_1^*L, c_2^!L) \arrow[r, "\mathrm{Tr}_B"] \arrow[d] & i_*^c KB_{X^c,\natural} \arrow[d, "\mathrm{Tr}_{B/A}"] \\ \delta^* c_* \mathrm{RHom}_A(c_1^*L, c_2^!L) \arrow[r, "\mathrm{Tr}_A"'] & i_*^c KA_{X^c,\natural}, \end{tikzcd} \tag{6.7.4}$$

where the left vertical arrow is restriction of scalars. Consequently, for $u \in \mathrm{Hom}_B(c_1^*L, c_2^!L)$, one has, in $H^0(X^c, KA_{X^c,\natural})$,

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_{B/A}(\mathrm{Tr}_B(u)). \tag{6.7.5}$$

6.8

Let A, B be flat Λ-algebras, X, Y S-schemes, $Z = X \times_S Y$, and let $c : C \to X^2$, $d : D \to Y^2$ be correspondences. Let

$$e = c \times_S d : E = C \times_S D \longrightarrow Z^2$$

be their product. Then

$$Z^e = X^c \times_S Y^d.$$

Put $R = A \otimes_\Lambda B$. As in (III 5.3), one defines the external tensor product

$$\operatorname{Hom}_A(c_1^* L, c_2^! L) \otimes \operatorname{Hom}_B(d_1^* M, d_2^! M) \xrightarrow{\overset{\mathrm{L}}{\otimes}_S} \operatorname{Hom}_R(e_1^*(L \overset{\mathrm{L}}{\otimes}_S M), e_2^!(L \overset{\mathrm{L}}{\otimes}_S M)), \tag{6.8.1}$$

and one deduces easily from the commutativity of the squares 5.12.1, 5.12.3 that, for

$$u \in \operatorname{Hom}_A(c_1^* L, c_2^! L), \qquad v \in \operatorname{Hom}_B(d_1^* M, d_2^! M),$$

one has, in $H^0(Z^e, KR_{Z^e, L_\natural})$,

$$\operatorname{Tr}_R(u \overset{\mathrm{L}}{\otimes}_S v) = \operatorname{Tr}_A(u) \operatorname{Tr}_B(v), \tag{6.8.2}$$

the product on the right being defined by the canonical arrow

$$H^0(X^c, KA_{X^c, L_\natural}) \otimes H^0(Y^d, KB_{Y^d, L_\natural}) \longrightarrow H^0(Z^e, KR_{Z^e, L_\natural}), \tag{6.8.3}$$

coming from the isomorphism

$$KA_{X^c, L_\natural} \overset{\mathrm{L}}{\otimes}_S KB_{Y^d, L_\natural} \longrightarrow K(A \otimes B)_{Z^e, L_\natural}$$

given by (III 1.7.6) and arrow (4) of 5.9.1.

6.9

Let G be a finite group. If X is a G-S-scheme, that is, an S-scheme endowed with an action of G by S-automorphisms, and A is a Λ-algebra, we denote by $D(G, {}_A X)$ the derived category of the category of sheaves of left G-A-modules on X. For the notions of G-sheaf of sets, rings, modules, and so on, on X for the étale topology, we refer the reader to (X 1). We denote by the subscript c (resp. ctf) the full subcategory formed by complexes of G-A-modules which, as complexes of A-modules, have constructible cohomology (resp. finite tor-dimension and constructible cohomology).

Similarly, if B is a Λ-algebra, we denote by $D(G, {}_A X_B)$ the derived category of sheaves of G-(A, B)-bimodules on X. If G acts trivially on X, a left G-A-module on X is just a left $A[G]$-module on X without operators, hence

$$D(G, {}_A X) = D({}_{A[G]} X).$$

Since G is finite, one also has

$$D_c({}_A X) = D_c({}_{A[G]} X),$$

but only an inclusion

$$D_{\mathrm{ctf}}({}_{A[G]} X) \subset D_{\mathrm{ctf}}(G, {}_A X).$$

The duality formalism of (SGA 4 XVII, XVIII) extends without difficulty to G-S-schemes. Since the functor forgetting the action of G is faithful, it is enough to extend naturally the definition of the "six operations" ($\overset{\mathrm{L}}{\otimes}, \mathrm{RHom}, f^*, f_*, f_!, f^!$) and the canonical arrows considered in loc. cit. (base change, Künneth, duality, etc.). This extension presents no difficulty: for the definition of the functors $f_!$, $f^!$, it is enough to use equivariant compactifications. We leave to the reader the task of paraphrasing the definitions and results 6.2 to 6.8 in the setting of G-S-schemes.

If, for $i = 1, 2$, X_i is a G-S-scheme, $c : C \to X = X_1 \times_S X_2$ is a G-S-morphism, and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}(G, {}_A X_i)$, the elements of

$$\mathrm{Hom}_A(c_1^* L_1, c_2^! L_2),$$

where Hom_A is taken in the category $D(G, {}_A-)$, will be called equivariant cohomological correspondences from L_1 to L_2 with support in c. Note that, when G acts trivially on X_i ($i = 1, 2$) and $L_i \in \operatorname{ob} D_{\mathrm{ctf}}({}_{A[G]} X_i)$, the isomorphism 6.4.1 in $D(G, X)$ refines to an isomorphism

$$D_{A[G]} L_1 \overset{\mathrm{L}}{\otimes}_{S, A[G]} L_2 \xrightarrow{\sim} \mathrm{RHom}_{A[G]}(p_1^* L_1, p_2^! L_2).$$

It follows that, in a G-equivariant situation of type 6.5, with trivial action of G on the spaces, and with $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{A[G]} X) \subset D_{\mathrm{ctf}}(G, {}_A X)$, one may, for

$$u \in \mathrm{Hom}_{A[G]}(c_1^* L, c_2^! L),$$

define a local term

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, K A[G]_{X^c, L_\natural}),$$

finer, in the evident sense, than the local term

$$\mathrm{Tr}_A(u) \in H^0(G, X^c, K A_{X^c, L_\natural})$$

obtained by using only finite tor-dimension of L with respect to A.

6.10

Let X, c be as in 6.5, and let $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]} X)$. For

$$u \in \mathrm{Hom}_{\Lambda[G]}(c_1^* L, c_2^! L),$$

6.5.9 gives a local trace

$$\mathrm{Tr}_{\Lambda[G]}(u) \in H^0(X^c, K\Lambda[G]_{X^c, \natural}) = H^0(X^c, K_{X^c} \otimes \Lambda[G]_\natural), \tag{6.10.1}$$

where $G_\natural$ is the set of conjugacy classes of G. For every $x \in G_\natural$, one may consider the "value of $\mathrm{Tr}_{\Lambda[G]}(u)$ at x,"

$$\mathrm{Tr}_{\Lambda[G]}(u)(x) \in H^0(X^c, K_{X^c}),$$

the image of $\mathrm{Tr}_{\Lambda[G]}(u)$ under the projection

$$H^0(X^c, K_{X^c} \otimes \Lambda[G]_\natural) \longrightarrow H^0(X^c, K_{X^c})$$

defined by x. It follows from 6.7.5 and 5.11.2 that

$$\mathrm{Tr}_\Lambda(u) = |G| \ \mathrm{Tr}_{\Lambda[G]}(u)(e), \tag{6.10.2}$$

where e denotes the identity element of G.

More generally, let $g \in G$, let Z_g be the centralizer of g, and let

$$gu \in \mathrm{Hom}_{\Lambda[Z_g]}(c_1^* L, c_2^! L)$$

be the correspondence obtained by composing u with left multiplication by $g : c_1^* L \to c_1^* L$. We have $L \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[Z_g]}X)$, and hence 6.10.2 implies

$$\mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[Z_g]}(gu)(e). \tag{6.10.3}$$

By the same argument as in 5.11.3, one deduces from 6.7.4 the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gu)(e) = \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}), \tag{6.10.4}$$

which allows 6.10.3 to be rewritten, if desired, in the form

$$\mathrm{Tr}_\Lambda(gu) = |Z_g|\ \mathrm{Tr}_{\Lambda[G]}(u)(g^{-1}). \tag{6.10.5}$$

6.11

Let $f : X \to Y$ be a proper G-S-morphism, and let

$$\begin{tikzcd} C \arrow[r, "c"] \arrow[d] & X \times_S X \arrow[d, "f \times_S f"] \\ D \arrow[r, "d"'] & Y \times_S Y \end{tikzcd} \tag{6.11.1}$$

be a commutative equivariant square of G-S-schemes, with c and d proper. Assume that G acts trivially on Y and D. We denote by

$$f^{c/d} \quad (\text{or } f^c) : X^c \longrightarrow Y^d \tag{6.11.2}$$

the morphism induced by 6.11.1 on the schemes of fixed points of c and d, with the notation of 6.5.1. Let $L \in \mathrm{ob}\, D_{\mathrm{ctf}}(G, {}_A X)$, and let

$$u \in \mathrm{Hom}(c_1^* L, c_2^! L)$$

be a G-equivariant correspondence with support in c. By direct image, it gives a G-equivariant correspondence

$$f_* u \in \mathrm{Hom}_{\Lambda[G]}(d_1^* f_* L, d_2^! f_* L).$$

For $g \in G$, let Z_g denote the centralizer of g, and let

$$gu \in \mathrm{Hom}((gc)_1^* L, c_2^! L) \tag{6.11.3}$$

be the Z_g-equivariant correspondence with support in

$$gc \overset{\mathrm{dfn}}{=} (gc_1 = c_1 g, c_2) : C \to X \times_S X, \tag{6.11.4}$$

defined as the composite of the arrows

$$g^*(c_1^* L) \xrightarrow{\alpha_g} c_1^* L \xrightarrow{u} c_2^! L,$$

where α_g is the arrow giving the action of G on $c_1^* L$. It is clear that

$$f_*(gu) = g(f_* u). \tag{6.11.5}$$

Proposition 6.12. Under the hypotheses of 6.11, assume that

$$f_* L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y).$$

Then, for every $g \in G$,

$$f_*^{gc} \operatorname{Tr}_\Lambda(gu) = |Z_g| \ \operatorname{Tr}_{\Lambda[G]}(f_* u)(g^{-1}), \tag{6.12.1}$$

where

$$f_*^{gc} : H^0(X^{gc}, K_{X^{gc}}) \longrightarrow H^0(Y^d, K_{Y^d})$$

is the Gysin morphism defined by the adjunction arrow

$$f_*^{gc} K_{X^{gc}} = f_*^{gc} f^{gc!} K_{Y^d} \longrightarrow K_{Y^d}.$$

Indeed, by 6.10.5 applied to $f_* u$, and by 6.11.5, one has

$$\operatorname{Tr}_\Lambda(f_*(gu)) = |Z_g| \ \operatorname{Tr}_{\Lambda[G]}(f_* u)(g^{-1}),$$

and 6.12.1 therefore follows from the Lefschetz formula (III 4.5.1).

6.13

Let ℓ be a prime number distinct from p. For every integer $n \geq 1$, put

$$\Lambda_n = \mathbb{Z}/\ell^n.$$

Under the hypotheses of 6.11, let $L = (L_n)$ be a projective system of objects of

$$D_{\mathrm{ctf}}(G, X, \Lambda_n)$$

such that

$$L_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n \xrightarrow{\sim} L_n \quad (m \geq n),$$

and let

$$u \in \operatorname{Hom}(c_1^* L, c_2^! L)$$

be a projective system of equivariant correspondences $u_n \in \operatorname{Hom}(c_1^* L_n, c_2^! L_n)$ such that, for $m \geq n$,

$$u_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n = u_n$$

modulo the isomorphism $L_m \overset{\mathrm{L}}{\otimes}_{\Lambda_m} \Lambda_n = L_n$. By compatibility of local traces with extension of scalars, the local terms, for $g \in G$,

$$\operatorname{Tr}_{\Lambda_n}(gu_n) \in H^0(X^{gc}, K\Lambda_{n,X^{gc}})$$

define an element

$$\operatorname{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(X^{gc}, K_{\ell,X^{gc}}) \overset{\mathrm{dfn}}{=} \varprojlim_n H^0(X^{gc}, K\Lambda_{n,X^{gc}}),$$

and the terms

$$f_*^{gc}\,\mathrm{Tr}_{\Lambda_n}(gu_n) = \mathrm{Tr}_{\Lambda_n}(gf_*u_n)$$

an element

$$f_*\,\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(Y^d, K_{\ell,Y^d}) \stackrel{\text{dfn}}{=} \varprojlim_n H^0(Y^d, K\Lambda_{n,Y^d}).$$

Assume moreover that

$$f_*M_n \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda_n[G]}Y)$$

for all n. Then the local traces

$$\mathrm{Tr}_{\Lambda_n[G]}(f_*u_n)$$

similarly define an element

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u) \in H^0(Y^d, K\mathbb{Z}_\ell[G]_{Y^d,\natural}) \stackrel{\text{dfn}}{=} \varprojlim_n H^0(Y^d, K\Lambda_n[G]_{Y^d,\natural}).$$

It follows from 6.12 that

$$f_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) = |Z_g|\ \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)(g^{-1}). \tag{6.13.1}$$

6.14

Let X be a G-S-scheme, let $j : U \to X$ be an equivariant open immersion, and let

$$c : C \to X \times_S X$$

be a morphism of G-S-schemes such that $c_2 = (p_2c) : C \to X$ is quasi-finite and of finite tor-dimension. Suppose that

$$c_1^{-1}(U) \subset c_2^{-1}(U).$$

Then there are commutative squares

$$\begin{array}{ccccc} U & \longleftarrow & c_1^{-1}(U) & \longleftarrow & c_2^{-1}(U) \longrightarrow U \\ \downarrow & & \downarrow & & \downarrow \\ X & \xleftarrow{c_1} & C & \xrightarrow{c_2} & X \\ \uparrow & & \uparrow & & \uparrow \\ X-U & \longleftarrow & c_2^{-1}(X-U) & \longrightarrow & X-U \end{array} \tag{6.14.1}$$

The trace morphism

$$\mathrm{Tr}_{c_2} : c_{2!}\Lambda_C \longrightarrow \Lambda_X$$

then defines a filtered G-equivariant correspondence

$$\underline{c} \in \mathrm{Hom}(c_1^*\Lambda_X, c_2^!\Lambda_X), \tag{6.14.2}$$

where Λ_X is endowed with the filtration defined by the submodule $i_!\Lambda_U$, that is,

$$F^n\Lambda_X = \Lambda_X \quad (n \leq 0), \qquad F^1\Lambda_X = i_!\Lambda_U, \qquad F^n\Lambda_X = 0 \quad (n \geq 1).$$

The associated graded consists of two G-equivariant correspondences

$$\mathrm{gr}^0\,\underline{c} = \underline{c}_{X-U,X} \in \mathrm{Hom}(c_1^*\Lambda_{X-U,X}, c_2^!\Lambda_{X-U,X}),$$

$$\mathrm{gr}^1\,\underline{c} = \underline{c}_{U,X} \in \mathrm{Hom}(c_1^*\Lambda_{U,X}, c_2^!\Lambda_{X-U,X}), \tag{6.14.3}$$

where we have put

$$\Lambda_{X-U,X} = j_*\Lambda_{X-U}, \qquad \Lambda_{U,X} = i_!\Lambda_U.$$

We have

$$\underline{c}_{X-U,U} = j_*\underline{c}_{X-U}, \tag{6.14.4}$$

where

$$\underline{c}_{X-U} \in \mathrm{Hom}(c_1^*\Lambda_{X-U}, c_2^!\Lambda_{X-U})$$

is the correspondence defined by $\mathrm{Tr}_{c_2,X-U}$. We write $\underline{c}$, $\underline{c}_{U,X}$, and so on, when we want to specify the base ring.

Forgetting the action of G, there are local terms

$$\mathrm{Tr}_\Lambda(\underline{c}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{U,X}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) \in H^0(X^c, K_{X^c}), \tag{6.14.5}$$

which, by additivity of traces (III 4.13.1), satisfy the relation

$$\mathrm{Tr}_\Lambda(\underline{c}) = \mathrm{Tr}_\Lambda(\underline{c}_{U,X}) + \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}). \tag{6.14.6}$$

Moreover, by 6.14.4,

$$\mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) = j_*^c\,\mathrm{Tr}_\Lambda(\underline{c}_{X-U}), \tag{6.14.7}$$

where

$$j_*^c : H^0((X-U)^c, K_{(X-U)^c}) \longrightarrow H^0(X^c, K_{X^c})$$

is the direct-image morphism defined by $j^c : (X-U)^c \to X^c$. Recall (III 4.3) that, if X is smooth, c a closed immersion, and X^c finite, $\mathrm{Tr}_\Lambda(\underline{c})$ is the image in Λ of the intersection multiplicity (C, X) of C with the diagonal. If, moreover, $X-U$ is smooth, $\mathrm{Tr}_\Lambda(\underline{c}_{X-U})$ is the image in Λ of $(c_2^{-1}(X-U), X-U)$. Note also that, if $X-U$ is proper over S and $c|X-U$ is the diagonal, then the Lefschetz formula (III 4.7) implies that

$$\int_{X-U} \mathrm{Tr}(\underline{c}_{X-U}) \in \Lambda$$

is the Euler–Poincaré characteristic of $X-U$ with values in Λ.

6.15

Let there be a G-equivariant diagram 6.11.1, with f, c, d proper, and an equivariant open immersion $U \to X$. Suppose that c_2 is quasi-finite and of finite tor-dimension, that

$$c_1^{-1}(U) \subset c_2^{-1}(U),$$

that G acts trivially on C and D, and that

$$f|U : U \longrightarrow f(U) = V$$

is an étale Galois covering with group G. This last hypothesis implies that

$$Rf_*(\Lambda_{U,X}) \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[G]}Y) :$$

indeed $Rf_*(\Lambda_{U,X})$ is concentrated in degree 0 and is extension by zero of a sheaf on V locally isomorphic to $\Lambda[G]$. Hence one may apply 6.12 to $L = \Lambda_{U,X}$, $u = \underline{c}_{U,X}$, and, in view of 6.14.6 and 6.14.7, one obtains:

Proposition 6.16. Under the hypotheses of 6.15, for every $g \in G$ one has

$$f_*^{gc} \operatorname{Tr}_\Lambda(g\underline{c}) - (fj)_*^{gc} \operatorname{Tr}_\Lambda(g\underline{c}_{X-U}) = |Z_g| \ \operatorname{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1}), \tag{6.16.1}$$

where $j : X - U \to X$ is the inclusion, and Z_g denotes the centralizer of g.

Applying 6.16 to the projective system of correspondences $\underline{c}_\ell = (\underline{c}_{\Lambda_n})$, where $\Lambda_n = \mathbb{Z}/\ell^n$ with ℓ prime distinct from p, and taking into account that 6.14.6 and 6.14.7 give, after passage to the limit and application of f_*^{gc},

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) = f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,U,X}) + (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}),$$

one deduces, by 6.13:

Corollary 6.17. Under the hypotheses of 6.15, for every $g \in G$ one has

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_\ell) - (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(g\underline{c}_{\ell,X-U}) = |Z_g| \ \operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_*\underline{c}_{\ell,U,X})(g^{-1}), \tag{6.17.1}$$

with the notation of 6.16.

In view of the compatibility of local traces with localization and of the remarks made in 6.14, one sees that 6.17 gives an affirmative solution, in a more general formulation, to Grothendieck's divisibility conjecture (XII 4.5), in the case where the endomorphism φ of G considered in loc. cit. is the identity.

In fact, by following 5.13, it is not difficult to give divisibility statements analogous to 6.12, 6.13, 6.16, 6.17, in the case where c is not G-equivariant, but where c_2 is and c_1 satisfies

$$c_1(xg) = c_1(x)\varphi(g)$$

for every point x of G, φ being a given endomorphism of G. The divisibility formulas are then written as above, with Z_g replaced by the φ-centralizer of g^{-1} (5.13.5).

6.18

Put

$$N_G = \sum_{g\in G} g \in \Lambda[G]. \tag{6.18.1}$$

Let X be an S-scheme. For $L \in \operatorname{ob} D({}_{\Lambda[G]}X)$, left multiplication by N_G induces an arrow in $D(X)$

$$N_G : \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} L \longrightarrow \operatorname{RHom}_{\Lambda[G]}(\Lambda, L), \tag{6.18.2}$$

where Λ is regarded as a $\Lambda[G]$-algebra by the natural augmentation sending every $g \in G$ to 1.

Lemma 6.18.3. If $L \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}X)$, the arrow 6.18.2 is an isomorphism.

By dévissage, one reduces to the case where L is perfect, then to $L = \Lambda[G]$, and the assertion follows from the fact that N_G (6.18.1) is a basis of the module of G-invariants of the left $\Lambda[G]$-module $\Lambda[G]$.

Proposition 6.19. In the situation of 6.11, let $M \in \operatorname{ob} D_{\mathrm{ctf}}(Y)$,

$$v \in \operatorname{Hom}(d_1^* M, d_2^! M),$$

and let $a : M \to f_* L$ be an arrow in $D({}_{\Lambda[G]}Y)$, where M is regarded as an object of $D({}_{\Lambda[G]}Y)$ by the augmentation $\Lambda[G] \to \Lambda$, such that the square

$$\begin{tikzcd} d_1^* M \arrow[r, "v"] \arrow[d] & d_2^! M \arrow[d] \\ d_1^* f_* L \arrow[r, "f_* u"'] & d_2^! f_* L \end{tikzcd}$$

is commutative. Suppose that $f_* M \in \operatorname{ob} D_{\mathrm{ctf}}({}_{\Lambda[G]}Y)$ and that a induces an isomorphism

$$M \xrightarrow{\sim} \operatorname{RHom}_{\Lambda[G]}(\Lambda, f_* L).$$

Then

$$\operatorname{Tr}_\Lambda(v) = \sum_{g \in G_\natural} \operatorname{Tr}_{\Lambda[G]}(f_* u)(g^{-1}). \tag{6.19.1}$$

By 6.18.3, a, composed with N_G^{-1}, gives an isomorphism

$$M \xrightarrow{\sim} \Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} f_* L,$$

under which v is identified with

$$\Lambda \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} f_* u.$$

By 6.7.2, $\operatorname{Tr}_\Lambda(v)$ is therefore the image of $\operatorname{Tr}_{\Lambda[G]}(f_* u)$ under the arrow induced by the augmentation

$$\Lambda[G] \longrightarrow \Lambda.$$

This augmentation, by definition, sends every conjugacy class to 1, whence the conclusion.

6.20

Let, for $i = 1, 2$, X_i be a G-S-scheme, and let

$$X = X_1 \times_S X_2, \qquad c : C \to X$$

be a G-S-morphism such that c_2 is quasi-finite and of finite tor-dimension. Let $L_i \in \operatorname{ob} D^+(G, {}_A X_i)$. The trace morphism

$$\operatorname{Tr}_{c_2} : c_{2!} c_2^! L_2 \longrightarrow L_2$$

defines by adjunction an arrow in $D(G, X)$, again denoted

$$\operatorname{Tr}_{c_2} : c_2^! L_2 \longrightarrow c_2'^{\,!} L_2, \tag{6.20.1}$$

which associates to every G-equivariant morphism

$$u \in \mathrm{Hom}(c_1^* L_1, c_2^* L_2)$$

a G-equivariant correspondence

$$u^! = \mathrm{Tr}_{c_2} \circ u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2). \tag{6.20.2}$$

Proposition 6.21. Let $f : X \to Y$ be a proper G-morphism, and let 6.11.1 be an equivariant commutative square, with trivial action of G on Y and D. Suppose c, d are proper, c_2, d_2 quasi-finite and of finite tor-dimension, and the square

$$\begin{tikzcd} C \arrow[r, "c_2"] \arrow[d, "h"'] & X \arrow[d, "f"] \\ D \arrow[r, "d_2"'] & Y \end{tikzcd}$$

is cartesian and tor-independent (SGA 6 III 1.5). Let $M \in \mathrm{ob}\, D_{\mathrm{ctf}}(Y)$,

$$v \in \mathrm{Hom}(d_1^* M, d_2^* M),$$

hence a G-equivariant homomorphism

$$h^* v \in \mathrm{Hom}(c_1^* f^* M, c_2^* f^* M),$$

and G-equivariant correspondences

$$(h^* v)^! \in \mathrm{Hom}(c_1^* f^* M, c_2^! f^* M), \qquad f_*((h^* v)^!) \in \mathrm{Hom}(d_1^* (f_* f^* M), d_2^! f_* f^* M).$$

Then there is a commutative square of arrows in $D({}_{\Lambda[G]}Y)$, where M is regarded as an object of $D({}_{\Lambda[G]}Y)$ by the trivial action,

$$\begin{tikzcd} d_1^* M \arrow[r, "v^!"] \arrow[d] & d_2^! M \arrow[d] \\ d_1^* f_* f^* M \arrow[r, "f_*((h^* v)^!)"'] & d_2^! f_* f^* M, \end{tikzcd}$$

where the vertical arrows are given by the adjunction arrow $M \to f_* f^* M$.

Put $h^* v = u$, $f^* M = L$, and consider the diagram

$$\begin{tikzcd} d_1^* M \arrow[r, "v"] \arrow[d] & d_2^* M \arrow[r, "\mathrm{Tr}_{d_2}"] \arrow[d] & d_2^! M \arrow[d] & \\ d_1^* f_* L \arrow[r] & h_* c_1^* L \arrow[r, "h_* \mathrm{Tr}_{c_2}"] & h_* c_2^! L \arrow[r] & d_2^! f_* L. \end{tikzcd}$$

The arrows (1) and (3) are given by the adjunction arrow $M \to f_* L$, (2) by the adjunction arrow $d_2^* M \to h_* h^* (d_2^* M) = h_* c_2^* L$, (4) by the base-change arrow, and (5) is the canonical isomorphism

(SGA 4 XVIII 3.1.12.3). By definition, $v^!$ is the composite of the upper row. On the other hand, it follows from the interpretation of the direct image of a correspondence (III 3.5 b), 3.7) that $f_*u^!$ is the composite of the lower row. It remains only to prove commutativity of the squares (A) and (B). The commutativity of (A) follows immediately from the definition of the base-change arrow. As for (B), it follows from the fact that, in view of the tor-independence hypothesis, Tr_{d_2} is compatible with base change by f (SGA $4^{1/2}$ Cycle 2.3.3).

Applying 6.19, one obtains:

Corollary 6.22. Under the hypotheses of 6.21, assume that

$$f_*f^*M \in \mathrm{ob}\, D_{\mathrm{ctf}}({}_{\Lambda[G]}Y),$$

and that the adjunction arrow

$$M \to f_*f^*M$$

induces an isomorphism

$$M \xrightarrow{\sim} \mathrm{RHom}_{\Lambda[G]}(\Lambda, f_*f^*M).$$

Then

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g\in G_\natural} \mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}). \tag{6.22.1}$$

Proposition 6.23. Under the hypotheses of 6.21, suppose moreover that there is an open immersion

$$i : V \to Y$$

such that

$$d_1^{-1}(V) = d_2^{-1}(V),$$

and that f induces a principal Galois covering with group G,

$$f_V : U = f^{-1}(V) \to V.$$

Suppose also that M is of the form $M = i_!E$, where $E \in \mathrm{ob}\, D(V)$, and that there is an isomorphism

$$\alpha : f^*E \xrightarrow{\sim} \Lambda_X \overset{\mathrm{L}}{\otimes}_\Lambda E_0$$

in $D(G, X)$, with $E_0 \in \mathrm{ob}\, D(\Lambda[G])$, perfect over Λ. Let

$$v \in \mathrm{Hom}(d_1^*M, d_2^*M)$$

be as in 6.21, and let

$$v_0 \in \mathrm{Hom}_{\Lambda[G]}(E_0, E_0)$$

be the homomorphism defined, thanks to α, by $f^*v|c_1^{-1}(U)$. Then, if $\underline{c}_{U,X}$ denotes the correspondence defined in 6.14.3, with the notation 6.20.2 one has

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g\in G_\natural} \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1})\, \mathrm{Tr}_\Lambda(gv_0). \tag{6.23.1}$$

Note that, by the central character of Tr_Λ, the right hand side does not depend on the choice of α.

By the projection formula

$$f_*\Lambda_Y \overset{\mathrm{L}}{\otimes} M \xrightarrow{\sim} f_*f^*M$$

and 6.15, M satisfies the hypotheses of 6.22. On the other hand, α defines an isomorphism

$$f^*M \xrightarrow{\sim} \Lambda_{U,X} \otimes E_0$$

under which $(f^*v)^!$ is identified with the external tensor product $\underline{c}_{U,X} \otimes v_0$. By the projection formula, f_*f^*M is therefore identified with $f_*(\Lambda_{U,X}) \otimes E_0$, and $f_*((f^*v)^!)$ with the external tensor product $f_*\underline{c}_{U,X} \otimes v_0$. For $g \in G$, the Z_g-equivariant correspondence $gf_*((f^*v)^!)$ is therefore identified with

$$gf_*\underline{c}_{U,X} \otimes gv_0.$$

Since, by 6.10.4,

$$\mathrm{Tr}_{\Lambda[G]}(f_*((f^*v)^!))(g^{-1}) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*((f^*v)^!))(e),$$

it is enough, in view of 6.22.1, to prove the formula

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[G]}(f_*\underline{c}_{U,X})(g^{-1})\,\mathrm{Tr}_{\Lambda}(gv_0),$$

or, equivalently by 6.10.4,

$$\mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{\Lambda[Z_g]}(gf_*\underline{c}_{U,X})(e)\,\mathrm{Tr}_{\Lambda}(gv_0). \tag{6.23.2}$$

Replacing G by Z_g if necessary, we may suppose that $g = e$. Returning to the definition of the traces, one sees that it is enough to establish the following lemma.

Lemma 6.23.3. (cf. SGA $4^{1/2}$ Report 4.7). Let P be a G_V-module locally a direct factor of a free module of finite type, let $P' = i_!P$, and let Q be a complex of $\Lambda[G]$-modules, perfect over Λ. Then $P \overset{\mathrm{L}}{\otimes}_\Lambda Q$, considered as an object of $D({}_{\Lambda[G]}V)$ by the diagonal action of G, is perfect over $\Lambda[G]$, and there is a commutative square

$$\begin{tikzcd}
\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q,Q) \arrow[r] \arrow[d] & K_Y \arrow[d, equal] \\
\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P' \overset{\mathrm{L}}{\otimes} Q, p_2^!(P' \overset{\mathrm{L}}{\otimes} Q)) \arrow[r] & K_Y,
\end{tikzcd}$$

where $\delta : Y \to Y \times_S Y$ is the diagonal, the upper horizontal arrow is the tensor product of the arrows

$$\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \xrightarrow{6.5.3} K\Lambda[G]_{Y,\natural} \xrightarrow{x \mapsto x(e)} K_Y$$

and

$$\mathrm{RHom}_{\Lambda[G]}(Q,Q) \xrightarrow{\mathrm{Tr}_\Lambda} \Lambda,$$

the vertical arrow is deduced from the external tensor product by restriction along the diagonal

$$\Lambda[G] \to \Lambda[G] \otimes \Lambda[G],$$

and the lower horizontal arrow is the composite of 6.5.3 and of $x \mapsto x(e)$.

The first assertion is evident, for one reduces to $P = \Lambda[G]_V$, and $P \overset{\mathrm{L}}{\otimes}_\Lambda Q$ is then isomorphic to the tensor product of P by Q considered with the trivial action of G. For the second, first note that, by 6.5.4 and 6.5.5, one has

$$\delta^*\mathrm{RHom}_{\Lambda[G]}(p_1^*P', p_2^!P') \xrightarrow{\sim} D_{\Lambda[G]}(i_!P) \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} i_!P \xrightarrow{\sim} i_!(D_{\Lambda[G]}P \overset{\mathrm{L}}{\otimes}_{\Lambda[G]} P),$$

so that we are reduced to checking that the square induced by the preceding square on V commutes. We may therefore suppose that $V = Y$. The square then reads

$$\begin{array}{ccc}
\mathrm{RHom}_{\Lambda[G]}(P, K_Y \overset{\mathrm{L}}{\otimes} P) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \longrightarrow & K_Y \\
\downarrow & & \| \\
\mathrm{RHom}_{\Lambda[G]}(P \overset{\mathrm{L}}{\otimes} Q, K_Y \overset{\mathrm{L}}{\otimes} P \overset{\mathrm{L}}{\otimes} Q) & \longrightarrow & K_Y,
\end{array}$$

where the vertical arrow is the external tensor product, the upper horizontal arrow is given by

$$\mathrm{Tr}_{\Lambda[G]}(-)(e) \otimes \mathrm{Tr}_\Lambda,$$

and the lower horizontal arrow by $\mathrm{Tr}_{\Lambda[G]}(-)(e)$. By descent, one reduces to $P = \Lambda[G]_Y$, and, by the initial remark, one may then suppose that G acts trivially on Q. The assertion is then an immediate consequence of 5.12.5.

6.24

If Λ is annihilated by a power of a prime number $\ell \neq p$, the right hand side of 6.23.1 may be rewritten in the form

$$\sum_{g \in G_\natural} \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_* \underline{c}_{\ell,U,X})(g^{-1}) \, \mathrm{Tr}_\Lambda(gv_0),$$

with the notation of 6.17. Formula 6.17.1 shows that the local terms defined by Grothendieck in (XII 6.2) coincide with Verdier's local terms, at least when the endomorphism φ of G considered in loc. cit. is the identity; but, as already observed, this restriction is not serious. If Y is proper over S, then by conjugating 6.23.1 with the Lefschetz formula for $Y \to S$ (III 4.7), one obtains a formula generalizing (XII 6.3), and a fortiori the Lefschetz formula for Frobenius proved in (SGA $4^{1/2}$ Report) and Part I of the present exposé.

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In SGA $4^{1/2}$:

Finiteness Theorem: finiteness theorems in ℓ-adic cohomology;

Cycle: the cohomology class associated with a cycle;

Report: report on the trace formula.

Exposé V. J-adic Projective Systems

by J.-P. Jouanolou

Throughout this exposé, the letter A denotes a ring with unit. The present exposé is intended to develop a formalism which will be used in the following exposé for the study of ℓ-adic cohomology.

1. Generalities on abelian A-categories

Definition 1.1. An A-category (resp. an abelian A-category) is a pair $(\mathcal{C}, \rho)$ formed by an additive (resp. abelian) category $\mathcal{C}$ and a homomorphism of rings

$$\rho : A \longrightarrow \mathrm{End}(\mathrm{id}_{\mathcal{C}})$$

from A to the ring of endomorphisms of the identity functor from $\mathcal{C}$ to $\mathcal{C}$.

Given two A-categories (resp. abelian A-categories) $(\mathcal{C}, \rho)$ and $(\mathcal{C}', \rho')$, a morphism of A-categories (resp. of abelian A-categories) from $(\mathcal{C}, \rho)$ to $(\mathcal{C}', \rho')$ is an additive (resp. exact) functor $F : \mathcal{C} \to \mathcal{C}'$ such that, for every object M of $\mathcal{C}$ and every element a of A, the equality below is satisfied:

$$F(\rho(a)_M) = \rho'(a)_{F(M)}.$$

Example 1.1.1

Let $\mathcal{C}$ be an additive (resp. abelian) category. Recall that the category of left $(A, \mathcal{C})$-modules, denoted $(A, \mathcal{C})$-mod, is the category described as follows:

(i) Its objects are pairs (X, α) consisting of an object X of $\mathcal{C}$ and a homomorphism of rings

$$\alpha : A \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X, X).$$

(ii) A morphism $(X,\alpha) \to (Y,\beta)$ is a $\mathcal{C}$-morphism $u : X \to Y$ such that, for every element a of A, the following diagram is commutative:

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[d, "\alpha(a)"'] & Y \arrow[d, "\beta(a)"] \\ X \arrow[r, "u"'] & Y. \end{tikzcd}$$

The category $(A,\mathcal{C})$-mod is evidently an $A^\natural$-category (resp. an abelian $A^\natural$-category), where $A^\natural$ is the centre of A. Moreover, it has the following properties:

(a) The functor $(A,\mathcal{C})\text{-mod} \to \mathcal{C}$ obtained by forgetting the A-module structure commutes with inductive and projective limits.

(b) Let I be a category. In order that the I-indexed inductive (resp. projective) limits of $(A,\mathcal{C})$-mod be representable, it is enough that this be so for the I-indexed inductive (resp. projective) limits of $\mathcal{C}$.

1.2. External tensor product

Let $(\mathcal{C},\rho)$ be an abelian A-category. If M is a right A-module and X an object of $\mathcal{C}$, the external tensor product of X by M, denoted $M \otimes_A X$, is the covariant functor from $\mathcal{C}$ to the category of sets, defined on objects Y of $\mathcal{C}$ by the equality below and in the evident way on arrows:

$$(M \otimes_A X)(Y) = \mathrm{Hom}_A(M, \mathrm{Hom}_{\mathcal{C}}(X,Y)),$$

where $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ is regarded as a right A-module via the action of A on X. Interpreting $M \otimes_A X$ and Y as objects of the category $\widehat{\mathcal{C}} = \mathrm{Hom}(\mathcal{C}, \mathrm{Ens})$, the preceding equality takes the more familiar form

$$\mathrm{Hom}_{\widehat{\mathcal{C}}}(M \otimes_A X, Y) = \mathrm{Hom}_A(M, \mathrm{Hom}_{\mathcal{C}}(X,Y)).$$

If the functor $M \otimes_A X$ is representable, a representative will be denoted and named in the same way.

If $u : M \to M'$ and $v : X \to X'$ are two morphisms, respectively in (A-mod) and in $\mathcal{C}$, one defines in the usual way a $\mathcal{C}$-morphism

$$u \otimes_A v : M \otimes_A X \longrightarrow M' \otimes_A X',$$

which allows us to speak of the external tensor-product bifunctor, covariant in both variables. Its properties are summarized in the following proposition.

Proposition 1.2.1. The tensor-product bifunctor is additive and commutes with inductive limits.

The proof, classical in the case of modules over a ring, is immediate.

Proposition 1.2.2 (Representability criteria).

(a) For every object X of $\mathcal{C}$, $A \otimes_A X$ is represented by X.

(b) For every finitely presented A-module M and every object X of $\mathcal{C}$, the functor $M \otimes_A X$ is representable.

(c) If the category $\mathcal{C}$ has arbitrary direct sums, the functor $M \otimes_A X$ is representable for every pair (M, X).

Proof. Assertion (a) is clear. The other two assertions then follow from the lemma below, whose proof is immediate.

Lemma 1.2.3. Let $M \to N \to P \to 0$ be an exact sequence of A-modules and let X be an object of $\mathcal{C}$. If the functors $M \otimes_A X$ and $N \otimes_A X$ are represented by objects Y and Z of $\mathcal{C}$, then the functor $P \otimes_A X$ is represented by the cokernel of the morphism $Y \to Z$ corresponding to $\alpha \otimes_A \mathrm{id}_X$.

Remark 1.2.4

Let $\mathfrak{G}$ be an ideal of finite type of A. For every object X of $\mathcal{C}$, the functor $(A/\mathfrak{G}) \otimes_A X$ is representable, and the arrow

$$p_X : X \longrightarrow (A/\mathfrak{G}) \otimes_A X$$

coming from the canonical epimorphism from A onto $A/\mathfrak{G}$ and from part (a) of 1.2.2 is an epimorphism. The same statement remains valid when the ideal $\mathfrak{G}$ is no longer assumed to be of finite type, but $\mathcal{C}$ has arbitrary direct sums. Put $\mathfrak{G}X = \mathrm{Ker}(p_X)$. When the category has filtered inductive limits, it is clear that

$$\mathfrak{G}X = \sum_{a \in \mathfrak{G}} aX.$$

If $u : X \to Y$ is an epimorphism, u maps $\mathfrak{G}X$ epimorphically onto $\mathfrak{G}Y$. Finally, for an object X to be annihilated by every element of $\mathfrak{G}$, it is necessary and sufficient that $\mathfrak{G}X = 0$. Then every morphism from an object Z to X factors, by means of p_Z, through a morphism from $(A/\mathfrak{G}) \otimes_A Z$ to X.

2. The Mittag-Leffler-Artin-Rees condition

Throughout this paragraph, the letter $\mathcal{C}$ denotes an abelian category and we put

$$P = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}),$$

the category of projective systems indexed by the ordered set $\mathbb{N}$ of positive integers and with values in $\mathcal{C}$.

2.1. The Mittag-Leffler-Artin-Rees condition

Let $X = (X_n, u_n)_{n \geq 0}$ be an object of the category P. If r denotes an integer ≥ 0, we agree to write $X[r]$ for the projective system $(X_{n+r}, u_{n+r})_{n \geq 0}$. A morphism $h : X \to Y$ of P defines in the evident way a morphism denoted $h[r]$ from $X[r]$ to $Y[r]$, functorially. If r and s are two integers satisfying $s \geq r \geq 0$, the transition morphisms $X_{n+s} \to X_{n+r}$ $(n \geq 0)$ define a morphism

$$w_{sr} : X[s] \longrightarrow X[r],$$

and it is clear that, if t is a third integer with $t \geq s \geq r$, the equality below is satisfied:

$$w_{sr} \circ w_{ts} = w_{tr}.$$

In other words, we have thus associated to X an object $\overline{X}$ of the category $\mathrm{Hom}(\mathbb{N}^\circ, P)$. Moreover, if $h : X \to Y$ is a morphism of P, there corresponds to it in the evident way a morphism $\overline{h}$ from $\overline{X}$ to $\overline{Y}$, functorially. Finally, if

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

is an exact sequence of P, the corresponding sequence below in $\overline{P} = \mathrm{Hom}(\mathbb{N}^\circ, P)$ is exact:

$$0 \longrightarrow \overline{X} \xrightarrow{\overline{u}} \overline{Y} \xrightarrow{\overline{v}} \overline{Z} \longrightarrow 0.$$

Definition 2.1.1. Let an object of the category P be given. It is said to satisfy the Mittag-Leffler-Artin-Rees condition (abbreviated MLAR) if there exists an integer r such that, for every integer $s \geq r$, the equality below is satisfied:

$$\mathrm{Im}(X[s] \xrightarrow{w_{s0}} X) = \mathrm{Im}(X[r] \xrightarrow{w_{r0}} X).$$

Equivalently, for every integer $n \geq 0$, the images of X_{n+r} and X_{n+s} in X_n by the transition morphisms are the same. Finally, a third interpretation is that the associated projective system $\overline{X}$ satisfies the Mittag-Leffler condition (abbreviated ML) (EGA 0_{III} 13.1.2). The following proposition is deduced from this by (EGA 0_{III} 13.2.1).

Proposition 2.1.2. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence of P.

(i) If Y satisfies (MLAR), then Z satisfies (MLAR).

(ii) If X and Z satisfy (MLAR), then Y satisfies (MLAR).

2.2. AR-null projective systems

Definition 2.2.1. Let X be an object of the category P. It is called AR-null if there exists an integer $r \geq 0$ such that the canonical morphism $X[r] \to X$ is zero.

It is clear in particular that an AR-null projective system satisfies the condition (MLAR).

Proposition 2.2.2. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence of P. The following assertions are equivalent:

(i) Y is AR-null;

(ii) X and Z are AR-null.

In other words, the full subcategory P_0 of P, whose objects are the AR-null projective systems, is thick in P.

Proof. It is clear that (i) implies (ii). Conversely, let r and t be integers satisfying the condition of Definition 2.2.1 for X and Z respectively. One sees immediately that the integer $s+t$ satisfies the conditions of Definition 2.2.1 for Y.

2.3. Reminders on the calculus of fractions and applications

Let $\mathcal{E}$ be a category and S a subset of $\mathrm{Fl}(\mathcal{E})$. A category of fractions of $\mathcal{E}$ with respect to S is a pair $(p, \mathcal{E}(S^{-1}))$ consisting of a category $\mathcal{E}(S^{-1})$ and a functor $p : \mathcal{E} \to \mathcal{E}(S^{-1})$ satisfying the following universal property: for a functor u from $\mathcal{E}$ to a category $\mathcal{F}$ to factor through p, it is necessary and sufficient that it make all arrows belonging to S invertible; the factorization is then unique. The uniqueness of such a pair up to isomorphism is clear. We refer the reader to Exposé I of the Séminaire Homotopique de Strasbourg for the proof of existence, and also for most of the results stated without proof in what follows in this paragraph. See also P. Gabriel and M. Zisman, *Calculus of fractions and homotopy theory*, Erg. der Math., 35, Springer-Verlag, 1967.

Certain properties of the multiplicative part S ensure a simple description of the category $\mathcal{E}(S^{-1})$.

Definition 2.3.1. A subset S of $\mathrm{Fl}(\mathcal{E})$ is said to admit a right calculus of fractions if the following conditions hold:

(a) The identity arrows of $\mathcal{E}$ belong to S.

(b) If $u : X \to Y$ and $v : Y \to Z$ belong to S, the composite $v \circ u$ belongs to S.

(c) Every diagram of type (D_1) below, in which t belongs to S, can be completed to a diagram of type (D_2), in which s belongs to S:

$$(D_1) \qquad X \xrightarrow{u} Y \xleftarrow{t} Y'$$

$$(D_2) \qquad \begin{tikzcd} X' \arrow[r, "u'"] \arrow[d, "s"'] & Y' \arrow[d, "t"] \\ X \arrow[r, "u"'] & Y. \end{tikzcd}$$

(d) For every commutative diagram

$$X \underset{g}{\overset{f}{\rightrightarrows}} Y \xrightarrow{s} Z,$$

in which s belongs to S, there exists a commutative diagram

$$T \xrightarrow[t]{} X \underset{g}{\overset{f}{\rightrightarrows}} Y,$$

in which t belongs to S.

Dually one defines the following notion.

Definition 2.3.2. A subset S of $\mathrm{Fl}(\mathcal{E})$ is said to admit a left calculus of fractions if the following conditions hold:

(a') The identity arrows of $\mathcal{E}$ belong to S.

(b') The composite of two arrows belonging to S belongs to S.

(c') Every diagram of type $(D_1)'$ below, in which s belongs to S, can be completed to a diagram of type $(D_2)'$, in which t belongs to S:

$$(D_1)' \qquad \begin{tikzcd} X' \arrow[r, "u'"] \arrow[d, "s"'] & Y' \\ X & \end{tikzcd} \qquad (D_2)' \qquad \begin{tikzcd} X' \arrow[r, "u'"] \arrow[d, "s"'] & Y' \arrow[d, "t"] \\ X \arrow[r, "u"'] & Y. \end{tikzcd}$$

(d') For every commutative diagram Δ of the form below, in which s belongs to S, there exists a morphism t with source Y and belonging to S such that $t \circ f = t \circ g$:

$$(\Delta) \qquad X' \xrightarrow{s} X \underset{g}{\overset{f}{\rightrightarrows}} Y.$$

2.3.3

Suppose that S is a subset of $\mathrm{Fl}(\mathcal{E})$ admitting a right calculus of fractions. It follows from part (d) of 2.3.1 that, for every object X of $\mathcal{E}$, the category $(S/X)^\circ$, opposite to the category of arrows over X belonging to S, is filtering (SGA 4 I 2.7). Consider then the category $\mathcal{E}_S$ described as follows:

- the objects of $\mathcal{E}_S$ are the objects of $\mathcal{E}$;
- if X and Y are two objects of $\mathcal{E}$,

$$\mathrm{Hom}_{\mathcal{E}_S}(X, Y) = \varinjlim_{(S/X)^\circ} \mathrm{Hom}_{\mathcal{E}}(\mathrm{source}(\cdot), Y).$$

If P_S denotes the evident functor from $\mathcal{E}$ to $\mathcal{E}_S$, the pair $(P_S, \mathcal{E}_S)$ is a solution of the universal problem described at the beginning of the paragraph. In this case we agree to write $\mathcal{E}(S^{-1})$ for this particular solution.

2.3.4

Let now S be a subset of $\mathrm{Fl}(\mathcal{E})$ admitting a left calculus of fractions. Then the functor $P_S : \mathcal{E} \to \mathcal{E}(S^{-1})$ commutes with finite inductive limits. In particular, if $\mathcal{E}$ is additive, $\mathcal{E}(S^{-1})$ is additive and P_S is an additive functor.

2.3.5

If $\mathcal{E}$ is an abelian category and if S is a subset of $\mathrm{Fl}(\mathcal{E})$ admitting both a right calculus of fractions and a left calculus of fractions, the category $\mathcal{E}(S^{-1})$ is abelian and the functor P_S is exact. Moreover, the pair $(P_S, \mathcal{E}(S^{-1}))$ is a quotient of $\mathcal{E}$ by the thick abelian subcategory $P_S^{-1}(0)$.

2.4. The category $P_{AR} = \mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$

Returning to the notation of 2.1, let AR be the set of canonical arrows (cf. 2.1)

$$v_{rX} : X[r] \longrightarrow X,$$

where X runs through the set of objects of P and r through the set of integers ≥ 0.

Proposition 2.4.1. The subset AR of $\mathrm{Fl}(P)$ admits both a right calculus of fractions and a left calculus of fractions.

Proof. The parts (a) $\Rightarrow$ (a') and (b) $\Rightarrow$ (b') of 2.3.1 and 2.3.2 are clear.

We prove (c). Let (D_1) be the commutative diagram below, in which v_{rY} denotes the canonical arrow from $Y[r]$ to Y:

$$\begin{tikzcd} & Y[r] \arrow[d, "v_{rY}"'] \\ X \arrow[r] & Y. \end{tikzcd}$$

The diagram (D_2) below is commutative, and v_{rX}, the canonical morphism from $X[r]$ to X, belongs to AR, whence the assertion:

$$\begin{tikzcd} X[r] \arrow[r, "u[r]"] \arrow[d, "v_{rX}"'] & Y[r] \arrow[d, "v_{rY}"] \\ X \arrow[r, "u"'] & Y. \end{tikzcd}$$

We prove (d). Let a commutative diagram be given:

$$X \underset{g}{\overset{f}{\rightrightarrows}} Z[r] \xrightarrow{v_{rZ}} Z.$$

Applying translation by r, one obtains a commutative diagram

$$X[r] \underset{g[r]}{\overset{f[r]}{\rightrightarrows}} Z[2r] \xrightarrow{v_{rZ}[r]} Z[r].$$

It is clear that the following diagram is commutative, so that $f \circ v_{rX} = g \circ v_{rX}$:

$$\begin{tikzcd} X[r] \arrow[r, shift left, "f[r]"] \arrow[r, shift right, "g[r]"'] \arrow[d, "v_{rX}"'] & Z[2r] \arrow[d, "v_{rZ}[r]"] \\ X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Z[r]. \end{tikzcd}$$

We prove (c'). First observe that, if Y denotes an object of P and r a positive integer, there exists an object T of P such that $Y = T[r]$ and such that, for every object X of P, the homomorphism given by translation by r

$$\mathrm{Hom}_P(X, T) \longrightarrow \mathrm{Hom}_P(X[r], Y)$$

is a bijection. The projective system (T_n, w_n) defined as follows plainly works:

$$T_n = 0 \quad \text{for } n \geq r-1,$$

and $T_n = Y_{n-r}$ otherwise, with $w_n = 0$ for $n \geq r-1$ and $w_n = u_{n-r}$ otherwise.

This said, suppose a diagram of the following type is given, and show that it can be completed as indicated in the statement of (c'):

$$\begin{tikzcd} X[r] \arrow[r, "u'"] \arrow[d, "v_{rX}"'] & Y' \\ X. & \end{tikzcd}$$

For this, let Y be a projective system such that $Y' = Y[r]$ and such that the canonical map

$$\mathrm{Hom}_P(X, Y) \longrightarrow \mathrm{Hom}_P(X[r], Y')$$

is a bijection. Then it is clear that the following diagram is commutative, where u denotes the unique morphism from X to Y such that $u' = u[r]$:

$$\begin{tikzcd} X[r] \arrow[r, "u'"] \arrow[d, "v_{rX}"'] & Y' = Y[r] \arrow[d, "v_{rY}"] \\ X \arrow[r, "u"'] & Y. \end{tikzcd}$$

We prove (d'). Consider the commutative diagram below, and show that there exists a morphism w from Y such that $w \circ f = w \circ g$:

$$\begin{tikzcd} X[r] \arrow[r, "v_{rX}"] & X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y. \end{tikzcd}$$

It is first of all clear that the following diagram is commutative:

$$\begin{tikzcd} X[r] \arrow[r, shift left, "{f[r]}"] \arrow[r, shift right, "{g[r]}"'] \arrow[d, "v_{rX}"'] & Y[r] \arrow[d, "v_{rY}"] \\ X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y. \end{tikzcd}$$

In particular $v_{rY} \circ f[r] = v_{rY} \circ g[r]$. Let now T be the system associated to Y and r as at the beginning of the proof of (c'), so that in particular $Y = T[r]$. The translates by r of the morphisms $v_{rT} \circ f$ and $v_{rT} \circ g$ are respectively equal to $v_{rY} \circ f$ and $v_{rY} \circ g[r]$, hence are equal. It follows from the fact that translation by r is injective on morphisms from X to T that the equality below holds, whence the assertion:

$$v_{rT} \circ f = v_{rT} \circ g.$$

Definition 2.4.2. Let $\mathcal{C}$ be an abelian category. The category of projective systems of $\mathcal{C}$ up to translation, denoted $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$, is the category described as follows:

(i) The objects of $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$ are the objects of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$.

(ii) If X and Y are two objects of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, a morphism from X to Y in the category $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$ is an element of the set

$$\mathrm{Hom}_{AR}(X, Y) = \varinjlim_{r} \mathrm{Hom}(X[r], Y),$$

where, for every pair (r, s) of integers with $s \geq r$, the map

$$\mathrm{Hom}(X[r], Y) \longrightarrow \mathrm{Hom}(X[s], Y)$$

comes from the canonical morphism

$$X[s] \longrightarrow X[r].$$

The composition of morphisms is defined in the evident way.

In what follows, the category $\mathcal{C}$ being fixed, we put, to simplify notation,

$$P = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}), \qquad P_{AR} = \mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C}).$$

2.4.3

We have seen (2.4.1) that the subset AR of $\mathrm{Fl}(P)$ admits a right calculus of fractions. This allows us (2.3.3) to consider the category

$$P'_{AR} = P(AR^{-1}),$$

which, recall, has the same objects as P. Let X and Y be two objects of P, and denote by $\mathrm{Hom}'_{AR}(X, Y)$ the set of isomorphisms from X to Y in P'_{AR}. An element of $\mathrm{Hom}_{AR}(X, Y)$ defines in the evident way an element of $\mathrm{Hom}'_{AR}(X, Y)$. More precisely, one thus defines a functor

$$m : P_{AR} \longrightarrow P'_{AR}.$$

Proposition 2.4.3. The functor m is an isomorphism of categories.

Proof. We must see that, for every pair (X, Y) of objects of P, the corresponding map

$$m_{XY} : \mathrm{Hom}_{AR}(X, Y) \longrightarrow \mathrm{Hom}'_{AR}(X, Y)$$

is a bijection. It is plainly a surjection. For injectivity, one is immediately reduced to seeing that, for every commutative diagram of the type below, there exists an integer t greater than or equal to r and s such that the composite of ρ and the canonical morphism from $X[t]$ to $X[s]$ is the canonical morphism from $X[t]$ to $X[r]$:

$$\begin{tikzcd}
 & X[s] \arrow[dl, "\rho"'] \\
X[r] \arrow[d, "v_{rX}"'] & \\
X &
\end{tikzcd}$$

But this follows from commutativity of the diagram below, in which the letter c denotes all canonical morphisms involved:

$$\begin{tikzcd}
X[r+s] \arrow[r, "\rho(s)"] \arrow[d, "c"'] & X[2s] \arrow[d, "c"] \arrow[dl, "c"'] \\
X[r] & X[s] \arrow[l, "\rho"]
\end{tikzcd}$$

Proposition 2.4.4.

(i) The category P_{AR} is abelian.

(ii) The functor $p_{AR} : P \to P_{AR}$ is exact.

(iii) For an object X of P to be AR-null, it is necessary and sufficient that $p_{AR}(X) = 0$.

(iv) The pair (p_{AR}, P_{AR}) is a quotient of the abelian category P by the thick abelian subcategory of AR-null objects.

Proof. If (iii) is admitted, the other assertions are consequences of (2.3.5). It follows from (2.3.4) that, for an object X of P to be AR-null, it is necessary and sufficient that $p_{AR}(\mathrm{id}_X) = 0$, whence (iii).

Corollary 2.4.5. Let $u : X \to Y$ be a morphism in the category P. The following assertions are equivalent:

(i) $p_{AR}(u)$ is an isomorphism.

(ii) The kernel and cokernel of u are AR-null.

A morphism $u : X \to Y$ in the category P of projective systems, satisfying the equivalent conditions (i) and (ii), will be called an AR-isomorphism.

Definition 2.4.6. Two objects X and Y of the category P are called AR-isomorphic if $p_{AR}(X)$ and $p_{AR}(Y)$ are isomorphic in the category P_{AR}. If X and Y are two AR-isomorphic objects of the category P, there need not exist in general an AR-isomorphism u from X to Y.

2.5. Characterization of projective systems satisfying the Mittag-Leffler-Artin-Rees condition

Proposition 2.5.1. The following assertions for an object X of P are equivalent:

(i) X satisfies (MLAR) (2.1.1).

(ii) There exists a strict projective system Y and an AR-isomorphism $u : Y \to X$ (2.4.5).

(iii) There exists a strict projective system Y such that X and Y are isomorphic in the category P_{AR}.

(iv) X satisfies the Mittag-Leffler condition and, denoting by X' the projective system of universal images of X, the projective system X/X' is AR-null.

Proof. We shall prove the chain of implications

$$(i) \Rightarrow (iv) \Rightarrow (iii) \Rightarrow (ii) \Rightarrow (i).$$

$(i) \Rightarrow (iv)$. There exists an integer m such that the canonical morphism w_{mX} from $X[m]$ to X factors through the canonical inclusion i of X' in X. It follows that $p_{AR}(i)$ is a direct epimorphism and therefore, since p_{AR} is exact, that $p_{AR}(X/X') = 0$. We conclude by (2.4.4) (iii).

$(iv) \Rightarrow (iii)$. Condition (iv) means, with the preceding notation, that $p_{AR}(i)$ is an epimorphism. But because p_{AR} is exact, it is also a monomorphism, hence an isomorphism. For Y one takes the projective system X'.

$(iii) \Rightarrow (ii)$. The P_{AR}-isomorphism from Y to X whose existence is assumed is the image under p_{AR} of a P-morphism $Y[r] \to X$, where r denotes a positive or zero integer. The conclusion follows from the fact that $Y[r]$ is strict.

$(ii) \Rightarrow (i)$. We may suppose Y contained in X and u the canonical injection. Let T be the cokernel of u. The hypothesis means that T is AR-null, hence there exists an integer r such that, for every $s \geq r$, the canonical morphism from $T[s]$ to T is zero. It follows that, for $s \geq r$, the image of $X[s]$ in X is contained in Y; since it also plainly contains the image, equal to Y, of $Y[s]$ in Y, it is equal to Y. Thus Y is the image of $X[s]$ for s sufficiently large, whence X satisfies (MLAR), and Y is the projective system of universal images of X.

Corollary 2.5.2. Let $u : Y \to X$ be an AR-isomorphism, with Y strict. Under these conditions, X satisfies (MLAR) and $u(Y)$ is the projective system of universal images of X.

3. J-adic and AR-J-adic projective systems

Let A be a commutative ring with unit, let J be an ideal of A, and let $(\mathcal{C}, \rho)$ be an abelian A-category. We suppose either that J is of finite type, or that $\mathcal{C}$ has infinite direct sums. The notation is the same as in the preceding paragraph. In particular, the symbols P and P_{AR} denote respectively the categories $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ and $\mathrm{Hom}_{AR}(\mathbb{N}^\circ, \mathcal{C})$.

3.1. J-adic projective systems

Definition 3.1.1. Let X be an object of P. It is called J-adic if it satisfies the following conditions:

(i) for every integer $n \geq 0$, $J^{n+1} X_n = 0$;

(ii) for every pair (m, n) of integers with $m \geq n \geq 0$, the morphism

$$A/J^{n+1} \otimes_A X_m \longrightarrow X_n$$

deduced from the transition morphism $X_m \to X_n$ is an isomorphism.

We shall write J-ad$(\mathcal{C})$ for the full subcategory of P generated by the J-adic projective systems.

Remark 3.1.2

Let I and J be two ideals of A and suppose either that I and J are of finite type, or that $\mathcal{C}$ has infinite direct sums. Then one can show that, if I and J define the same topology on A, the categories I-ad$(\mathcal{C})$ and J-ad$(\mathcal{C})$ are equivalent.

Proposition 3.1.3 (Stability under exact sequences). Let

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

be an exact sequence of P.

(i) If X is strict and Y is J-adic, then Z is J-adic.

(ii) If Y is strict and Z is J-adic, then X is strict.

(iii) If X and Z are J-adic, then Y is J-adic.

Proof. If $m \geq n$, it is clear that the diagram (T) below, in which the vertical arrows denote the transition morphisms, is commutative and exact:

$$\begin{tikzcd} A/J^{n+1} \otimes_A X_m \arrow[r, "\mathrm{id} \otimes u_m"] \arrow[d, "f_{nm}"'] & A/J^{n+1} \otimes_A Y_m \arrow[r, "\mathrm{id} \otimes v_m"] \arrow[d, "g_{nm}"] & A/J^{n+1} \otimes_A Z_m \arrow[r] \arrow[d, "h_{nm}"] & 0 & \\ 0 \arrow[r] & X_n \arrow[r] & Y_n \arrow[r] & Z_n \arrow[r] & 0. \end{tikzcd}$$

In case (i), f_{nm} is an epimorphism and g_{nm} an isomorphism. In case (ii), g_{nm} is an epimorphism and h_{nm} an isomorphism. In case (iii), f_{nm} and h_{nm} are isomorphisms. In all three cases, the snake lemma applied to the diagram (T) gives the conclusion.

3.2. AR-J-adic projective systems

Proposition 3.2.1. The following assertions, for an object X of P such that $J^{n+1}X_n = 0$ for every $n \geq 0$, are equivalent:

(i) There exists a J-adic projective system Y and an AR-isomorphism $u : Y \to X$ (2.4.5).

(ii) There exists a J-adic projective system Y, isomorphic to X in the category P_{AR}.

Proof. We have to show that (ii) implies (i). A P_{AR}-isomorphism from Y to X is represented, for some integer r, by an AR-isomorphism $v : Y[r] \to X$. Since $J^{n+1}X_n = 0$ for every $n \geq 0$ and Y is J-adic, v is the composite of a P-morphism $u : Y \to X$ and the canonical morphism from $Y[r]$ to Y. It is then clear that u is an AR-isomorphism, whence the assertion.

Definition 3.2.2. A projective system indexed by $\mathbb{N}$ with values in $\mathcal{C}$ is called AR-J-adic if:

(a) $J^{n+1}X_n = 0$ for every $n \geq 0$;

(b) it satisfies the equivalent conditions (i) and (ii) of Proposition 3.2.1.

Proposition 3.2.3. Let X be a projective system indexed by $\mathbb{N}$ with values in $\mathcal{C}$. Suppose that $J^{n+1}X_n = 0$ for every $n \geq 0$. Then, in order that X be AR-J-adic, it is necessary and sufficient that it satisfy (MLAR) and that, denoting by X' its projective system of universal images, there exist an integer $r \geq 0$ such that, for every pair (m, n) of integers with $m \geq n + r$, the transition morphism below is an isomorphism:

$$X'_m/J^{n+1}X'_m \longrightarrow X'_{n+r}/J^{n+1}X'_{n+r}.$$

Proof.

(a) The conditions of the statement are sufficient. If X satisfies (MLAR), it is AR-isomorphic to its projective system of universal images X'. Moreover, the hypothesis made on X' means that the projective system

$$(X'_{n+r}/J^{n+1}X'_{n+r})_{n\geq 0},$$

with the evident transition morphisms, is J-adic. It is clear that this projective system is AR-isomorphic to X'.

(b) The conditions of the statement are necessary. If X is AR-J-adic, it is AR-isomorphic to a strict projective system, hence satisfies (MLAR). The projective system X' of universal images of X is AR-isomorphic to X, hence is AR-J-adic. To prove the second point, it remains to see that, if X is a strict AR-J-adic projective system, there exists an integer r such that the projective system $\ell_r(X)$ defined by

$$\ell_r(X)_n = X_{n+r}/J^{n+1}X_{n+r}$$

is J-adic. For this, let $u : Y \to X$ be an AR-isomorphism, with Y J-adic. Since the kernel T of u is AR-null, let r be an integer such that the canonical morphism $T[r] \to T$ is zero. It is immediate that the canonical morphism

$$\ell_r(u) : \ell_r(Y) \longrightarrow \ell_r(X)$$

is an isomorphism, whence the assertion.

Proposition 3.2.4 (Stability under exact sequences). Let

$$0 \longrightarrow X \xrightarrow{u} Y \xrightarrow{v} Z \longrightarrow 0$$

be an exact sequence of P, the projective systems X, Y, and Z being such that

$$J^{n+1}X_n = J^{n+1}Y_n = J^{n+1}Z_n = 0 \qquad (n \geq 0).$$

(i) If X satisfies (MLAR) and Y is AR-J-adic, then Z is AR-J-adic.

(ii) If Y satisfies (MLAR) and Z is AR-J-adic, then X satisfies (MLAR).

Proof. We use freely, for every projective system X and every integer r, the notation $\ell_r(X)$ already made explicit.

Proof of (i). If X' and Y' denote the projective systems of universal images of X and Y, then u maps X' into Y', whence the evident exact commutative diagram below, in which the arrows are evident:

$$\begin{tikzcd} 0 \arrow[r] & X' \arrow[r] \arrow[d] & Y' \arrow[r] \arrow[d] & \widetilde{Z} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & X \arrow[r] & Y \arrow[r] & Z \arrow[r] & 0. \end{tikzcd}$$

Since the category P_{AR} is abelian and the functor $P \to P_{AR}$ is exact, it follows from this diagram, and from the fact that the inclusions of X' and Y' into X and Y respectively are AR-isomorphisms (2.5.1), that the morphism $\widetilde{Z} \to Z$ is an AR-isomorphism.

This allows us to reduce to the case where X and Y, and therefore Z, are strict. Then there exists an integer r such that the projective system $\ell_r(Y)$ is J-adic. The functor ℓ_r is plainly right exact, whence an exact sequence

$$\ell_r(X) \xrightarrow{\ell_r(u)} \ell_r(Y) \xrightarrow{\ell_r(v)} \ell_r(Z) \longrightarrow 0.$$

After replacing $\ell_r(X)$ by its image under $\ell_r(u)$, we are under the hypotheses for applying (3.1.3 (i)), which shows that $\ell_r(Z)$ is J-adic, whence the assertion.

(ii) Let Y' and Z' be the projective systems of universal images of Y and Z. Consider the evident exact commutative diagram below, in which the two vertical arrows on the right are the canonical inclusions:

$$\begin{tikzcd} 0 \arrow[r] & \widetilde{X} \arrow[r] \arrow[d] & Y' \arrow[r] \arrow[d] & Z' \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & X \arrow[r, "u"'] & Y \arrow[r, "v"'] & Z \arrow[r] & 0. \end{tikzcd}$$

It shows that the morphism $\widetilde{X} \to X$ is an AR-isomorphism, which allows us to reduce to the case where Y and Z are strict. In this case, since Z is strict and AR-J-adic, it follows from 3.2.3 that there exists an integer r such that $\ell_r(Z)$ is J-adic. Considering the exact commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \widetilde{X} \arrow[r] \arrow[d] & \ell_r(Y) \arrow[r] \arrow[d] & \ell_r(Z) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & X \arrow[r, "u"'] & Y \arrow[r, "v"'] & Z \arrow[r] & 0, \end{tikzcd}$$

in which, as before, $\widetilde{X} \to X$ is an AR-isomorphism, allows us to reduce to the case where Y is strict and Z is J-adic. Then (3.1.3 (ii)) gives the conclusion.

4. Filtrations and graduations

4.1. Generalities

4.1.1

Let $\mathcal{C}$ be an abelian category and let $X = (X_n)_{n \in \mathbb{Z}}$ be a projective system indexed by $\mathbb{Z}$ with values in $\mathcal{C}$. Suppose that

$$X_i = 0 \quad (i < 0).$$

For every element p of $\mathbb{Z}$, define a subobject F^pX of X in the category $P_1 = \mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{C})$ by setting

$$(F^pX)_n = F^pX_n = \begin{cases} \mathrm{Ker}(X_n \to X_{p-1}), & n \geq p-1, \\ 0, & n < p-1, \end{cases}$$

with the transition morphisms defined in the evident way. Thus one obtains a decreasing filtration $(F^pX)_{p\in\mathbb{Z}}$ of X satisfying

$$F^pX = X \quad \text{if } p \leq 0,$$

$$(F^pX)_{p-1} = 0 \quad \text{for all } p.$$

Moreover, the snake lemma applied to the exact commutative diagram (D) below proves that, as soon as $n \geq p$, the canonical morphism

$$F^pX_n/F^{p+1}X_n \longrightarrow \mathrm{Im}(X_n \to X_p) \cap \mathrm{Ker}(X_p \to X_{p-1})$$

is an isomorphism:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Ker}(X_n \to X_p) \arrow[r] \arrow[d] & X_n \arrow[r] \arrow[d] & \mathrm{Im}(X_n \to X_p) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & 0 \arrow[r] & X_{p-1} \arrow[r, "\sim"'] & X_{p-1} \arrow[r] & 0.
\end{tikzcd}$$

If now X is a projective system indexed by $\mathbb{N}$ with values in $\mathcal{C}$, we extend it to a projective system indexed by $\mathbb{Z}$ by putting $X_n = 0$ for $n < 0$, with the evident transition morphisms, which allows the preceding constructions to be applied to it. This allows us also to consider the category

$$P = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

as a full subcategory of P_1.

Definition 4.1.2. Let X be a projective system indexed by $\mathbb{Z}$. We say that X is essentially constant if there exists an integer p such that, for every pair (r,s) of integers greater than or equal to p and satisfying $s \geq r$, the transition morphism

$$X_s \longrightarrow X_r$$

is an isomorphism. It is clear that it is enough to require this when $r = p$.

Proposition 4.1.3 (The property (MLAR) and associated gradeds). The following assertions for an object X of the category P are equivalent:

(i) X satisfies (MLAR).

(ii) The projective system with values in the category of objects of $\mathcal{C}$ graded by $\mathbb{N}$,

$$r \longmapsto (\mathrm{gr}^\alpha(X_{\alpha+r}))_{\alpha\in\mathbb{N}},$$

with the evident transition morphisms, is essentially constant.

(iii) The projective system with values in the category of objects of $\mathcal{C}$ graded by $\mathbb{N}$,

$$n \longmapsto (\mathrm{gr}^p(X_n))_{p\in\mathbb{N}},$$

with the evident transition morphisms, satisfies (MLAR).

Proof. $(i) \Rightarrow (ii)$ follows from the expression of $F^pX_n/F^{p+1}X_n$ given in 4.1.1.

$(ii) \Rightarrow (iii)$. Let r be an integer such that, for every $s \geq r$ and every p, the canonical morphism

$$\mathrm{gr}^p(X_{p+s}) \longrightarrow \mathrm{gr}^p(X_{p+r})$$

is an isomorphism. We shall see that, for every pair (m,n) of integers with $m \geq n+r$, the equality below is satisfied for every integer p:

$$\mathrm{Im}(\mathrm{gr}^p(X_m) \to \mathrm{gr}^p(X_n)) = \mathrm{Im}(\mathrm{gr}^p(X_{n+r}) \to \mathrm{gr}^p(X_n)).$$

Indeed, if $p \geq n+1$, then $\mathrm{gr}^p(X_n) = 0$; if $p \leq n$, the chain of inequalities $m \geq n+r \geq p+r$ shows that the canonical morphism

$$\mathrm{gr}^p(X_m) \longrightarrow \mathrm{gr}^p(X_{p+r})$$

is an isomorphism, whence the assertion.

$(iii) \Rightarrow (i)$. Denote by Y the projective system of (iii), and let r be an integer such that

$$\mathrm{Im}(Y[s] \to Y) = \mathrm{Im}(Y[r] \to Y) \qquad (s \geq r).$$

If n and ℓ are two integers, with $\ell \geq n$, the filtration induced on $\mathrm{Im}(X_\ell \to X_n)$ by that of X_n is also the quotient filtration of that of X_ℓ, and consequently, endowing $\mathrm{Im}(X_\ell \to X_n)$ with this filtration, the equality below is satisfied:

$$\mathrm{gr}^p(\mathrm{Im}(X_\ell \to X_n)) = \mathrm{Im}(\mathrm{gr}^p(X_\ell) \to \mathrm{gr}^p(X_n)).$$

In particular, condition (iii) is expressed by saying that, as soon as $\ell \geq n+r$, the canonical monomorphism

$$j : \mathrm{Im}(X_\ell \to X_n) \longrightarrow \mathrm{Im}(X_{n+r} \to X_n)$$

induces an isomorphism on associated gradeds. Since the gradings are finite, it follows that j itself is an isomorphism, whence (i).

4.1.4

If X is an object of $\mathcal{P}$ satisfying (ML), it follows from the expression of the associated graded recalled in (4.1.1) that, for every integer p, the projective system

$$\mathrm{gr}^p(X) = \left(\mathrm{gr}^p(X_n)\right)_{n\in\mathbb{N}}$$

is essentially constant.

Its projective limit, which exists without any hypothesis on the abelian category $\mathcal{C}$, will be called the p-th component of the strict associated graded of X, and will be denoted

$$\mathrm{grs}^p(X).$$

4.2. Projective systems adapted to J and the Artin-Rees lemma

We suppose given a commutative ring A, an ideal J of A, and an abelian A-category $\mathcal{C}$.

Definition 4.2.1. An object X of the category $\mathcal{P} = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ is said to be adapted to J if its filtration $(F^pX)_{p\in\mathbb{N}}$ as a projective system (4.1.1) is compatible with the J-adic filtration of X, i.e. if for every pair (p, q) of non-negative integers the inclusion below holds:

$$J^pF^q(X) \subset F^{p+q}(X).$$

Lemma 4.2.2. Let X be an object of the category $\mathcal{P}$. The following assertions are equivalent:

(i) X is adapted to J.

(ii) For every pair (n, p) of integers with $p \leq n + 1$, one has

$$J^pF^{n-p+1}(X_n) = 0.$$

Proof. $(i) \Rightarrow (ii)$: then

$$J^pF^{n-p+1}(X_n) \subset F^{n+1}(X_n) = 0.$$

$(ii) \Rightarrow (i)$: if $n < p + q$, $F^{p+q}(X_n) = 0$ by definition. Moreover

$$J^pF^q(X_n)$$

is contained in $J^{n+1-q}F^q(X_n)$, and the latter is zero by hypothesis. If $n > p + q$, the image of $J^pF^q(X_n)$ in X_{p+q-1} under the transition morphism is contained in $J^pF^q(X_{p+q-1})$, which is zero by hypothesis. The inclusion

$$J^pF^q(X_n) \subset F^{p+q}(X_n)$$

therefore follows by definition of the filtration of a projective system.

In particular, it follows from (4.2.2) that every projective subsystem of a projective system adapted to J is adapted to J.

Examples. a) A J-adic projective system is adapted to J. It follows that every projective subsystem of a J-adic projective system is adapted to J.

b) If a projective system X satisfies condition (ML) and is adapted to J, then its projective system of universal images is also adapted to J.

4.2.3. The associated graded of a projective system adapted to J

Let X be a projective system adapted to J. Since the filtration of X is compatible with J, it is clear that the graded object

$$\left(\mathrm{gr}^p(X)\right)_{p\in\mathbb{N}}$$

is canonically endowed with the structure of a graded $\operatorname{gr}_J(A)$-module. In particular, by passage to the limit, one deduces that when X satisfies condition (ML), the graded object

$$\operatorname{grs}(X) = \left(\operatorname{grs}^p(X)\right)_{p\in\mathbb{N}}$$

is canonically endowed with the structure of a graded $\operatorname{gr}_J(A)$-module.

Let us briefly spell out how, when moreover X is strict, one defines the corresponding morphisms:

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q}).$$

Since the projective system X is strict, we know (4.1.1 and 4.1.2) that the following canonical morphism is an isomorphism:

$$\varphi : F^q(X_{p+q})/F^{q+1}(X_{p+q}) \longrightarrow F^q(X_q).$$

Moreover, the fact that the filtration of X is adapted to J gives a morphism

$$J^p \otimes_A F^q(X_{p+q}) \longrightarrow F^{p+q}(X_{p+q})$$

which factors, since

$$J^{p+1}F^q(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0, \qquad J^pF^{q+1}(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0,$$

through

$$J^p/J^{p+1} \otimes_A \left(F^q(X_{p+q})/F^{q+1}(X_{p+q})\right)$$

as a morphism

$$\beta : (J^p/J^{p+1}) \otimes_A \left(F^q(X_{p+q})/F^{q+1}(X_{p+q})\right) \longrightarrow F^{p+q}(X_{p+q}).$$

The morphism ω_{pq} is then given by the following expression:

$$\omega_{pq} = \beta \circ (\operatorname{id}_{J^p/J^{p+1}} \otimes_A \varphi)^{-1}.$$

Lemma 4.2.4. Let X be a J-adic projective system. The canonical morphism

$$\operatorname{gr}_J(A) \otimes_A X_0 \longrightarrow \operatorname{grs}(X),$$

coming from the graded $\operatorname{gr}_J(A)$-module structure of $\operatorname{grs}(X)$, is an epimorphism.

Proof. This follows immediately from the description of the morphisms ω_{pq} given in (4.2.3).

Proposition 4.2.5. Characterization of strict AR-J-adic projective systems by means of their associated gradeds. Let X be a strict projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. Suppose that for every integer n, one has

$$J^{n+1}X_n = 0.$$

The following assertions are equivalent:

(i) X is AR-J-adic.

(ii) There exists an integer $r > 0$ such that, for every integer n, the following inclusion holds:

$$F^n(X_n) \subset J^{n-r} X_n.$$

(iii) There exists an integer r such that, for every integer m, the following canonical morphism is an isomorphism whenever $n \geq m + r$:

$$\mathrm{gr}^n(J^m X) \longrightarrow \mathrm{gr}^n(X).$$

In the statement of (ii), we agree to set $J^{n-r} X_n = X_n$ for $n < r$.

Proof. The equivalence of (ii) and (iii) is clear, since X and $J^m X$ are strict. Let us prove the equivalence of (i) and (ii). We know (3.2.3) that (i) is equivalent to the existence of an integer r such that, for every pair (m, n) of integers with $m \geq n + r$, the transition morphism below is an isomorphism:

$$X_{m+1}/J^{n+1} X_{m+1} \longrightarrow X_m/J^{n+1} X_m.$$

Consider the following exact commutative diagram:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & J^{n+1} X_{m+1} & \longrightarrow & J^{n+1} X_m & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & & & \\
0 & \longrightarrow & F^{m+1}(X_{m+1}) & \longrightarrow & X_{m+1} & \longrightarrow & X_m & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & & & \\
 & & X_{m+1}/J^{n+1} X_{m+1} & \longrightarrow & X_m/J^{n+1} X_m & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & 0 & & 0. & & & &
\end{array}$$

It shows the existence of a canonical isomorphism between

$$\mathrm{Ker}\big(X_{m+1}/J^{n+1} X_{m+1} \longrightarrow X_m/J^{n+1} X_m\big)$$

and

$$F^{m+1}(X_{m+1})/\big(F^{m+1}(X_{m+1}) \cap J^{n+1} X_{m+1}\big).$$

In particular, for the transition morphism after reduction modulo J^{n+1} to be an isomorphism, it is necessary and sufficient that $F^{m+1}(X_{m+1})$ be contained in $J^{n+1} X_{m+1}$, which is precisely assertion (ii).

Theorem 4.2.6. Artin-Rees lemma. Let X be a projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. Suppose that:

(i) X satisfies (MLAR);

(ii) X is adapted to J;

(iii) $\mathrm{grs}(X)$ is a graded $(\mathrm{gr}_J(A), \mathcal{C})$-module of finite type, i.e. for every increasing sequence $(M_n)_{n\in\mathbb{N}}$ of graded sub-$\mathrm{gr}_J(A)$-modules of $\mathrm{grs}(X)$ whose union is $\mathrm{grs}(X)$, there exists an integer p such that

$$\mathrm{grs}(X) = M_p.$$

Then X is AR-J-adic.

Proof. Recall that if X' is the projective system of universal images of X, we have set

$$\operatorname{grs}(X) = \operatorname{gr}(X').$$

Moreover, X' is adapted to J, so it remains to show that X' is AR-J-adic. Suppose that the following lemma has been proved.

Lemma 4.2.7. Let X be a strict projective system adapted to J. For every integer s, the graded subobject $\big((M_s)_n\big)_{n\in\mathbb{N}}$ of $\operatorname{grs}(X)$ defined by the equalities

$$(M_s)_n = J^{\,n-s}X_n \cap F^n(X_n) \qquad (J^i = A \text{ if } i < 0),$$

is a graded $\operatorname{gr}_J(A)$-submodule of $\operatorname{grs}(X)$.

To prove the theorem, we may suppose X strict. Then $\operatorname{grs}(X)$ is the union of the increasing sequence of the M_s, hence is equal to one of them. This means that there exists an integer s_0 such that

$$F^n(X_n) \subset J^{\,n-s_0}X_n$$

for every integer n. We conclude by (4.2.5).

Proof of 4.2.7. Return to the notation of (4.2.3). We have to see that the morphism

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q})$$

sends

$$J^p/J^{p+1} \otimes_A \big(J^{q-s}X_q \cap F^q(X_q)\big)$$

into

$$J^{p+q-s}X_{p+q} \cap F^{p+q}(X_{p+q}).$$

Now the canonical morphism

$$\rho : X_{p+q}/F^{q+1}(X_{p+q}) \longrightarrow X_q$$

is an isomorphism and sends

$$\big(J^{q-s}X_{p+q} + F^{q+1}(X_{p+q})\big)/F^{q+1}(X_{p+q})$$

onto $J^{q-s}X_q$. It follows that the inverse image under $\operatorname{id}(J^p/J^{p+1}) \otimes \varphi$ (4.2.3) of

$$J^p/J^{p+1} \otimes_A \big(J^{q-s}X_q \cap F^q(X_q)\big)$$

is

$$J^p/J^{p+1} \otimes_A \frac{(J^{q-s}X_{p+q} \cap F^q(X_{p+q})) + F^{q+1}(X_{p+q})}{F^{q+1}(X_{p+q})}.$$

It is now clear that the image under β (4.2.3) of this last object is contained in

$$J^{p+q-s}X_{p+q} \cap J^pF^q(X_{p+q}),$$

hence a fortiori in

$$J^{p+q-s}X_{p+q} \cap F^{p+q}(X_{p+q}).$$

This proves the lemma.

5. Noetherian J-adic and AR-J-adic projective systems

Throughout this paragraph, a noetherian commutative ring A, an ideal J of A, and an abelian A-category $\mathcal{C}$ are fixed.

5.1. Generalities

Definition 5.1.1. An object X of the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ is said to be noetherian J-adic (resp. artinian) if it is J-adic and if its components are noetherian (resp. artinian) objects of the category $\mathcal{C}$.

We shall denote by J-adn($\mathcal{C}$) (resp. J-adf($\mathcal{C}$)) the full subcategory of $\mathcal{P} = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are noetherian (resp. artinian) J-adic projective systems. It is clear (3.1.3) that an extension of noetherian (resp. artinian) J-adic projective systems is of the same type.

Proposition 5.1.2. Let X be a J-adic object of $\mathcal{P}$. In order for X to be noetherian (resp. artinian) J-adic, it is necessary and sufficient that it be J-adic and that X_0 be noetherian (resp. artinian).

Proof. Indeed, let X be a J-adic projective system such that X_0 is noetherian (resp. artinian). Consider the canonical epimorphism (4.2.4):

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X).$$

Since X_0 is noetherian (resp. artinian) and J^n is of finite type for every n, the existence of this epimorphism shows that for every integer n, the kernel of the canonical morphism

$$X_n \longrightarrow X_{n-1}$$

is noetherian (resp. artinian). It follows at once by induction on n that the X_n are noetherian (resp. artinian).

Definition 5.1.3. Let X be a projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. It is said to be noetherian (resp. artinian) AR-J-adic if it is AR-J-adic and if its components are noetherian (resp. artinian) objects.

We shall denote by AR-J-adn($\mathcal{C}$) (resp. AR-J-adf($\mathcal{C}$)) the full subcategory of $\mathrm{Hom}_{\mathrm{AR}}(\mathbb{N}^\circ, \mathcal{C})$ (2.4) generated by the images of noetherian (resp. artinian) AR-J-adic projective systems.

It is then clear that the canonical functors

$$P^n_{\mathrm{AR}} : J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{C}),$$

$$P^f_{\mathrm{AR}} : J\text{-adf}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adf}(\mathcal{C}),$$

obtained by restriction of P_{AR} (2.4), are equivalences of categories.

Let us briefly recall the statement and proof of Hilbert's theorem.

Theorem 5.1.4. Hilbert's theorem. Let A be a commutative ring, $\mathcal{C}$ an abelian A-category, and M an object of finite type (resp. noetherian) of $\mathcal{C}$. For every A-algebra of finite type B, graded by $\mathbb{N}$, the graded $(B, \mathcal{C})$-module

$$B \otimes_A M$$

is of finite type (resp. noetherian).

N.B. If $B = \bigoplus_{n\geq 0} B_n$, then $B \otimes_A M$ denotes the graded object $\left(B_n \otimes_A M\right)_{n\in\mathbb{N}}$.

Proof. One reduces immediately to the case where B is the polynomial ring $A[T]$, endowed with its usual grading. The $(B, \mathcal{C})$-module $A[T] \otimes_A M = (M_n)_{n\in\mathbb{N}}$ is then obtained by putting $M_n = M$ for every n, with T sending M_n to M_{n+1} by the identity of M, for every integer n.

Let N be a graded sub-$(A[T], \mathcal{C})$-module of $A[T] \otimes_A M$. The components $(N_n)_{n\in\mathbb{N}}$ of N define an increasing sequence

$$N_0 \subset N_1 \subset \cdots \subset N_n \subset \cdots$$

of subobjects of M. Suppose now that M is noetherian (resp. of finite type), and let

$$N^1 \subset N^2 \subset \cdots \subset N^p \subset \cdots$$

be an increasing sequence of graded sub-$(A[T], \mathcal{C})$-modules of $A[T] \otimes_A M$, whose union, in the first hypothesis, when M is of finite type, is $A[T] \otimes_A M$.

Consider the following double-entry table:

$$\begin{array}{ccccccc}
N_0^1 & \subset & N_0^2 & \subset \cdots \subset & N_0^p & \subset \cdots \\
\cap & & \cap & & \cap & \\
N_1^1 & & N_1^2 & & N_1^p & \\
\vdots & & \vdots & & \vdots & \\
\cap & & \cap & & \cap & \\
N_n^1 & \subset & N_n^2 & \subset \cdots \subset & N_n^p & \subset \cdots .
\end{array}$$

It is clear by hypothesis on M that there exists a pair (p_0, n_0) such that $N_{n_0}^{p_0}$ is maximal (resp. equal to M). If p_1 is an integer $\geq p_0$ such that the $N_n^{p_1}$ $(n \leq n_0)$ are the largest elements of the sequences $(N_n^p)_{p\in\mathbb{N}}$, it is clear that $N^p = N^{p_1}$ as soon as $p \geq p_1$, which allows one to conclude in both cases.

Corollary 5.1.5. Suppose moreover that the category $\mathcal{C}$ has denumerable direct sums. Then for every object M of finite type (resp. noetherian) of $\mathcal{C}$, and every A-algebra of finite type B, the $(B, \mathcal{C})$-module $B \otimes_A M$ is of finite type (resp. noetherian).

Proof. One reduces to the case where $B = A[T]$. Let N be a sub-$(A[T], \mathcal{C})$-module of $A[T] \otimes_A M$, considered as an object of $\mathcal{C}$. We shall denote by $\mathrm{gr}(N)$ the associated graded of N for the filtration induced by the natural filtration of the graded object $A[T] \otimes_A M$: $\mathrm{gr}(N)$ is a graded sub-$(A[T], \mathcal{C})$-module of the graded $(A[T], \mathcal{C})$-module $A[T] \otimes_A M$.

Now let

$$N^0 \subset N^1 \subset \cdots \subset N^p \subset \cdots$$

be an increasing sequence of sub-$(A[T], \mathcal{C})$-modules of $A[T] \otimes_A M$, whose union is $A[T] \otimes_A M$ in the first hypothesis. In the first hypothesis (M of finite type), there exists an integer i such that $\mathrm{gr}(N^i) = A[T] \otimes_A M$, whence

$$N^i = A[T] \otimes_A M.$$

In the second hypothesis (M noetherian), the sequence of the $\mathrm{gr}(N^i)$ is stationary, hence the sequence of the N^i is stationary as well.

In what follows, Hilbert's theorem will be useful to us especially through the following proposition.

Proposition 5.1.6. Let A be a noetherian commutative ring, J an ideal of A, and $\mathcal{C}$ an abelian A-category. For every noetherian J-adic projective system X (5.1.1), the graded object $\mathrm{grs}(X)$ is a noetherian $(\mathrm{gr}_J(A), \mathcal{C})$-module.

Proof. Since $\mathrm{gr}_J(A)$ is an A-algebra of finite type, Hilbert's theorem (5.1.4) implies that

$$\mathrm{gr}_J(A) \otimes_A X_0$$

is a noetherian graded $(\mathrm{gr}_J(A), \mathcal{C})$-module. The assertion then follows from the existence of the epimorphism (4.2.4):

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X).$$

5.2. Structure of J-adn($\mathcal{C}$) and AR-J-adn($\mathcal{C}$)

In this section, we shall denote by $\mathcal{E}_{\mathcal{C}}$, or by $\mathcal{E}$ if there is no possible confusion, the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are noetherian AR-J-adic projective systems.

Proposition 5.2.1. The category $\mathcal{E}_{\mathcal{C}}$ is stable under kernels and cokernels in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$. In other words, $\mathcal{E}_{\mathcal{C}}$ is an abelian category and the inclusion functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

is exact.

Proof. Let $u : X \to Y$ be a morphism of $\mathcal{E}_{\mathcal{C}}$. We shall successively see that the cokernel, the image, and the kernel of u are noetherian AR-J-adic. Of course, the proof that the image of u is noetherian AR-J-adic is only a technical intermediate step.

a) $\mathrm{Coker}(u)$ is noetherian AR-J-adic. Since X satisfies (MLAR) and Y is AR-J-adic, $\mathrm{Coker}(u)$ is AR-J-adic (3.2.4 (ii)).

b) $\mathrm{Im}(u)$ is noetherian AR-J-adic. From the exact sequence

$$0 \longrightarrow \mathrm{Im}(u) \longrightarrow Y \longrightarrow \mathrm{Coker}(u) \longrightarrow 0,$$

the assertion will follow from Lemma 5.2.1.1 below.

Lemma 5.2.1.1. Let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be an exact sequence in the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$. If M and N are noetherian AR-J-adic, then L is noetherian AR-J-adic.

Proof. It is plainly enough to see that it is AR-J-adic. One reduces, in the same way as in the proof of (3.2.4), to the case where M and N are strict, and then even noetherian J-adic. In this case, it follows from 3.1.3 (ii) that L is strict. The projective system L is strict, adapted to J, and its associated graded, as a graded subobject of $\mathrm{grs}(M)$, is noetherian: $\mathrm{grs}(M)$, a quotient of $\mathrm{gr}_J(A) \otimes_A M_0$, is noetherian (5.1.6). The assertion then follows from the Artin-Rees lemma (4.2.6).

c) $\mathrm{Ker}(u)$ is noetherian AR-J-adic. It is enough to apply (5.2.1.1) to the evident exact sequence below:

$$0 \longrightarrow \mathrm{Ker}(u) \longrightarrow X \longrightarrow \mathrm{Im}(u) \longrightarrow 0.$$

Proposition 5.2.2. The objects of the abelian category AR-J-adn$(\mathcal{C})$ are noetherian.

Proof. One reduces immediately to proving the following. Suppose we have a noetherian J-adic object X, a sequence of noetherian J-adic objects $(Y^n)_{n\in\mathbb{N}}$, and morphisms in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$

$$v_n : Y^n \longrightarrow X$$

such that

(i) the v_n become monomorphisms in AR-J-adn$(\mathcal{C})$;

(ii) for every integer n, v_n factors through v_{n+1} in AR-J-adn$(\mathcal{C})$,

then the v_n remain stationary, up to isomorphism, for sufficiently large n, in AR-J-adn$(\mathcal{C})$.

Now, to say that v_n factors through v_{n+1} in AR-J-adn$(\mathcal{C})$ means that there exists an integer r and a morphism

$$f : Y^n[r] \longrightarrow Y^{n+1}$$

such that, with a denoting the canonical morphism

$$Y^n[r] \longrightarrow Y^n,$$

the following equality holds:

$$v_n \circ a = v_{n+1} \circ f.$$

Since the morphism a is strict, it follows in particular that the image of v_n is contained in that of v_{n+1}. Replacing each Y^n by $\mathrm{Im}(v_n)$, we are reduced to seeing that every increasing sequence of strict projective subsystems of X is stationary. But if $(Z^n)_{n\in\mathbb{N}}$ is such a sequence, then, since $\mathrm{grs}(X)$ is noetherian, the increasing sequence of the $\mathrm{grs}(Z^n)$ is stationary, hence so is the sequence of the Z^n.

Summarizing:

Theorem 5.2.3. The categories J-adn$(\mathcal{C})$ and AR-J-adn$(\mathcal{C})$ are abelian and noetherian.

As for AR-J-adn$(\mathcal{C})$, this is what we have just proved. As for J-adn$(\mathcal{C})$, it is equivalent to it.

Proposition 5.2.4. Stability under extension. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. Suppose that:

(i) Z is AR-J-adic;

(ii) X is artinian AR-J-adic.

Then Y is AR-J-adic.

Proof. As in the proof of (3.2.4), one reduces to the case where Y and Z are strict and Z is J-adic. Then X is strict by (3.1.3 (ii)). There then exists an integer r such that the projective system $(X_{n+r}/J^{n+1}X_{n+r})_{n\in\mathbb{N}}$ is artinian J-adic. Since X is a quotient of it and is AR-isomorphic to it, we are reduced to proving the following lemma.

Lemma 5.2.4.1. Let

$$0 \longrightarrow T \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^{\circ}, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. If X is artinian J-adic, Z is J-adic, and T is AR-zero, then Y is AR-J-adic.

Proof. To prove the lemma, we first show that, for every integer n, the evident projective system

$$\left(Y_{n+k}/J^{n+1}Y_{n+k}\right)_{k\in\mathbb{N}}$$

is essentially constant. Let S_n^k denote the kernel of the evident morphism

$$X_{n+k}/J^{n+1}X_{n+k} \longrightarrow Y_{n+k}/J^{n+1}Y_{n+k}.$$

Consider the following evident exact commutative diagram:

$$\begin{array}{ccccccccc} 0 & \to & S_n^k & \to & X_{n+k}/J^{n+1}X_{n+k} & \to & Y_{n+k}/J^{n+1}Y_{n+k} & \to & Z_{n+k}/J^{n+1}Z_{n+k} \to 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \to & T_n & \to & X_n & \to & Y_n & \to & Z_n \qquad\quad \to 0. \end{array}$$

With n fixed, the objects S_n^k are identified with subobjects of T_n and form a decreasing sequence. Since T_n is artinian, there exists an integer k_0 such that $S_n^k = S_n^{k_0}$ for every integer $k \geq k_0$; it follows at once that the canonical morphisms

$$Y_{n+k}/J^{n+1}Y_{n+k} \longrightarrow Y_{n+k_0}/J^{n+1}Y_{n+k_0}$$

are isomorphisms.

This being said, denote by Y' the projective system defined by

$$Y'_n = \varprojlim_r Y_{n+r}/J^{n+1}Y_{n+r}$$

on objects, and in the evident way on morphisms. The projective system Y' is J-adic. Moreover, by passage to the limit one obtains an exact commutative diagram of the following type:

$$\begin{array}{ccccccccccc} 0 & \to & S & \to & X & \to & Y' & \to & Z & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow \mathrm{id} & & \downarrow \mathrm{id} & & \\ 0 & \to & T & \to & X & \to & Y & \to & Z & \to & 0. \end{array}$$

Since T is AR-zero, the snake lemma applied to this diagram shows that the evident morphism $Y' \to Y$ is an AR-isomorphism, whence Y is AR-J-adic.

Corollary 5.2.5. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^{\circ}, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. Suppose that:

(i) X is artinian AR-J-adic (resp. artinian and noetherian);

(ii) Z is artinian AR-J-adic (resp. noetherian).

Then Y is artinian AR-J-adic (resp. noetherian).

5.3. Noetherian AR-J-adic projective systems and δ-functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories, and let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor. The functor F extends in the clear way to a functor, again denoted F,

$$F : \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{D}),$$

which is additive and "commutes with translations".

Proposition 5.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories and

$$T = (T^i)_{i\in\mathbb{N}}$$

an exact A-linear δ-functor from $\mathcal{C}$ to $\mathcal{D}$. Suppose that:

(i) For every noetherian object M of $\mathcal{C}$, annihilated by a power of J, and every integer n, $T^n(M)$ is noetherian.

(ii) For every graded A-algebra B of finite type, annihilated by J, and every noetherian graded $(B, \mathcal{C})$-module N, the graded $(B, \mathcal{D})$-modules $T^n(N)$, $n \in \mathbb{N}$, are noetherian.

Then, for every noetherian AR-J-adic object X of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, the $T^n(X)$ are noetherian AR-J-adic.

Proof. Since the functors T^n commute with translations, it is enough to prove the assertion when X is noetherian J-adic. In that case, since $\mathrm{grs}(X)$ is a noetherian graded $\mathrm{gr}_J(A)$-module (5.1.6), it follows from hypothesis (ii) and from a variant of a corollary of Shih's theorem (cf. Appendix A.3) that for every integer n the following assertions hold:

$$\text{(a)} \quad T^n(X) \text{ satisfies (MLAR)},$$

$$\text{(b)} \quad \mathrm{grs}\ T^n(X) \text{ is a noetherian graded } (\mathrm{gr}_J(A), \mathcal{D})\text{-module.}$$

The Artin-Rees lemma (4.2.6) allows us to conclude.

Corollary 5.3.2. The δ-functor

$$T = (T^i)_{i\in\mathbb{N}} : \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{D})$$

induces an exact A-linear δ-functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathcal{E}_{\mathcal{D}},$$

commuting with translations, and hence, by passage to the quotient, a new exact A-linear δ-functor, again denoted $(T^i)_{i\in\mathbb{N}}$,

$$\text{AR-}J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{D}).$$

Appendix: Shih's theorem

The purpose of the present appendix is to give a proof of Shih's theorem in the form used in the proof of (5.3.1). This form being more general than the one appearing in EGA 0_{III}, a direct proof, adapted from that of loc. cit., has proved necessary.

Terminology. Let r_0 be an integer ≥ 1. The datum, for every integer $r \geq r_0$, of a bigraded object

$$E_r = (E_r^{pq})_{p\in\mathbb{Z},\ q\in\mathbb{Z}},$$

endowed with a differential d_r of bidegree $(r, 1-r)$, and, for every r, of an isomorphism

$$H(E_r) \xrightarrow{\sim} E_{r+1},$$

will be called below a spectral filter. The notion of a morphism of spectral filters is defined in the evident way.

A.1. Recollections on the spectral sequences associated with δ-functors

Let $\mathcal{C}$ be an abelian category and X an object of $\mathcal{C}$, endowed with a decreasing filtration $(F^pX)_{p\in\mathbb{Z}}$. For every exact δ-functor T from $\mathcal{C}$ to an abelian category $\mathcal{D}$, the method of "exact couples" makes it possible to describe a spectral sequence

$$E(X): \quad E_1^{pq} = T^{p+q}(F^pX/F^{p+1}X) \Rightarrow E^n = T^n(X),$$

for which the filtration of E^n is defined as follows:

$$F^p(E^n) = \mathrm{Im}(T^n(F^pX) \longrightarrow T^n(X)) = \mathrm{Ker}(T^n(X) \longrightarrow T^n(X/F^pX)).$$

Moreover, if the filtration of X is finite, the spectral sequence so obtained is weakly convergent (EGA 0_{III} 11.1.3).

Functoriality. If $f : X \to Y$ is a morphism of filtered objects of $\mathcal{C}$, one constructs functorially a morphism of spectral sequences

$$E(f) : E(X) \longrightarrow E(Y),$$

coinciding with $T^{p+q}(\mathrm{gr}^p f)$ (resp. $T^n(f)$) on $E_1^{pq}(X)$ (resp. $E^n(X)$). In particular, if $u : X \to X$ is an endomorphism of X such that

$$u(F^pX) \subset F^{p+h}(X)$$

for an integer h independent of p, then $E(u)$ is an endomorphism of bidegree $(h, -h)$ of the spectral filter associated with $E(X)$, commuting with the differentials.

A.2. The case of projective systems

Let now $X = (X_n)_{n\in\mathbb{Z}}$ be a projective system of objects of an abelian category $\mathcal{C}$, such that

$$X_i = 0 \qquad (i < 0).$$

If T denotes an exact δ-functor from $\mathcal{C}$ to an abelian category $\mathcal{D}$, T extends naturally to an exact δ-functor, again denoted T,

$$\mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{D}).$$

Thus, endowing X with its natural filtration as a projective system (EGA 0_{III} 13.4.1), one obtains a spectral sequence

$$E(X) = E(X_n)_{n \in \mathbb{Z}},$$

where the $E(X_n)$ denote the spectral sequences associated with T and the X_n endowed with the induced filtration.

Properties of $E(X)$. a) For every integer n, the filtration of the abutment $E^n(X)$ in the spectral sequence $E(X)$ is identical with its natural filtration as a projective system when X is strict.

Indeed, denoting by the letter F the first filtration and by the letter Φ the second, one has for every integer α:

$$\begin{aligned} {}^pF\,E^n(X)_\alpha &= \mathrm{Ker}(T^n(X_\alpha) \longrightarrow T^n(X_\alpha / F^p X_\alpha)) \\ &= \begin{cases} 0, & \alpha \le p-1, \\ \mathrm{Ker}(T^n(X_\alpha) \longrightarrow T^n(X_{p-1})), & \alpha \ge p-1. \end{cases} \end{aligned}$$

In both cases one recognizes the expression of $\Phi^p E^n(X)$.

This shows in particular that, for $\beta \ge \alpha \ge p$, the transition morphism

$$E_\infty^{pq}(X_\beta) \longrightarrow E_\infty^{pq}(X_\alpha)$$

which is identified with

$$\mathrm{gr}_T^p\, T^{p+q}(X_\beta) \longrightarrow \mathrm{gr}_T^p\, T^{p+q}(X_\alpha),$$

is a monomorphism, by EGA 0_{III} 13.4.1.3.

b) Fixing $\alpha \in \mathbb{Z}$, it is clear that, since $F^i(X_\alpha) = 0$ for $i \ge \alpha + 1$, one has

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \quad \text{for } r > p,$$

$$Z_r^{pq}(X_\alpha) = Z_{r+1}^{pq}(X_\alpha) \quad \text{for } r > \alpha - p.$$

c) When X satisfies the Mittag-Leffler condition, the various projective systems $\mathrm{gr}^p(X)$ are essentially constant (EGA 0_{III} 13.4.3). It follows at once that, for $r < +\infty$, the projective systems $E_r^{pq}(X)$ are essentially constant. We shall denote by

$$\mathcal{E}_r^{pq}(X)$$

their projective limits. Since the differential operators pass to the limit, one thus obtains a spectral filter $\mathcal{E}_r^{pq}(X)$, $p, q \in \mathbb{Z}$, $r \ge 1$. The objects

$$\mathcal{Z}_r^{pq}(X), \quad \mathcal{B}_r^{pq}(X) \quad (r < +\infty)$$

associated with this spectral filter are precisely the limits of the essentially constant projective systems $Z_r^{pq}(X)$ and $B_r^{pq}(X)$.

It also follows from part b) that

$$B_\infty^{pq}(X) = B_{p+1}^{pq}(X),$$

which allows one to put

$$\mathcal{B}_{\infty}^{pq}(X) = \mathcal{B}_{p+1}^{pq}(X).$$

Still supposing that X satisfies (ML), since, for every α,

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \qquad (r > p),$$

one has a sequence of canonical monomorphisms

$$\cdots \subset \mathcal{E}_r^{pq}(X) \subset \cdots \subset \mathcal{E}_{p+2}^{pq}(X) \subset \mathcal{E}_{p+1}^{pq}(X). \tag{S}$$

Suppose from now on that X is strict. The following expressions for $Z_r^{pq}(X_n)$ and $B_r^{pq}(X_n)$:

$$Z_r^{pq}(X_n) = \mathrm{Im}\Big[T^{p+q-1}(F^{p-r+1}X_n/F^pX_n) \xrightarrow{\partial} T^{p+q}(F^pX_n/F^{p+1}X_n)\Big],$$

$$B_r^{pq}(X_n) = \mathrm{Im}\Big[T^{p+q}(F^pX_n/F^{p+r}X_n) \longrightarrow T^{p+q}(F^pX_n/F^{p+1}X_n)\Big],$$

show that

$$\begin{cases} Z_r^{pq}(X_m) \xrightarrow{\sim} Z_r^{pq}(X_n) & \text{for } m \geq n \geq p, \\ B_r^{pq}(X_m) \xrightarrow{\sim} B_r^{pq}(X_n) & \text{for } m \geq n \geq p+r-1, \end{cases} \tag{d_1}$$

where the canonical morphisms are isomorphisms. In particular, the canonical morphism

$$\mathcal{E}_r^{pq}(X) \longrightarrow E_r^{pq}(X_{p+r-1})$$

is an isomorphism. Moreover, it follows from b) that

$$\begin{cases} Z_\infty^{pq}(X_{p+r-1}) = Z_r^{pq}(X_{p+r-1}) & (r \geq 1), \\ B_\infty^{pq}(X_{p+r-1}) = B_r^{pq}(X_{p+r-1}) & (r > p). \end{cases} \tag{d_2}$$

This makes it possible to define, for every r, a canonical epimorphism

$$\theta_r^{pq} : \mathcal{E}_r^{pq}(X) \longrightarrow E_\infty^{pq}(X_{p+r-1})$$

which is an isomorphism when $r > p$. In addition, if a is an integer such that

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \qquad (r \geq a),$$

for example $a = p+1$ suffices, so that

$$\cdots \hookrightarrow \mathcal{E}_r^{pq}(X) \hookrightarrow \cdots \hookrightarrow \mathcal{E}_{a+1}^{pq}(X) \hookrightarrow \mathcal{E}_a^{pq}(X), \tag{S bis}$$

then, for $s \geq r \geq a$, the diagrams

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow & & \downarrow \text{can} \\ \mathcal{E}_r^{pq}(X) & \xrightarrow{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array} \tag{d_3}$$

are commutative.

A.3. A variant of Shih's theorem

Suppose given a ring S filtered by ideals $F^i(S)$, $i \geq 0$, acting on X compatibly with its filtration as a projective system.

It follows from the generalities of A.1 that, for every integer i, $\mathrm{gr}^i(S)$ acts on the bigraded objects $E_r(X)$ by morphisms of bidegree $(i, -i)$. In particular, for every integer n,

$$E_r^{(n)}(X) = \big(E_r^{p,q}(X)\big)_{p+q=n}$$

is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module. The same holds by passage to the limit for

$$\mathcal{E}_r^{(n)}(X) = \big(\mathcal{E}_r^{p,q}(X)\big)_{p+q=n}.$$

The ring S, acting on X compatibly with its filtration as a projective system, likewise acts on $T^n(X)$ for every integer n. Consequently, the strict associated graded $\mathrm{grs}(X)$ is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{C})$-module, and similarly, when $T^n(X)$ satisfies (ML), the strict associated graded grs $T^n(X)$ is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module.

Theorem A.3. Suppose that, for an integer n, the graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules

$$\mathcal{E}_1^{(n)}(X) = T^n(\mathrm{grs}^\bullet X) \quad \text{and} \quad \mathcal{E}_1^{(n-1)}(X) = T^{n-1}(\mathrm{grs}^\bullet X)$$

are noetherian. Then:

(i) $T^n(X)$ satisfies (MLAR);

(ii) grs $T^n(X)$ is a noetherian $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module.

Proof. i) For every pair (m, r) of integers,

$$\mathcal{B}_r^{(m)}(X) = \big(\mathcal{B}_r^{p,m-p}(X)\big)_{p\in\mathbb{N}}$$

is a graded sub-$(\mathrm{gr}^\bullet(S), \mathcal{D})$-module of $\mathcal{E}_1^{(m)}(X)$. In particular, for every integer m, the $\mathcal{B}_r^{(m)}(X)$ form an increasing sequence, indexed by r, of graded sub-$(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules of $\mathcal{E}_1^{(m)}(X)$. Using the fact that $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ are noetherian graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules, one deduces that the sequences

$$\big[\mathcal{B}_r^{(n)}(X)\big]_{r\in\mathbb{N}} \qquad \text{and} \qquad \big[\mathcal{B}_r^{(n-1)}(X)\big]_{r\in\mathbb{N}}$$

are stationary.

Thus there exists an integer $a \geq 1$ such that, for every (p, q), with $p + q = n$ or $n - 1$,

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \qquad (r \geq a). \tag{A.3.1}$$

Consequently, for $r \geq a + 1$, the arrows d_r with source and target $\mathcal{E}_r^{pq}(X)$, $p + q = n$, are zero. In other words, the monomorphisms

$$\mathcal{E}_s^{pq}(X) \hookrightarrow \mathcal{E}_r^{pq}(X) \qquad (s \geq r \geq a + 1)$$

deduced from (A.3.1) with $p+q=n$ are in fact isomorphisms. For $s \geq r \geq a+1$, the commutativity of diagram (d_3), with $p+q=n$,

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow\wr & & \downarrow \\ \mathcal{E}_r^{pq}(X) & \xrightarrow{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array}$$

shows that the transition morphism

$$E_\infty^{pq}(X_{p+s-1}) \longrightarrow E_\infty^{pq}(X_{p+r-1}),$$

which is in any case a monomorphism (cf. a)), is an epimorphism, hence an isomorphism. In other words, the projective system

$$r \longmapsto \left[\bigoplus_p \operatorname{gr}^p T^n(X_{p+r})\right]_{r\geq 1}$$

is constant from $r=a$ on, and therefore $T^n(X)$ satisfies (MLAR), by (4.1.3).

ii) Denoting by

$$\mathcal{E}_\infty^{(n)}(X), \quad \mathcal{Z}_\infty^{(n)}(X), \quad \mathcal{B}_\infty^{(n)}(X)$$

the "limits" of the various projective systems indexed by r considered above, we see that

$$\mathcal{E}_\infty^{(n)}(X) = \mathcal{Z}_\infty^{(n)}(X)/\mathcal{B}_\infty^{(n)}(X)$$

is a noetherian $(\operatorname{gr}^\bullet(S), \mathcal{D})$-module, since $\mathcal{Z}_\infty^{(n)}(X)$ is contained in $\mathcal{E}_1^{(n)}(X)$. Moreover, it follows from A.2 a) that

$$\mathcal{E}_\infty^{(n)}(X) = \operatorname{grs} T^n(X).$$

Exposé VI

ℓ-ADIC COHOMOLOGY

by J.-P. Jouanolou

Throughout this exposé, a prime number ℓ is fixed.

As usual, $X_{\text{ét}}$ denotes the étale site of a prescheme X.

1. Constructible ℓ-adic sheaves

Throughout this paragraph, fix once and for all a locally noetherian prescheme X.

1.1. Definition and generalities

Definition 1.1.1. A constructible ℓ-adic sheaf on X is a projective $(\ell\mathbb{Z})$-adic system of constructible abelian sheaves on $X_{\text{ét}}$ (V 3.2.1).

For a projective $(\ell\mathbb{Z})$-adic system

$$F = (F_n)_{n\in\mathbb{N}}$$

of abelian sheaves on $X_{\text{ét}}$ to be a constructible ℓ-adic sheaf, it is enough that F_0 be constructible. Indeed, the assertion is local and follows, in the noetherian case, from (V 5.1.2).

Let $\mathrm{Abc}(X)$ denote the category of constructible abelian sheaves on X. We shall denote by

$$\ell\text{-}\operatorname{adc}(X)$$

the full subcategory of $\operatorname{Hom}(\mathbb{N}^\circ, \mathrm{Abc}(X))$ generated by the constructible ℓ-adic sheaves. With the notation of Exposé V,

$$\ell\text{-}\operatorname{adc}(X) = (\ell\mathbb{Z})\text{-}\operatorname{ad}(\mathrm{Abc}(X)).$$

Letting $\mathrm{Ab}(X)$ denote the category of abelian sheaves on $X_{\text{ét}}$, one knows (SGA 4 IX 2.9) that, when X is noetherian, the constructible abelian sheaves on X are the noetherian abelian sheaves. In this case, the category $\ell\text{-}\operatorname{adc}(X)$ is clearly identified with the category denoted

$$(\ell\mathbb{Z})\text{-}\operatorname{adn}(\mathrm{Ab}(X))$$

in Exposé V.

Proposition 1.1.2. Let $F = (F_n)_{n\in\mathbb{N}}$ be an object of the category $\operatorname{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$. For F to be constructible ℓ-adic, it is necessary and sufficient that it be so locally for the étale topology of X.

Proof. It is clear that the property of F of being ℓ-adic may be read above a covering of the final object of the site. Moreover, it follows from (SGA 4 IX 2.8) that the F_n are constructible if they are so locally.

Proposition 1.1.3. The category $\ell\text{-}\operatorname{adc}(X)$ of constructible ℓ-adic sheaves on X is abelian. It is noetherian when X is noetherian.

Proof. When X is noetherian, the two assertions are consequences of (V 5.2.3). In the general case, one must see that every morphism

$$u : F \longrightarrow G$$

of constructible ℓ-adic sheaves on X has a kernel and a cokernel.

As for the cokernel, this follows from (V 3.1.3). Let us prove the existence of the kernel of u. For every noetherian open V of X, one knows that the restriction of u to V,

$$u_V : F_V \longrightarrow G_V,$$

has a kernel in the category $\ell\text{-}\operatorname{adc}(V)$, admitting the following simple description. If K is the kernel of u_V in the category $\operatorname{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(V))$, K satisfies the Mittag-Leffler-Artin-Rees condition and, denoting by K' its projective system of universal images, there exists an integer r such that the projective system

$$m_r(K') = (K'_{n+r}/\ell^{n+1}K'_{n+r})_{n\in\mathbb{N}}$$

(with the evident transition morphisms) is constructible ℓ-adic. The evident composite below is a kernel of u_V in the category $\ell\text{-}\operatorname{adc}(V)$:

$$m_r(K') \longrightarrow K' \longrightarrow K.$$

The preceding construction shows in particular that, if W is an open contained in V, the restriction to W of a kernel for u_V is a kernel for u_W. Let $(V_i)_{i\in I}$ be a covering of X by noetherian opens. If i and j are two elements of I, the restrictions to $V_i \cap V_j$ of the kernels of u_{V_i} and u_{V_j} are canonically isomorphic, which allows them to be glued to construct a kernel of u.

1.1.4

Let $F = (F_n)_{n\in\mathbb{N}}$ be a constructible ℓ-adic sheaf. Since the ring $\mathbb{Z}_\ell$ acts evidently on each sheaf F_n, it also acts on F. Moreover, if F and G are two constructible ℓ-adic sheaves and

$$u : F \longrightarrow G$$

is a morphism, u is compatible with the actions of $\mathbb{Z}_\ell$ on F and G, respectively. This shows that $\ell\text{-}\operatorname{adc}(X)$ is canonically endowed with the structure of an abelian $\mathbb{Z}_\ell$-category.

This justifies the following terminology.

Terminology. In what follows, constructible ℓ-adic sheaves will be called constructible $\mathbb{Z}_\ell$-sheaves.

One should take care not to confuse this notion with that of a $(\mathbb{Z}_\ell)_X$-module, which will not be used below.

The category $\ell\text{-}\operatorname{adc}(X)$ will also be denoted

$$\mathbb{Z}_\ell\text{-}\operatorname{fc}(X).$$

1.2. Constructible twisted constant $\mathbb{Z}_\ell$-sheaves

Definition 1.2.1. A constructible $\mathbb{Z}_\ell$-sheaf

$$F = (F_n)_{n\in\mathbb{N}}$$

is said to be twisted constant if and only if the sheaves F_n are locally constant on $X_{\text{ét}}$.

The category of constructible twisted constant $\mathbb{Z}_\ell$-sheaves is the full subcategory of $\mathbb{Z}_\ell\text{-}\operatorname{fc}(X)$ generated by the constructible twisted constant $\mathbb{Z}_\ell$-sheaves.

Example 1.2.2. The $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$.

Assume that the integer ℓ is prime to the residual characteristics of X. For every integer n, raising to the ℓ-th power defines a morphism of locally constant sheaves:

$$\mu_{\ell^{n+1}} \longrightarrow \mu_{\ell^n}.$$

This evidently defines a projective system:

$$\mathbb{Z}_\ell(1) = (\mu_{\ell^{n+1}})_{n\in\mathbb{N}}.$$

Reducing, for each integer n, to the case where μ_{ℓ^n} and $\mu_{\ell^{n+1}}$ are constant, one sees that the projective system $\mathbb{Z}_\ell(1)$ is ℓ-adic.

When $X = \mathrm{Spec}(K)$, where K is a field of finite type over the prime field, the $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ is an example of a constructible twisted constant ℓ-adic sheaf (1.2.1) for which there is no finite étale extension K' of K such that the inverse images of the F_n on $X' = \mathrm{Spec}(K')$ are constant. Indeed, K' would contain, for every integer n, a primitive ℓ^n-th root of unity, which is not possible for a finite-type extension of the prime field.

Reminders 1.2.3. (SGA 1 V 7). One knows that the category of constructible locally constant abelian torsion sheaves on a prescheme T is equivalent to the category of étale coverings of T endowed with the structure of an abelian X-group.

Now let X be a connected locally noetherian prescheme. Let R_X denote the category of étale coverings of X, and let Ensf denote the category of finite sets. If a is a geometric point of X, let F_a be the functor

$$F_a : R_X \longrightarrow \mathrm{Ensf},$$

which associates to every étale covering of X its geometric fibre at a. The fundamental group

$$\pi_1(X) = \pi_1(X, a)$$

of X at a, being by definition the group of automorphisms of F_a, suitably topologized, allows the functor F_a to be interpreted as a functor

$$R_X \longrightarrow \mathrm{Ensf}(\pi_1)$$

from the category R_X to the category of finite sets endowed with a continuous action of $\pi_1 = \pi_1(X, a)$.

Recall the following proposition.

Proposition 1.2.3.1. The functor

$$F_a : R_X \longrightarrow \mathrm{Ensf}(\pi_1)$$

is an equivalence of categories. It commutes with inductive limits and finite projective limits.

Let $Lc(X)$ and $\mathrm{Abf}(\pi_1)$ denote respectively the category of constructible locally constant abelian torsion sheaves and the category of finite abelian groups endowed with a continuous operation of the fundamental group. The functor F_a induces an exact functor, still denoted F_a,

$$F_a : Lc(X) \longrightarrow \mathrm{Abf}(\pi_1),$$

which is an equivalence of categories.

1.2.4. Twisted constant $\mathbb{Z}_\ell$-sheaves on a connected prescheme

The reminders in (1.2.3) make the lemma below immediate.

Lemma 1.2.4.1. The extension $\widehat{F}_a$ of F_a to the category of projective systems indexed by $\mathbb{N}$ induces an equivalence from the category of constructible twisted constant ℓ-adic sheaves to the category

$$(\ell\mathbb{Z})\text{-}\,\mathrm{ad}(\mathrm{Abf}(\pi_1)).$$

Now let $F = (F_n)_{n\in\mathbb{N}}$ be an object of the category

$$(\ell\mathbb{Z})\text{-}\operatorname{ad}(\operatorname{Abf}(\pi_1)).$$

Its projective limit

$$\varprojlim_{\mathbb{N}} F_n$$

is naturally endowed with a structure of $\mathbb{Z}_\ell$-module of finite type on which π_1 acts continuously for the ℓ-adic topology. Recall that if G and M denote respectively a profinite group and a $\mathbb{Z}_\ell$-module of finite type, a continuous operation of G on M means a continuous map

$$G \times M \longrightarrow M, \qquad (g, m) \longmapsto g \cdot m,$$

such that

$$\text{(i)} \quad g \cdot (m + n) = g \cdot m + g \cdot n \quad (g \in G,\ m, n \in M),$$

and

$$\text{(ii)} \quad (gh) \cdot m = g \cdot (h \cdot m) \quad (g, h \in G,\ m \in M).$$

It is then clear that, for every ℓ-adic integer a,

$$g \cdot (am) = a(g \cdot m) \qquad (g \in G,\ m \in M).$$

Moreover, the group of invertible $\mathbb{Z}_\ell$-linear maps from M to M, denoted $\operatorname{GL}(M)$, is identified with the projective limit of the groups $\operatorname{GL}(M/\ell^{n+1}M)$. Since these groups are finite, the group $\operatorname{GL}(M)$ is thus naturally endowed with the structure of a profinite group, and in particular with the structure of a compact topological group. One checks easily that to give a continuous operation of the profinite group G on M is the same as to give a morphism of topological groups

$$G \longrightarrow \operatorname{GL}(M),$$

where the latter group is endowed with the topology just described. This said, one evidently defines a functor

$$\varprojlim_{\mathbb{N}} : (\ell\mathbb{Z})\text{-}\operatorname{ad}(\operatorname{Abf}(\pi_1)) \longrightarrow \mathbb{Z}_\ell\text{-}\operatorname{mod}(\pi_1),$$

where $\mathbb{Z}_\ell\text{-}\operatorname{mod}(\pi_1)$ denotes the category of finite-type $\mathbb{Z}_\ell$-modules endowed with a continuous operation of π_1.

This allows us to state the following lemma.

Lemma 1.2.4.2. The preceding functor

$$\varprojlim_{\mathbb{N}} : (\ell\mathbb{Z})\text{-}\operatorname{ad}(\operatorname{Abf}(\pi_1)) \longrightarrow \mathbb{Z}_\ell\text{-}\operatorname{mod}(\pi_1)$$

is an equivalence of categories.

Proof. It is clear that it is essentially surjective: a finite-type $\mathbb{Z}_\ell$-module E on which π_1 acts continuously is the image of the projective system

$$(E/\ell^{n+1}E)_{n\in\mathbb{N}},$$

with the evident operations of π_1. Now let

$$F = (F_n)_{n \in \mathbb{N}}, \qquad G = (G_n)_{n \in \mathbb{N}}$$

be two objects of the category $(\ell\mathbb{Z})\text{-}\operatorname{ad}(\operatorname{Abf}(\pi_1))$, and, for simplicity, write $\operatorname{Hom}_{\pi_1}(F, G)$ for the set of morphisms from F to G in this category. Forgetting the action of π_1, one may regard F and G as objects of the category of ℓ-adic projective systems of finite abelian groups, and write $\operatorname{Hom}_0(F, G)$ for the set of morphisms from F to G in this category. The group π_1 acts on $\operatorname{Hom}_0(F, G)$ as follows: an element s of π_1 clearly defines two automorphisms, denoted in the same way, of F and G respectively in the category of ℓ-adic projective systems of finite abelian groups, and if u belongs to $\operatorname{Hom}_0(F, G)$, one puts

$$s \cdot u = sus^{-1}.$$

Under these conditions, it is clear that $\operatorname{Hom}_{\pi_1}(F, G)$ is canonically identified with the set of π_1-invariants in $\operatorname{Hom}_0(F, G)$, and similarly that $\operatorname{Hom}_{\pi_1}(F_n, G_n)$ is identified for every n with

$$\operatorname{Hom}_{\mathbb{Z}}(F_n, G_n)^{\pi_1},$$

the operation of π_1 being defined analogously. It follows, for example, from (Serre, Groupes proalgébriques, Prop. 12) that, denoting by M and N the projective limits of F and G respectively, one has

$$\operatorname{Hom}_{\mathbb{Z}_\ell}(M, N) = \varprojlim_n \operatorname{Hom}_{\mathbb{Z}}(F_n, G_n).$$

Taking the invariants of both sides under π_1 and using the fact that this operation, being associated with a projective limit, commutes with projective limits, one obtains the assertion.

Lemmas 1.2.4.1 and 1.2.4.2 allow one to state the following less precise proposition.

Proposition 1.2.5. The category of constructible twisted constant $\mathbb{Z}_\ell$-sheaves on a connected locally noetherian prescheme X is equivalent to the category of finite-type $\mathbb{Z}_\ell$-modules on which $\pi_1(X)$ acts continuously for the ℓ-adic topology.

Remark. Let F be a constructible twisted constant $\mathbb{Z}_\ell$-sheaf, and let M be the corresponding finite-type $\mathbb{Z}_\ell$-module endowed with a continuous operation of π_1. For F to be locally constant, it is clear that it is enough for π_1 to act on M through a finite quotient; this is not always the case (1.2.2). It is even necessary when X is normal. After decomposing X into connected components, one may, in order to see this, reduce to the case where X is integral. I claim that then, if K denotes its field of rational functions, and if there exists a finite separable extension K' of K such that the inverse image $F_{K'}$ of F above $\operatorname{Spec}(K')$ is constant, then the group $\pi_1(X)$ acts on M through a finite group. Indeed (SGA 1 V 8.2), $\pi_1(K) \to \pi_1(X)$ is surjective, and the hypothesis means that the kernel of the composite

$$\pi_1(K) \longrightarrow \pi_1(X) \longrightarrow \operatorname{Aut}(M)$$

has finite index. The same is therefore true for the kernel of $\pi_1(X) \to \operatorname{Aut}(M)$.

Proposition 1.2.6. Let X be a locally noetherian prescheme and let F be a projective system, indexed by $\mathbb{N}$, of abelian sheaves on $X_{\text{ét}}$. The following assertions are equivalent:

(i) F is a constructible $\mathbb{Z}_\ell$-sheaf;

(ii) every noetherian open U of X is a finite union of locally closed subsets $(Z_i)_{1\leq i\leq q}$ above which the inverse image of F is a constructible twisted constant $\mathbb{Z}_\ell$-sheaf;

(iii) X admits a covering by opens, each a finite union of locally closed subsets, above which the inverse image of F is a constructible twisted constant $\mathbb{Z}_\ell$-sheaf.

Proof. It is clear that (ii) implies (iii). If F satisfies (iii), its geometric fibres are ℓ-adic, hence F is ℓ-adic by (SGA 4 VIII 3.6). Moreover, its components are constructible (SGA 4 IX 2.8).

To see that (i) implies (ii), one immediately reduces to the case where X is noetherian. If F is constructible ℓ-adic, its strict graded object $\mathrm{gr}^s(F)$ is a noetherian $(\mathrm{gr}\,\ell\mathbb{Z}, \mathrm{Ab}(X))$-module (V 5.1.6). It follows (SGA 4 IX 2.9) that X is a finite union of locally closed subsets $(Z_i)_{1\leq i\leq q}$ above which the inverse image of $\mathrm{gr}^s(F)$ is a locally constant sheaf. Since, for a sheaf with finite fibres, the property of being locally constant is stable under extensions, one deduces that, for every integer n, the inverse images on each Z_i of the component F_n of F are locally constant sheaves.

1.3. Operations $\otimes$ and Hom on $\mathbb{Z}_\ell$-sheaves

1.3.1. Tensor product. Let F and G be two objects of the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$. If

$$F = (F_n)_{n\in\mathbb{N}} \quad \text{and} \quad G = (G_n)_{n\in\mathbb{N}},$$

the tensor product $F \otimes G$ of F and G is defined by the following formula on objects, and in the evident way on arrows:

$$(F \otimes G)_n = F_n \otimes G_n.$$

More precisely, one clearly defines a right exact additive bifunctor, called the tensor product:

$$\otimes : \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \times \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)).$$

Proposition 1.3.2. If F and G are two constructible $\mathbb{Z}_\ell$-sheaves, then the tensor product $F \otimes G$ is a constructible $\mathbb{Z}_\ell$-sheaf.

Proof. The ℓ-adic property is clear. Moreover, it follows immediately from the definition of constructible sheaves (SGA 4 IX 2.3) that a tensor product of two constructible sheaves is constructible.

By restriction one therefore defines a right exact additive bifunctor, again called tensor product:

$$\otimes : \mathbb{Z}_\ell\text{-}\mathrm{fc}(X) \times \mathbb{Z}_\ell\text{-}\mathrm{fc}(X) \longrightarrow \mathbb{Z}_\ell\text{-}\mathrm{fc}(X).$$

1.3.3. Sheaf of homomorphisms

Let $F = (F_n)_{n\in\mathbb{N}}$ and $G = (G_n)_{n\in\mathbb{N}}$ be two constructible $\mathbb{Z}_\ell$-sheaves. Suppose that F is locally free, meaning that, for every integer n, the sheaf F_n is locally free over the constant sheaf

$$(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X.$$

For every integer n, the structural exact sequence

$$G_{n+1} \xrightarrow{\ell^{n+1}} G_{n+1} \longrightarrow G_n \longrightarrow 0$$

gives, since F_{n+1} is locally free over $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$, an exact sequence

$$\mathcal{H}om(F_{n+1}, G_{n+1}) \xrightarrow{\ell^{n+1}} \mathcal{H}om(F_{n+1}, G_{n+1}) \longrightarrow \mathcal{H}om(F_{n+1}, G_n) \longrightarrow 0. \tag{1}$$

Since $\ell^{n+1} G_n = 0$, the structural exact sequence

$$F_{n+1} \xrightarrow{\ell^{n+1}} F_{n+1} \longrightarrow F_n \longrightarrow 0$$

clearly defines an isomorphism

$$\mathcal{H}om(F_{n+1}, G_n) \xleftarrow{\sim} \mathcal{H}om(F_n, G_n). \tag{2}$$

Combining (1) and (2), one obtains, for every integer n, a morphism

$$\mathcal{H}om(F_{n+1}, G_{n+1}) \longrightarrow \mathcal{H}om(F_n, G_n).$$

Hence a projective system

$$\mathcal{H}om(F, G) = \big(\mathcal{H}om(F_n, G_n)\big)_{n \in \mathbb{N}},$$

which is clearly a $\mathbb{Z}_\ell$-sheaf.

1.3.4. Generalities on invertible ℓ-adic sheaves. Application to the sheaves $\mathbb{Z}_\ell(r)$ $(r \in \mathbb{Z})$

Let L be a constructible $\mathbb{Z}_\ell$-sheaf. We shall say that it is invertible if, for every integer n, its n-th component L_n is an invertible $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$-module, i.e. locally free of rank one. This is equivalent to saying that L is locally free in the sense of the preceding paragraph and that the geometric fibres of the L_n are nonzero cyclic abelian groups.

If L is an invertible $\mathbb{Z}_\ell$-sheaf, we agree to put

$$L^{-1} = \mathcal{H}om(L, \mathbb{Z}_\ell).$$

It is clear that there is an isomorphism

$$L \otimes L^{-1} \xrightarrow{\sim} \mathbb{Z}_\ell,$$

deduced componentwise from the analogous isomorphism (EGA 0_I 5.4.3.1). Following (EGA 0_I 5.4.4), we agree to put $L^{\otimes 0} = \mathbb{Z}_\ell$ and, for $n \geq 1$,

$$L^{\otimes -n} = (L^{-1})^{\otimes n}.$$

With this notation, there is then a canonical functorial isomorphism

$$L^{\otimes m} \otimes L^{\otimes n} \xrightarrow{\sim} L^{\otimes (m+n)}$$

for all rational integers m and n. Moreover, this isomorphism has the usual associativity properties.

In particular, the graded objects

$$(L^{\otimes i})_{i \in \mathbb{N}} \quad \text{and} \quad (L^{\otimes i})_{i \in \mathbb{Z}}$$

are $\mathbb{Z}_\ell$-algebras, graded respectively by $\mathbb{N}$ and $\mathbb{Z}$.

Example. The $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ defined in (1.2.2) is invertible, and in what follows we shall agree to put

$$\mathbb{Z}_\ell(r) = \big(\mathbb{Z}_\ell(1)\big)^{\otimes r} \qquad (r \in \mathbb{Z}).$$

1.4. Various categories constructed from the category of constructible $\mathbb{Z}_\ell$-sheaves

1.4.1. Constructible A-sheaves (A a finite $\mathbb{Z}_\ell$-algebra). Let A be a finite $\mathbb{Z}_\ell$-algebra. Define a category denoted

$$A\text{-}\mathrm{fc}(X),$$

whose objects are called constructible A-sheaves, as follows.

(i) An object of $A\text{-}\mathrm{fc}(X)$ is a pair (F, u) consisting of a constructible $\mathbb{Z}_\ell$-sheaf F and a morphism of $\mathbb{Z}_\ell$-algebras

$$u : A \longrightarrow \mathrm{End}(F).$$

(ii) If (F, u) and (F', u') are two constructible A-sheaves, an arrow $(F, u) \to (F', u')$ is a morphism of $\mathbb{Z}_\ell$-sheaves

$$f : F \longrightarrow F'$$

such that, for every element a of A, the diagram below is commutative:

$$\begin{tikzcd} F \arrow[r, "f"] \arrow[d, "u(a)"'] & F' \arrow[d, "u'(a)"] \\ F \arrow[r, "f"'] & F'. \end{tikzcd}$$

A constructible twisted constant A-sheaf is a constructible A-sheaf whose underlying constructible $\mathbb{Z}_\ell$-sheaf is twisted constant. The category of constructible twisted constant A-sheaves is the full subcategory of the category of constructible A-sheaves generated by the constructible twisted constant A-sheaves.

It follows immediately from (1.2.4) that, if X is connected, the category of constructible twisted constant A-sheaves is equivalent to the category of finite-type A-modules endowed with a continuous A-linear operation of $\pi_1(X)$ on the underlying finite-type $\mathbb{Z}_\ell$-module (1.2.4).

1.4.2. Constructible $\mathbb{Q}_\ell$-sheaves. It is clear that the full subcategory of $\mathbb{Z}_\ell\text{-}\mathrm{fc}(X)$ generated by the constructible $\mathbb{Z}_\ell$-sheaves of torsion, i.e. killed by a power of ℓ, is thick.

Definition 1.4.3. The category of constructible $\mathbb{Q}_\ell$-sheaves on X, denoted

$$\mathbb{Q}_\ell\text{-}\mathrm{fc}(X),$$

is the quotient abelian category, in the sense of Gabriel's thesis, of $\mathbb{Z}_\ell\text{-}\mathrm{fc}(X)$ by the thick abelian subcategory of $\mathbb{Z}_\ell$-sheaves of torsion. An object of $\mathbb{Q}_\ell\text{-}\mathrm{fc}(X)$ is called a constructible $\mathbb{Q}_\ell$-sheaf.

The category $\mathbb{Q}_\ell\text{-}\mathrm{fc}(X)$ admits the following simple description. It has the same set of objects as $\mathbb{Z}_\ell\text{-}\mathrm{fc}(X)$, and, if F and G are two such objects,

$$\mathrm{Hom}_{\mathbb{Q}_\ell}(F, G) = \mathbb{Q}_\ell \otimes_{\mathbb{Z}_\ell} \mathrm{Hom}_{\mathbb{Z}_\ell}(F, G),$$

where $\mathrm{Hom}_{\mathbb{Z}_\ell}(F, G)$ and $\mathrm{Hom}_{\mathbb{Q}_\ell}(F, G)$ denote respectively the morphisms in $\mathbb{Z}_\ell\text{-}\mathrm{fc}(X)$ and $\mathbb{Q}_\ell\text{-}\mathrm{fc}(X)$. Indeed, if

$$u : \mathbb{Z}_\ell\text{-}\mathrm{fc}(X) \longrightarrow D$$

denotes an exact functor from $\mathbb{Z}_\ell\text{-}\operatorname{fc}(X)$ to an abelian category D, then to say that u makes multiplication by the elements of $\mathbb{Z}_\ell$ invertible is the same as to say that u vanishes on the torsion objects; this immediately shows that the abelian category described above by objects and morphisms solves the universal problem of passage to the quotient.

A constructible $\mathbb{Q}_\ell$-sheaf is said to be twisted constant if it is the image of a constructible twisted constant $\mathbb{Z}_\ell$-sheaf. The category of constructible twisted constant $\mathbb{Q}_\ell$-sheaves is the full subcategory of

$$\mathbb{Q}_\ell\text{-}\operatorname{fc}(X)$$

generated by the constructible twisted constant $\mathbb{Q}_\ell$-sheaves.

It follows from (1.2.4) that, if X is connected, the category of constructible twisted constant $\mathbb{Q}_\ell$-sheaves is equivalent to the category of finite-dimensional $\mathbb{Q}_\ell$-vector spaces V endowed with a continuous operation of $\pi_1(X)$, i.e. a continuous map

$$u : \pi_1(X) \times V \longrightarrow V$$

satisfying the properties made explicit in (1.2.4), which in particular implies that $u(g, v)$ is $\mathbb{Q}_\ell$-linear in v for every $g \in \pi_1(X)$.

It is immediate to see that this is equivalent to giving a morphism of topological groups

$$\pi_1(X) \longrightarrow \mathrm{GL}(V),$$

the latter being endowed with its natural topology as an ℓ-adic analytic group.

1.4.3. Constructible L-sheaves (L a finite $\mathbb{Q}_\ell$-algebra). Let L be a finite $\mathbb{Q}_\ell$-algebra. In a manner analogous to what was done in (1.4.1) starting from constructible $\mathbb{Z}_\ell$-sheaves, one defines, starting from the category of constructible $\mathbb{Q}_\ell$-sheaves, the category of constructible L-sheaves, and then the category of constructible twisted constant L-sheaves. If X is connected, the category of constructible twisted constant L-sheaves is equivalent to the category of finite-dimensional L-vector spaces endowed with a continuous operation of $\pi_1(X)$, i.e. a continuous operation of $\pi_1(X)$ on the underlying $\mathbb{Q}_\ell$-vector space, commuting with the operation of L.

1.4.4

If A is a finite commutative $\mathbb{Z}_\ell$-algebra and F, G are two constructible A-sheaves, the projective system

$$F \otimes_A G = (F_n \otimes_A G_n)_{n \in \mathbb{N}},$$

with the evident transition morphisms, is a $\mathbb{Z}_\ell$-sheaf. Moreover A acts evidently on each of its components, which allows it to be endowed naturally with the structure of an A-sheaf. Thus one defines a right exact bifunctor, called tensor product,

$$A\text{-}\operatorname{fc}(X) \times A\text{-}\operatorname{fc}(X) \longrightarrow A\text{-}\operatorname{fc}(X).$$

Now let F and G be two A-sheaves and suppose that F is locally free, i.e. that, for every integer n, F_n is a locally free $(A/\ell^{n+1}A)_X$-module. Then, as in (1.3.2), one sees that the projective system

$$\mathcal{H}om_A(F, G) = \left(\mathcal{H}om_{A_n}(F_n, G_n)\right)_{n \in \mathbb{N}},$$

with the evident transition morphisms, is a $\mathbb{Z}_\ell$-sheaf. Moreover, since A acts naturally on its components, it is canonically endowed with the structure of an A-sheaf.

As in (1.3.3), one defines the notion of an invertible A-sheaf and, if L is such an A-sheaf, gives a meaning to the expression $L^{\otimes r}$ for every natural integer r. Paragraph (1.3.3) adapts without difficulty to this situation.

1.4.5

Let F be a constructible $\mathbb{Z}_\ell$-sheaf, and let p be the canonical functor

$$\mathbb{Z}_\ell\text{- fc}(X) \longrightarrow \mathbb{Q}_\ell\text{- fc}(X).$$

The composite of p with the functor $G \mapsto F \otimes G$ makes multiplication by the elements of $\mathbb{Z}_\ell$ invertible and hence defines a functor

$$\mathbb{Q}_\ell\text{- fc}(X) \longrightarrow \mathbb{Q}_\ell\text{- fc}(X),$$

which is right exact. Applying the same argument to the first variable, one thus defines a right exact bifunctor, called tensor product,

$$\mathbb{Q}_\ell\text{- fc}(X) \times \mathbb{Q}_\ell\text{- fc}(X) \longrightarrow \mathbb{Q}_\ell\text{- fc}(X).$$

If X is connected and F and G are two constructible twisted constant $\mathbb{Q}_\ell$-sheaves corresponding to representations of $\pi_1(X)$ on vector spaces V and W, respectively, then the tensor product $F \otimes G$ is again locally constant and corresponds to the diagonal representation of $\pi_1(X)$ on $V \otimes W$.

1.4.6

Let F be a locally free constructible $\mathbb{Z}_\ell$-sheaf (1.3.3). The composite of p with the functor $G \mapsto \mathcal{H}om(F, G)$ (1.3.2) makes multiplication by elements of $\mathbb{Z}_\ell$ invertible and hence defines a functor

$$\mathcal{H}om(F, \cdot) : \mathbb{Q}_\ell\text{- fc}(X) \longrightarrow \mathbb{Q}_\ell\text{- fc}(X).$$

If F and F' are two locally free constructible $\mathbb{Z}_\ell$-sheaves defining isomorphic $\mathbb{Q}_\ell$-sheaves, then there exists a morphism of $\mathbb{Z}_\ell$-sheaves

$$u : F \longrightarrow F'$$

whose kernel and cokernel are torsion. This immediately shows that the functors $\mathcal{H}om(F, \cdot)$ and $\mathcal{H}om(F', \cdot)$ from $\mathbb{Q}_\ell\text{- fc}(X)$ to itself are isomorphic.

Now let F be a constructible twisted constant $\mathbb{Q}_\ell$-sheaf. If we again denote by F the corresponding $\mathbb{Z}_\ell$-sheaf, it follows from the interpretation of the restrictions of F to the connected components Y of X as continuous $\pi_1(Y)$-modules that F may be written uniquely, up to isomorphism, in the form

$$F = P \otimes T,$$

where P is locally free and T is torsion. This allows one to define, for every $\mathbb{Q}_\ell$-sheaf G, a $\mathbb{Q}_\ell$-sheaf

$$\mathcal{H}om(F, G) = \mathcal{H}om(P, G).$$

An isomorphism $u : F \to F'$ of $\mathbb{Q}_\ell$-sheaves defines, for every G, an isomorphism

$$\mathcal{H}om(F', G) \xrightarrow{\sim} \mathcal{H}om(F, G).$$

More precisely, one thereby defines a bifunctor

$$\mathcal{H}om : (\mathbb{Q}_\ell\text{- fct}(X))^\circ \times \mathbb{Q}_\ell\text{- fc}(X) \longrightarrow \mathbb{Q}_\ell\text{- fc}(X),$$

where $\mathbb{Q}_\ell$- fct(X) denotes the category of constructible twisted constant $\mathbb{Q}_\ell$-sheaves. This bifunctor has the usual exactness properties.

1.4.7

Let L now be a finite-type extension of $\mathbb{Q}_\ell$, and let A denote the integral closure of $\mathbb{Z}_\ell$ in L, so that A is a free $\mathbb{Z}_\ell$-algebra of finite type and a Dedekind ring. The category of constructible L-sheaves is isomorphic to the quotient category of the category of constructible A-sheaves by the category of constructible torsion A-sheaves. This allows one, as in (1.4.6), to define from the tensor product of (1.4.4) a tensor product:

$$L\text{- fc}(X) \times L\text{- fc}(X) \longrightarrow L\text{- fc}(X).$$

Since A is a Dedekind ring, the argument used in (1.4.6) shows that every twisted constant A-sheaf is the direct sum of a locally free A-sheaf and a torsion A-sheaf. Denoting by L- fct(X) the category of constructible twisted constant L-sheaves, one defines, again as in (1.4.6), a bifunctor

$$\mathcal{H}om : (L\text{- fct}(X))^\circ \times L\text{- fc}(X) \longrightarrow L\text{- fc}(X),$$

with the usual exactness properties.

1.5. Constructible $(AR, \mathbb{Z}_\ell)$-sheaves

Definition 1.5.1. Let X be a locally noetherian prescheme. An object F of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ is called a constructible $(AR, \mathbb{Z}_\ell)$-sheaf if X admits a covering

$$(U_i)_{i \in I}$$

by noetherian opens such that the restriction of F above each U_i is a $(AR, \ell\mathbb{Z})$-noetherian object (V 1.3).

It follows immediately from (V 5.2.1) that the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves is stable under kernels and cokernels. In particular, it is an abelian category.

1.5.2. Constructible $(AR, \mathbb{Z}_\ell)$-sheaves which are locally AR-null

It is clear that an AR-null object (V 2.2.1) of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ is a constructible $(AR, \mathbb{Z}_\ell)$-sheaf if and only if its components are constructible. It follows in particular from (V 2.2.2) and (SGA 4 IX 2.6) that the category of AR-null constructible $(AR, \mathbb{Z}_\ell)$-sheaves is a thick abelian subcategory of the full category generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves in $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$.

Agreeing to say that an object P of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ is locally AR-null if there exists an open covering of X for the étale topology (or, equivalently, for the Zariski topology) above each open of which P is AR-null, one sees, by reducing through localization to the preceding case, that the category of locally AR-null constructible $(AR, \mathbb{Z}_\ell)$-sheaves is a thick abelian subcategory of that of constructible $(AR, \mathbb{Z}_\ell)$-sheaves.

Observe that, if X is noetherian, the constructible $(AR, \mathbb{Z}_\ell)$-sheaves are the noetherian $(AR, \ell\mathbb{Z})$-adic objects of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$, and locally AR-null is equivalent to AR-null.

Definition 1.5.3. Let X be a locally noetherian prescheme. The category of constructible $(AR, \mathbb{Z}_\ell)$-sheaves on X, denoted

$$(AR, \mathbb{Z}_\ell)\text{-}\,\mathrm{fc}(X),$$

is the quotient abelian category of the full subcategory generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves in

$$\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$$

by the full thick subcategory generated by the locally AR-null constructible $(AR - \mathbb{Z}_\ell)$-sheaves.

Notation 1.5.4. Whenever useful, $U(X)$ will denote the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X))$ generated by the constructible $(AR, \mathbb{Z}_\ell)$-sheaves.

Since every constructible $\mathbb{Z}_\ell$-sheaf is in particular a constructible $(AR, \mathbb{Z}_\ell)$-sheaf, one defines in the evident way a functor

$$u : \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X) \longrightarrow (AR, \mathbb{Z}_\ell)\text{-}\,\mathrm{fc}(X).$$

Proposition 1.5.5. The above functor u is an equivalence of categories.

Proof. To prove that u is fully faithful and essentially surjective, one reduces "by gluing" as in the proof of 1.1.3 to the analogous statement (V 5.1.3).

2. Formalism of ℓ-adic cohomology

2.1. Inverse image

Let

$$f : X \longrightarrow Y$$

be a morphism of locally noetherian preschemes. The inverse-image functor

$$f^* : \mathrm{Ab}(Y) \longrightarrow \mathrm{Ab}(X)$$

is exact and therefore extends in the evident way to an exact functor, denoted in the same way,

$$f^* : \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(Y)) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)).$$

This latter functor evidently carries constructible $\mathbb{Z}_\ell$-sheaves to constructible $\mathbb{Z}_\ell$-sheaves, and locally AR-null projective systems to locally AR-null projective systems. With the notation of (1.5.4), it therefore induces two "inverse image" functors

$$f^* : \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(Y) \longrightarrow \mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X),$$

$$f^* : U(Y) \longrightarrow U(X),$$

the second of which is exact.

The equivalence of categories (1.5.5), together with the fact that f^* carries locally AR-null projective systems to locally AR-null projective systems, then shows that the inverse-image functor

$$f^* : \mathbb{Z}_\ell\text{- fc}(Y) \longrightarrow \mathbb{Z}_\ell\text{- fc}(X)$$

is also exact.

If $g : Y \to Z$ is another morphism of locally noetherian preschemes, the functors f^* and g^* are evidently related by an isomorphism

$$(g \circ f)^* \xrightarrow{\sim} f^* \circ g^*,$$

with the usual associativity property (SGA 1 VI 7 ff.).

2.2. Higher direct images

Let

$$f : X \longrightarrow Y$$

be a separated morphism of finite type between locally noetherian preschemes. Recall the following result of étale cohomology (SGA 4 XVII; for the proper case, cf. SGA 4 XIV 1.1).

Lemma 2.2.1. Let A be a commutative ring annihilated by a nonzero integer n, and let M be a constructible $(A, \mathrm{Ab}(X))$-module. For every integer i, the sheaf

$$R^i f_!(M)$$

is a constructible $(A, \mathrm{Ab}(Y))$-module.

The following lemma follows from this by applying (V 5.3.1) after restriction over the noetherian opens of Y.

Lemma 2.2.2. For every integer i and every constructible $(AR, \mathbb{Z}_\ell)$-sheaf F on X, the projective system

$$R^i f_!(F)$$

is a constructible $(AR, \mathbb{Z}_\ell)$-sheaf.

This said, the exact δ-functor

$$(R^i f_!)_{i\in\mathbb{Z}} : \mathrm{Ab}(X) \longrightarrow \mathrm{Ab}(Y)$$

extends in the evident way to an exact δ-functor, denoted in the same way,

$$\mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathrm{Ab}(Y)),$$

which by restriction gives an exact δ-functor

$$U(X) \longrightarrow U(Y). \tag{1}$$

Moreover, the functors $R^i f_!$ carry locally AR-null projective systems to locally AR-null projective systems; hence, after passage to the quotient, the δ-functor (1) gives an exact δ-functor

$$(R^i f_!)_{i\in\mathbb{Z}} : (AR, \mathbb{Z}_\ell)\text{- fc}(X) \longrightarrow (AR, \mathbb{Z}_\ell)\text{- fc}(Y).$$

Using now the equivalence of categories (1.5.5), this latter δ-functor is interpreted as an exact δ-functor

$$\mathbb{Z}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Z}_\ell\text{-}\operatorname{fc}(Y).$$

Definition 2.2.3. The functor

$$R^i f_! : \mathbb{Z}_\ell\text{-}\operatorname{fc}(X) \longrightarrow \mathbb{Z}_\ell\text{-}\operatorname{fc}(Y)$$

is called the i-th direct image functor with proper supports of f.

Remark. Assuming the analogue of Lemma (2.2.1) proved for the ordinary higher direct images when X and Y are excellent, which is done in characteristic zero (SGA 4 XIX 5.1), one defines in the same way higher direct images in ℓ-adic cohomology in that case. One could also develop the result of the following paragraph in this framework.

Proposition 2.2.4 (Leray spectral sequence). Let $f : X \to Y$ and $g : Y \to Z$ be two separated morphisms of finite type between locally noetherian preschemes, and let F be a constructible $\mathbb{Z}_\ell$-sheaf, respectively a constructible $(AR, \mathbb{Z}_\ell)$-sheaf. There exists in $\mathbb{Z}_\ell\text{-}\operatorname{fc}(Z)$ a biregular spectral sequence (EGA 0_{III} 11.1.3), functorial in F:

$$R^i g_!(R^j f_!(F)) \Longrightarrow R^{i+j}(g \circ f)_!(F).$$

Proof. By the equivalence of categories (1.5.5) and the way in which higher direct images are defined for constructible $\mathbb{Z}_\ell$-sheaves, it is enough to show the analogous statement for constructible $(AR, \mathbb{Z}_\ell)$-sheaves. For any abelian category C, let us agree to denote by $\operatorname{Sp}(C)$ the category of spectral sequences beginning at degree 2 with values in C. Associating to every object F of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$ the ordinary Leray spectral sequence, obtained by extension to projective systems,

$$R^i g_!(R^j f_!(F)) \Longrightarrow R^{i+j}(g \circ f)_!(F), \tag{S}$$

it is clear that one obtains a functor

$$\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X)) \longrightarrow \operatorname{Sp}(\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(Z))),$$

which by restriction gives, by (2.2.1) and because $U(Z)$ is thick in $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(Z))$, a functor

$$U(X) \longrightarrow \operatorname{Sp}(U(Z)). \tag{1}$$

Since the canonical functor

$$U(Z) \longrightarrow (AR, \mathbb{Z}_\ell)\text{-}\operatorname{fc}(Z)$$

is exact, (1) yields an additive functor

$$E : U(X) \longrightarrow \operatorname{Sp}((AR, \mathbb{Z}_\ell)\text{-}\operatorname{fc}(Z)). \tag{2}$$

For every object F of $\operatorname{Hom}(\mathbb{N}^\circ, \operatorname{Ab}(X))$, the associated spectral sequence (S) is biregular: this follows from the analogous assertion for the components of the projective system F and from the fact that the spectral sequence under consideration is supported in the first quadrant. This shows that, for every object of $U(X)$, the corresponding spectral sequence is biregular.

Moreover, if $u : F \to G$ is a morphism of $U(X)$ whose kernel and cokernel are locally AR-null, then, because the higher direct images carry locally AR-null projective systems to locally AR-null projective systems, $E(u)$ is an isomorphism on the terms E_2^{pq}; it is therefore an isomorphism, since the spectral sequences considered are biregular. It follows that (2) defines, by passage to the quotient, a functor

$$(AR, \mathbb{Z}_\ell)\text{-}\,\mathrm{fc}(X) \longrightarrow \mathrm{Sp}((AR, \mathbb{Z}_\ell)\text{-}\,\mathrm{fc}(Z)), \tag{3}$$

which associates in a precise way to every constructible $(AR, \mathbb{Z}_\ell)$-sheaf F a biregular spectral sequence

$$R_!^i g(R_!^j f(F)) \Longrightarrow R_!^{i+j}(g \circ f)(F).$$

2.2.3. Other properties

A) Cohomological dimension. Let $f : X \to Y$ be a separated morphism of finite type between locally noetherian preschemes. Suppose that the relative dimension of f is at most an integer d. Then, for every constructible $(AR, \mathbb{Z}_\ell)$-sheaf F,

$$R^i f_!(F) = 0 \qquad \text{for } i > 2d.$$

This follows from the analogous assertion for the components of F (SGA 4 X 4.1 and 5.2).

B) Proper base change. Consider a cartesian square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y. \end{tikzcd}$$

Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms of locally noetherian preschemes, with f separated of finite type. Let $X' = X \times_Y Y'$, which is locally noetherian since f is of finite type, and let

$$f' : X' \to Y', \qquad g' : X' \to X$$

be the canonical morphisms. For every integer i and every constructible $\mathbb{Z}_\ell$-sheaf F, there exists an isomorphism in $\mathbb{Z}_\ell\text{-}\,\mathrm{fc}(Y')$, functorial in F:

$$w_i : g^*(R^i f_!(F)) \xrightarrow{\sim} R^i f'_!(g'^* F).$$

Moreover, if

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

is an exact sequence of $\mathbb{Z}_\ell\text{-}\,\mathrm{fc}(X)$ (resp. of $(AR, \mathbb{Z}_\ell)\text{-}\,\mathrm{fc}(X)$), then the $(w_i)_{i \in \mathbb{N}}$ commute with boundary operators; in other words, the diagrams below are commutative:

$$\begin{tikzcd} g^*(R^i f_!(F'')) \arrow[r, "w_i"] \arrow[d, "g^*(\delta)"'] & R^i f'_!(g'^* F'') \arrow[d, "\delta"] \\ g^*(R^{i+1} f_!(F')) \arrow[r, "w_{i+1}"'] & R^{i+1} f'_!(g'^* F'). \end{tikzcd}$$

Proof. It is enough to prove the analogous assertion for constructible $(AR, \mathbb{Z}_\ell)$-sheaves. But it is deduced by passage to the quotient from the analogous assertion for projective systems of torsion sheaves, which is an immediate consequence of the proper base-change theorem (cf. SGA 4 XII 5.1 in the proper case and SGA 4 XVII in the general case).

C) Exact sequence relative to an open and its complement. Let $f : X \to Y$ be a compactifiable morphism of locally noetherian preschemes, let U be an open of X, and let Z be the closed complement. For every constructible $\mathbb{Z}_\ell$-sheaf F, there exists an exact sequence in $\mathbb{Z}_\ell\text{-}\operatorname{fc}(Y)$, functorial in F:

$$\cdots \longrightarrow R^i f_{!U}(F/U) \longrightarrow R^i f_!(F) \longrightarrow R^i f_{!Z}(F/Z) \longrightarrow R^{i+1} f_{!U}(F/U) \longrightarrow \cdots .$$

Proof. Analogous to that of B).

2.2.5

The formalism developed in this paragraph for $\mathbb{Z}_\ell$-sheaves extends without difficulty to A-sheaves, where A is a finite $\mathbb{Z}_\ell$-algebra, and to L-sheaves, where L is a finite extension of $\mathbb{Q}_\ell$. For example, in the case of $\mathbb{Q}_\ell$-sheaves, the fundamental point in the proofs is that all the functors considered for $\mathbb{Z}_\ell$-sheaves carry $\mathbb{Z}_\ell$-sheaves of torsion to $\mathbb{Z}_\ell$-sheaves of torsion.

3. ℓ-adic cohomology class associated with a cycle

3.1

If X is a proper scheme, or a smooth finite-type scheme, over an algebraically closed field k, one knows that, for every constructible abelian torsion sheaf M prime to the characteristic of k, the cohomology groups

$$H^i(X, M)$$

are abelian groups of finite length. This permits (2.2.1, Remark) the definition, for every $\mathbb{Z}_\ell$-sheaf F, with ℓ prime to the characteristic of k, of a noetherian (AR, ℓ)-adic projective system of abelian groups

$$H^i(X, F)$$

with the usual properties. In particular, the method used in number 2 makes it possible to define the cup-product, the inverse-image homomorphism for a morphism of schemes over k, and the direct-image homomorphism for a proper morphism of smooth schemes over k (cf. Exposé IV for the definitions in the case of torsion sheaves).

Moreover, one knows (1.2.4) that the category of $\mathbb{Z}_\ell$-sheaves on an algebraically closed field is equivalent to the category of finite-type $\mathbb{Z}_\ell$-modules, the equivalence being supplied by the projective-limit functor. In practice it will be convenient to identify the finite-type $\mathbb{Z}_\ell$-module $H^i(X, F)$ with the corresponding $\mathbb{Z}_\ell$-sheaf.

3.2

Let ℓ be a prime number $\neq \operatorname{car} k$, and let X be a smooth scheme of finite type over k. If Z is an algebraic cycle of codimension d on X, then in Exposé IV one associated to it, for every integer n, a cohomology class

$$\gamma_X^{(n)}(Z) \in H^{2d}(X, (\mu_X^{(n)})^{\otimes d}),$$

where $\mu_X^{(n)}$ denotes the locally free $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$-module of the ℓ^{n+1}-st roots of unity. Moreover, one saw that, if $n' \geq n$, $\gamma_X^{(n)}(Z)$ is the image of $\gamma_X^{(n')}(Z)$ in

$$H^{2d}(X, (\mu_X^{(n)})^{\otimes d})$$

by the morphism

$$H^{2d}(X, (\mu_X^{(n')})^{\otimes d}) \longrightarrow H^{2d}(X, (\mu_X^{(n)})^{\otimes d})$$

deduced from raising to the power $\ell^{n'-n}$:

$$\mu_X^{(n')} \longrightarrow \mu_X^{(n)}.$$

Thus one associates to Z an element, denoted $\gamma_X(Z)$:

$$\gamma_X(Z) \in H^{2d}(X, \mathbb{Z}_\ell(d)),$$

where $\mathbb{Z}_\ell(d)$ is defined in (1.3.4). More generally, if Z is a cycle on X, one associates to it by linearity an ℓ-adic cohomology class in

$$\bigoplus_d H^{2d}(X, \mathbb{Z}_\ell(d)),$$

which is in fact the projective limit of the cohomology classes associated with Z in

$$\bigoplus_d H^{2d}(X, (\mu_X^{(n)})^{\otimes d}).$$

3.3

By passage to the limit, it is evident that the ℓ-adic cohomology classes of the preceding paragraph satisfy the same formal properties as the cohomology classes studied in Exposé IV, namely:

(i) If $f : X \to Y$ is a proper morphism of smooth preschemes over k and Z is a cycle on X, then

$$f_*(\gamma_X(Z)) = \gamma_Y(f_*(Z)). \tag{*}$$

(ii) If $f : X \to Y$ is a morphism of smooth preschemes over k and Z is a cycle on Y, then

$$\gamma_X(f^*Z) = f^*(\gamma_Y(Z)). \tag{**}$$

(iii) If X is a smooth prescheme over k and Z_1 and Z_2 are two cycles on X, then

$$\gamma_X(Z_1 \cdot Z_2) = \gamma_X(Z_1) \cdot \gamma_X(Z_2), \tag{**}$$

where the dot on the left-hand side denotes the intersection product of cycles, and the one on the right-hand side the cup-product in cohomology.

(iv) Two algebraically equivalent cycles on a smooth k-prescheme have the same ℓ-adic cohomology class.

($*$) This compatibility is not proved in (SGA $4^{1/2}$ Cycle), but is not difficult to deduce from the results of loc. cit.

($**$) In the non-excessive case, see (SGA $4^{1/2}$ Cycle 2.3.8) for a precise statement.

Exposé VII

COHOMOLOGY OF SOME CLASSICAL SCHEMES
AND
THE COHOMOLOGICAL THEORY OF CHERN CLASSES

by

J.-P. Jouanolou

Throughout the text, an integer $\nu > 0$ is fixed and we put

$$A = \mathbb{Z}/\nu\mathbb{Z};$$

the residual characteristics of all schemes considered are assumed to be prime to the integer ν. If X is such a scheme, μ_X denotes the sheaf of ν-th roots of unity and $A^\bullet(X)$ denotes the cohomology ring

$$\bigoplus_i H^{2i}(X, \mu_X^{\otimes i}),$$

with multiplication given by cup-product.

1. Vector bundles

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module, understood to be of finite type. Put $X = \mathrm{V}(E)$ and $U = \mathrm{V}(E)^*$, the complement of the zero section, and denote by

$$p : X \to S, \qquad q : U \to S$$

the canonical projections.

Proposition 1.1. Let L be an A-module.

(i) a) For every integer $j > 0$, one has

$$R^j p_*(p^* L) = 0.$$

b) The adjunction morphism

$$L \longrightarrow p_* p^*(L)$$

is an isomorphism.

(ii) Suppose that E has constant rank r.

a)

$$R^i_! p(p^* L) = 0 \quad \text{if } i \neq 2r.$$

b) The trace morphism (SGA 4 XVIII)

$$R^{2r}_! p(p^* L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

Proof. Part (i) follows immediately by localization for the Zariski topology from (SGA 4 XV 2.2). Assertion (ii) is essentially contained in (SGA 4 XVIII).

Corollary 1.2. Let L be an A_S-module. For every integer i, the inverse-image morphism

$$H^i(S, L) \longrightarrow H^i(X, p^* L)$$

is an isomorphism.

Proof. By (1.1), the adjunction morphism $L \to Rp_*(p^* L)$ is an isomorphism. Since $H^*(X, p^* L) = H^*(S, Rp_* p^*(L))$, the assertion follows.

Proposition 1.3. Suppose that E has constant rank r. For every A_S-module L, the following properties hold.

(i) a)

$$R^i q_*(q^* L) = 0 \quad \text{if } i \neq 0, 2r-1.$$

b) The adjunction morphism

$$L \longrightarrow q_* q^*(L)$$

is an isomorphism.

c) If Y denotes the complement of U in X, then the morphism

$$R^{2r-1} q_*(q^* L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

obtained as the composite of the boundary morphism

$$R^{2r-1} q_*(q^* L) \longrightarrow R^{2r}_Y p_*(p^* L) \tag{1}$$

and the purity isomorphism (SGA 4 XVIII)

$$R^{2r}_Y p_*(p^* L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

(ii) a)

$$R^i_! q(q^* L) = 0 \quad \text{if } i \neq 1, 2r.$$

b) The boundary morphism

$$L \longrightarrow R^1_! q(q^* L)$$

coming from the unlimited exact sequence, where $f : Y \to S$ is the canonical projection,

$$\cdots \longrightarrow R^i_! q(q^* L) \longrightarrow R^i_! p(p^* L) \longrightarrow R^i f_*(f^* L) \longrightarrow R^{i+1}_! q(q^* L) \longrightarrow \cdots$$

is an isomorphism.

c) The trace morphism

$$R^{2r}_! q(q^* L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

Proof. Let $\alpha : U \to X$ and $\beta : Y \to X$ be the canonical immersions. One knows (SGA 4 XVI 3.7) that, putting $F = p^*(L)$,

$$F \xrightarrow{\sim} \alpha_* \alpha^*(F) \quad \text{and} \quad (R^j \alpha_*)\alpha^*(F) = 0 \quad \text{if } j \neq 0, 2r-1.$$

Moreover, the sheaves $(R^j \alpha_*)\alpha^* F$ for $j \geq 1$ have support in Y, and hence are acyclic for the functor p_*. From these remarks one obtains assertions (i) a) and b) by using the Leray spectral sequence

$$R^i p_*(R^j \alpha_*)(q^* L) \Longrightarrow R^n q_*(q^* L).$$

Assertion (i) c) follows from (1.1 (i)) and from the unlimited exact sequence

$$\cdots \longrightarrow R^i_Y p_*(p^* L) \longrightarrow R^i p_*(p^* L) \longrightarrow R^i q_*(q^* L) \longrightarrow R^{i+1}_Y p_*(p^* L) \longrightarrow \cdots .$$

The sheaf $R^{2r}_Y p_*(p^* L)$ is associated with the presheaf

$$T \longmapsto H^{2r}_{Y \times_S T}(X \times_S T, p^* L).$$

We prove (ii). Assertion c) comes from the fact that q is smooth and has connected geometric fibres of dimension r. The exact sequence mentioned in b) is obtained by applying the functor $R_! p$ to the canonical exact sequence

$$0 \longrightarrow \alpha_!(q^* L) \longrightarrow p^* L \longrightarrow \beta_!(\beta^* p^* L) \longrightarrow 0.$$

Using (1.1 (i)), one obtains assertions (ii) a) and b) without difficulty from this unlimited exact sequence.

Remark 1.4. For $S = \mathrm{Spec}(\mathbb{C})$, (1.3) says that the complement of the origin in $\mathbb{C}^r$ has the same cohomology as the real sphere of dimension $2r-1$, a deformation retract of $\mathbb{C}^r - 0$.

Corollary 1.5 (Gysin exact sequence). The Leray spectral sequence

$$H^i(S, R^j q_*(q^* L)) \Rightarrow H^{i+j}(U, q^* L)$$

defines an unlimited exact sequence, called the Gysin exact sequence,

$$\cdots \longrightarrow H^i(S, L) \longrightarrow H^i(U, q^* L) \longrightarrow H^{i-2r+1}(S, L \otimes \mu_S^{\otimes -r}) \longrightarrow H^{i+1}(S, L) \longrightarrow \cdots .$$

Proof. Lemma of two lines.

2. Projective schemes

In what follows, the mode of exposition adopted is based on a suggestion of L. Illusie, and the method followed is close to a method of P. Deligne for proving the degeneration of certain spectral sequences [6].

2.1

Let X and Y be two schemes and let $f : X \to Y$ be a morphism. We shall define an arrow

$$Rf_*(A_X) \otimes Rf_*(A_X) \longrightarrow Rf_*(A_X) \tag{2.1.1}$$

in $D(Y)$, having the usual associativity and commutativity properties, and therefore endowing $Rf_*(A_X)$ with a structure of generalized algebra. For this, write

$$C^*(X, A_X)$$

for the Godement resolution of A_X (SGA 4 XVII). Proceeding as in (TF 6.6), one defines a morphism of resolutions

$$C^*(X, A_X) \otimes C^*(X, A_X) \longrightarrow C^*(X, A_X \otimes A_X).$$

Composing it with the morphism

$$C^*(X, A_X \otimes A_X) \longrightarrow C^*(X, A_X)$$

deduced from the multiplication of A_X, one finally obtains a morphism

$$C^*(X, A_X) \otimes C^*(X, A_X) \longrightarrow C^*(X, A_X). \tag{2.1.2}$$

Applying the functor f_*, one obtains without difficulty a morphism of complexes of A_Y-modules

$$f_*C^*(X, A_X) \otimes f_*C^*(X, A_X) \longrightarrow f_*C^*(X, A_X),$$

and hence (2.1.1), using flat resolutions of the components of the expression on the left.

2.2

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of rank $r+1$. We denote by

$$P = P_S(E) \xrightarrow{p} S$$

the corresponding projective bundle. The canonical $\mathcal{O}_P$-module $\mathcal{O}_P(1)$ defines, as is known, by means of the Kummer exact sequence, an element

$$\xi \in H^2(P, \mu),$$

and we shall denote by η the image of ξ under the canonical morphism

$$H^2(P, \mu) \longrightarrow H^0(S, R^2p_*(\mu))$$

coming from the Leray spectral sequence

$$H^i(S, R^jp_*(\mu)) \Rightarrow H^{i+j}(P, \mu).$$

The class ξ is identified with an arrow $A_P \to \mu_P[2]$ in $D(P)$, which by transposition defines $\mu_P^{\otimes -1}[-2] \to A_P$, and hence by the usual adjunction an arrow

$$c : \mu_S^{\otimes -1}[2] \longrightarrow Rp_*(A_P),$$

coinciding with η on cohomology in dimension 2.

By tensor product, the arrow c defines, by means of (2.2.1), arrows

$$c_i : \mu_S^{\otimes -i}[-2i] \longrightarrow Rp_*(A_P),$$

for every integer $i \geq 1$, and we agree to denote by c_0 the adjunction arrow

$$A_S \longrightarrow Rp_*(A_P).$$

Thus one defines an arrow

$$\gamma = \bigoplus_{i=0}^{r} c_i : \bigoplus_{i=0}^{r} \mu_S^{\otimes -i}[-2i] \longrightarrow Rp_*(A_P)$$

of $D(S)$.

Theorem 2.2.1. The arrow γ is an isomorphism.

Once the arrow γ has been defined, (2.2.1) is equivalent to the following corollary.

Proposition 2.2.2. Under the preceding hypotheses, one has:

a)
$$R^{2i+1}p_*(A_P) = 0 \quad \text{for every } i \in \mathbb{N}.$$

b) When $\bigoplus_i R^{2i}p_*(\mu_P^{\otimes i})$ is endowed with its A_S-algebra structure deduced from cup-product, the morphism of A_S-algebras
$$A_S[T] \longrightarrow \bigoplus_i R^{2i}p_*(\mu_P^{\otimes i})$$
defined by η has kernel the ideal (T^{r+1}) and hence defines an isomorphism of A_S-algebras
$$\theta : A_S[T]/(T^{r+1}) \xrightarrow{\sim} \bigoplus_i R^{2i}p_*(\mu_P^{\otimes i}).$$

c) The morphism
$$A_S \longrightarrow R^{2r}p_*(\mu^{\otimes r})$$
obtained by composing θ with the restriction to A_S of multiplication by T^r on $A_S[T]/(T^{r+1})$ is the inverse of the trace morphism (SGA 4 XVIII)
$$R^{2r}p_*(\mu^{\otimes r}) \longrightarrow A_S;$$
in other words,
$$\mathrm{Tr}(\eta^r) = 1.$$

Proof. Since the data are stable under base change, the proper base-change theorem (SGA 4 XII 5.1) and compatibility of the trace morphism with base change (SGA 4 XVIII) allow one to reduce to checking the analogous assertion on the fibres of P. In other words, one may suppose that S is the spectrum of a separably closed field k and that p is the canonical morphism $\mathbb{P}_k^r \to k$. In this case, the proof uses the following lemma.

Lemma 2.2.3. Let k be a separably closed field and let $p : \mathbb{P}_k^r \to k$ be the canonical projection. Then:

a)
$$R^{2i+1}p_*(A_P) = 0 \quad \text{for every } i.$$

b)
$$R^{2i}p_*(A_P) \simeq A_k \quad \text{for } i \in [0,r], \qquad R^{2i}p_*(A_P) = 0 \quad \text{if } i > r.$$

The lemma, manifestly true for $r = 0$, is proved by induction on r. Suppose therefore that it is true for $r-1$ and prove it for r. The projective space $\mathbb{P}_k^{r-1}$ embeds as a closed subspace in $\mathbb{P}_k^r$ with complement the open $\mathbb{E}_k^r$, whence an exact sequence of cohomology with proper supports

$$\cdots \longrightarrow H_!^i(\mathbb{E}_k^r) \longrightarrow H^i(\mathbb{P}_k^r) \longrightarrow H^i(\mathbb{P}_k^{r-1}) \longrightarrow H_!^{i+1}(\mathbb{E}_k^r) \longrightarrow \cdots .$$

The computation of cohomology with proper supports of $\mathbb{E}_k^r$ (1.1) shows that, if $i \neq (2r, 2r-1)$, the inverse-image map

$$H^i(\mathbb{P}_k^r) \longrightarrow H^i(\mathbb{P}_k^{r-1})$$

is an isomorphism, whence the assertion in this case. If $i = 2r-1$, one similarly sees that $H^{2r-1}(\mathbb{P}_k^r)$ is contained in $H^{2r-1}(\mathbb{P}_k^{r-1})$, hence is zero by the induction hypothesis. Finally, if $i = 2r$, the relations

$$H^{2r-1}(\mathbb{P}_k^{r-1}) = H^{2r}(\mathbb{P}_k^{r-1}) = 0$$

show that

$$H_!^{2r}(\mathbb{E}_k^r) \xrightarrow{\sim} H^{2r}(\mathbb{P}_k^r),$$

whence the assertion by (1.1).

The lemma implies part a) of 2.2.1. We now prove that θ is an isomorphism. The morphism θ may be interpreted as a morphism of rings

$$A[T]/(T^{r+1}) \longrightarrow \bigoplus_i H^{2i}(P, \mu^{\otimes i}).$$

Thus, by (2.2.3), the assertion will be proved if one shows that, for every $i \in [0,r]$, ξ^i is a free element of the A-module $H^{2i}(P, \mu^{\otimes i})$. It is enough to check this for ξ^r, which will be an immediate consequence of 2.2.2 c).

We now prove part c). In practice, this is the definition of the trace morphism in this situation; in any case, it expresses the compatibility of the trace morphism and the fundamental class in the diagram

$$\begin{tikzcd} x_0 \arrow[r, "\mathrm{inj}"] \arrow[dr] & \mathbb{P}_k^r \arrow[d, "p"] \\ & k, \end{tikzcd}$$

where $\mathrm{inj} : x_0 \to \mathbb{P}^r_k$ is the injection of a closed point of $\mathbb{P}^r_k$.

Corollary 2.2.4. Let L be a complex of A_S-modules. The arrow

$$\mathrm{R}\Gamma(S, Y \overset{\mathrm{L}}{\otimes}_A \mathrm{id}(L)) : \bigoplus_{i=0}^{r} \mathrm{R}\Gamma(S, L \otimes \mu_S^{\otimes -i})[-2i] \longrightarrow \mathrm{R}\Gamma(P, p^*(L))$$

is an isomorphism.

Indeed, it is enough to apply the functor $\mathrm{R}\Gamma_S$ to the isomorphism $Y \overset{\mathrm{L}}{\otimes}_A \mathrm{id}_L$.

2.2.5

To state the next corollary, for every scheme X we shall put

$$A^*(X) = \bigoplus_i H^{2i}(X, \mu_X^{\otimes i})$$

and shall use this notation systematically in what follows.

Corollary 2.2.6. The inverse-image homomorphism

$$p^* : A^*(S) = \bigoplus_i H^{2i}(S, \mu_S^{\otimes i}) \longrightarrow \bigoplus_i H^{2i}(P, \mu_P^{\otimes i}) = A^*(P)$$

is an injective morphism of unital rings, and the elements $1, \xi, \ldots, \xi^r$ form a basis of the $A^*(S)$-module $A^*(P)$. In particular, $A^*(P)$ is a free $A^*(S)$-module of the same rank as E.

3. Chern Classes

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of constant rank r. Denoting by $\check{E}$ the dual of E, put $P = P_S(\check{E})$. We have seen (2.2.6) that, if ξ denotes the class of $\mathcal{O}_P(1)$ in $H^2(P, \mu)$, then the $A^*(S)$-module $A^*(P)$ is free with basis $1, \xi, \ldots, \xi^{r-1}$. In particular, it is immediate for reasons of homogeneity that there exists a unique relation of the form

$$\xi^r + c_1\xi^{r-1} + \cdots + c_i\xi^{r-i} + \cdots + c_r = 0, \tag{3.1}$$

with $c_i \in H^{2i}(S, \mu^{\otimes i})$ for every $i \in [0, r]$.

Now let E be any locally free $\mathcal{O}_S$-module. For every integer r, the set S_r of points of S at which E has rank r is open. Thus one obtains a partition

$$S = \coprod_r S_r$$

of S into disjoint opens; in particular, for every A_S-module M and every integer i, the product of the inverse-image morphisms

$$H^i(S, M) \longrightarrow \prod_r H^i(S_r, M/S_r)$$

is a bijection.

The preceding considerations justify the following definition.

Definition 3.2. Let S be a scheme and E a locally free $\mathcal{O}_S$-module. For every integer $i \geq 0$, the i-th Chern class of E, denoted $c_i(E)$, is the element of $H^{2i}(S, \mu^{\otimes i})$ whose restriction, for every integer $r \geq 0$, to the open subset S_r of S consisting of the points where E has rank r, is the element c_i defined by relation 3.1. If the rank of E is bounded, the total Chern class of E is the element

$$c(E) = \sum_i c_i(E)$$

of $A^*(S)$; in the general case, this denotes the corresponding element of

$$\widehat{A}^*(S) = \prod_i A^i(S).$$

3.3

It is evident that, for all S and E, $c_0(E) = 1$, and that, if the rank of E is bounded by d, then $c_i(E) = 0$ for $i > d$.

Proposition 3.4. The Chern classes just defined satisfy the following properties.

(i) *Normalization.* Let L be an invertible $\mathcal{O}_S$-module. If $[L]$ denotes the image of the class of L under the boundary morphism

$$H^1(S, \mathbb{G}_m) \longrightarrow H^2(S, \mu),$$

then

$$c(L) = 1 + [L].$$

(ii) *Stability under base change.* Let $f : S' \to S$ be a morphism of schemes and let E be a locally free $\mathcal{O}_S$-module. For every integer i,

$$f^*(c_i(E)) = c_i(f^*E).$$

(iii) *Additivity.* Let

$$0 \to E' \to E \to E'' \to 0$$

be an exact sequence of locally free $\mathcal{O}_S$-modules. For every integer n, one has

$$c_n(E) = \sum_{i+j=n} c_i(E')c_j(E'').$$

Before beginning the proof, we extract the following corollary.

Corollary 3.5. For every exact sequence $0 \to E' \to E \to E'' \to 0$ of locally free $\mathcal{O}_S$-modules, one has

$$c(E) = c(E')c(E'')$$

in $\widehat{A}^*(S)$.

Proof of 3.4. Following A. Grothendieck: *La théorie des classes de Chern*, Bull. S.M.F. (1958), pp. 137–154.

(i) Let E be a locally free $\mathcal{O}_S$-module; put $P = P_S(\check{E})$ and let $p : P \to S$ be the canonical projection. Recall that there is a canonical surjective homomorphism

$$p^*(\check{E}) \longrightarrow \mathcal{O}_P(1).$$

In particular, if E is invertible and we put $E = L$, in agreement with the notation of (i), then $P = S$ and there is an isomorphism

$$L \xrightarrow{\sim} \mathcal{O}_P(1).$$

Therefore $\xi + [L] = 0$, and the assertion follows from the definition of Chern classes.

(ii) Immediate verification, left to the reader.

(iii) By using a partition of S into opens on which the modules in question have constant rank, we may suppose that E, E', E'' have constant ranks r, r', r'', respectively. Suppose that the following lemma has been proved; we show how it implies the assertion.

Lemma 3.6. Let E be a locally free $\mathcal{O}_S$-module endowed with a finite filtration

$$0 = E_0 \subset E_1 \subset \cdots \subset E_i \subset \cdots \subset E_r = E$$

whose consecutive quotients $L_i = E_i/E_{i-1}$ $(1 \leq i \leq r)$ are invertible. Then

$$c(E) = (1 + [L_1])(1 + [L_2]) \cdots (1 + [L_r]).$$

Let $h : Z \to S$ be the fibre product of the flag schemes of E' and E''. By the definition of flag schemes, it is clear that $h^*(E')$ and $h^*(E'')$ admit finite filtrations with invertible quotients (L'_i) $(1 \leq i \leq r')$ and (L''_j) $(1 \leq j \leq r'')$, respectively. It follows that $h^*(E)$ admits a filtration with consecutive quotients the L'_i and the L''_j. Applying Lemma 3.6 then shows that

$$c(h^*(E)) = \prod_{1 \leq i \leq r'} c(L'_i) \prod_{1 \leq j \leq r''} c(L''_j) = c(h^*(E'))c(h^*(E'')),$$

hence, by (ii),

$$h^*(c(E)) = h^*(c(E')c(E'')). \tag{3.5.1}$$

But h is a composite of morphisms of the form

$$P_T(F) \longrightarrow T \qquad (F \text{ locally free } \mathcal{O}_T\text{-module}),$$

so by (2.2.5) the inverse-image homomorphism $h^* : A^*(S) \to A^*(Z)$ is injective. This gives the desired assertion by (3.5.1).

We prove Lemma 3.6. By the preceding argument, it is enough to prove the analogous relation

$$c(p^*E) = \prod_i c(p^*L_i),$$

where, recall, $p : P = P_S(\check{E}) \to S$ is the canonical projection. For every $i \in [1, r]$, the monomorphism $E_i \to E$ defines by duality an epimorphism $\check{E} \to \check{E}_i$; putting $Y_i = P_S(\check{E}_i)$, one obtains a sequence of closed subschemes of P:

$$S = Y_1 \subset Y_2 \subset \cdots \subset Y_i \subset \cdots \subset Y_r = P.$$

For every pair (i, j) of integers with $1 \leq i \leq j \leq r$, let

$$\alpha_{ji} : Y_i \to Y_j$$

be the canonical closed immersion, and put $\alpha_i = \alpha_{r,i}$. For every integer $i \in [1, r-1]$, it follows from the evident lemma (3.6.1) below that Y_i is defined in Y_{i+1} by an invertible ideal isomorphic to

$$\alpha_{i+1}^*(p^*(\check{L}_{i+1}) \otimes \mathcal{O}_P(-1)).$$

Lemma 3.6.1. Let $0 \to L \to E \to F \to 0$ be an exact sequence of locally free $\mathcal{O}_S$-modules. Then the sequence

$$0 \to L \otimes_S S(E) \xrightarrow{u} S(E) \xrightarrow{v} S(F) \to 0,$$

where v denotes the evident morphism and u is the degree-one morphism corresponding to the inclusion of L in E, is an exact sequence of graded $\mathcal{O}_S$-modules.

Moreover, Y_1 is identified with S and we have an isomorphism

$$\alpha_1^*(p^*(\check{L}_1)) \simeq \mathcal{O}_{Y_1}(1),$$

i.e.

$$\alpha_1^*(p^*(\check{L}_1) \otimes \mathcal{O}_P(-1)) \simeq \mathcal{O}_{Y_1}. \tag{3.6.2}$$

We shall be able to conclude by using the following two properties of the Gysin homomorphism (SGA 4 XVIII) associated with a closed immersion $u : Z \to X$ of smooth S-schemes defined by an ideal J of $\mathcal{O}_X$:

$$u_*(x\, u^*(y)) = u_*(x)y, \tag{3.6.3}$$

and, if the ideal J is invertible, the relation

$$u_*(1_Z) = -[J] = [\check{J}]. \tag{3.6.4}$$

Indeed, from this we shall prove, by induction increasing on the integer i, that

$$(\alpha_i)_*(\alpha_i)^*\Big(\prod_{j \leq i}[p^*(\check{L}_j) \otimes \mathcal{O}_P(-1)]\Big) = 0. \tag{3.6.5}$$

For $i = r$, assertion (3.6.5) says

$$\prod_i [p^*(\check{L}_i) \otimes \mathcal{O}_P(-1)] = 0,$$

or equivalently

$$\prod_i (\xi + c_1(p^*(\check{L}_i))) = 0,$$

which immediately gives (3.6). Assertion (3.6.5) for $i = 1$ follows from (3.6.2). Suppose it true for i and prove it for $i + 1$. For simplicity put

$$N = \prod_{j \leq i}[p^*(\check{L}_j) \otimes \mathcal{O}_P(-1)].$$

The equality (3.6.5) for i can also be written

$$(\alpha_{i+1})_*(\alpha_{i+1,i})_*(\alpha_{i+1,i})^*(\alpha_{i+1})^*(N) = 0,$$

hence, in view of the projection formula for $\alpha_{i+1,i}$,

$$(\alpha_{i+1})_*\big[(\alpha_{i+1})^*(N)(\alpha_{i+1})^*(p^*(\check{L}_{i+1}) \otimes \mathcal{O}_P(-1))\big] = 0,$$

which is exactly the desired result.

Corollary 3.7.

(i) For every locally free $\mathcal{O}_S$-module E and every integer i,

$$c_i(\check{E}) = (-1)^i c_i(E).$$

(ii) If E is a locally free $\mathcal{O}_S$-module of rank r, then

$$c_1(E) = c_1(\Lambda^r E).$$

(iii) More generally, the Chern classes of a tensor product or of an exterior power are expressed by means of the universal polynomials made explicit in (SGA 6 V 6.1).

Proof. We prove, for example, (ii); the other assertions are seen similarly. Using the flag scheme of E, one reduces to the case where E admits a filtration with invertible quotients L_i. The assertion then follows from the well-known isomorphism

$$\Lambda^r(E) \simeq \bigotimes_i L_i.$$

3.8

Given a scheme X locally of finite type over $\mathrm{Spec}(\mathbb{C})$, the set $X(\mathbb{C})$ of its rational points is canonically endowed with the structure of a locally compact topological space. Thus, by algebraic topology, one has a theory of Chern classes [3], assigning to every continuous complex vector bundle V on $X(\mathbb{C})$ classes $c_i(V) \in H^{2i}(X(\mathbb{C}), \mathbb{Z})$. Now let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of continuous complex-valued functions on $X(\mathbb{C})$, and let $\mathcal{E}$ be a locally free $\mathcal{O}_{\mathbb{C}}$-module of finite type. One defines Chern classes

$$c_i(\mathcal{E}) \in H^{2i}(X(\mathbb{C}), \mathbb{Z}),$$

namely those of the complex vector bundle whose sheaf of continuous sections is $\mathcal{E}$. Finally, using the evident map $\mathbb{Z} \to \mathbb{Z}/\nu\mathbb{Z}$, one obtains Chern classes, denoted by the same symbol when no confusion is possible,

$$c_i(\mathcal{E}) \in H^i(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z}). \tag{3.8.1}$$

On the other hand, when $X_{\text{ét}}$ is equipped with its structural sheaf $\mathcal{O}_a$ and $X(\mathbb{C})$ with the ring $\mathcal{O}_{\mathbb{C}}$, there is a morphism of ringed topoi (SGA 4 XI)

$$\varepsilon : X(\mathbb{C}) \to X_{\text{ét}},$$

the structural morphism of rings

$$\varepsilon^*(\mathcal{O}_a) \to \mathcal{O}_{\mathbb{C}} \tag{3.8.2}$$

coming from the fact that every algebraic function is continuous. The morphism ε induces a morphism of abelian sheaves

$$\varepsilon^*(\mu_{X_{\text{ét}}}) \to \mu_{X(\mathbb{C})}, \tag{3.8.3}$$

where the letter μ denotes, in each of the two contexts, the sheaf of ν-th roots of unity. Finally, we shall canonically identify $(\mathbb{Z}/\nu\mathbb{Z})_{X(\mathbb{C})}$ and $\mu_{X(\mathbb{C})}$ by means of the isomorphism

$$(\mathbb{Z}/\nu\mathbb{Z})_{X(\mathbb{C})} \xrightarrow{\sim} \mu_{X(\mathbb{C})} \tag{3.8.4}$$

sending the section 1 to the section $e^{2i\pi/\nu}$.

Proposition 3.8.5. Let E be a locally free $\mathcal{O}_X$-module. For every integer $i \geq 0$, let

$$\varepsilon^* : H^{2i}(X_{\text{ét}}, \mu_X^{\otimes i}) \to H^{2i}(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z})$$

be the inverse-image morphism deduced from 3.8.3 and 3.8.4. Then

$$c_i(\varepsilon^* E) = \varepsilon^* c_i(E).$$

Proof. It is enough to check this when E is invertible. Then, with the usual identifications, $\varepsilon^*(E)$ is the image of E under the canonical map

$$\varepsilon^* : H^1(X_{\text{ét}}, \mathbb{G}_m) \to H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*), \tag{3.8.6}$$

and one has to see that the diagram

$$\begin{tikzcd} H^1(X, \mathbb{G}_m) \arrow[r, "\varepsilon^*"] \arrow[d, "\partial"'] & H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*) \arrow[d, "c_1"] \\ H^2(X, \mu_X) \arrow[r, "\varepsilon^*"'] & H^2(X(\mathbb{C}), \mathbb{Z}/\nu\mathbb{Z}) \end{tikzcd} \tag{3.8.7}$$

where ∂ is the boundary morphism coming from the canonical exact sequence

$$0 \to \mu_X \to \mathbb{G}_m \xrightarrow{\nu} \mathbb{G}_m \to 0 \tag{3.8.8}$$

and c_1 is the first Chern class (3.8.1), is commutative. In fact, using that the morphism "first Chern class"

$$H^1(X(\mathbb{C}), \mathcal{O}_{\mathbb{C}}^*) \to H^2(X, \mathbb{Z})$$

is the boundary morphism of the canonical exact sequence

$$0 \to \mathbb{Z} \to \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \to 0, \qquad t \mapsto e^{2i\pi t},$$

one easily sees that the morphism c_1 of (3.8.7) is nothing other than the boundary morphism associated with the exact sequence

$$0 \to \mathbb{Z}/\nu\mathbb{Z} \to \mathcal{O}_{\mathbb{C}}^* \xrightarrow{\nu} \mathcal{O}_{\mathbb{C}}^* \to 0, \qquad t \mapsto e^{2i\pi t/\nu},$$

which, in view of (3.8.4), is the analogue of (3.8.8). The assertion follows at once.

3.9

Let X be a scheme and E a locally free $\mathcal{O}_X$-module. Immediate passage to the limit allows one to define, for every prime number ℓ prime to the residual characteristics of X, ℓ-adic Chern classes

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)) \stackrel{\text{def}}{=} \varprojlim_{\nu} H^{2i}(X, \mu_{\ell^\nu}^{\otimes i}),$$

which have the usual properties made explicit in 3.4 and 3.7 and which, in the context of 3.8, are manifestly compatible, in an evident sense, with the Chern classes defined topologically.

Tensoring by $\mathbb{Q}_\ell$, one obtains ℓ-adic Chern classes modulo torsion

$$c_i(E) \in H^{2i}(X, \mathbb{Q}_\ell(i)) = H^{2i}(X, \mathbb{Z}_\ell(i)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell,$$

which manifestly satisfy the same properties.

Now suppose that X is a quasi-projective smooth scheme over a field k, and denote here by $C^*(X)$ its Chow ring (Sém. Chevalley: Anneaux de Chow et applications III 4). Recall that in IV we defined a morphism of graded rings

$$\varphi_X : C^*(X) \to A^*(X)$$

compatible with direct-image and inverse-image morphisms. Moreover, the canonical isomorphism

$$\varepsilon_X : \operatorname{Pic}(X) \xrightarrow{\sim} C^1(X)$$

which sends every invertible $\mathcal{O}_X$-module L to the class of the divisor of a rational section of L not vanishing on any connected component of X, gives, on the model of the aforementioned article of Grothendieck, where k was assumed algebraically closed, a theory of Chern classes with values in $C^*(X)$. The composite

$$\varphi_X \circ \varepsilon_X : \operatorname{Pic}(X) \longrightarrow H^2(X, \mathbb{Z}_\ell(1))$$

is just the first ℓ-adic Chern class. Indeed, it is enough to check this on elements of $\operatorname{Pic}(X)$ of the form $\check{J}$, with J an invertible ideal of $\mathcal{O}_X$, and then this follows from (3.6.4). By the uniqueness properties of theories of Chern classes, it follows that the morphism φ_X sends Chern classes to Chern classes.

4. Self-intersection Formula and Applications

Let S be a scheme, X an S-scheme, and Y a closed subscheme of X, smooth over S and everywhere of codimension d. Finally suppose that X is smooth over S in a neighbourhood of Y. Let $i : Y \to X$ be the canonical closed immersion and J the ideal of $\mathcal{O}_X$ defining Y. Recall (SGA $4^{1/2}$ Cycle) that in this case one has Gysin isomorphisms

$$i_* : H^{2p}(Y, \mu_Y^{\otimes p}) \to H_Y^{2p+2d}(X, \mu_X^{\otimes(p+d)}),$$

namely cup-products with the fundamental class. If no confusion is possible, we shall denote by the same symbol the Gysin morphisms

$$i_* : H^{2p}(Y, \mu_Y^{\otimes p}) \to H^{2p+2d}(X, \mu_X^{\otimes(p+d)})$$

obtained from the preceding ones by forgetting supports.

Theorem 4.1. In the preceding situation, if

$$N = J/J^2$$

denotes the normal sheaf of i, then for every $y \in A^*(Y)$ one has

$$i^* i_*(y) = y\, c_d(\check{N}).$$

Before beginning the proof of this theorem, we indicate a few variants, proved in exactly the same way.

a) If ℓ is a prime number prime to the residual characteristics of S, the analogous formula is true in $H^*(Y, \mathbb{Z}_\ell(*))$.

b) One may put in supports. If, for example, T is a closed subset of Y and $y \in H_T^{2p}(Y, \mu_Y^{\otimes p})$, formula 4.1 is still true in $H_T^{2p+2d}(Y, \mu_Y^{\otimes(p+d)})$. Of course, i_* then denotes the Gysin morphism

$$H_T^{2p}(Y, \mu_Y^{\otimes p}) \to H_T^{2p+2d}(X, \mu_X^{\otimes(p+d)}),$$

which is again an isomorphism (SGA $4^{1/2}$ Cycle).

c) Finally, the two cases may be combined.

Since the fundamental tool in the proof of 4.1 is the theory of blown-up schemes, we now recall a few facts about it. Let

$$f : X' \to X$$

be the X-scheme obtained by blowing up Y, and let

$$\begin{tikzcd} Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\ Y \arrow[r, "i"'] & X \end{tikzcd}$$

be the cartesian diagram constructed from f and i. Recall (cf. SGA 6 VII 3.1, whose notation we adopt) that j is a regular immersion of codimension 1 defined by an ideal

$$J' \simeq \mathcal{O}_{X'}(1)$$

and that $Y' = P_Y(N)$. Moreover, one has an exact sequence of $\mathcal{O}_{Y'}$-modules

$$0 \to F \to g^*(N) \to \mathcal{O}_{Y'}(1) \to 0.$$

Lemma 4.2. For every A_X-module M, the homomorphisms

$$f^* : H_Y^*(X, M) \to H_{Y'}^*(X', f^*(M))$$

and

$$f_* : H_{Y'}^*(X', f^*(M)) \to H_Y^*(X, M)$$

are related by $f_*f^* = \mathrm{id}$. Moreover, f_* has degree 0.

Proof. Since f is a complete-intersection morphism of relative dimension zero (SGA 6 VII 1.8), the last assertion follows from the definition of the Gysin homomorphism (SGA $4^{1/2}$ Cycle). For the first, the projection formula for f,

$$f_*(x\, f^*(x')) = f_*(x)x',$$

where $x \in H^*(X, A)$ and $x' \in H^*_Y(X, M)$, reduces the claim to proving that $f_*(1) = 1$. It is enough to check the analogous assertion over $U = X - Y$, which is evident because f induces an isomorphism from $U' = X' - Y'$ onto U.

Remark 4.3. Of course, the analogue of 4.2 obtained by eliminating supports is also true and is proved in the same way.

We now come to the proof of 4.1. First, by a general result of (SGA $4^{1/2}$ Cycle), which is proved in an essentially formal way, one has

$$i^*i_*(y) = y\, i^*i_*(1),$$

so it is enough to prove

$$i^*i_*(1_Y) = c_d(\check{N}),$$

where i_* here denotes the Gysin isomorphism

$$H^0(Y, A_Y) \to H^{2d}_Y(X, \mu_X^{\otimes d}).$$

Since $f^*i_*1_Y \in H^{2d}_{Y'}(X', \mu_{X'}^{\otimes d})$, and since the Gysin morphism

$$j_* : H^{2d-2}(Y', \mu_{Y'}^{\otimes(d-1)}) \to H^{2d}_{Y'}(X', \mu_{X'}^{\otimes d})$$

is an isomorphism, there exists

$$u \in H^{2d-2}(Y', \mu_{Y'}^{\otimes(d-1)})$$

such that

$$f^*i_*(1_Y) = j_*(u). \tag{4.1.1}$$

Applying j^* to both sides of 4.1.1, one obtains

$$j^*f^*(1_Y) = j^*j_*(u).$$

Putting $\xi = c_1(\mathcal{O}_{Y'}(1))$, it follows from 3.6.4 that

$$g^*i^*i_*(1_Y) = -u\,\xi. \tag{4.1.2}$$

The element u admits a unique decomposition of the form

$$u = a_0 + a_1\xi + \cdots + a_{d-1}\xi^{d-1}, \qquad a_i \in H^{2d-2-2i}(Y, \mu_Y^{\otimes(d-1-i)}). \tag{4.1.3}$$

In view of the defining relation for the Chern classes of $\check{N}$,

$$\xi^d + c_1(\check{N})\xi^{d-1} + \cdots + c_{d-1}(\check{N})\xi + c_d(\check{N}) = 0, \tag{4.1.4}$$

we deduce the equality

$$u\xi = -a_{d-1}c_d(\check{N}) + (a_0 - a_{d-1}c_{d-1}(\check{N}))\xi + \cdots + (a_{d-2} - a_{d-1}c_1(\check{N}))\xi^{d-1}.$$

Comparing this last expression with (4.1.2), and using the injectivity of g^* (2.2.6), one obtains

$$i^*i_*(1_Y) = a_{d-1}c_d(\check{N}). \tag{4.1.5}$$

It remains to determine a_{d-1}. First, the homomorphism g_* lowers degrees by $2d-2$, hence

$$g_*(\xi^i) = 0 \quad (0 \le i \le d-2),$$

and moreover (2.2.2)

$$g_*(\xi^{d-1}) = 1.$$

Applying g_* to both sides of 4.1.3, we obtain

$$g_*(u) = a_{d-1}. \tag{4.1.6}$$

On the other hand, applying f_* to both sides of 4.1.1, we obtain, in view of 4.2,

$$i_*(1_Y) = i_*g_*(u). \tag{4.1.7}$$

But the homomorphism i_* in question is injective; hence

$$a_{d-1} = g_*(u) = 1_Y,$$

which completes the proof.

Remarks 4.4. a) The essential ingredient in what precedes is the purity theorem. Once it is available in full generality, statement 4.1 will be valid with Y regular and purely of codimension d in X, and X excellent and regular in a neighbourhood of Y. In this form, the result is in any case established in characteristic 0, by (SGA 4 XIX).

b) When S is the spectrum of a separably closed field and X and Y are quasi-projective and smooth over S, the analogous statement in the Chow ring of Y was established by Mumford (cf. 9.2).

Corollary 4.5. Under the hypotheses of 4.1, let x and y belong to $A^*(Y)$. Then

$$i_*(x)i_*(y) = i_*(xy\, c_d(\check{N})).$$

Proof. The projection formula for the morphism i shows that

$$i_*(x)i_*(y) = i_*(x\, i^*i_*(y)),$$

and the conclusion follows from 4.1.

Corollary 4.6. Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of constant rank r, and let $g : X = V(\check{E}) \to S$ be the vector bundle whose sheaf of sections is E. For every section f of X over S, one has

$$f_*(1_S) = g^*(c_r(E))$$

in $H^{2r}(X, \mu_X^{\otimes r})$.

Proof. Since g^* is an isomorphism (1.2) and $g \circ f = \mathrm{id}$, f^* is bijective. It is therefore enough to check that

$$f^*f_*(1_S) = f^*g^*(c_r(E)).$$

If N denotes the normal sheaf of f, this last equality is equivalent, by 4.1, to

$$c_r(\check{N}) = c_r(E).$$

In fact one even has an isomorphism $\check{N} \simeq E$, since the exact sequence of (EGA IV 17.2.5), applied to the situation

$$S \xrightarrow{f} X \xrightarrow{g} S,$$

gives an isomorphism $N = f^*\Omega^1_{X/S}$, and moreover $\Omega^1_{X/S} \simeq g^*\check{E}$ (EGA IV 16.4.8).

Corollary 4.7. Under the hypotheses of 4.6, assume in addition that S is smooth and quasi-projective over a separably closed field. Let σ_X denote the zero-section cycle of X, and let $Y = f^{-1}(\sigma_X)$ be the cycle of zeros of f. Then

$$\mathrm{cl}_S(Y) = c_r(E)$$

in $H^{2r}(S, \mu_S^{\otimes r})$.

Proof. By IV,

$$\mathrm{cl}_S(Y) = f^* \,\mathrm{cl}_X(\sigma_X).$$

Let s_0 be the zero-section morphism of X. Then

$$\mathrm{cl}_X(\sigma_X) \overset{\mathrm{dfn}}{=} (s_0)_*(1_S) \overset{(4.6)}{=} g^*(c_r(E)),$$

and the assertion follows by applying f^* to the extreme terms.

Remark 4.8. Let ℓ be a prime number distinct from the residual characteristics of S. By simple passage to the projective limit, one deduces from 4.6 and 4.7 the corresponding statements in ℓ-adic cohomology.

Now let ℓ be a prime number and let X be a proper scheme over a separably closed field k of characteristic prime to ℓ. Recall (IV) that the ℓ-adic Euler-Poincaré characteristic of X is the integer

$$\mathrm{EP}_\ell(X) = \sum_i (-1)^i \dim H^i(X, \mathbb{Q}_\ell).$$

Corollary 4.9. Let X be a smooth, proper, connected scheme of dimension n over a separably closed field k, with $\mathrm{char}(k) \neq \ell$. If $p : X \to k$ denotes the canonical projection, then

$$\mathrm{EP}_\ell(X) = p_* c_n(\check{\Omega}^1_X)$$

in $\mathbb{Q}_\ell = H^0(k, \mathbb{Q}_\ell)$.

Proof. Let $j : X \to X \times X$ be the diagonal immersion of X, and let $q : X \times X \to k$ be the canonical projection. Applying 4.1 to the morphism j gives

$$c_n(\check{\Omega}^1_X) = j^* j_*(1_X),$$

hence

$$p_*(c_n(\check{\Omega}^1_X)) = p_* j^* j_*(1_X) = q_* j_* j^* j_*(1_X).$$

Using the projection formula for j, one deduces

$$p_*(c_n(\check{\Omega}^1_X)) = q_*(j_*(1_X) \cup j_*(1_X)),$$

where, by IV, the second member is equal to the ℓ-adic Euler-Poincaré characteristic of X.

4.10

When X is projective over k, the result recalled from IV shows that $\mathrm{EP}_\ell(X)$ is, in the Chow ring of $X \times X$, the self-intersection of the diagonal cycle, and consequently is independent of the prime number $\ell \neq \mathrm{char}(k)$. We shall therefore be able to speak of the Euler-Poincaré characteristic $\mathrm{EP}(X)$ of X, without reference to the integer ℓ.

Proposition 4.11. Let X be a smooth, projective, connected scheme of dimension n over a separably closed field k. Then

$$\mathrm{EP}(X) = \sum_{p,q} (-1)^{p+q} \dim H^p(X, \Omega_X^q).$$

Proof. We use the notation of the proof of 4.9, and let J be the ideal of the diagonal immersion of X. As observed in 4.10, $\mathrm{EP}(X)$ is, in the Chow ring of $X \times X$, the intersection product $\Delta \cdot \Delta$ of the diagonal cycle Δ with itself. To compute it, one may neglect torsion and therefore work in $\mathrm{Gr}^n_{\mathrm{top}}(X)$. This shows that $\mathrm{EP}(X)$ is the image under the direct-image morphism

$$q_* : K(X \times X) \to K(k) = \mathbb{Z}$$

of the square of the class of $\mathcal{O}_{X\times X}/J$. But, by (SGA 6 VII 2.5), this square is equal to

$$\sum_i (-1)^i \lambda^i([\Omega_X^1]),$$

and the desired formula follows immediately, by the definition of q_*.

5. Flag Schemes

Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of rank r, and let $(p_i)_{1\leq i\leq m}$ be integers > 0 satisfying

$$p_1 + p_2 + \cdots + p_m = r.$$

Recall that a flag of type $(p_1, \ldots, p_m)$ of E is a finite filtration

$$0 = E_m \subset E_{m-1} \subset \cdots \subset E_0 = E$$

of E such that, for every integer $i \in [1, m]$, the quotient E_{i-1}/E_i is locally free of rank p_i.

One defines a functor $(\mathrm{Sch}\,/S)^\circ \to (\mathrm{Ens})$ by assigning to an S-scheme $u : T \to S$ the set of flags of type $(p_1, \ldots, p_m)$ of $u^*(E)$, and to a morphism of S-schemes the inverse image, in the evident sense, on flags. It is proved (EGA I, 2nd edition) that this functor is representable, and any representative is called an S-scheme of flags of type $(p_1, \ldots, p_m)$ of E.

In what follows, $f : X \to S$ will denote the S-scheme of flags of type $(p_1, \ldots, p_m)$ of E. Moreover, the scheme of flags of type $(1, 1, \ldots, 1)$ of E will be denoted D and called the flag scheme of E, without further specification, if no confusion is possible.

It is clear from the definition that the $\mathcal{O}_X$-module $f^*(E)$ admits a canonical finite filtration

$$0 = F_m \subset F_{m-1} \subset \cdots \subset F_0 = f^*(E),$$

for which the quotient $E_i = F_{i-1}/F_i$ $(1 \leq i \leq m)$ is locally free of rank p_i. For every integer $i \in [1, m]$, let

$$g_i : D_i \to X$$

be the X-scheme of flags of E_i, and let $g : D_1 \times_X \cdots \times_X D_m \to X$ be the fibre product of the X-schemes D_i. It is clear that the composite

$$h = f \circ g : \prod_i D_i \to S \tag{5.1}$$

is S-isomorphic to the S-scheme of flags of E; up to S-isomorphism, one may therefore write

$$D = \prod_i D_i.$$

Proposition 5.2. In $D^b(S, A)$ there is an isomorphism

$$\mathrm{R}f_*(A) \simeq \bigoplus_i R^{2i} f_*(A)[-2i].$$

Proof. Since h is a composite of finitely many morphisms of the form

$$P_T(F) \to T \qquad (F \text{ locally free } \mathcal{O}_T\text{-module}),$$

it follows from (2.2.1) that

$$\mathrm{R}h_*(A) \simeq \bigoplus_j R^j h_*(A)[-j].$$

For the same reason, one also has

$$\mathrm{R}g_*(A) \simeq \bigoplus_j R^j g_*(A)[-j].$$

Using ([6] 2.16), one obtains an isomorphism

$$\mathrm{R}f_*(g_*A) \simeq \bigoplus_j R^j f_*(g_*A)[-j].$$

Since g is smooth with geometrically connected fibres, $A_X \simeq g_*(A_D)$; hence it remains to show that

$$R^{2i+1} f_*(A) = 0 \qquad (i \in \mathbb{N}). \tag{5.2.1}$$

For this, one immediately reduces, using conservativity of the system of fibre functors (SGA 4 VIII 3.5), to the case where S is the spectrum of a separably closed field k. Then, since g is a composite of projective bundles, (2.2.4) shows that the inverse-image map

$$H^{2i+1}(X, A_X) \to H^{2i+1}(D, A_D)$$

is a monomorphism, so one may suppose that $X = D$. In this case, since the projection $D \to k$ is itself a composite of projective bundles, the assertion follows by induction from (2.2.4).

Corollary 5.3. For every morphism of schemes $u : S \to T$, the Leray spectral sequence

$$E_2^{pq} = R^p u_* \, R^q f_*(A) \Rightarrow R^{p+q}(u \circ f)_*(A)$$

is degenerate.

Proof. Cf. [6] 1.2.

We shall now specify the multiplicative structure of the cohomology of X, first examining a particular case. With the preceding notation, one obtains a morphism of rings

$$\alpha : A^*(S)[T_{ij_i}]_{1\leq i\leq m,\, 1\leq j_i\leq p_m} \to A^*(X)$$

by sending $A^*(S)$ into $A^*(X)$ by inverse image f^* and, for every pair (i,j), as above, sending the element T_{ij} to the j-th Chern class $c_j(E_i)$. For every integer p, one has, by (3.4 (iii)),

$$c_p(f^*E) = \sum_{i_1+i_2+\cdots+i_m=p} c_{i_1}(E_1)\cdots c_{i_m}(E_m),$$

so that, by functoriality of Chern classes, the morphism vanishes on the ideal J_X generated by the elements

$$c_p(E) - \sum_{i_1+\cdots+i_m=p} T_{1,i_1}\cdots T_{m,i_m}$$

for varying p. Hence there is a morphism of rings

$$\varphi_X : \big(A^*(S)[T_{ij_i}]_{(i,j_i)}\big)/J_X \to A^*(X).$$

Proposition 5.4. The above morphism of rings φ_X is an isomorphism. The $A^*(S)$-module $A^*(X)$ is free of rank

$$\frac{r!}{p_1!p_2!\cdots p_m!}.$$

Proof. We follow Grothendieck's exposé (Séminaire Chevalley 1958, Anneaux de Chow et applications, Exp. IV). We shall need the following lemma (cf. Lemma 1, p. 19 of loc. cit.).

Lemma 5.4.1. a) Let R be a commutative ring, and let $(a_i)_{1\leq i\leq r}$ be elements of R. Let σ_i be the i-th elementary symmetric function of the elements $(U_i)_{1\leq i\leq r}$, and let I be the ideal of the polynomial ring $R[U_1, U_2, \ldots, U_r]$ generated by the elements $\sigma_i - a_i$. Then the R-algebra

$$R[U_1, U_2, \ldots, U_r]/I$$

is a free R-module of rank $r!$.

b) Let $r, p_1, p_2, \ldots, p_m$ be integers > 0 satisfying $p_1+\cdots+p_m = r$. Then the morphism of polynomial rings

$$V = R[T_{i,j_i}]_{1\leq i\leq m,\, 1\leq j_i\leq p_i} \longrightarrow R[U_i]_{1\leq i\leq r} = W$$

obtained by sending T_{i,j_i} to the j_i-th elementary symmetric function of the elements

$$(U_{p_1+\cdots+p_{i-1}+s})_{1\leq s\leq p_i}$$

is injective and makes W a free V-module of rank $p_1!\cdots p_m!$.

In particular, the morphism of R-algebras

$$R[T_1, \ldots, T_r] \to R[U_1, \ldots, U_r]$$

obtained by sending T_i to the i-th elementary symmetric function of the U_j is injective and makes the second member a free module of rank $r!$ over the first.

We show how (5.4.1) implies (5.4). First suppose that X is the scheme D of flags of E, and for simplicity put $T_{i1} = U_i$. Since the canonical projection $D \to S$ is a composite of $r-1$ morphisms of the form

$$P_{Y_i}(M_i) \to Y_i \quad (1 \le i \le r-1),$$

where M_i is a locally free $\mathcal{O}_{Y_i}$-module of rank $r-i+1$, it follows from (2.2.6) that $A^*(D)$ is a free $A^*(S)$-module of rank $r!$, and that the morphism φ_D is surjective. But Lemma 5.4.1 a) shows that the source of φ_D is also a free $A^*(S)$-module of rank $r!$; hence the assertion in the case $X = D$.

In the general case, it is clear that the diagram

$$\begin{tikzcd} A^*(X) \arrow[rr, "g^*"] & & A^*(D) \\ C = A^*(S)[T_{i,j_i}]/J_X \arrow[u, "\varphi_X"] \arrow[rr, "\gamma"'] & & A^*(S)[U_k]/J_D \arrow[u, "\varphi_D"'] \end{tikzcd}$$

commutes; here the map γ is obtained by passage to the quotient from the morphism of $A^*(S)$-algebras

$$A^*(S)[T_{i,j_i}] \to A^*(S)[U_k]$$

defined by sending T_{i,j_i} to the j_i-th elementary symmetric function of the elements

$$(U_{p_1+\cdots+p_{i-1}+s})_{1\le s\le p_i}.$$

Since φ_D is an isomorphism, 5.4.1 b) implies that $A^*(D)$ is a free C-module of rank $p_1!\cdots p_m!$. On the other hand, expressing g as a fibre product of X-schemes of flags shows that $A^*(D)$ is also a free $A^*(X)$-module of rank $p_1!\cdots p_m!$. It follows at once that φ_X is an isomorphism.

In the statement of the next proposition, we agree to put, for every scheme Y and every A_Y-module M,

$$Q(Y, M) = \bigoplus_{n,i} H^n(Y, M \otimes \mu_Y^{\otimes i}).$$

The A-module $Q(Y, M)$ is canonically endowed with a structure of $A^*(Y)$-module, coming from cup-product. If $u : Z \to Y$ is a morphism of schemes, the inverse-image morphism

$$u^* : Q(Y, M) \to Q(Z, u^*M)$$

is compatible with the $A^*(Y)$-module structure on $Q(Y, M)$ and the $A^*(Z)$-module structure on $Q(Z, u^*M)$. One deduces a morphism of $A^*(Z)$-modules, called canonical,

$$A^*(Z) \otimes_{A^*(Y)} Q(Y, M) \to Q(Z, u^*M).$$

Proposition 5.5. Let S be a scheme, and let E be a locally free $\mathcal{O}_S$-module of rank r. Let $f : X \to S$ be the S-scheme of flags of type $(p_1, \ldots, p_m)$ of E. Then, for every A_S-module L, the canonical morphism

$$\alpha : A^*(X) \otimes_{A^*(S)} Q(S, L) \to Q(X, f^*L)$$

is an isomorphism.

Proof. When $X = P_S(E)$, this is an immediate consequence of (2.2.4) and (2.2.6). Hence it follows that it is true for $X = D$. In the general case, since the canonical morphisms $g : D \to X$ and $h : D \to S$ are composites of projective bundles, the canonical morphisms

$$A^*(D) \otimes_{A^*(S)} Q(S, L) \to Q(D, h^*L)$$

and

$$A^*(D) \otimes_{A^*(X)} Q(X, g^*L) \to Q(D, g^*L)$$

are isomorphisms. Consequently $\mathrm{id}_{A^*(D)} \otimes \alpha$ is an isomorphism, whence the result, since $A^*(D)$ is free over $A^*(X)$.

We shall now give a relative variant of (5.4) and (5.5). With the preceding notation, let $u : S \to T$ be a morphism of schemes. Put

$$A^*(X/T) = \bigoplus_i R^{2i}(u \circ f)_*(A_X)$$

and similarly

$$A^*(S/T) = \bigoplus_i R^{2i}u_*(A_S).$$

Finally, for every A_S-module L, put

$$Q(S/T, L) = \bigoplus_{n,i} R^{2i}u_*(L \otimes \mu_S^{\otimes i})$$

and

$$Q(X/T, f^*L) = \bigoplus_{n,i} R^n(u \circ f)_*(f^*L \otimes \mu_X^{\otimes i}).$$

For every pair (i, j) of integers, with $1 \leq i \leq m$ and $1 \leq j \leq p_i$, the image of the Chern class $c_j(E_i)$ by the canonical morphism

$$H^{2j}(X, \mu_X^{\otimes j}) \to H^0(T, R^{2j}(u \circ f)_*(\mu_X^{\otimes j}))$$

defines an A_T-morphism

$$c_{ij} : A_T \to R^{2j}(u \circ f)_*(\mu_X^{\otimes j}).$$

The data of the c_{ij} and the inverse image f^* allow one to define a morphism of $A^*(S/T)$-algebras

$$\theta_1 : A^*(S/T)[T_{i,j_i}]_{(i,j_i)} \to A^*(X/T).$$

Denoting by c_i the sections of $A^*(S/T)$ defined by the Chern classes $c_i(E)$, one sees as in (5.4) that θ_1 vanishes on the ideal J locally generated by the relations

$$\sum_{i_1+\cdots+i_m=p} T_{1,i_1} \cdots T_{m,i_m} = c_p \qquad (p \in \mathbb{N}).$$

Hence there is a morphism of $A^*(S/T)$-algebras

$$\theta : A^*(S/T)[T_{i,j_i}]_{(i,j_i)}/J \to A^*(X/T).$$

Proposition 5.6. With the preceding notation:

a) The morphism of $A^*(S/T)$-algebras

$$\theta : A^*(S/T)[T_{i,j_i}]_{(i,j_i)}/J \to A^*(X/T)$$

is an isomorphism.

b) For every A_S-module L, the canonical morphism of $A^*(X/T)$-modules

$$A^*(X/T) \otimes_{A^*(S/T)} Q(S/T, L) \to Q(X/T, f^*L)$$

is an isomorphism.

Proof. Simple sheafification of (5.4) and (5.5), respectively.

6. Group Schemes

We begin with a general property of objects twisted by torsors.

Proposition 6.1. Let S be a scheme, G a quasi-compact smooth S-group scheme with connected fibres, and X an S-scheme endowed with a left action of G. If $f : E \to S$ is a right S-torsor under G, denote by X_E the S-scheme obtained from X by twisting by means of E and the action of G on X. Let $p : X \to S$ and $q : X_E \to S$ be the canonical projections. Under these conditions, for every A_S-module M and every integer i, the A_S-module $R^i q_*(q^*M)$ is canonically isomorphic to $R^i p_*(p^*M)$.

Proof.

$$\begin{tikzcd} E \times_S X \arrow[r, "\varphi"] \arrow[d, "\mathrm{pr}_1"'] & X_E \arrow[d, "q"] \\ E \arrow[r, "f"'] & S \end{tikzcd} \qquad \begin{tikzcd} E \times_S X \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & X \arrow[d, "p"] \\ E \arrow[r, "f"'] & S. \end{tikzcd}$$

It is known that diagram (D_1), where φ denotes the canonical passage-to-the-quotient map, is cartesian. Since f is smooth, the smooth base-change theorem gives an isomorphism

$$u : f^*R^i q_*(q^*M) \xrightarrow{\sim} R^i(\mathrm{pr}_1)_*(\varphi^* q^* M).$$

Similarly, the cartesian diagram (D_2) gives an isomorphism

$$v : f^*R^i p_*(p^*M) \xrightarrow{\sim} R^i(\mathrm{pr}_1)_*((\mathrm{pr}_2)^* p^* M).$$

But $p \circ \mathrm{pr}_2 = q \circ \varphi$, so after the evident identification of $\varphi^* q^*$ with $(\mathrm{pr}_2)^* p^*$ one obtains an isomorphism

$$v^{-1} \circ u : f^*R^i q_*(q^*M) \xrightarrow{\sim} f^*R^i p_*(p^*M).$$

Since the geometric fibres of f are non-empty and connected, it follows from (SGA 4 XII 6.5 (i)) that the adjunction morphism

$$\mathrm{adj} : \mathrm{id} \to f_* f^*$$

is an isomorphism, and therefore

$$(\mathrm{adj})^{-1} f_*(v^{-1} \circ u) \circ (\mathrm{adj})$$

is the desired isomorphism.

Proposition 6.2. (Mme Raynaud [5].) Let S be a scheme and let $p : G \to S$ be a reductive S-group scheme (SGA 3 XIX). For every locally constant constructible A_S-module M and every integer i, the A_S-module $R^i p_*(p^* M)$ is locally constant and constructible. Moreover, for every morphism $u : S' \to S$ defining a cartesian diagram

$$\begin{tikzcd} G' \arrow[r, "u'"] \arrow[d, "p'"'] & G \arrow[d, "p"] \\ S' \arrow[r, "u"'] & S, \end{tikzcd}$$

the base-change morphism

$$u^* R^i p_*(p^* M) \to R^i p'_*(p \circ u')^*(M)$$

is an isomorphism.

Proof. The two assertions of the statement are local for the étale topology of S. By (SGA 3 XII 2.3), we may therefore suppose that G is split. Let B be a Borel subgroup of G of the type indicated in (SGA 3 XXII 5.1.1), so that the quotient G/B exists by (SGA 3 XXII 5.8.5), and let $q : G \to G/B$ and $r : G/B \to S$ be the canonical morphisms. Since r is proper and smooth, it satisfies the proper base-change theorem and the functors $R^i r_*$ send locally constant constructible abelian sheaves to sheaves of the same type (SGA 4 XVI 2.2). Moreover, by (SGA 4 XXII 5.5.1), the S-scheme B is isomorphic to a finite product of vector bundles and punctured vector bundles, and consequently it follows immediately from (1.1) and (1.2) that, if $t : B \to S$ denotes the canonical morphism, the sheaves $R^i t_*(t^* M)$, with M a locally constant constructible A_S-module, are locally constant constructible and stable under base change. Now let M be a locally constant constructible A_S-module. The preceding considerations and (6.1) imply that the sheaves $R^i q_*(p^* M)$ are locally constant constructible; using the Leray spectral sequence

$$R^i r_*(R^j q_*(p^* M)) \Rightarrow R^{i+j} p_*(p^* M),$$

one deduces the same for the sheaves $R^i p_*(p^* M)$. Finally, by (6.1), the sheaves $R^i q_*(p^* M)$ are stable under base change; using the proper base-change theorem for r, it follows that the sheaves $R^i p_*(p^* M)$ are also stable under base change.

Remark 6.3. Since every reductive S-group is locally, for the étale topology, an inverse image by the canonical morphism $S \to \mathrm{Spec}(\mathbb{Z})$ of a reductive $\mathrm{Spec}(\mathbb{Z})$-scheme (SGA 3 XXIII 5.7 and XXV 1.3), (6.2) shows that the computation of the cohomology, with coefficients of torsion prime to the characteristic, of reductive group schemes over an algebraically closed field reduces to the transcendental problem of computing the cohomology of reductive group schemes of the same type over the field of complex numbers; this is known by the methods of algebraic topology, in view of the comparison theorem (SGA 4 XI 4.4).

Corollary 6.4. Let S be a scheme and let $p : G \to S$ be an S-group scheme which is an extension of a finite étale S-group scheme F by a reductive S-group scheme H. Then the conclusions of (6.2) are valid for every locally constant constructible A_S-module M.

Proof. After the base change $F \to S$, we may suppose that F is of the form Γ_S, where Γ is an ordinary finite group. Then there is an isomorphism of S-schemes

$$G \simeq \coprod_{\gamma \in \Gamma} H,$$

and the conclusion follows from (6.2).

Corollary 6.5. Let S be a scheme, G a reductive S-group scheme, and H a reductive sub-S-group scheme of G, such that the quotient sheaf of G by H for the faithfully flat quasi-compact topology is representable. Then, if

$$p : G/H \to S$$

is the canonical morphism, the conclusions of (6.2) are valid for every locally constant constructible A_S-module M.

Proof. Let $f : G \to G/H$ be the canonical morphism; it is smooth, hence of descent for étale sheaves (SGA 4 VIII 9.1). For every integer $r \geq 0$, put

$$G_r = G \times_{G/H} G \times_{G/H} \cdots \times_{G/H} G \quad (r+1 \text{ times}),$$

which, by SGA 3 V 1.5, is canonically identified with

$$G \times_S H \times_S \cdots \times_S H,$$

where H occurs r times, and let $f_r : G_r \to G/H$ and $g_r : G_r \to S$ be the canonical morphisms, so that in particular $f_0 = f$. Finally, for every integer $q \geq 0$, let $H^q(M)$ denote the complex

$$R^q(g_0)_*(f_0^* M) \to R^q(g_1)_*(f_1^* M) \rightrightarrows \cdots \to R^q(g_r)_*(f_r^* M) \cdots .$$

By (SGA), one has a spectral sequence of descent, or Čech spectral sequence,

$$E(M) : E_2^{pq} = H^p(H^q(M)) \Rightarrow R^{p+q} p_*(M).$$

Since the schemes G_r are reductive group schemes, it follows from 6.2 that the terms E_2^{pq} are locally constant constructible and compatible with every base change. It follows at once that the same is true for the E^n.

7. Complete Intersections

In this paragraph a separably closed field k is given. If ℓ is a prime number $\neq \operatorname{char}(k)$, we denote, for every proper k-scheme X and every integer $i \geq 0$, by

$$b_\ell^i(X) \quad \text{or} \quad b^i(X)$$

when no confusion is possible, the i-th Betti number of X for coefficients $\mathbb{Z}_\ell$. If X is projective over k, a hyperplane section of X means the zero locus of a section transverse to the zero section of a vector bundle $V(\check{\mathcal{L}})$, with $\mathcal{L}$ an ample invertible $\mathcal{O}_X$-module.

Theorem 7.1. (Lefschetz theorem.) Let X be a smooth, connected, projective k-scheme of dimension n, let Y be a subscheme of X which is the intersection of d hyperplane sections of X, and let L be a locally constant constructible A_X-module. Then:

(i) The inverse-image homomorphism

$$H^i(X,L) \to H^i(Y,L|Y)$$

is an isomorphism for $i \le n-d-1$ and a monomorphism for $i = n-d$.

(ii) Suppose Y is smooth over k, of dimension $n-d$. Then the Gysin homomorphism

$$H^i(Y,L|Y) \otimes \mu_Y^{\otimes d} \to H^{i+2d}(X,L)$$

is an isomorphism for $i \ge n-d+1$ and an epimorphism for $i = n-d$.

Proof. It is well known that the complement of a hyperplane section is an affine open; consequently $U = X - Y$ is a union of d affine opens. Assertion (i) then follows immediately from the exact cohomology sequence

$$\cdots \to H_!^i(U,L|U) \to H^i(X,L) \to H^i(Y,L|Y) \xrightarrow{\partial} H_!^{i+1}(U,L|U) \to \cdots$$

and from the following lemma.

Lemma 7.1.1. Let U be a k-scheme of finite type of dimension n, which is a union of d affine open subsets, and let F be an A_U-module. Then:

a) if F is constructible,
$$H_c^i(U,F) = 0 \qquad \text{for } i \ge n+d;$$

b) if F is locally constant and U is smooth over k,
$$H_c^i(U,F) = 0 \qquad \text{for } i \le n-d.$$

For $d = 1$, assertion a) is just (SGA 4 XIV 3.2). In the general case, writing $\mathfrak{U}$ for a covering of U by d affine open subsets, one deduces a) from the Čech spectral sequence

$$H^p(\mathfrak{U}, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(U,F).$$

We prove b). Since the functors $H_c^i(U,\cdot)$ commute with filtered inductive limits (SGA 4 VII 3.3), one may suppose F constructible; the assertion then follows from a), by Poincaré duality (SGA 4 XVIII).

Finally, in view of the definition of the Gysin morphism through the canonical isomorphism

$$H^{i-2d}(Y,(L|Y) \otimes \mu_Y^{\otimes -d}) \simeq H_Y^i(X,L), \tag{7.1.2}$$

assertion (ii) follows from 7.1.1 a) and from the exact cohomology sequence associated with an open subset and its complement

$$\cdots \to H^{i-1}(U,L|U) \xrightarrow{\partial} H_Y^i(X,L) \to H^i(X,L) \to H^i(U,L|U) \to \cdots.$$

Remark 7.1.3. In fact, it follows from excision that the canonical isomorphism (7.1.2) generalizes to the case where the smoothness hypothesis on X, and the hypothesis that L is constructible locally constant, are satisfied only in a neighbourhood of Y; hence 7.1 (ii) generalizes in this setting.

For substantially more general formulations, see (SGA 2 XIV 4.6), which, however, is proved only by means of resolution of singularities, and hence provisionally in characteristic 0.

Corollary 7.2. Let X be a smooth connected projective k-scheme of dimension n, let Y be a subscheme of X which is the intersection of d hyperplane sections of X, and let L be a constructible twisted constant $\mathbb{Z}_\ell$-sheaf ($\ell \neq \mathrm{car}(k)$). Then:

(i) the inverse-image homomorphism

$$\alpha_i : H^i(X, L) \to H^i(Y, L|Y)$$

is an isomorphism for $i \leq n-d-1$, and a monomorphism for $i = n-d$. If moreover L is constructible locally free, the cokernel of α_{n-d} is torsion-free;

(ii) suppose that Y is smooth over k, of dimension $n-d$. Then the Gysin homomorphism

$$\beta_i : H^i(Y, (L|Y)(-d)) \to H^{i+2d}(X, L)$$

is an isomorphism for $i \geq n-d+1$ and an epimorphism for $i = n-d$. If moreover L is constructible locally free, the epimorphism β_{n-d} is split and its kernel is torsion-free.

Proof. We prove (i), assertion (ii) following from it by Poincaré duality. In the general case, a simple passage to the limit starting from 7.1 (i) gives the conclusion. When L is a locally free $\mathbb{Z}_\ell$-sheaf, apply the following lemma to the inverse-image morphism

$$R\Gamma(X, L) \to R\Gamma(Y, L|Y)$$

in $D(\mathbb{Z}_\ell)$; hypothesis (iii) is satisfied by 7.1 (i).

Lemma 7.2.1. Let V be a discrete valuation ring, let k_V be its residue field, and let $u : K \to L$ be a morphism in $D(V)$. Suppose that K and L have cohomology of finite type, and denote by M the mapping-cylinder of u. Then for every $m \in \mathbb{Z}$ the following conditions are equivalent:

(i) $H^p(M) = 0$ for $p \leq m-1$, and $H^m(M)$ is torsion-free;

(ii) for every V-module N, the morphism

$$H^p(u \otimes \mathrm{id}_N) : H^p(K \otimes N) \to H^p(L \otimes N)$$

is an isomorphism for $p \leq m-1$ and a monomorphism for $p = m$;

(ii bis) for every V-module N, one has

$$H^p(M \otimes N) = 0 \qquad (p \leq m-1);$$

(iii) the morphism

$$H^p(u \otimes \mathrm{id}) : H^p(K \otimes k_V) \to H^p(L \otimes k_V)$$

is an isomorphism for $p \leq m-1$ and a monomorphism for $p = m$;

(iii bis) one has

$$H^p(M \otimes k_V) = 0 \qquad (p \leq m-1).$$

Moreover these equivalent conditions imply:

(i bis) the morphism

$$H^p(u) : H^p(K) \to H^p(L)$$

is an isomorphism for $p \leq m-1$, and a monomorphism with free cokernel for $p = m$.

Proof. The following diagram of implications is evident:

$$\begin{array}{ccc} \text{(ii)} & \Leftrightarrow & \text{(ii bis)} \\ \Downarrow & & \Downarrow \\ \text{(i)} & \Rightarrow & \text{(i bis)} \\ \Uparrow & & \Uparrow \\ \text{(iii)} & \Leftrightarrow & \text{(iii bis).} \end{array}$$

We shall prove separately that (i)⇒(ii bis) and that (iii bis)⇒(i), which will suffice. The implication (i)⇒(ii bis) is immediate from the Künneth exact sequence

$$0 \to H^p(M) \otimes_V N \to H^p(M \otimes N) \to \operatorname{Tor}_1^V(H^{p+1}(M), N) \to 0.$$

The same exact sequence, for $N = k_V$, shows that if (iii bis) holds, then

$$H^p(M) \otimes k_V = 0 \quad (p \leq m-1), \qquad \operatorname{Tor}_1^V(H^m(M), k_V) = 0.$$

One deduces (i) by Nakayama's lemma and the local criterion of flatness (EGA 0_{III} 10.1.3).

Under the preceding hypotheses, one sees that knowledge of the cohomology groups of X with coefficients in $\mathbb{Q}_\ell$ implies knowledge of those of Y, except in dimension $n-1$. The calculation of that remaining group is immediately reduced, under these conditions, to the calculation of the Euler-Poincaré characteristic of Y. This is the calculation to which we now turn, in a slightly more general setting.

Proposition 7.3. Let X be a proper smooth connected k-scheme of dimension n, let E be a locally free $\mathcal{O}_X$-module of constant rank r, and let Y be the zero-locus of a section of $V(\check{E})$ transverse to the zero-section. Let $p : X \to k$ be the canonical morphism and p_* the corresponding Gysin morphism. Then

$$\mathrm{EP}(Y) = p_*\big(c(\Omega^1_X)c(E)^{-1}c_r(E)\big)$$

in $H^0(k, \mathbb{Q}_\ell) = \mathbb{Q}_\ell$.

Proof. By (EGA IV 17.13.2), Y is smooth and purely of dimension $n-r$. If $i : Y \to X$ denotes the canonical immersion, the conormal sheaf of i is the inverse image by i of the conormal sheaf $\check{E}$ to the zero-section of $V(\check{E})$. Thus one has an exact sequence

$$0 \to i^*(\check{E}) \to i^*(\Omega^1_X) \to \Omega^1_Y \to 0, \tag{7.3.1}$$

whence immediately

$$c(\Omega^1_Y) = i^*\big(c(\Omega^1_X)c(E)^{-1}\big). \tag{7.3.2}$$

By (3.10) we therefore get

$$\mathrm{EP}(Y) = p_* i_* i^*\big(c(\Omega^1_X)c(E)^{-1}\big),$$

or, using the projection formula for i,

$$\mathrm{EP}(Y) = p_*\big(c(\Omega^1_X)c(E)^{-1}i_*(1_Y)\big).$$

The stated formula follows immediately from (4.7).

Corollary 7.4. Let X be a proper smooth connected k-scheme of dimension n, and let Y be a smooth closed subscheme of dimension $n-1$ of X, hence defined by a section of an invertible $\mathcal{O}_X$-module L. Let p be the canonical morphism from X to k and p_* the corresponding Gysin morphism. Then

$$\mathrm{EP}(X) - \mathrm{EP}(Y) = p_*\big(c(\Omega^1_X)c(L)^{-1}\big) \tag{7.4.1}$$

in $H^0(k, \mathbb{Q}_\ell) = \mathbb{Q}_\ell$. Moreover,

$$b^{n-1}(Y) = 2b^{n-1}(X) - b^n(X) + (-1)^n p_*\big(c(\Omega^1_X)c(L)^{-1}\big). \tag{7.4.2}$$

Proof. Formula (7.4.1) follows from 7.3 and 4.2, taking into account the evident identity

$$1 - c(L)^{-1}c_1(L) = c(L)^{-1}.$$

Using (7.2), one sees that

$$b^{n-1}(Y) = b^{n-1}(X) - b^n(X) + b^{n+1}(X) + (-1)^n(\mathrm{EP}(X) - \mathrm{EP}(Y)).$$

But $b^{n-1}(X) = b^{n+1}(X)$ by Poincaré duality, and (7.4.2) follows.

Corollary 7.5. Let X be a smooth connected subscheme of dimension n of $\mathbb{P}^{n+r}_k$, a global complete intersection of r divisors $X_1, \ldots, X_r$ of $\mathbb{P}^{n+r}_k$. For each $i \in \{1, \ldots, r\}$, let d_i be the degree of X_i. Then:

(i) the cohomology groups $H^p(X, \mathbb{Z}_\ell)$ are free $\mathbb{Z}_\ell$-modules of finite type;

(ii) if p is an integer $< 2n$ and $\neq n$, then

$$H^p(X, \mathbb{Z}_\ell) = \begin{cases} 0, & \text{if } p \text{ is odd,} \\ \mathbb{Z}_\ell, & \text{if } p \text{ is even;} \end{cases}$$

(iii) the Euler-Poincaré characteristic of X is given by

$$\mathrm{EP}(X) = d_1 \cdots d_r \sum_{\substack{0 \le i \le n \\ j_1 + \cdots + j_r = n-i}} (-1)^{n-i} \binom{n+r+1}{i} d_1^{j_1} \cdots d_r^{j_r}.$$

Proof. Assertion (i) follows immediately from (7.2), and so does (ii), in view of the cohomology of projective spaces (no. 2). We prove (iii). Let P be the ambient projective space, let $p : P \to k$ be the canonical projection, let p_* be the corresponding Gysin morphism, and let ξ be the class of $\mathcal{O}_P(1)$. For every $i \in \{1, \ldots, r\}$, the scheme X_i is defined by a section of an invertible $\mathcal{O}_P$-module L_i, so that X is the zero-locus of a section of $V(\check{L}_1 \oplus \cdots \oplus \check{L}_r)$ transverse to the zero-section. On the other hand, one has an exact sequence

$$0 \to \Omega^1_P \otimes \mathcal{O}_P(1) \to \mathcal{O}_P^{n+r+1} \to \mathcal{O}_P(1) \to 0,$$

hence

$$c(\Omega^1_P) = (1+\xi)^{n+r+1}.$$

Formula (7.3) gives

$$\begin{aligned}\mathrm{EP}(X) &= p_*(c(\Omega^1_P)c(L_1 \oplus \cdots \oplus L_r)^{-1}c_r(L_1 \oplus \cdots \oplus L_r)) \\ &= p_*(d_1 \cdots d_r\, \xi^r (1+\xi)^{n+r+1}(1+d_1\xi)^{-1} \cdots (1+d_r\xi)^{-1}).\end{aligned}$$

Using the fact that $p_*(\xi^{n+r}) = 1$ (2.2.2 c)), the formula of the statement follows without difficulty.

Remark 7.6. One may also obtain the expression for the Euler-Poincaré characteristic by using (4.11) and by transcribing Hirzebruch's calculations ([3], pp. 160–161), which use the Riemann-Roch theorem. For a more complete calculation and the determination of the Hodge cohomology of complete intersections, the reader is referred to (SGA 7 XI).

7.7

By analogy with what happens in the transcendental case, where for constant coefficients one has another well-known theorem of Lefschetz ([4]), one is led to conjecture the following statement. If X is a smooth connected projective k-scheme of dimension n, and if ξ is the class in $H^2(X, \mathbb{Q}_\ell(1))$ of a very ample invertible $\mathcal{O}_X$-module, then for every twisted constant $\mathbb{Q}_\ell$-sheaf L and every integer $p < n$, multiplication by ξ^{n-p}:

$$H^p(X, L) \to H^{2n-p}(X, L(n-p))$$

is an isomorphism. As Deligne observed, the statements (7.7) are false without an additional hypothesis on L. However, Deligne proved them in *La conjecture de Weil II* when, in addition, L is pure.

This conjectural statement has the following consequences, likewise not proved in car $\neq 0$.

Let X be a smooth connected projective k-scheme of dimension n, let Y be a smooth hyperplane section of X, and let L be a twisted constant $\mathbb{Q}_\ell$-sheaf on X.

(i) The inverse-image morphism

$$H^p(X, L) \to H^p(Y, L|Y)$$

is an epimorphism for $p \geq n$.

(ii) The Gysin morphism

$$H^p(Y, (L|Y)(-1)) \to H^{p+2}(X, L)$$

is a monomorphism for $p \leq n-2$.

Proof. Let $j : X \to \mathbb{P}^r$ be a closed immersion and let ξ_X (resp. ξ_Y) be the class in $H^2(X, \mathbb{Q}_\ell(1))$ (resp. $H^2(Y, \mathbb{Q}_\ell(1))$) of $\mathcal{O}_X(1)$ (resp. $\mathcal{O}_Y(1)$). Let $\alpha : Y \to X$ denote the canonical injection. For $q \leq n-1$, the diagram

$$\begin{array}{ccc} H^q(X, L) & \xrightarrow{\alpha^*} & H^q(Y, L|Y) \\ \cup \xi_X^{n-1-q} \downarrow & & \downarrow \cup \xi_Y^{n-1-q} \\ H^{2n-2-q}(X, L(n-1-q)) & \xrightarrow{\alpha^*} & H^{2n-2-q}(Y, (L|Y)(n-1-q)) \end{array}$$

is commutative and the right column is conjecturally an isomorphism. Put $p = 2n-2-q$. For $q \leq n-2$, hence $p \geq n$, the top row is an isomorphism by (7.2), giving (i). Assertion (ii) is proved similarly by considering, this time, the following diagram, with $p \leq n-1$:

$$\begin{array}{ccc} H^p(Y,(L|Y)(-1)) & \xrightarrow{\text{Gysin}} & H^{p+2}(X,L) \\ \cup\xi_Y^{n-1-p}\downarrow & & \downarrow\cup\xi_X^{n-1-p} \\ H^{2n-2-p}(Y,(L|Y)(n-2-p)) & \xrightarrow{\text{Gysin}} & H^{2n-p}(X,L(n-1-p)). \end{array}$$

In this diagram, the bottom row is an isomorphism for $2n-2-p \geq n$, i.e. for $p \leq n-2$ (7.2 (ii)).

For a discussion of these conjectures and their consequences, the reader is referred to Kleiman's exposé, *Algebraic cycles and the Weil conjectures*, in *Dix exposés sur la cohomologie des schémas*.

8. Blown-up varieties

In this section we give a cohomological variant of the calculations, for the group K and for the "group of cycle classes" of a blown-up scheme, given in SGA 6 VII 3.7 and 4.8. The hypotheses and notations are those of the beginning of paragraph 4 (4.1 to 4.3). Under these conditions, the "fundamental class" morphism

$$\gamma : j_*(\mu_{Y'}^{\otimes -1})[-2] \to A_{X'}$$

gives, after applying the functor Rf_*, a morphism

$$Rf_*(\gamma) : i_*Rg_*(\mu_{Y'}^{\otimes -1})[-2] \to Rf_*(A_{X'}).$$

Theorem 8.1. The morphism in $D^+(X,A)$

$$A_X \oplus i_*Rg_*(\mu_{Y'}^{\otimes -1})[-2] \to Rf_*(A_{X'}) \oplus i_*(\mu_Y^{\otimes -d})[-2d] \tag{8.1.1}$$

defined by the matrix

$$\begin{pmatrix} \text{adj} & Rf_*(\gamma) \\ 0 & i_*(\text{trace}) \end{pmatrix}$$

is an isomorphism.

Proof. By conservativity of the system of fibre functors, it is enough to verify the assertion above each point x of X. It is evident above $U = X - Y$. For $x \in Y$, proper base change reduces us to proving the following lemma, with $r = d-1$.

Lemma 8.2. Let k be a separably closed field, let $r \geq 1$ be an integer, and let $P = \mathbb{P}_k^r$. Then:

a) the canonical map $A \to H^0(P,A)$ is a bijection;

b) $H^1(P,A) = H^{2r-1}(P,A) = 0$;

c) for $0 \leq p \leq 2r-2$, multiplication by $\xi = c_1(\mathcal{O}_P(1)) \in H^2(P,\mu)$ is an isomorphism in the corresponding degree;

d) the trace morphism

$$H^{2r}(P,\mu^{\otimes -1}) \to H^0(k,\mu^{\otimes -(r+1)})$$

is a bijection.

The lemma is an immediate consequence of 2.2.2.

For the statement of the following corollary, we agree that, for every X-scheme $h : T \to X$, every closed subset Z of T, and every complex M of A_X-modules, we write

$$H^*(T, M) = H^*(T, h^*(M)), \qquad H^*_Z(T, M) = H^*_Z(T, h^*(M)).$$

Corollary 8.3. For every complex M of A_X-modules and every $p \in \mathbb{Z}$, the maps

$$H^p(X, M) \oplus H^{p-2}(Y', M \otimes \mu^{\otimes -1}) \to H^p(X', M) \oplus H^{p-2d}(Y, M \otimes \mu^{\otimes -d}) \tag{8.3.1}$$

and

$$H^p_Y(X, M) \oplus H^{p-2}(Y', M \otimes \mu^{\otimes -1}) \to H^p_{Y'}(X', M) \oplus H^{p-2d}(Y, M \otimes \mu^{\otimes -d}) \tag{8.3.2}$$

defined by the matrix

$$\begin{pmatrix} f^* & j_* \\ 0 & g_* \end{pmatrix}$$

are bijections.

Proof. Temporarily denote by u the morphism (8.1.1). By the projection formula, the arrows (8.3.1) and (8.3.2) are those induced on the p-th cohomology objects by the morphisms $R\Gamma(X, u \otimes \mathrm{id}_M)$ and $R\Gamma_Y(X, u \otimes \mathrm{id}_M)$ respectively. The assertion follows.

8.4

We now make explicit the inverse isomorphism of (8.3.2), and for this we need a few lemmas.

Lemma 8.4.1. Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of constant rank d, and let $p : P = \mathbb{P}_S(E) \to S$ be the associated projective bundle. Let F be the $\mathcal{O}_P$-module defined by the canonical exact sequence

$$0 \to F \to p^*(E) \to \mathcal{O}_P(1) \to 0.$$

Then

$$p_*(c_{d-1}(\check{F})) = 1.$$

Proof. Let c_i denote the Chern classes of E, and put $\xi = c_1(\mathcal{O}_P(1))$. From

$$p^*(c(\check{E})) = c(\check{F})(1 + \xi)$$

it follows that

$$c(\check{F}) = p^*(c(\check{E}))(1 + \xi)^{-1},$$

hence

$$c_{d-1}(\check{F}) = \sum_{0 \le i \le d-1} (-1)^{d-1-i} c_i\, \xi^{d-1-i}. \tag{8.4.2}$$

Now p_* lowers degrees by $2d - 2$, and $p_*(\xi^{d-1}) = 1$ (2.2.2). Applying p_* to both sides of (8.4.2) and using the projection formula gives the lemma.

Lemma 8.4.3. Let F denote the locally free $\mathcal{O}_{Y'}$-module defined by the canonical exact sequence

$$0 \to F \to g^*(N) \to \mathcal{O}_{Y'}(1) \to 0.$$

Let M be an A_X-module which is locally constant constructible in a neighbourhood of Y, and let $p \geq 0$ be an integer. For every $y \in H^p(Y, M)$ one has

$$f^*(i_*(y)) = j_*(g^*(y)c_{d-1}(\check{F}))$$

in $H^{p+2d}_{Y'}(X', M \otimes \mu^{\otimes d})$.

Proof. Put

$$z = f^*(i_*(y)) - j_*(g^*(y)c_{d-1}(\check{F})).$$

Then

$$j^*(z) = g^*i^*(y) - j^*j_*(g^*(y)c_{d-1}(\check{F})),$$

so, by (4.1) and the linearity lemma of (SGA 4 XVIII),

$$j^*(z) = g^*(y)\big(c_d(g^*\check{N}) - c_{d-1}(\check{F})c_1(\mathcal{O}_{Y'}(-1))\big) = 0. \tag{8.4.4}$$

Writing $U = X - Y$ and $U' = X' - Y'$, one has an evident exact commutative diagram

$$\begin{array}{ccccc} H^{p+2d-1}(U') & \xrightarrow{\partial} & H^{p+2d}_{Y'}(X') & \xrightarrow{j^*} & H^{p+2d}(Y') \\ \uparrow & & \uparrow f^* & & \uparrow g^* \\ H^{p+2d-1}(U) & \xrightarrow{\partial} & H^{p+2d}_{Y}(X) & \xrightarrow{i^*} & H^{p+2d}(Y), \end{array} \tag{8.4.5}$$

with evident simplifications of notation. Considering (8.4.5) and (8.4.4), one sees that z is of the form

$$z = f^*(t),$$

with $t \in H^{p+2d}_Y(X, M \otimes \mu^{\otimes d})$. By (4.2), $t = f_*(z)$, so it is enough to see that $f_*(z) = 0$. But

$$f_*(z) = f_*f^*i_*(y) - f_*j_*(g^*(y)c_{d-1}(\check{F})),$$

and using (4.2) gives

$$f_*(z) = i_*(y) - i_*g_*(g^*(y)c_{d-1}(\check{F})).$$

The conclusion follows from the projection formula for g and from (8.4.2).

Proposition 8.5. Let M be an A_X-module which is locally constant constructible in a neighbourhood of Y.

a) Let

$$\gamma : H^p_{Y'}(X', M \otimes \mu^{\otimes d}) \to H^{p-2}(Y', M \otimes \mu^{\otimes(d-1)})$$

be the unique map defined by the relation

$$f^* \circ f_* = j_* \circ \gamma + \mathrm{id}\,. \tag{8.5.1}$$

Then

$$g_* \circ \gamma = 0, \tag{8.5.2}$$

and the inverse matrix of (8.3.2) is

$$\begin{pmatrix} f_* & -i_* \\ -\gamma & g^*(\cdot)c_{d-1}(\check{F}) \end{pmatrix}. \tag{8.5.3}$$

b) For every integer $p \geq 0$, the sequence

$$0 \to H^{p-2d}(Y, M(-d)) \xrightarrow{\lambda} H^{p-2}(Y', M(-1)) \oplus H^p(X, M) \xrightarrow{\mu} H^p(X', M) \to 0,$$

where

$$\lambda(x) = (g^*(x)c_{d-1}(\check{F}), -i_*(x)), \qquad \mu(x, y) = j_*(x) + f^*(y),$$

is exact. Moreover, the map λ has the left inverse

$$\lambda'(x, y) = g_*(x).$$

Proof. We prove a). In view of 4.2 ($f_* f^* = \mathrm{id}$), in order to see that

$$\begin{pmatrix} f_* & -i_* \\ -\gamma & g^*(\cdot)c_{d-1}(\check{F}) \end{pmatrix} \circ \begin{pmatrix} f^* & j_* \\ 0 & g_* \end{pmatrix} = \mathrm{id},$$

it is enough to prove

$$\gamma j_* + \mathrm{id} = g^*(g_*(\cdot)c_{d-1}(\check{F})), \tag{8.5.4}$$

and

$$\gamma f^* = 0. \tag{8.5.5}$$

From (8.5.1) one deduces

$$f^* f_* j_* = j_* \gamma j_* + j_*,$$

and hence, by (4.2), $j_* \gamma f^* = 0$; this gives (8.5.5), since j_* is an isomorphism. We prove (8.5.4). Again (8.5.1) gives

$$f^* f_* j_* = j_* \gamma j_* + j_*,$$

so, since $f_* j_* = i_* g_*$ and using (8.4.3),

$$j_* g^*(g_*(\cdot)c_{d-1}(\check{F})) = j_* \gamma j_* + j_*.$$

Multiplying on the left by the inverse of j_* gives (8.5.4). Finally, (8.5.2) is obtained by writing that the product of the two matrices in the other order is the identity.

We prove b). From 8.4.3 and 8.4.1 respectively one gets $\mu \circ \lambda = 0$ and $\lambda' \circ \lambda = \mathrm{id}$. Moreover, μ is surjective by (8.3). It remains to prove that $\mathrm{Ker}(\mu) \subset \mathrm{Im}(\lambda)$. But $\mu \circ (\lambda \circ \lambda') = 0$ and $\lambda' \circ (\lambda \circ \lambda') = \mathrm{id}$, so $\mu(x) = 0$ implies

$$(\mu \oplus \lambda')(\lambda \circ \lambda'(x)) = (\mu \oplus \lambda')(x),$$

and hence $x = \lambda \circ \lambda'(x)$ by 8.3.

8.6

We shall now try to determine the multiplicative structure of the cohomology of X'. For every $p \in \mathbb{Z}$ and every A_X-module M which is locally constant constructible in a neighbourhood of Y, put

$$\nabla^p(X', M) = H^{p-2}(Y', M(-1)) \oplus H^p(X, M).$$

For $p, q \in \mathbb{Z}$ and two such modules M, N, define a composition law

$$\cup : \nabla^p(X', M) \otimes_A \nabla^q(X', N) \to \nabla^{p+q}(X', M \otimes_A N) \tag{8.6.1}$$

by

$$(y_1', x_1) \cup (y_2', x_2) = \big(-y_1'y_2'\xi + y_1'g^*i^*(x_2) + (-1)^{pq}y_2'g^*i^*(x_1),\, x_1x_2\big). \tag{8.6.2}$$

This composition law is clearly anticommutative, and an easy but tedious calculation shows that it is associative. In particular,

$$\nabla^*(X', A) = \bigoplus_{p\in\mathbb{Z}} \nabla^p(X', A)$$

is also endowed with a structure of graded-commutative A-algebra, in the sense of algebras graded by $\mathbb{Z}$, and

$$\nabla^*(X', M) = \bigoplus_{p\in\mathbb{Z}} \nabla^p(X', M)$$

is endowed with a graded module structure over the graded ring $\nabla^*(X', A)$.

Proposition 8.6.3.

a) For every pair (M, N) of A_X-modules locally constant constructible in a neighbourhood of Y, the diagram

$$\begin{array}{ccc} \nabla^*(X', M) \otimes \nabla^*(X', N) & \xrightarrow{\mu\otimes\mu} & H^*(X', M) \otimes H^*(X', N) \\ \cup\big\downarrow & & \big\downarrow\cup \\ \nabla^*(X', M\otimes N) & \xrightarrow{\quad\mu\quad} & H^*(X', M\otimes N) \end{array}$$

is commutative. In particular, the surjective map

$$\mu_A = \mu : \nabla^*(X', A) \to H^*(X', A)$$

is a ring morphism, and

$$\mu_M = \mu : \nabla^*(X', M) \to H^*(X', M)$$

is a morphism of modules over variable rings.

b) If $\nabla^*(X', A)$ is made to act on $H^*(Y, M(-d))(-2d)$ by the map

$$\nabla^p(X', A) \otimes H^{q-2d}(Y, M(-d)) \to H^{p+q-2d}(Y, M(-d)), \qquad (y', x)\otimes y \mapsto i^*(x)y,$$

then the map λ of 8.5 b) becomes a morphism of $\nabla^*(X', A)$-modules.

Proof. We prove a). With the notation of 8.6.2, one must prove the equality

$$\begin{aligned} j_*\big(-y_1'y_2'\xi + y_1'g^*i^*(x_2) + (-1)^{pq}y_2'g^*i^*(x_1)\big) + f^*(x_1x_2) \\ = \big(j_*(y_1') + f^*(x_1)\big)\big(j_*(y_2') + f^*(x_2)\big). \end{aligned}$$

It follows at once from

$$\begin{cases} f^*(x_1)f^*(x_2) = f^*(x_1x_2), & \text{multiplicativity of } f^*, \\ j_*(-y_1'y_2'\xi) = j_*(y_1')j_*(y_2'), & (4.5), \\ j_*(y'g^*i^*(x)) = j_*(y')f^*(x), & \text{projection formula.} \end{cases}$$

We prove b). The equality to be proved,

$$\lambda(i^*(x)y) = (y', x)\lambda(y),$$

breaks up into the two following equalities:

$$g^*(i^*(x)y)c_{d-1}(\check{F}) = -y' g^*(y)c_{d-1}(\check{F})\xi + y' g^* i^*(-i_*(y)) + g^*(i^*(x))g^*(y)c_{d-1}(\check{F}),$$

and

$$-i_*(i^*(x)y) = -x\, i_*(y).$$

The second is just the projection formula for i. The first follows immediately from

$$-\xi\, c_{d-1}(\check{F}) = g^*(c_d(\check{N})) \quad \text{(exact sequence of 8.4.3)}$$

and

$$i^* i_*(y) = y\, c_d(\check{N}) \qquad (4.1).$$

9. Chow ring of a blown-up variety and the self-intersection formula in the Chow ring

In this paragraph we propose to give Mumford's proof of the analogue of 4.1 in the Chow ring. We shall then show how one can deduce from it the "key formula" (8.4.3), and hence determine the structure of the Chow ring of a blown-up variety.

In accordance with the conventions of 3.9, $C^*(X)$ denotes the Chow ring of a quasi-projective smooth scheme X over a field k. The theory of Chern classes used is that of [1]. We shall use the following lemma several times.

Lemma 9.1. Let S be a quasi-projective smooth scheme over a separably closed field k, let E be a locally free $\mathcal{O}_S$-module of constant rank $r+1$, and let $p : P = \mathbb{P}_S(E) \to S$ be the associated projective bundle. Let F be the $\mathcal{O}_P$-module defined by the canonical exact sequence

$$0 \to F \to p^*(E) \to \mathcal{O}_P(1) \to 0, \tag{9.1.0}$$

and put $\xi = c_1(\mathcal{O}_P(1))$. Then:

(i) for every $y \in C^*(S)$,

$$y = p_*(p^*(y)c_r(\check{F})); \tag{9.1.1}$$

(ii) suppose that $E = N \oplus \mathcal{O}_S$, with N a locally free $\mathcal{O}_S$-module of rank r, so that P identifies with the projective completion $\widehat{N}$ of $V(N)$, and let

$$s : S \to \mathbb{P}_S(E)$$

be the section obtained by composing the zero-section of $V(N)$ with the canonical inclusion $V(N) \hookrightarrow \widehat{N}$. Then:

(a) for every $y \in C^*(S)$ one has

$$s_*(y) = p^*(y)c_r(\check{F}) = p^*(y)\left(\sum_{i=0}^{r} p^*(c_i(\check{N}))\xi^{r-i}\right) \quad \text{in } C^*(\widehat{N}), \tag{9.1.2}$$

$$s^* s_*(y) = y\, c_r(\check{N}) \quad \text{in } C^*(S); \tag{9.1.3}$$

(b) for every $z \in C^*(\widehat{N})$,
$$s^*(z) = p_*(z\, c_r(\check{F})) \quad \text{in } C^*(S). \tag{9.1.4}$$

Proof. We prove (9.1.1). By the projection formula for p, it is enough to show that $p_*(c_r(\check{F})) = 1$. For this it suffices to copy the proof of 8.4.1, using the relations ([CH] 4 (17)). The equality of the first and third members of (9.1.2) is just Lemma 3 of [1]. Paraphrasing the proof of (8.4.1), one sees that
$$c_r(\check{F}) = \sum_{i=0}^{r} p^*(c_i(\check{N}))\xi^{r-i}, \tag{9.1.2 bis}$$
which proves (9.1.2). Applying s^* to the two extreme members of (9.1.2), and using $p \circ s = \mathrm{id}$, one sees that (9.1.3) will follow from
$$s^*\left(\sum_{i=0}^{r-1} p^*(c_i(\check{N}))\xi^{r-i}\right) = 0,$$
and a fortiori from
$$s^*(\xi^i) = 0 \qquad (i > 0). \tag{9.1.5}$$
We prove (9.1.5). If $\alpha : V(N) \hookrightarrow \widehat{N}$ denotes the canonical inclusion, then evidently $\alpha^*(\mathcal{O}_P(1)) = \mathcal{O}_{V(N)}$, whence
$$\alpha^*(\xi) = \alpha^*(c_1(\mathcal{O}_P(1))) = c_1(\mathcal{O}_{V(N)}) = 0$$
by functoriality of Chern classes. The assertion follows immediately from the definition of s. We prove (9.1.4). We shall use the equality
$$\xi\, c_r(\check{F}) = c_{r+1}(p^*(\check{E}) \oplus \mathcal{O}_P) = 0, \tag{9.1.6}$$
which follows at once from the exact sequence (9.1.0). The element z has a unique decomposition
$$z = \sum_{i=0}^{r} p^*(a_i)\xi^i, \qquad a_i \in C^*(S),$$
([CH] 4 Th. 1). It follows from (9.1.5) that $s^*(z) = a_0$. On the other hand, (9.1.6) shows that $z\, c_r(\check{F}) = p^*(a_0)c_r(\check{F})$, and the conclusion follows from (9.1.1).

We now state the self-intersection formula in the Chow ring.

Theorem 9.2 (Mumford). Let k be a separably closed field and let $i : Y \hookrightarrow X$ be a closed immersion, purely of codimension d, of quasi-projective smooth k-schemes. If N denotes the normal sheaf of i, then for every $y \in C^*(Y)$ one has
$$i^* i_*(y) = y\, c_d(\check{N})$$
in $C^*(Y)$.

For the proof we shall need a number of notations and auxiliary schemes. Embed X in $Z = X \times_k \mathbb{P}^1_k$ by the morphism $u : t \mapsto (t, 0)$, so that there is a cartesian diagram
$$\begin{tikzcd} \widehat{N} \arrow[r, "j_1"] \arrow[d, "g_1"'] & Z' \arrow[d, "f_1"] \\ Y \arrow[r, "i_1"'] & Z, \end{tikzcd}$$

where Z' denotes the blowing-up of Z along Y, and $\widehat{N} = \mathbb{P}_Y(N \oplus \mathcal{O}_Y)$ is the projective completion of the vector bundle $V(N)$. As in the statement of 9.1, let $s : Y \to \widehat{N}$ be the morphism obtained by composing the zero-section of $V(N)$ with the canonical inclusion $V(N) \hookrightarrow \widehat{N}$.

The inclusion $Y \subset X$ identifies $Y \times \mathbb{P}^1$ with a closed subscheme of $X \times \mathbb{P}^1$. Let W' be the strict transform of $Y \times \mathbb{P}^1$ by f_1, i.e. the subscheme of Z' obtained by blowing up $Y \times \mathbb{P}^1$ along $Y = Y \times 0$. It is clear that $W' \cap \widehat{N}$ identifies with the zero-section of $\widehat{N}$ and that, since Y has codimension 1 in $Y \times \mathbb{P}^1$, the canonical morphism $f_2 : W' \to Y \times \mathbb{P}^1$ is an isomorphism. The remaining notation relative to W' is fixed by the canonical diagram (D_1).

Similarly, if X' denotes the closed subscheme of Z' which is the blowing-up of $X = X \times 0$ along Y, it is clear that $\mathbb{P}(N) = X' \cap \widehat{N}$ identifies with the hyperplane at infinity H of $\widehat{N}$. The remaining notation relative to X' is fixed by the canonical diagram (D_2).

We now give, in the form of a lemma, a number of useful relations. Let E be the locally free $\mathcal{O}_{\widehat{N}}$-module defined by the canonical exact sequence

$$0 \to E \to g_1^*(N \oplus \mathcal{O}_Y) \to \mathcal{O}_{\widehat{N}}(1) \to 0, \tag{9.3.0}$$

and put

$$\bar{\xi} = c_1(\mathcal{O}_{\widehat{N}}(1)). \tag{9.3.1}$$

Lemma 9.3. With the preceding notation, one has

$$j_1^*(\beta_*(1)) = c_d(\check{E}) \quad \text{in } C^*(\widehat{N}), \tag{9.3.2}$$

$$j_1^*(v_*(1)) = \bar{\xi} \quad \text{in } C^*(\widehat{N}), \tag{9.3.3}$$

$$f_1^* u_*(1) = (j_1)_*(1) + v_*(1) \quad \text{in } C^*(Z'). \tag{9.3.4}$$

Proof. All the k-schemes under consideration are smooth, so it is immediate for dimension reasons that the morphisms j_1 and β (resp. j_1 and v) are transverse. From ([CH] 4 Cor. 1 to Prop. 2) one obtains

$$\begin{cases} j_1^* \beta_*(1) = s_*(1), \\ j_1^* v_*(1) = k_*(1). \end{cases}$$

Equations (9.3.2) and (9.3.3) then come from (9.1.2) and from the normalization of Chern classes ([CH] 4–18), respectively. We prove (9.3.4). Since the cycle X' of Z' is the strict transform of $X \times 0$ by f_1, one has a priori an equality of the form

$$f_1^* u_*(1) = (j_1)_* g_1^*(\mu) + v_*(1), \tag{9.3.5}$$

with $\mu \in \Gamma(Y, \mathbb{N})$; it remains to show that $\mu = 1$. Applying j_1^* to both sides of (9.3.5), one obtains, cf. (D_2),

$$g_1^* i^* u^* u_*(1) = j_1^*(j_1)_* g_1^*(\mu) + j_1^* v_*(1). \tag{9.3.6}$$

A very special case of (9.1.3), or a direct verification, shows that $u^* u_*(1) = 0$. Using (9.3.3), it follows from (9.3.6) that

$$0 = j_1^*(j_1)_* g_1^*(\mu) + \bar{\xi}. \tag{9.3.7}$$

After restricting to a neighbourhood of a connected component of Y, one may suppose Y connected, so that (9.3.7) takes the form

$$0 = \mu\, j_1^*(j_1)_*(1) + \bar{\xi}, \qquad \mu \in \mathbb{N}. \tag{9.3.7 bis}$$

Since $j_1^*(j_1)_*(1)$ is homogeneous of degree 1, it is of the form

$$j_1^*(j_1)_*(1) = g_1^*(\rho) + \nu\bar{\xi},$$

with $\rho \in C^1(Y)$ and $\nu \in \mathbb{Z}$. Substituting this expression into (9.3.7 bis) gives at once $\rho = 0$ and $\mu\nu = -1$, with $\mu \in \mathbb{N}$; hence $\mu = 1$ and $\nu = -1$, completing the proof of (9.3.4).

The key to the proof of 9.2 lies in the proof of the following lemma, which would be evident if one knew, as in cohomology, that

$$j_1^*(j_1)_*(y) = -y\bar{\xi}$$

for every $y \in C^*(\widehat{N})$.

Lemma 9.4. For every $y \in C^*(\widehat{N})$, one has

$$c_d(\check{E})\, j_1^*(j_1)_*(y) = 0.$$

Proof. The element y has a unique decomposition in $C^*(\widehat{N})$ of the form

$$y = \sum_{0 \le p \le d} g_1^*(a_p)\bar{\xi}^p,$$

or, by (9.3.3),

$$y = \sum_{0 \le p \le d} g_1^*(a_p)\big(j_1^* v_*(1)\big)^p. \tag{9.4.1}$$

Using the projection formula ([CH] 4 I 9) for j_1 and (9.1.6), one is reduced to proving 9.4 when y is of the form $g_1^*(a)$, with $a \in C^*(Y)$. Then $j_1^*(j_1)_* g_1^*(a)$ has in turn a unique decomposition in $C^*(\widehat{N})$ of the form

$$j_1^*(j_1)_* g_1^*(a) = \sum_{0 \le p \le d} g_1^*(b_p)\bar{\xi}^p, \tag{9.4.2}$$

and, using (9.1.6) once again, it remains to prove that $g_1^*(b_0)c_d(\check{E}) = 0$. In fact, we shall prove the stronger statement $b_0 = 0$. In view of (9.3.3), we may rewrite (9.4.2) as

$$j_1^*(j_1)_* g_1^*(a) = g_1^*(b_0) + z\, j_1^* v_*(1), \tag{9.4.3}$$

with $z \in C^*(\widehat{N})$. Applying $(j_1)_*$ to both sides of (9.4.3) and using the projection formula for j_1, one gets

$$(j_1)_* g_1^*(b_0) = \big((j_1)_* g_1^*(a)\big)\big((j_1)_*(1) - v_*(1)\big) - (j_1)_*(z). \tag{9.4.4}$$

Substituting in this equality the expression for $(j_1)_*(1)$ deduced from (9.3.4) gives

$$(j_1)_* g_1^*(b_0) = (j_1)_* g_1^*(a) f_1^* u_*(1) - v_*(1)\big[(j_1)_* g_1^*(a) + (j_1)_*(z)\big]. \tag{9.4.5}$$

Since $X \times 0$ is linearly equivalent to $X \times 1$ in $X \times \mathbb{P}^1$, one has

$$(j_1)_* g_1^*(a) f_1^* u_*(1) = 0. \tag{9.4.6}$$

Comparing with (9.4.5), we get

$$(j_1)_* g_1^*(b_0) = -v_*(1)\big[(j_1)_* g_1^*(a) + (j_1)_*(z)\big]. \tag{9.4.7}$$

Moreover, it is clear that $W' \cap X' = \emptyset$: they could meet at most above Y, where one gives the zero-section and the other the hyperplane at infinity of $\widehat{N}$. Hence $\beta^* v_*(1) = 0$, and from (9.4.7)

$$\beta^*(j_1)_* g_1^*(b_0) = 0. \tag{9.4.8}$$

But, as already observed in the proof of 9.3.2, β and j_1 are transverse; thus (9.4.8) gives

$$0 = \delta_* g_1^*(b_0) = \delta_*(b_0). \tag{9.4.9}$$

Finally, applying $(f_2)_*$ to (9.4.9), one obtains $\gamma_*(b_0) = 0$, whence $b_0 = 0$, since γ is injective, Y having an evident retraction.

9.5

We finally show how the preceding results imply the theorem. Let $p : X \times \mathbb{P}^1 \to X$ be the first projection, so that $i = p \circ i_1$. If $y \in C^*(Y)$, then evidently

$$i^* i_*(y) = i_1^*(\bar{x}), \qquad \bar{x} = p^* i_*(y). \tag{9.5.1}$$

By (9.1.1) it follows that

$$i^* i_*(y) = (g_1)_* g_1^* i_1^*(\bar{x}) c_d(\check{E}),$$

hence

$$i^* i_*(y) = (g_1)_* \big[j_1^* f_1^*(\bar{x}) c_d(\check{E}) \big]. \tag{9.5.2}$$

To prove the theorem, by linearity we may suppose that y is the irreducible cycle associated with a subvariety T of Y, so that $\bar{x}$ corresponds to the cycle $T \times \mathbb{P}^1$ of $Z = X \times \mathbb{P}^1$. Let θ denote the strict transform of $T \times \mathbb{P}^1$ by f_1. Then

$$f_1^*(\bar{x}) = \theta + (j_1)_*(w), \qquad w \in C^*(\widehat{N}). \tag{9.5.3}$$

Substituting this expression into (9.5.2), 9.4 gives

$$i^* i_*(y) = (g_1)_* \big[j_1^*(\theta) c_d(\check{E}) \big], \tag{9.5.4}$$

and, by (9.1.4),

$$i^* i_*(y) = s^* j_1^*(\theta). \tag{9.5.5}$$

But the reduced scheme underlying θ is the blowing-up of $T \times \mathbb{P}^1$ along $T = (T \times \mathbb{P}^1) \cap Y$; it meets the zero-section of $\widehat{N}$ transversely along the image of T by that zero-section. Hence

$$j_1^*(\theta) = s_*(y),$$

which, together with (9.5.5) and (9.1.3), completes the proof of the theorem.

We now turn to the “key formula” mentioned in the introduction. The adapted notation, recalled for the most part, is that of 4. Observe that it is compatible with the notation introduced in the present paragraph, except that the scheme denoted Y' in 4 is here denoted H (for “hyperplane”). We shall use both notations concurrently.

Proposition 9.6. Let k be a separably closed field and let $i : Y \hookrightarrow X$ be a closed immersion, purely of codimension d, of quasi-projective smooth k-schemes. Let $f : X' \to X$ be the X-scheme obtained by blowing up X along Y, and let

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[i]{} & X \end{array}$$

be the cartesian diagram obtained from f and i. If F denotes the locally free $\mathcal{O}_{Y'}$-module defined by the canonical exact sequence

$$0 \to F \to g^*(N) \to \mathcal{O}_{Y'}(1) \to 0, \tag{9.6.1}$$

then

$$f^* i_*(y) = j_*(g^*(y) c_{d-1}(\check{F})) \tag{9.6.2}$$

in $C^*(X')$, for every $y \in C^*(Y)$.

Before proving 9.6, we isolate in a lemma some properties of the diagrams (D_1) and (D_2) which have not yet been established.

Lemma 9.7. One has

$$c_q(\check{E}) = k_*(c_{q-1}(\check{F})) + (g_1)^*(c_q(\check{N})), \tag{9.7 i}$$

$$\beta^* f_2^* \gamma_*(y) = f_1^* \gamma_* i_*(y) \qquad \text{for every } y \in C^*(Y), \tag{9.7 ii}$$

$$f_1^* u_*(t) = (j_1)_*((i \circ g_1)^*(t)) + v_* f^*(t) \qquad \text{for every } t \in C^*(X). \tag{9.7 iii}$$

Proof. We prove (i). Put $\xi = c_1(\mathcal{O}_{Y'}(1)) = k^*(\bar{\xi})$. Paraphrasing the proof of (8.4.1), one obtains

$$c_q(\check{E}) = \sum_{p=0}^{d} g_1^*(c_p(\check{N}))\xi^{d-p}, \tag{9.7.1}$$

$$c_{d-1}(\check{F}) = \sum_{q=0}^{d-1} g^*(c_q(\check{N}))\xi^{d-1-q}. \tag{9.7.2}$$

From (9.7.2) one gets

$$k_*[c_{d-1}(\check{F})] = \sum_{q=0}^{d-1} k_* k^*[g_1^*(c_q(\check{N}))\bar{\xi}^{d-1-q}],$$

and hence, by the projection formula for k and the fact that $k_*(1) = \bar{\xi}$,

$$k_*[c_{d-1}(\check{F})] = \sum_{q=0}^{d-1} g_1^*(c_q(\check{N}))\bar{\xi}^{d-q}. \tag{9.7.3}$$

Assertion (i) follows at once by comparing (9.7.1) and (9.7.3). We prove (ii). We may suppose that y is an irreducible cycle of Y, corresponding to a subvariety denoted by the same letter. Then the cycle $\gamma_*(y) = y \times 0$ is rationally equivalent to $y \times 1$, so it is enough to verify the equality of cycles

$$\beta^* f_2^*(y \times 1) = f_1^* \gamma_*(y \times 1). \tag{9.7.4}$$

Since $(y \times 1) \cap (y \times 0) = \emptyset$, f_1 is an isomorphism above an open neighbourhood of $y \times 1$, which gives (9.7.4). We prove (iii). If $p : X \times \mathbb{P}^1 \to X$ is the first projection, then (9.1.2) gives

$$u_*(t) = p^*(t)u_*(1), \tag{9.7.5}$$

and comparing with (9.3.4) yields

$$f_1^* u_*(t) = (p \circ f_1)^*(t)\big[(j_1)_*(1) + v_*(1)\big]. \tag{9.7.6}$$

The projection formulas for j_1 and v then give

$$f_1^* u_*(t) = (j_1)_*\big[(p \circ f_1 \circ j_1)^*(t)\big] + v_*\big[(p \circ f_1 \circ v)^*(t)\big], \tag{9.7.7}$$

which is (iii), since $p \circ f_1 \circ j_1 = i \circ g_1$ and $p \circ f_1 \circ v = f$.

9.8

We now prove 9.6. We shall first prove

$$v^* f^* i_*(y) = v_* j_*(g^*(y) c_{d-1}(\check{F})), \tag{9.8.1}$$

and then prove that v_* is injective. From (9.7)(iii) one gets

$$v^* f^* i_*(y) = f_1^* u_* i_*(y) - (j_1)_*[g_1^* i^* i_*(y)],$$

and therefore, by (9.2),

$$v^* f^* i_*(y) = f_1^* u_* i_*(y) - (j_1)_*[g_1^*(y\, c_d(\check{N}))]. \tag{9.8.2}$$

On the other hand,

$$v_* j_*(g^*(y) c_{d-1}(\check{F})) = (j_1)_*\big[k_* k^* g_1^*(y) c_{d-1}(\check{F})\big],$$

or, by the projection formula for k,

$$v_* j_*(g^*(y) c_{d-1}(\check{F})) = (j_1)_*\big[g_1^*(y) k_*(c_{d-1}(\check{F}))\big].$$

Using (9.7)(i), this gives

$$v_* j_*(g^*(y) c_{d-1}(\check{F})) = (j_1)_*[g_1^*(y) c_d(\check{E})] - (j_1)_*[g_1^*(y\, c_d(\check{N}))]. \tag{9.8.3}$$

Comparing (9.8.2) and (9.8.3), one sees that (9.8.1) is equivalent to

$$f_1^* u_* i_*(y) = (j_1)_*[g_1^*(y) c_d(\check{E})], \tag{9.8.4}$$

i.e. to the equality corresponding to (9.6.2) when X is replaced by $X \times \mathbb{P}^1$. By (9.1.2),

$$(j_1)_*[g_1^*(y) c_d(\check{E})] = (j_1)_* s_*(y) = \beta_* f_2^* \gamma_*(y). \tag{9.8.5}$$

Also, evidently,

$$f_1^* u_* i_*(y) = f_1^* \gamma_* i_*(y), \tag{9.8.6}$$

and (9.8.4) follows by comparing (9.8.5), (9.8.6), and (9.7)(ii).

Lemma 9.8.7. The map v_* is injective.

Proof. Let Z'' be the blowing-up of Z' along W', or equivalently along $\widehat{N} \cup W' = f_1^{-1}(Y \times \mathbb{P}^1)$. The strict transform X'' of X in Z'' is also the strict transform of X' in Z''. The notation is determined by the canonical diagram

$$\begin{array}{ccc} X'' & \xrightarrow{v''} & Z'' \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f_1'} \\ X' & \xrightarrow[v]{} & Z'. \end{array}$$

Since $X' \cap W' = \emptyset$, the morphism f_1' is an isomorphism above an open neighbourhood of X', and in particular f' is an isomorphism. Moreover, since f_1' is transverse to v, one has

$$(f_1')^* \circ v_* = v_*'' \circ (f')^*, \tag{9.8.8}$$

so it is enough to show that v_*'' is injective. But, by the commutativity property of blow-ups, Z'' may also be obtained by first blowing up $X \times \mathbb{P}^1$ along $Y \times \mathbb{P}^1$, giving a scheme Z_1', and then blowing up Z_1' along the inverse image of Y. Clearly $Z_1' = X' \times \mathbb{P}^1$, with projection $Z_1' \to X \times \mathbb{P}^1$ given by $f \times \mathrm{id}_{\mathbb{P}^1}$. The strict transform of $X \times 0$ by $f \times \mathrm{id}$ identifies with $X' \times 0$, which contains $H = H \times 0 = (f \times \mathrm{id})^{-1}(Y)$. If one now blows up Z_1' along H, since H is purely of codimension 1 in X', the strict transform of X' is canonically isomorphic to X'. We recall that we want to prove that v_* is injective. Comparing with (D_2), one sees that it is equivalent to prove that v_* is injective in the case where, in addition, Y has codimension 1 in X; this is what we now do. But then $(f_1')^* \circ v_* = v_*'' \circ (f')^*$ gives the conclusion, since f' is an isomorphism and v_* is a direct monomorphism, v admitting a retraction.

We now state the theorem corresponding to (8.5 b).

Theorem 9.9. For every integer $p \geq 0$, the sequence

$$0 \to C^{p-d}(Y) \xrightarrow{\lambda} C^{p-1}(Y') \oplus C^p(X) \xrightarrow{\mu} C^p(X') \to 0$$

in which λ and μ are the maps

$$\lambda : x \mapsto (g^*(x) c_{d-1}(\check{F}), -i_*(x)), \qquad \mu : (x, y) \mapsto j_*(x) + f^*(y)$$

is exact. Moreover, the map λ has the left inverse

$$\lambda' : (x, y) \mapsto g_*(x).$$

Proof. The last assertion is just (9.1.1). Likewise, (9.6.2) says that $\mu \circ \lambda = 0$. We prove that μ is surjective. We shall need

$$f_* f^*(z) = z \qquad (z \in C^*(X)), \tag{9.9.1}$$

which follows immediately from the projection formula for f and the evident fact that $f_*(1) = 1$. Let $t \in C^p(X')$. By (9.9.1),

$$f_*(t - f^* f_*(t)) = 0.$$

Since f induces an isomorphism from $U' = X' - Y'$ to $U = X - Y$, the restriction of $t - f^* f_*(t)$ to U' is zero, and hence ([CH] 4 th. 2) there is an equality

$$t - f^* f_*(t) = j_*(a) \qquad \text{with } a \in C^*(Y').$$

We now prove exactness in the middle. Let $a \in C^{p-1}(Y')$ and $b \in C^p(X)$ satisfy

$$j_*(a) + f^*(b) = 0. \tag{9.9.2}$$

Applying f_* to both sides gives

$$f_*j_*(a) + f_*f^*(b) = 0,$$

and in view of (9.9.1),

$$b = -i_*g_*(a). \tag{9.9.3}$$

Putting $x = g_*(a)$, it remains only to prove that

$$a = g^*(x)c_{d-1}(\check{F}). \tag{9.9.4}$$

Put

$$z = a - g^*(x)c_{d-1}(\check{F}) \qquad (x = g_*(a)),$$

and prove $z = 0$. We shall first show that $j_*(z) = 0$ and $g_*(z) = 0$, and then that an element z satisfying these two relations is necessarily zero. By definition,

$$j_*(z) = j_*(a) - j_*(g^*(x)c_{d-1}(\check{F})),$$

so, by (9.6.2),

$$j_*(z) = j_*(a) - f^*i_*(x).$$

Replacing $i_*(x)$ by $-b$ using (9.9.3), one deduces from (9.9.2) that $j_*(z) = 0$. By the definition of x,

$$g_*(z) = x - g_*(g^*(x)c_{d-1}(\check{F})) = 0$$

by the last assertion of the theorem. It remains to explain why the vanishing of z follows from $j_*(z) = 0$ and $g_*(z) = 0$. Since $Y' = \mathbb{P}_Y(N)$, z has a unique expression

$$z = g^*(a_0) + g^*(a_1)\xi + \cdots + g^*(a_{d-1})\xi^{d-1},$$

where $a_i \in C^{p-1-i}(Y)$. Since $g_*(z) = 0$, the projection formula for g, together with the fact that g_* lowers degrees by $d-1$, gives $a_{d-1} = 0$. On the other hand, by (9.2),

$$z\,\xi = j^*j_*(z) = 0,$$

hence

$$g^*(a_0)\xi + \cdots + g^*(a_{d-1})\xi^{d-1} = 0.$$

Thus all a_i are zero and $z = 0$.

9.10

With the notation of (9.9), the map

$$\omega = \lambda \circ \lambda'$$

is a projector satisfying $\mathrm{Im}(\omega) = \mathrm{Im}(\lambda)$. Therefore μ induces an isomorphism

$$\bar{\mu} : \mathrm{Ker}(\omega) \xrightarrow{\sim} C^p(X').$$

But

$$\mathrm{Ker}(\omega) = \mathrm{Ker}(g_*) \oplus C^p(X) \hookrightarrow C^{p-1}(Y') \oplus C^p(X). \tag{9.10.1}$$

Indeed, an element (x, y) of $C^{p-1}(Y') \oplus C^p(X)$ belongs to $\mathrm{Ker}(\omega)$ if and only if

$$g^*(g_*(x))c_{d-1}(\check{F}) = 0,$$

that is, in view of (9.1.1), if and only if $g_*(x) = 0$. On the other hand, the structure of the Chow ring of a projective bundle, and in particular the relations

$$g_*(\xi^i) = 0 \quad \text{for } i < d-1, \qquad g_*(\xi^{d-1}) = 1,$$

show that

$$\mathrm{Ker}(g_*) = g^*C^{p-1}(Y) \oplus g^*C^{p-2}(Y)\xi \oplus \cdots \oplus g^*C^{p-d+1}(Y)\xi^{d-2}.$$

In other words, $\bar{\mu}$ may be interpreted as an isomorphism

$$C^{p-1}(Y) \oplus C^{p-2}(Y) \oplus \cdots \oplus C^{p-d+1}(Y) \oplus C^p(X) \xrightarrow{\sim} C^p(X'), \tag{9.10.2}$$

whose inverse is easy to exhibit.

As Murre pointed out to me, the statement (9.10.2) was proved for $d = 2$ by Samuel in *Relations d'équivalence*, Proceedings of the International Congress, Edinburgh 1958.

Of course, in the framework of étale or ℓ-adic cohomology one has an analogue of (9.10.2), deduced this time from (8.5).

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Exposé VIII

GROUPS OF CLASSES OF ABELIAN AND TRIANGULATED CATEGORIES. PERFECT COMPLEXES.

by A. Grothendieck, written up by I. Bucur

In the present exposé, we quickly outline the notions and results which will be indispensable to us in the following exposés, concerning in particular pseudo-coherent and perfect complexes. A more complete exposé on the latter, with more detailed proofs, will appear in the first exposés of the seminar on the Riemann-Roch theorem for schemes, which will follow the present seminar.[11]

1. The case of abelian categories

Let $\mathcal{C}$ be an abelian category and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$. A function f from $\mathcal{D}$ to an abelian group G is said to be additive if, for every exact sequence in $\mathcal{C}$

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

such that E', E'', and E are objects of $\mathcal{D}$, one has

$$f(E) = f(E') + f(E'').$$

Let f be an additive function from $\mathcal{D}$ to an abelian group G. The following properties are easy consequences of the definition:

$$\alpha)\quad f(0) = 0, \qquad \beta)\quad f(X) = f(Y) \quad \text{if } X \text{ and } Y \text{ are isomorphic},$$

$$\gamma)\quad f(X \oplus Y) = f(X) + f(Y).$$

$\delta)$ Let $E = (E^i, d^i)$ be a finite complex in $\mathcal{D}$

such that the objects $B^{i+1} = \operatorname{Im}(d^i)$, $Z^i = \operatorname{Ker}(d^i)$ and $H^i(E) = Z^i/B^i$ are in $\mathcal{D}$. Then

$$\sum_i (-1)^i f(E^i) = \sum_i (-1)^i f(H^i(E)).$$

$\varepsilon)$ If $F \in \operatorname{ob}\mathcal{D}$ has a finite filtration

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = F,$$

and if one assumes that, for $0 < i \leq n$, F_i and F_i/F_{i-1} are objects of $\mathcal{D}$, then

$$f(F) = \sum_{i=1}^{n} f(F_i/F_{i-1}).$$

$\psi)$ If $\mathcal{D}$ is stable under infinite direct sums, then $f = 0$.

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To prove δ), one uses the exact sequences

$$0 \longrightarrow Z^i \longrightarrow E^i \longrightarrow B^{i+1} \longrightarrow 0,$$

$$0 \longrightarrow B^i \longrightarrow Z^i \longrightarrow H^i(E) \longrightarrow 0.$$

The proof of ε) is by induction on n, using the exact sequence

$$0 \longrightarrow F_{n-1} \longrightarrow F \longrightarrow F/F_{n-1} \longrightarrow 0.$$

To prove ψ), let $E \in \operatorname{ob}\mathcal{D}$ and

$$F = \bigoplus_{n \in \mathbb{Z}^+} F_n, \qquad F_n = E, \quad n \in \mathbb{Z}^+.$$

Then $E \oplus F \simeq F$, and applying γ) gives

$$f(F) = f(F) + f(E),$$

hence $f(E) = 0$.

If, for every abelian group G, one considers the abelian group $A_{\mathcal{D}}(G)$ of additive functions from $\mathcal{D}$ to G, one obtains in an evident way a covariant functor

$$A_{\mathcal{D}} : \mathrm{Ab} \longrightarrow \mathrm{Ab}\,.$$

The functor $A_{\mathcal{D}}$ is representable. A representing pair $(K_{\mathcal{D}}(\mathcal{C}), \varphi)$ may be obtained as follows. Let L be the free abelian group generated by the objects of $\mathcal{D}$, and let R be the subgroup of L generated by the elements of the form $E - E' - E''$ for each exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

of $\mathcal{C}$. Put

$$K_{\mathcal{D}}(\mathcal{C}) = L/R.$$

The map φ is obtained by composing the two canonical maps

$$\operatorname{ob}\mathcal{D} \longrightarrow L \longrightarrow L/R.$$

When there is no danger of confusion, one may abbreviate the notation:

$$K_{\mathcal{D}}(\mathcal{C}) = K(\mathcal{D}).$$

2. The case of triangulated categories

Let $\mathcal{C}$ be a triangulated category and let G be an abelian group. A function f from $\mathcal{C}$ to G is said to be additive if, for every distinguished triangle

$$\begin{tikzcd} & Z \arrow[dl, "w"'] & \\ X \arrow[rr, "u"'] & & Y \arrow[ul, "v"'] \end{tikzcd}$$

one has

$$f(Y) = f(X) + f(Z).$$

Given an additive function f, the following relations are immediate consequences of the definition:

$$\text{a) } f(0) = 0, \qquad \text{b) } f(X[n]) = (-1)^n f(X),$$

$$\text{c) } f(X) = f(Y) \text{ if } X \text{ and } Y \text{ are isomorphic}, \qquad \text{d) } f(X \oplus Y) = f(X) + f(Y).$$

Indeed the following triangles are distinguished:

$$\begin{tikzcd} & 0 \arrow[dl] & \\ X \arrow[rr, "1_X"'] & & X \arrow[ul] \end{tikzcd} \qquad \begin{tikzcd} & X[1] \arrow[dl] & \\ X \arrow[rr] & & 0 \arrow[ul] \end{tikzcd}$$

$$\begin{tikzcd} & 0 \arrow[dl] & \\ X \arrow[rr, "u"'] & & Y \arrow[ul] \end{tikzcd} \qquad \begin{tikzcd} & Y[1] \arrow[dl] & \\ X \oplus Y \arrow[rr] & & X \arrow[ul] \end{tikzcd}$$

(u an isomorphism). To form the last triangle, we used the fact that the sum of two distinguished triangles is distinguished.

As in the case of abelian categories, one obtains in an evident way a functor

$$A : \mathrm{Ab} \longrightarrow \mathrm{Ab}$$

which associates to each abelian group G the abelian group $A(G)$ of additive functions on $\mathcal{C}$ with values in G. This functor is representable, and it is easy to construct a representing pair $(K(\mathcal{C}), \mathrm{cl}_{\mathcal{C}})$. Namely, one first considers the free abelian group $\mathbb{Z}^{(\mathrm{ob}\,\mathcal{C})}$ generated by the objects of $\mathcal{C}$, and then the subgroup R generated by the elements of the form $X - Y - Z$, where

$$\begin{tikzcd} & Z \arrow[dl] & \\ Y \arrow[rr] & & X \arrow[ul] \end{tikzcd}$$

is a distinguished triangle. The quotient group $\mathbb{Z}^{(\mathrm{ob}\,\mathcal{C})}/R$ is $K(\mathcal{C})$, and $\mathrm{cl}_{\mathcal{C}}$ is obtained by composing the two canonical maps

$$\mathrm{ob}\,\mathcal{C} \longrightarrow \mathbb{Z}^{(\mathrm{ob}\,\mathcal{C})} \longrightarrow K(\mathcal{C}).$$

3. Functorial character

Let $\mathcal{C}$ and $\mathcal{C}'$ be two triangulated categories, and let $T : \mathcal{C} \to \mathcal{C}'$ be an exact functor from $\mathcal{C}$ to $\mathcal{C}'$, that is, an additive graded functor carrying distinguished triangles to distinguished triangles. Denote, as above, by

$$A, A' : \mathrm{Ab} \longrightarrow \mathrm{Ab}$$

the functors associated with the categories $\mathcal{C}$ and $\mathcal{C}'$. There results a functorial morphism $A' \to A$ given by composition with T, hence a homomorphism

$$K(T) : K(\mathcal{C}) \longrightarrow K(\mathcal{C}')$$

uniquely determined by the commutativity of the diagram

$$\begin{tikzcd} \mathrm{ob}\,\mathcal{C} \arrow[r, "T"] \arrow[d, "\mathrm{cl}_{\mathcal{C}}"'] & \mathrm{ob}\,\mathcal{C}' \arrow[d, "\mathrm{cl}_{\mathcal{C}'}"] \\ K(\mathcal{C}) \arrow[r, "K(T)"'] & K(\mathcal{C}'). \end{tikzcd}$$

It follows that, if the functors $T, T_1 : \mathcal{C} \to \mathcal{C}'$ are isomorphic, then $K(T) = K(T_1)$. In particular, if T is an equivalence of categories, then $K(T)$ is an isomorphism.

If $\mathcal{C}''$ is a third triangulated category and

$$F : \mathcal{C} \times \mathcal{C}' \longrightarrow \mathcal{C}''$$

is a bifunctor exact in each variable, then it is clear that there results a bilinear map

$$K(F) : K(\mathcal{C}) \times K(\mathcal{C}') \longrightarrow K(\mathcal{C}'')$$

such that the following relation holds:

$$K(F)(\mathrm{cl}_{\mathcal{C}}(X), \mathrm{cl}_{\mathcal{C}'}(X')) = \mathrm{cl}_{\mathcal{C}''}(F(X, X')).$$

Proposition 3.1. Let $\mathcal{A}$ be a triangulated category, $\mathcal{B}$ a thick triangulated subcategory of $\mathcal{A}$, $\mathcal{A}/\mathcal{B}$ the quotient category of $\mathcal{A}$ by $\mathcal{B}$, $Q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$ the canonical quotient functor, and $i : \mathcal{B} \to \mathcal{A}$ the inclusion functor. Then the following sequence is exact:

$$K(\mathcal{B}) \xrightarrow{K(i)} K(\mathcal{A}) \xrightarrow{K(Q)} K(\mathcal{A}/\mathcal{B}) \longrightarrow 0.$$

Proof. It is clear that there is a natural homomorphism

$$\mathrm{Coker}(K(i)) \xrightarrow{\alpha} K(\mathcal{A}/\mathcal{B}).$$

On the other hand, there is an evident map

$$\mathrm{ob}(\mathcal{A}/\mathcal{B}) \xrightarrow{\gamma} \mathrm{Coker}(K(i)),$$

since $\mathrm{ob}(\mathcal{A}/\mathcal{B}) = \mathrm{ob}(\mathcal{A})$. Let $X, Y \in \mathrm{ob}(\mathcal{A}/\mathcal{B})$ and suppose that they are isomorphic. We want to show that $\gamma(X) = \gamma(Y)$. The category $\mathcal{A}/\mathcal{B}$ is obtained by a calculus of fractions on the right or on the left, the multiplicative system S being formed by the morphisms f which are contained in a distinguished triangle (X', Y', Z', f, g, h) in which Z' is an object of $\mathcal{B}$. (S is a saturated multiplicative system.) It follows that there exist X_1 and morphisms

$$X_1 \xrightarrow{s} X, \qquad X_1 \xrightarrow{t} Y$$

such that $s, t \in S$. It is therefore enough to prove that $\gamma(X) = \gamma(X_1)$, and this follows from the fact that there is a distinguished triangle of the form

$$(X_1, X, Z, s, \ldots),$$

with $Z \in \mathrm{ob}\,\mathcal{B}$.

The map γ is an additive function. Indeed, if (X, Y, Z, f, g, h) is a distinguished triangle of $\mathcal{A}/\mathcal{B}$, one must prove that

$$\gamma(X) + \gamma(Z) = \gamma(Y).$$

But the relation now follows because the triangle (X, Y, Z, f, g, h) is isomorphic to a distinguished triangle of $\mathcal{A}$. Consequently one obtains

$$\beta : K(\mathcal{A}/\mathcal{B}) \longrightarrow \mathrm{Coker}(K(i)),$$

and it is immediate to check that β is the inverse of α.

4. Comparison with the case of abelian categories

Let $\mathcal{C}$ be an abelian category and $D(\mathcal{C})$ the derived category of $\mathcal{C}$. As usual, denote by $D^b(\mathcal{C})$ the full subcategory of $D(\mathcal{C})$ generated by the bounded complexes. By associating with every object X of $\mathcal{C}$ the complex whose components are all zero except the component in dimension zero, which is equal to X, one obtains a homomorphism

$$K(\mathcal{C}) \xrightarrow{\mu} K(D^b(\mathcal{C})).$$

This homomorphism is in fact an isomorphism. Indeed, we may use Proposition 3.1 to construct an inverse to μ, taking into account that

$$D^b(\mathcal{C}) \simeq K^{b,b}(\mathcal{C})/K^{b,\phi}(\mathcal{C}),$$

where $K^{b,\phi}(\mathcal{C})$ is the thick subcategory of $K^{b,b}(\mathcal{C})$ consisting of the acyclic complexes. Let $X^\bullet = (X_i) \in \mathrm{ob}(K^{b,b}(\mathcal{C}))$. If one sets

$$f(X^\bullet) = \sum_i (-1)^i \operatorname{cl}_{\mathcal{C}}(X_i),$$

one obtains an additive function from $K^{b,b}(\mathcal{C})$ with values in $K(\mathcal{C})$. Indeed, one first observes that, if $X^\bullet$ and $Y^\bullet$ are homotopic, then $f(X^\bullet) = f(Y^\bullet)$, since $f(X^\bullet) = f(H(X^\bullet))$. It follows that there is a homomorphism

$$K(D^b(\mathcal{C})) \xrightarrow{\nu} K(\mathcal{C}),$$

and it is immediate to check that it is the inverse of μ.

Since the invariant $K(\mathcal{C})$ is often of no interest when $\mathcal{C}$ is too large, it is useful to relativize it with respect to a subcategory $\mathcal{C}_o$ of $\mathcal{C}$. Thus let $\mathcal{C}_o$ be a full subcategory of $\mathcal{C}$. We may define, on the one hand, the group

$$K_o(\mathcal{C}) = K_{\mathcal{C}_o}(\mathcal{C}).$$

Suppose, on the other hand, that the following condition is satisfied:

$$(A) \qquad \mathcal{C}_o \text{ is stable under finite direct sums.}$$

It is then easy to define the triangulated category $D^b_o(\mathcal{C})$ by using complexes in $\mathcal{C}$ with bounded cohomology and with components in the subcategory $\mathcal{C}_o$, and passing to the quotient by quasi-isomorphisms of such complexes. As above, there results a homomorphism

$$(*) \qquad K_o(\mathcal{C}) \longrightarrow K(D^b_o(\mathcal{C})).$$

Suppose that the subcategory $\mathcal{C}_o$ also satisfies the following condition (B):

$$(B) \qquad \text{If } X, Y \in \mathrm{ob}\,\mathcal{C}_o \text{ and } X \xrightarrow{u} Y \text{ is an epimorphism in } \mathcal{C},$$

then $\mathrm{Ker}\, u \in \mathrm{ob}\,\mathcal{C}_o$. Reasoning as above, one can show that the homomorphism $(*)$ is an isomorphism.

The canonical inclusion functor

$$D^b_o(\mathcal{C}) \longrightarrow D^b(\mathcal{C})$$

is not fully faithful in general. However:

Proposition 4.1. Suppose that the subcategory $\mathcal{C}_o$ satisfies conditions (A), (B), and the following condition (C):

$$(C) \qquad \text{Every diagram} \qquad \begin{tikzcd} A \arrow[r, "u"] & B \\ & L \arrow[u, "v"'] \end{tikzcd}$$

where u is an epimorphism and $L \in \operatorname{ob}\mathcal{C}_o$, can be completed to a commutative diagram

$$\begin{tikzcd} A \arrow[r, "u"] & B \\ L_1 \arrow[u] \arrow[r, "w"'] & L \arrow[u, "v"'] \end{tikzcd}$$

such that $L_1 \in \operatorname{ob}\mathcal{C}_o$ and w is an epimorphism. Under these conditions, the canonical functors

$$D^b_o(\mathcal{C}) \longrightarrow D^b(\mathcal{C}), \qquad D^-_o(\mathcal{C}) \longrightarrow D^-(\mathcal{C})$$

are fully faithful. Moreover, the essential image of $D^b_o(\mathcal{C})$ (resp. $D^-_o(\mathcal{C})$) is stable under direct factors.

The proof of this proposition essentially uses a criterion of Verdier ([2], 4–2 Theorem) and will be given in the first exposé of the seminar on Riemann-Roch (SGA 6 I 2.7).

Corollary. Suppose, in addition, that the subcategory $\mathcal{C}_o$ is stable under Ker and Coker, hence is an abelian category, and denote by $D^b(\mathcal{C})_o$ the full triangulated subcategory of $D^b(\mathcal{C})$ generated by the complexes of $D^b(\mathcal{C})$ whose cohomology objects are isomorphic to objects of $\mathcal{C}_o$. Then the inclusion functor

$$D^b(\mathcal{C}_o) \longrightarrow D^b(\mathcal{C})_o$$

is an equivalence between the two categories, and consequently one obtains the sequence of isomorphisms

$$K(\mathcal{C}_o) \simeq K_o(\mathcal{C}) \simeq K(D^b(\mathcal{C}_o)) \simeq K(D^b(\mathcal{C})_o). \tag{*}$$

Observation. Suppose that the subcategory $\mathcal{C}_o$ satisfies conditions (A) and (B), and that every object of $\mathcal{C}$ admits a finite resolution on the left by elements of $\mathcal{C}_o$. In this case, $\mathcal{C}_o$ also satisfies condition (C), and the canonical functor

$$D^b_o(\mathcal{C}) \longrightarrow D^b(\mathcal{C})$$

is in fact an equivalence of categories. Consequently, one has the isomorphism

$$K_{\mathcal{C}_o}(\mathcal{C}) \simeq K(\mathcal{C}).$$

[12]

[12] cf. SGA 6 IV 1.6.

5. Pseudo-coherent complexes

Let $\mathcal{C}$ be an abelian category and $\mathcal{C}_o$ a full subcategory of $\mathcal{C}$ satisfying conditions (A), (B), (C). We shall denote by $D_o^-(\mathcal{C})_{\mathrm{coh}}$ the essential image of $D_o^-(\mathcal{C})$ under the canonical functor

$$D_o^-(\mathcal{C}) \longrightarrow D(\mathcal{C}).$$

The objects of $D_o^-(\mathcal{C})_{\mathrm{coh}}$ will be called $\mathcal{C}_o$-pseudo-coherent complexes. Thus a complex of $\mathcal{C}$ is $\mathcal{C}_o$-pseudo-coherent if it is isomorphic in $D(\mathcal{C})$ to a bounded-above complex whose components are objects of $\mathcal{C}_o$.

The category $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is a triangulated subcategory of $D(\mathcal{C})$, in the sense that every distinguished triangle of $D(\mathcal{C})$ with two vertices in the subcategory $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is isomorphic to a triangle whose three objects are in the subcategory $D_o^-(\mathcal{C})_{\mathrm{coh}}$.

First, it is clear that the subcategory $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is stable under the translation automorphism $T : D(\mathcal{C}) \to D(\mathcal{C})$. Therefore, to prove that $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is a triangulated subcategory of $D(\mathcal{C})$, it is enough to check that if $U^\bullet$, $V^\bullet$ are objects of $D_o^-(\mathcal{C})_{\mathrm{coh}}$ and if the triangle

$$\begin{tikzcd} & W^\bullet \arrow[dl, "w"'] & \\ U^\bullet \arrow[rr, "u"'] & & V^\bullet \arrow[ul, "v"'] \end{tikzcd}$$

is distinguished for the triangulated category structure of $D(\mathcal{C})$, then $W^\bullet$ is also an object of $D_o^-(\mathcal{C})_{\mathrm{coh}}$. By hypothesis, there exist complexes $X^\bullet$, $Y^\bullet$, isomorphic in $D(\mathcal{C})$ to $U^\bullet$ and $V^\bullet$ respectively, whose components are objects of $\mathcal{C}_o$. We obtain a commutative square in $D(\mathcal{C})$:

$$\begin{tikzcd} U^\bullet \arrow[r, "u"] \arrow[d, "\sim"'] & V^\bullet \arrow[d, "\sim"] \\ X^\bullet \arrow[r, "f"'] & Y^\bullet. \end{tikzcd}$$

But, after replacing $X^\bullet$ by another isomorphic complex, we may suppose, using the fact that the inclusion functor

$$D_o^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C})$$

is fully faithful (Prop. 4.1), that f is an actual morphism of complexes. Let $Z^\bullet$ be the mapping cylinder of f, and let $g : Y^\bullet \to Z^\bullet$ be the canonical morphism. By the construction itself, the components of $Z^\bullet$ are objects of $\mathcal{C}_o$ (condition (A)). On the other hand,

$$\begin{tikzcd} & Z^\bullet \arrow[dl, "h"'] & \\ X^\bullet \arrow[rr, "f"'] & & Y^\bullet \arrow[ul, "g"'] \end{tikzcd}$$

being a distinguished triangle in $D(\mathcal{C})$, it follows that $Z^\bullet$ is isomorphic to $W^\bullet$.

The category $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is therefore endowed with a structure of triangulated category: a triangle in $D_o^-(\mathcal{C})_{\mathrm{coh}}$ is distinguished if and only if it is distinguished in the triangulated category $D(\mathcal{C})$.

The intersection of $D_o^-(\mathcal{C})_{\mathrm{coh}}$ with $D^b(\mathcal{C})_{\mathrm{coh}}$ will be denoted by $D_o^b(\mathcal{C})_{\mathrm{coh}}$.

An object M of $\mathcal{C}$ will be called pseudo-coherent if it is so as a complex reduced to degree zero; one proves that it is necessary and sufficient for this that it admit a resolution on the left whose components are objects of $\mathcal{C}_o$.

Proposition 5.1. A complex X is $\mathcal{C}_o$-pseudo-coherent if and only if it is isomorphic to a complex which is bounded on the right and whose components are $\mathcal{C}_o$-pseudo-coherent.

The proof of this proposition will appear in the first exposé of the seminar on Riemann-Roch (SGA 6 I 2.7).

6. Perfect complexes

An object of the category $D(\mathcal{C})$ will be called a $\mathcal{C}_o$-perfect complex if it is isomorphic to an object of $D^b_o(\mathcal{C})$, that is, to a bounded complex all of whose components are in $\mathcal{C}_o$. We shall denote by $D_o(\mathcal{C})_{\text{parf}}$ the full subcategory of $D(\mathcal{C})$ generated by the perfect complexes.

Thus the category $D_o(\mathcal{C})_{\text{parf}}$ is nothing other than the essential image of $D^b_o(\mathcal{C})$ under the inclusion functor

$$D^b_o(\mathcal{C}) \longrightarrow D(\mathcal{C}).$$

It is easy to see that $D_o(\mathcal{C})_{\text{parf}}$ is a triangulated subcategory of $D(\mathcal{C})$, and hence is endowed with a structure of triangulated category.

7. An important special case

Let A be a ring and let $\mathcal{C}$ be the category of A-modules, on the left for example. Denote by $\mathcal{C}_o$ the full subcategory of $\mathcal{C}$ generated by the projective A-modules of finite type. The subcategory $\mathcal{C}_o$ satisfies conditions (A), (B), (C). In this case we use the following notation:

$$D^-(A)_{\text{coh}} = D^-_o(\mathcal{C})_{\text{coh}}, \qquad D^b(A)_{\text{coh}} = D^b_o(\mathcal{C})_{\text{coh}},$$

$$D(A)_{\text{parf}} = D_o(\mathcal{C})_{\text{parf}}.$$

The complexes of $D(\mathcal{C})$ which are $\mathcal{C}_o$-pseudo-coherent (resp. $\mathcal{C}_o$-perfect) will simply be called pseudo-coherent (resp. perfect) complexes.

It will be useful to consider the following groups:

$$K_\bullet(A) = K(D^b(A)_{\text{coh}}),$$

$$K'(A) = K(D(A)_{\text{parf}}).$$

As for the group $K'(A)$, we have, by Proposition 4.1 and its corollary,

$$K'(A) = K(D(A)_{\text{parf}}) \simeq K(D^b_o(\mathcal{C})) \simeq K_{\mathcal{C}_o}(\mathcal{C}),$$

which shows that it is also the classical group of classes of projective modules of finite type over A.

In the special case where A is noetherian, the group $K_\bullet(A)$ also admits a classical description. For this, let $\mathcal{C}_1$ be the abelian category of modules of finite type. Then

$$K_\bullet(A) \simeq K(\mathcal{C}_1).$$

Indeed, using Proposition 4.1, one has

$$K(\mathcal{C}_1) \simeq K(D^b(\mathcal{C}_1)) \simeq K(D^b_{\mathcal{C}_1}(\mathcal{C})),$$

because $D^b(\mathcal{C}_1) \simeq D^b_{\mathcal{C}_1}(\mathcal{C})$ immediately.

In fact, if A is noetherian and $L^\bullet \in D^-(A)$, then it follows from Proposition 5.1 that $L^\bullet$ is pseudo-coherent if and only if the $H^i(L^\bullet)$ are of finite type.

For an arbitrary ring A, one has a homomorphism

$$K'(A) \xrightarrow{\gamma_A} K_\bullet(A),$$

which comes from the inclusion functor

$$(*) \qquad D(A)_{\mathrm{parf}} \longrightarrow D^b(A)_{\mathrm{coh}}.$$

In general, the homomorphism γ_A is neither injective nor surjective. In the special case of a regular ring A, every A-module M of finite type has finite projective dimension and it follows, thanks to 5.1, that the inclusion functor $(*)$ is in fact an equivalence between the two categories. In that case, γ_A is therefore an isomorphism.

8. Tor of complexes

Let A be a not necessarily commutative ring, and let ${}_A\mathcal{C}$ (resp. $\mathcal{C}_A$) be the abelian category of left (resp. right) A-modules. By $K^-({}_A\mathcal{C})$ we shall denote the triangulated category of complexes of ${}_A\mathcal{C}$ bounded above, and by ${}_A\mathcal{D}$ the full triangulated subcategory of $K^-({}_A\mathcal{C})$ consisting of complexes whose components are flat left A-modules.

Let $X^\bullet$ be an object of $K^-({}_A\mathcal{C})$, and consider the functor

$$\Phi_{X^\bullet} = X^\bullet \otimes_A \cdot \;:\; K^-(\mathcal{C}_A) \longrightarrow K^-(\mathrm{Ab}).$$

Under these conditions, the following propositions hold.

1) Every acyclic object of ${}_A\mathcal{D}$ is transformed by $\Phi_{X^\bullet}$ into an acyclic object of $K^-(\mathrm{Ab})$. Indeed, if $N^\bullet$ is an object of ${}_A\mathcal{D}$ such that $Z^i(N^\bullet)$, $B^i(N^\bullet)$, and $H^i(N^\bullet)$ are flat, one has the classical isomorphism

$$\bigoplus_{p+q=n} H_p(X^\bullet) \otimes H_q(N^\bullet) \longrightarrow H_n(X^\bullet \otimes_A N^\bullet).$$

2) Every object $Y^\bullet$ of $K^-(\mathcal{C}_A)$ is the target of a quasi-isomorphism whose source is an object of ${}_A\mathcal{D}$. It is enough for this to take a Cartan-Eilenberg resolution. We can therefore apply Theorem 2 of §2, no. 2 of [2] to conclude that $\Phi_{X^\bullet}$ admits a left derived functor:

$$\mathbf{L}\Phi_{X^\bullet} : D^-(\mathcal{C}_A) \longrightarrow D^-(\mathrm{Ab}).$$

The object $\mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$ may be obtained, up to isomorphism, as follows. Let $P^\bullet$ be an object of ${}_A\mathcal{D}$ and let $P^\bullet \to Y^\bullet$ be a quasi-isomorphism. Then $\mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$ is isomorphic to

$$Q(X^\bullet \otimes_A P^\bullet),$$

where, as usual, Q denotes the canonical functor

$$K^-(\mathrm{Ab}) \longrightarrow D^-(\mathrm{Ab}).$$

It follows that for each object $Y^\bullet$ of $D^-(\mathcal{C}_A)$, the functor

$$K^-({}_A\mathcal{C}) \longrightarrow D^-(\mathrm{Ab}), \qquad X^\bullet \longmapsto \mathbf{L}\Phi_{X^\bullet}(Y^\bullet)$$

is exact and vanishes on acyclic complexes, hence gives an exact bifunctor

$$D^-({}_A\mathcal{C}) \times D^-(\mathcal{C}_A) \longrightarrow D^-(\mathrm{Ab}),$$

$$(X^\bullet, Y^\bullet) \longmapsto X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet.$$

Passing to homology, one defines the $\mathrm{Tor}_i^A(X^\bullet, Y^\bullet)$ (cf. also [1], 6.3). In the particular case of a commutative ring A, one obtains in fact an exact bifunctor

$$D^-(\mathcal{C}) \times D^-(\mathcal{C}) \longrightarrow D^-(\mathcal{C}),$$

the tensor product in the sense of derived categories ($\mathcal{C} = {}_A\mathcal{C} = \mathcal{C}_A$).

Definition. We shall say that the complex X of $D(\mathcal{C}_A)$ has finite Tor-dimension if it lies in D^- and if there exists $n_o \in \mathbb{Z}^+$ such that

$$\mathrm{Tor}_n^A(X, Y) = 0$$

for every $Y \in \mathrm{ob}({}_A\mathcal{C})$ and every $n > n_o$.

We shall use the notion of Tor-dimension to obtain a characterization of perfect complexes.

Proposition 8.1. Let A be a not necessarily commutative ring and let $L^\bullet$ be a complex of A-modules. The following conditions are equivalent:

a) $L^\bullet$ is perfect;

b) $L^\bullet$ is pseudo-coherent, has bounded cohomology, and has finite Tor-dimension.[13]

For the proof, one uses the following lemma.

Lemma. (Bourbaki, Alg. Comm. I, §2, ex. 15). Every A-module E which is flat and of finite presentation is projective.

Proof of the lemma. Let $u : F \to F'$ be an epimorphism. We must prove that

$$\mathrm{Hom}_A(E, F) \longrightarrow \mathrm{Hom}_A(E, F')$$

is an epimorphism. It is enough to prove that, for every divisible group G,

$$\mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E, F'), G) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E, F), G)$$

[13] cf. SGA 6 I 5.8.1.

is a monomorphism. But the diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E,F'),G) \arrow[r] \arrow[d, "\sim"'] & \mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_A(E,F),G) \arrow[d, "\sim"] \\
\mathrm{Hom}_{\mathbb{Z}}(F',G)\otimes_A E \arrow[r] & \mathrm{Hom}_{\mathbb{Z}}(F,G)\otimes_A E
\end{tikzcd}$$

is commutative, and the vertical arrows are isomorphisms, by the hypothesis on E.

Proof of Proposition 8.1. The implication a) ⇒ b) is immediate. To prove b) ⇒ a), let

$$\cdots \longrightarrow P_n \xrightarrow{d_n} \cdots \longrightarrow P_i \longrightarrow 0 \longrightarrow \cdots$$

be a complex isomorphic to $L^\bullet$ in $D(\mathcal{C})$, where the P_j are projective A-modules of finite type. Since $L^\bullet$ has finite Tor-dimension and bounded cohomology, it follows that there exists $n_o \in \mathbb{Z}$ such that:

I) The sequence

$$\cdots \longrightarrow P_{n_o-p} \xrightarrow{d_{n_o-p}} \cdots \longrightarrow P_{n_o} \longrightarrow \mathrm{Ker}\, d_{n_o+1} \longrightarrow 0$$

is exact.

II) $\mathrm{Ker}\, d_{n_o+1}$ is flat.

III) The complex

$$0 \longrightarrow \mathrm{Ker}\, d_{n_o+1} \longrightarrow P_{n_o+1} \longrightarrow \cdots \longrightarrow P_i \longrightarrow 0 \longrightarrow \cdots$$

is isomorphic to $L^\bullet$ in $D(\mathcal{C})$.

But $\mathrm{Ker}\, d_{n_o+1}$ is of finite presentation, hence projective of finite type by the lemma, which proves 8.1.

If A is a commutative ring, and if $X^\bullet$, $Y^\bullet$ are two perfect complexes, then their tensor product

$$X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet$$

in the derived category is again a perfect complex. We obtain a bifunctor

$$D(A)_{\mathrm{parf}} \times D(A)_{\mathrm{parf}} \longrightarrow D(A)_{\mathrm{parf}}$$

which is exact in each variable. Consequently, one defines a product on $K'(A)$, and it is easy to see that $K'(A)$ is thereby endowed with the structure of a commutative ring with unit. In fact, $K'(A)$ can be endowed with the structure of a λ-ring; for details, see the seminar on Riemann-Roch following this one.[14]

The tensor product

$$X^\bullet \overset{\mathbf{L}}{\otimes}_A Y^\bullet$$

of a perfect complex $X^\bullet$ and a bounded pseudo-coherent complex $Y^\bullet$ is a bounded pseudo-coherent complex (A still a commutative ring). Consequently, one can introduce on $K_\bullet(A)$ the structure of a $K'(A)$-module.

[14] SGA 6 V 2.

9. Functorial properties

a) Let A and B be two not necessarily commutative rings, and let $u : A \to B$ be a homomorphism. Every left (resp. right) B-module M can be endowed with the structure of a left (resp. right) A-module, denoted $M_{[u]}$. In particular, $B_{[u]}$ is a right A-module.

The functor

$$E \longmapsto B_{[u]} \otimes_A E$$

allows one to define the exact functor

$$D(A)_{\text{parf}} \longrightarrow D(B)_{\text{parf}}$$

and consequently the homomorphism

$$K'(A) \xrightarrow{u^*} K'(B).$$

Thus one obtains a covariant functor from the category of rings (resp. commutative rings) to the category of abelian groups (resp. commutative rings).

b) Suppose now that the left A-module $B_{[u]}$ is pseudo-coherent. It follows that the functor

$$F \longmapsto F_{[u]}$$

transforms a free B-module of finite type into a pseudo-coherent A-module. By using Proposition 5.1, one obtains an exact functor

$$D^b(B)_{\text{coh}} \longrightarrow D^b(A)_{\text{coh}}$$

and consequently a canonical homomorphism

$$K_\bullet(B) \xrightarrow{u_*} K_\bullet(A).$$

The homomorphisms u_* and u^* satisfy the following identity, the projection formula:

$$u_*(u^*(\alpha) \cdot \beta) = \alpha\,(u_*(\beta)) \quad \text{for } \alpha \in K'(A),\ \beta \in K_\bullet(B).$$

This easy formula will be proved in a more general form in the exposés of the seminar on Riemann-Roch (SGA 6 IV 2.11.1.2).

10. Relative construction[15]

Let $u : A \to B$ be a homomorphism of rings, and let

$$D(B) \longrightarrow D(A)$$

be the associated exact functor. In $D(B)$, we can consider the complexes that are pseudo-coherent relative to A, those of finite Tor-dimension relative to A, and also the complexes perfect relative to A. In particular, it is natural to consider the triangulated subcategories

$$D(B)_{A-\text{parf}}, \qquad D^b(B)_{A-\text{coh}}$$

[15] cf. SGA 6 III, IV.

of $D(B)$ generated by the complexes that are perfect, respectively pseudo-coherent and bounded, relative to A. These two triangulated categories allow one to define the following groups:

$$K'(B \text{ rel } A) = K(D(B)_{A-\text{parf}}),$$

$$K_\bullet(B \text{ rel } A) = K(D^b(B)_{A-\text{coh}}).$$

Examples. Let G be a group and let Λ be a commutative ring. The homomorphism

$$\Lambda \xrightarrow{\varepsilon} \Lambda[G]$$

therefore allows one to define the groups

$$K'(\Lambda[G] \text{ rel } \Lambda), \qquad K_\bullet(\Lambda[G] \text{ rel } \Lambda),$$

which will be denoted by

$$R'_\Lambda(G), \qquad R^\Lambda_\bullet(G).$$

In fact, $R'_\Lambda(G)$ is a commutative ring and $R^\Lambda_\bullet(G)$ is a module over it.

Remarks.

a) If the ring Λ is not necessarily commutative, one may set, by definition,

$$\Lambda[G] = \Lambda \otimes_{\mathbb{Z}} \mathbb{Z}G$$

and consequently obtains a homomorphism

$$\Lambda \xrightarrow{\varepsilon} \Lambda[G].$$

In this way, one may give a meaning to the groups $R'_\Lambda(G)$ and $R^\Lambda_\bullet(G)$ even when Λ is non-commutative.

b) The preceding construction can be generalized to the case of a group prescheme, or of a pro-prescheme in groups, G over $\operatorname{Spec}\Lambda$, by starting from the abelian category of Λ-modules on which G operates; this is no longer made explicit in terms of modules over a ring. One then uses the natural inclusion functor into the category of Λ-modules. If Λ is a field, it can be made explicit in terms of modules over a pro-ring, and in the general case, in terms of comodules, at least if G is affine over Λ; cf. SGA 3 I 4.7.2.

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[2] J. L. Verdier, *Catégories Dérivées*, I.H.E.S., 1963.[16]

[16] Published in SGA $4^{1/2}$.

Exposé X — Euler–Poincaré Formula in Étale Cohomology

EULER–POINCARÉ FORMULA IN ÉTALE COHOMOLOGY

by A. Grothendieck

(written up by I. Bucur)

The purpose of the present exposé is to state and prove formula (7.2), of Euler–Poincaré type. This formula generalizes a formula obtained independently by Ogg and Chafarevitch, whose results are presented, from the viewpoint of étale cohomology, in M. Raynaud's exposé [1].

The Euler–Poincaré formula uses in an essential way the formalism of derived categories. As in loc. cit., the proof is made through a variant (5.1) of a well-known formula of Weil.

1. Sheaves on a scheme with operators

Let X be a scheme endowed with a group G acting by automorphisms. Proceeding as in ([1], chap. V), one may define the notion of an étale sheaf on X with group of operators G, or simply a "sheaf on $(X_{\text{ét}}, G)$".

By definition, to give an étale sheaf on X with group of operators G is to give a sheaf $F \in \mathrm{ob}(\widetilde{X}_{\text{ét}})$ and a morphism

$$S_g : F \longrightarrow g^*(F)$$

for each $g \in G$, in such a way that the following conditions are satisfied:

$$\text{a)} \quad S_e = 1_F,$$

and, for $g, h \in G$, the diagram

$$\begin{tikzcd} F \arrow[r, "S_{hg}"] \arrow[d, "S_h"'] & (hg)^*(F) \arrow[d, "C_{h,g}(F)"] \\ h^*(F) \arrow[r, "h^*(S_g)"'] & h^*(g^*(F)) \end{tikzcd}$$

is commutative, where

$$C_{h,g}(F) : (hg)^*(F) \longrightarrow h^*(g^*(F))$$

is the canonical isomorphism.

Morphisms between two sheaves (F, S_g), (F', S'_g) on X with group of operators G are defined in the evident way, and one obtains a category $\widetilde{X}_{\text{ét},G}$ which is a topos, as follows readily from Giraud's criterion (SGA 4 II 4.12).

Given a ring A of $\widetilde{X}_{\text{ét},G}$, one may consider the abelian category of A-modules, which will be denoted $\mathrm{Mod}(A)$.

For example, given a ring Λ, one may equip the constant sheaf Λ_X associated with Λ with the trivial structure of a sheaf with group of operators G, using the canonical transitivity isomorphisms

$$S_g : \Lambda_X \xrightarrow{\sim} g^*(\Lambda_X).$$

All the results proved in [1], chap. V, can be extended to this setting, and we leave the details to the reader.

2. Where one proves that a certain complex of $\Lambda[G]$-modules is perfect

Recall the following result of SGA 4 XVII.

Lemma 2.1. Let X and Y be two schemes and let $f : X \to Y$ be a proper morphism, with Y quasi-compact. Let A be a torsion ring, and let $F^\bullet$ be a complex of sheaves of A-modules on X of finite Tor-dimension (SGA 4 XVII). Then $Rf_*(F^\bullet)$ is of finite Tor-dimension.

Proposition 2.2. Let X be a scheme proper over a separably closed field k, let G be a finite group acting on X, and let V be an open subset of X stable under G and such that G acts admissibly (SGA 1 V 1.7) and freely on V. Let $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ a prime number different from the characteristic of k, and let Λ_V be the constant sheaf on V associated with Λ, endowed with its trivial structure of a sheaf with group of operators G. Let

$$i : V \longrightarrow X$$

be the immersion of V into X, and put $\Lambda_{V,X} = i_!(\Lambda_V)$. Under these conditions the complex of $\Lambda[G]$-modules

$$R\Gamma_X(\Lambda_{V,X})$$

is a perfect complex (VIII 7).

Proof. If $p : X \to Y = X/G$ is the canonical projection, it is well known that there is a canonical isomorphism

$$R\Gamma_X(\Lambda_{V,X}) \simeq R\Gamma_Y(Rp_*(\Lambda_{V,X}))$$

in the category $D(\Lambda[G])$. On the other hand, the complex $Rp_*(\Lambda_{V,X})$ of $\Lambda[G]$-modules on Y is of finite Tor-dimension. Indeed, since $p : X \to Y$ is a finite morphism of schemes, one has

$$R^q p_*(\Lambda_{V,X}) = 0 \qquad (q \neq 0)$$

(SGA 4 VIII 5.5). Hence

$$Rp_*(\Lambda_{V,X}) = p_*(\Lambda_{V,X}).$$

SGA 4 VIII 5.5 also gives the possibility of determining the fibres of the sheaf $p_*(\Lambda_{V,X})$, and, taking account of the fact that G acts freely on V, one finds

$$\big(p_*(\Lambda_{V,X})\big)_{\bar{y}} \simeq \begin{cases} \prod\limits_{x \in p^{-1}(y)} \Lambda_{V,X,\bar{x}} \simeq \Lambda[G], & y \in p(V), \\ 0, & y \notin p(V), \end{cases}$$

where $\bar{y}$ is the geometric point of Y localized at y.

Consequently $p_*(\Lambda_{V,X})$ is flat over $\Lambda[G]$, fibre by fibre (SGA 4 XVII), and therefore is of finite Tor-dimension. Applying 2.1 to the canonical morphism $Y \to \mathrm{Spec}(k)$, we deduce that

$$R\Gamma_X(\Lambda_{V,X}) = R\Gamma_Y(Rp_*(\Lambda_{V,X}))$$

is of finite Tor-dimension.

To conclude that $R\Gamma_X(\Lambda_{V,X})$ is perfect, it remains to show that it is pseudo-coherent and has bounded cohomology (VIII 8.1). Since $\Lambda[G]$ is noetherian, to verify that $R\Gamma_X(\Lambda_{V,X})$ is pseudo-coherent it is enough to show that the groups $H^q(X, \Lambda_{V,X})$ are of finite type over $\Lambda[G]$ (VIII 7.2), or equivalently over Λ; this follows from (SGA 4 XIV 1.1), applied to the Λ-module $\Lambda_{V,X}$.

3. Reminders on linear representations of finite groups

In this number, Λ denotes a commutative ring, and G a group.

3.1. Characters

Consider the full subcategory $\mathcal{C}$ of $\mathrm{Mod}(\Lambda[G])$ consisting of $\Lambda[G]$-modules M whose underlying Λ-module is projective of finite type. For such a module M and for $g \in G$, consider the endomorphism, for the Λ-module structure,

$$\varphi_M : M \longrightarrow M, \qquad \varphi_M(x) = gx.$$

One obtains a function

$$\chi_M : G \longrightarrow \Lambda, \qquad \chi_M(g) = \mathrm{Tr}^*_M(\varphi_g),$$

which is a central function on G and is called the character of the $\Lambda[G]$-module M.

Associating to each $M \in \mathrm{ob}(\mathcal{C})$ its character χ_M gives an additive function (VIII 1). Using the formalism of traces for endomorphisms of perfect complexes, one even obtains an additive function Tr on the category of complexes of $\Lambda[G]$-modules which are perfect as complexes of Λ-modules. Hence an homomorphism

$$\mathrm{Tr} : R^*_\Lambda(G) \longrightarrow \mathcal{F}(G, \Lambda),$$

where $R^*_\Lambda(G)$ is the ring of representations of G over Λ (VIII 10), and $\mathcal{F}(G, \Lambda)$ is the module of central functions $G \to \Lambda$.

When G is finite, the composite of the canonical homomorphism

$$K^*(\Lambda[G]) \longrightarrow R^*_\Lambda(G)$$

with Tr will be denoted by tr.

Recall the following result (IX th. 1, cor. 3).

Proposition 3.2. Let $\mathbb{Z}_\ell$ be the ring of ℓ-adic integers and let G be a finite group. Then the homomorphism

$$\mathrm{tr} : K^*(\mathbb{Z}_\ell[G]) \longrightarrow \mathcal{F}(G, \mathbb{Z}_\ell)$$

is a monomorphism. Two projective $\mathbb{Z}_\ell[G]$-modules having the same trace function in $\mathcal{F}(G, \mathbb{Z}_\ell)$ are isomorphic.

3.3. Restriction and induction

Let $\alpha : G \to G'$ be a homomorphism from a group G to a group G'. By restriction of operators one obtains an exact functor

$$J : \mathrm{Mod}(\Lambda[G']) \longrightarrow \mathrm{Mod}(\Lambda[G]).$$

This functor has a left adjoint

$$I : \mathrm{Mod}(\Lambda[G]) \longrightarrow \mathrm{Mod}(\Lambda[G']),$$

namely

$$I(F) = \Lambda[G'] \otimes_{\Lambda[G]} F.$$

If F is projective of finite type, respectively of finite type, then so is $I(F)$.

If $\varphi \in \mathcal{F}(G', \Lambda)$ is a central function on G', the function $\varphi \circ \alpha$ is a central function on G, and therefore one obtains an application

$$\alpha^* : \mathcal{F}(G', \Lambda) \longrightarrow \mathcal{F}(G, \Lambda).$$

If φ is the character of a $\Lambda[G']$-module M in $\mathcal{C}(G')$, then $\alpha^*(\varphi)$ is the character of the $\Lambda[G]$-module $J(M)$.

Now suppose that $\alpha : G \to G'$ is injective and that G has finite index in G'. If F is a $\Lambda[G]$-module in $\mathcal{C}(G)$ whose character is $\psi : G \to \Lambda$, then $I(F)$ lies in $\mathcal{C}(G')$ and the character $\psi^* : G' \to \Lambda$ of $I(F)$ is the character induced by ψ, i.e. the function on G' deduced from ψ by the formula

$$\psi^*(g') = \sum_{s \in G'/G} \psi(s^{-1} g' s), \tag{3.3.1}$$

with the convention that $\psi(s^{-1}g's) = 0$ if $s^{-1}g's \notin G$.

3.4. Additional structures on complexes

Let $X^\bullet$ and $Y^\bullet$ be two complexes of Λ-modules. As in the case of modules, if one of the complexes $X^\bullet$, $Y^\bullet$ has an additional structure of a Λ-B-bimodule, it is possible to introduce on $\mathrm{Hom}_\Lambda(X^\bullet, Y^\bullet)$ a structure of complex of B-modules, and consequently a corresponding structure on $\mathrm{RHom}_\Lambda(X^\bullet, Y^\bullet)$.

For example, suppose that $X^\bullet$ is an object of $D^-(\Lambda)$ and that $Y^\bullet$ is a complex of Λ-B-bimodules. In particular, $Y^\bullet$ is an object of $D(\Lambda)$. Let $P^\bullet \to X^\bullet$ be a projective resolution of $X^\bullet$. Then $\mathrm{Hom}_\Lambda(P^\bullet, Y^\bullet)$ is evidently a complex of left B-modules, i.e. an object of $D(B)$. If another projective resolution $P_1^\bullet \to X^\bullet$ is used, the complex of B-modules $\mathrm{Hom}_\Lambda(P_1^\bullet, Y^\bullet)$ is canonically isomorphic in $D(B)$ to $\mathrm{Hom}_\Lambda(P^\bullet, Y^\bullet)$. It follows that one may define $\mathrm{RHom}_\Lambda(X^\bullet, Y^\bullet)$ as an object of $D(B)$, determined up to unique isomorphism, and functorial in $X^\bullet$ in $D^-(\Lambda)$ and in $Y^\bullet$ in $D(\Lambda, B)$.

Analogous considerations apply to the complex $X^\bullet \otimes^L_\Lambda Y^\bullet$. For example, if $Y^\bullet$ is a complex of Λ-B-bimodules as above and $X^\bullet$ is a right bounded complex of right Λ-modules, one may define $X^\bullet \otimes^L_\Lambda Y^\bullet$ as an object of $D(B)$ determined up to unique isomorphism.

3.5

Let Λ be a commutative ring, let A and B be two algebras over Λ, and put $C = A \otimes_\Lambda B$. Thus C-modules are A-B-bimodules.

If $\mathcal{A}$ is a full triangulated subcategory of $D(\Lambda)$, denote by $D(C)_{\mathcal{A}}$ the full triangulated subcategory of $D(C)$ generated by the complexes which, when regarded as objects of $D(\Lambda)$, belong to $\mathcal{A}$.

Proposition 3.6. With the notation of 3.4 and 3.5, let $\mathcal{A}$ be a full triangulated subcategory of $D(\Lambda)$ stable under direct factors. If $P^\bullet$ is an object of $D(B)_{\mathrm{parf}}$ (VIII 7) and $M^\bullet$ is an object of

$D(C)_{\mathcal{A}}$, then the complex of Λ-modules $\mathrm{RHom}_B(P^\bullet, M^\bullet) \in D(\Lambda)$ defined in 3.4 is an object of $\mathcal{A}$. In particular, there is a pairing of triangulated categories

$$\mathrm{RHom}_B : D(B)_{\mathrm{parf}} \times D(C)_{\mathcal{A}} \longrightarrow \mathcal{A}, \tag{3.6.1}$$

that is, a bifunctor exact in each variable, and consequently (VIII 3) a bilinear map

$$K^*(B) \times K^*(D(C)_{\mathcal{A}}) \longrightarrow K^*(\mathcal{A}). \tag{3.6.2}$$

3.7

Suppose in particular that $B = \Lambda[G]$ and $C = A[G] = A \otimes_\Lambda \Lambda[G]$. Given a complex $X^\bullet$ of $A[G]$-modules, denote by $X^{\bullet G}$ the complex of A-modules which is the image of $X^\bullet$ under the functor that associates to each $A[G]$-module X the A-module of elements of X invariant under G.

Proposition 3.8. Let G be a finite group, let $P^\bullet$ be a bounded complex of $A[G]$-modules all of whose components are projective of finite type, and let $M^\bullet$ be a complex of $A[G]$-modules. Under these conditions there are canonical isomorphisms of complexes of A-modules

$$\mathrm{Hom}_{A[G]}(P^\bullet, M^\bullet) = \mathrm{Hom}_A(P^\bullet, M^\bullet)^G = (P^{\bullet *} \otimes_A M^\bullet)^G = P^{\bullet *} \otimes_{A[G]} M^\bullet. \tag{3.8.1}$$

The proof uses the well-known isomorphisms in the case of modules and is left to the reader. Note that, in the second and third terms, the operations of G on $\mathrm{Hom}_A(P^\bullet, M^\bullet)$, respectively on $P^{\bullet *} \otimes_A M^\bullet$, are defined by transport of structure from its operations on $P^\bullet$ and $M^\bullet$, respectively $P^{\bullet *}$ and $M^\bullet$.

Corollary 3.9. If $P^\bullet \in \mathrm{ob}(D(A[G])_{\mathrm{parf}})$ and $M^\bullet \in D(A[G])$, then in the derived category $D(A)$ one has the isomorphism

$$\mathrm{RHom}_{A[G]}(P^\bullet, M^\bullet) = \mathrm{RHom}_A(P^\bullet, M^\bullet)^G. \tag{3.9.1}$$

Indeed, since $P^\bullet \in D(A[G])_{\mathrm{parf}}$, we may suppose that $P^\bullet$ is bounded and that all its components are projective of finite type. Hence

$$\mathrm{RHom}_{A[G]}(P^\bullet, M^\bullet) = \mathrm{Hom}_{A[G]}(P^\bullet, M^\bullet),$$

and one can use the isomorphisms (3.8.1). But $P^{\bullet *}$ is, like $P^\bullet$, bounded with projective finite-type components, hence

$$P^{\bullet *} \otimes^L_A M^\bullet = P^{\bullet *} \otimes_A M^\bullet.$$

Similarly,

$$P^{\bullet *} \otimes^L_A M^\bullet = P^{\bullet *} \otimes_A M^\bullet.$$

Moreover the components of the complex $P^{\bullet *} \otimes_A M^\bullet$ are cohomologically trivial ([3] IX 3), and therefore

$$R\Gamma^G(P^{\bullet *} \otimes^L_A M^\bullet) = (P^{\bullet *} \otimes_A M^\bullet)^G.$$

4. The Swan representation

Let E be a smooth connected algebraic curve over the algebraically closed field k. If $x \in E$ is a closed point of E, its local ring $\mathcal{O}_{E,x}$, or simply $\mathcal{O}_x$, is a discrete valuation ring whose field of fractions is isomorphic to the field of rational functions on E and whose residue field is k. The corresponding valuation will be denoted v_x.

Lemma 4.1. Let E be a smooth algebraic curve over the algebraically closed field k, and let $g : E \to E$ be an endomorphism of E different from the identity. Suppose that $x \in E$ is an isolated fixed point of g, and let π be a uniformizer of $\mathcal{O}_{E,x}$. If v_x is the corresponding valuation of $\mathcal{O}_{E,x}$, then the multiplicity $v_x(g)$ of the fixed point x of the map g (IV) is equal to $v_x(g(\pi) - \pi)$.

In the product $E \times E$, the diagonal and the graph of the endomorphism g will be denoted respectively by Δ and Γ_g. Let

$$\gamma : E \longrightarrow E \times E, \qquad \delta : E \longrightarrow E \times E$$

be the corresponding immersions. On the completed local rings at x and at (x, x), they are given by homomorphisms

$$\gamma^*, \delta^* : \widehat{A} \widehat{\otimes} \widehat{A} \longrightarrow \widehat{A} \qquad (A = \widehat{\mathcal{O}}_x),$$

where

$$\delta^*(a \otimes b) = ab, \qquad \gamma^*(a \otimes b) = a\, g^*(b).$$

We have to evaluate the multiplicity $v_x(g)$ of the point $\delta(x)$ in the intersection $\Delta\Gamma_g$ on $E \times E$. The divisor Δ corresponds to the ideal generated by the element $1 \otimes \pi - \pi \otimes 1$ of $\widehat{A} \widehat{\otimes} \widehat{A}$. Hence the multiplicity of $\delta(x)$ in the intersection $\Delta\Gamma_g$ is

$$v_x(g) = v_x\big(\gamma^*(1 \otimes \pi - \pi \otimes 1)\big) = v_x(g(\pi) - \pi),$$

as required.

4.2

Let C be a smooth, proper, connected algebraic curve over the algebraically closed field k. If η is its generic point, let $K = k(\eta)$ be its field of rational functions.

Let

$$K' \supset K, \qquad [K' : K] = n$$

be a finite Galois extension of degree n with Galois group G. Let C' be the normalization of C in K', and let

$$p : C' \longrightarrow C$$

be the canonical projection. The Galois group G acts naturally on C'. The stabilizer of $y' \in C'$ in G will be denoted by $G_{y'}$.

If π is a uniformizer of $\mathcal{O}_{C',y'}$ and if $s \in G_{y'}$, $s \neq e$, put

$$v_{y'}(s(\pi) - \pi) = v_{y'}(s) \tag{4.2.1}$$

(the multiplicity of the fixed point y' for the map $s : C' \to C'$ (Lemma 4.1)).

The Swan representation associated with y' is defined by its character $\sigma_{y'}$, given by

$$\sigma_{y'}(s) = \begin{cases} 1 - v_{y'}(s), & s \neq e, \\ \sum\limits_{s \neq e} (v_{y'}(s) - 1), & s = e. \end{cases} \tag{4.2.2}$$

The results of Artin–Serre–Swan (IX Th. 4) show that, for a prime number ℓ distinct from the characteristic of k, there exists a projective $\mathbb{Z}_\ell[G_{y'}]$-module of finite type $Sw_{y'}$, determined up to isomorphism, whose character is $\sigma_{y'}$.

It is also useful to consider the $\mathbb{Z}_\ell[G_{y'}]$-module

$$Sw'_{y'} = Sw_{y'} \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell[G_{y'}]. \tag{4.2.3}$$

The character $\sigma'_{y'}$ of $Sw'_{y'}$ is given by

$$\sigma'_{y'}(s) = \begin{cases} 1 - v_{y'}(s), & s \neq e, \\ 1 + \sum\limits_{s \neq e} v_{y'}(s) = 1 + v_{y'}(\mathfrak{D}_{C'/C}), & s = e, \end{cases} \tag{4.2.4}$$

where $\mathfrak{D}_{C'/C}$ is the different of the covering C'/C; the last formula is just [3, Chap. VI 2, prop. 4] with $H = \{1\}$.

Finally, the function

$$a_{y'} = \sigma_{y'} + u_{G_{y'}} = \sigma'_{y'} - 1_{G_{y'}}, \tag{4.2.5}$$

where $u_{G_{y'}}$ denotes the character of the augmentation ideal of $\mathbb{Z}_\ell[G_{y'}]$, is also the character of a $\mathbb{Q}_\ell[G_{y'}]$-module, the Artin representation of $G_{y'}$, which will be denoted by $Art_{y'}$.

Thus

$$a_{y'}(s) = \begin{cases} -v_{y'}(s), & s \neq e, \\ \sum\limits_{s \neq e} v_{y'}(s) = v_{y'}(\mathfrak{D}_{C'/C}), & s = e. \end{cases} \tag{4.2.6}$$

Now consider the inclusion $G_{y'} \subset G$. We put

$$Sw_y = Sw_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \qquad Sw'_y = Sw'_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \tag{4.2.7}$$

$$Art_y = Art_{y'} \otimes_{\mathbb{Q}_\ell[G_{y'}]} \mathbb{Q}_\ell[G]. \tag{4.2.8}$$

The characters of the preceding representations are denoted by σ_y, σ'_y, and a_y.

4.3

The modules so obtained depend, up to isomorphism, only on the point $y \in C$, and not on the choice of the point y' of C' above y. Let y'' be another such point. There exists $g \in G$ such that $y'' = gy'$. The situation (C', y'', G) is deduced from the situation (C', y', G) by the isomorphism g on C' and by $\operatorname{int}(g) : h \mapsto ghg^{-1}$ on G. It follows by transport of structure that the modules Sw_y, etc., associated with y'' are deduced from those associated with y' by extension of the operators by means of $\operatorname{int}(g) : G \to G$. This extension is known to transform a module into an isomorphic module.

Proposition 4.4. With the notation of 3.7, with ($\Lambda = \mathbb{Z}_\ell$) and with $M^\bullet$ an object of $D^+(A[G])$, one has isomorphisms

$$(*) \qquad Sw_y \simeq \check{Sw}_y,$$

$$(**) \qquad \mathrm{Hom}^\bullet_{\mathbb{Z}_\ell[G]}(Sw_y, M^\bullet) \simeq Sw_y \otimes_{\mathbb{Z}_\ell[G]} M^\bullet.$$

Proof. For two projective $\mathbb{Z}_\ell[G]$-modules of finite type to be isomorphic, it is enough that they have the same character (3.2). If $\check{\sigma}_y$ denotes the character of $\check{Sw}_y$, then by (3.7.2)

$$\check{\sigma}_y(s) = \sigma_y(s^{-1}).$$

On the other hand

$$v_{y'}(s^{-1}(\pi) - \pi) = v_{y'}(s^{-1}(s(\pi) - \pi)) = v_{y'}(\pi - s(\pi)) = v_{y'}(s(\pi) - \pi),$$

so $Sw_y \simeq \check{Sw}_y$, whence formula $(*)$ by definition (4.2.7). Formula $(**)$ follows from 3.8.

4.5

We shall need below an explicit formula for the character of the Swan representation. In fact one has the formulas

$$\sigma_y(s) = \begin{cases} \displaystyle\sum_{\substack{p(z')=y \\ s(z')=z'}} (1 - v_{z'}(s)), & s \neq e, \\ \displaystyle\sum_{p(z')=y} \big(1 + v_{z'}(\mathfrak{D}_{C'/C})\big) - n, & s = e, \end{cases} \tag{4.5.1}$$

$$\sigma'_y(s) = \begin{cases} \displaystyle\sum_{\substack{p(z')=y \\ s(z')=z'}} (1 - v_{z'}(s)), & s \neq e, \\ \displaystyle\sum_{p(z')=y} \big(1 + v_{z'}(\mathfrak{D}_{C'/C})\big), & s = e, \end{cases} \tag{4.5.2}$$

$$a_y(s) = \begin{cases} \displaystyle -\sum_{\substack{p(z')=y \\ s(z')=z'}} v_{z'}(s), & s \neq e, \\ \displaystyle\sum_{p(z')=y} v_{z'}(\mathfrak{D}_{C'/C}), & s = e. \end{cases} \tag{4.5.3}$$

To deduce these formulas one uses (3.3.1) and takes into account the following fact: if one fixes a point $y'_0 \in C'$, then for every $y' \in C'$ such that $p(y') = p(y'_0)$ one can find an element $g' \in G$ such that $g'(y') = y'_0$. The elements g' form a system of representatives for the right cosets of $G_{y'_0}$ in G, and moreover

$$G_{y'_0} = g' G_{y'} g'^{-1}.$$

This system of representatives can be used in the second member of formula (3.3.1), and one obtains the formulas displayed above.

4.6

The invariants Sw_y, Sw'_y, Art_y can be defined under more general conditions. Let C' be a smooth algebraic curve over k, not necessarily connected. Suppose that the finite group G acts on C' and that there exists an open subset U of $C = C'/G$, dense in C, such that $C'|U$ is a principal covering of U with group G. For a point $y \in C$, one can define the modules Sw_y, Sw'_y, Art_y by proceeding as above. One can reduce to the case where C' is connected in the following way. Let $y' \in C'$ be such that $p(y') = y$, let C'_1 be the irreducible component of C' containing y', and let C_1 be the irreducible component of C containing y. If $G_{C'_1}$ is the stabilizer of the component C'_1, and if η'_1 (resp. η_1) is the generic point of C'_1 (resp. C_1), then the extension

$$K'_1 = k(\eta'_1) \supset k(\eta_1) = K_1$$

is Galois with Galois group $G_{C'_1}$ and C'_1 is the normalization of C_1 in K'_1 (SGA 1 I 10). One can therefore apply the preceding considerations to the curves C'_1, C_1 and the group $G_{C'_1}$. In this way one obtains projective $\mathbb{Z}_\ell[G_{C'_1}]$-modules of finite type

$$Sw_{G_{C'_1},y}, \qquad Sw'_{G_{C'_1},y}, \qquad Art_{G_{C'_1},y},$$

and Sw_y, Sw'_y, Art_y are the induced modules associated with the inclusion $G_{C'_1} \subset G$.

Remark 4.6. The modules Sw_y, Sw'_y, Art_y are purely local invariants: they are known from the group $G_{y'}$ acting on the completion of the local ring $\mathcal{O}_{C',y'}$. Moreover $Sw_y = 0$ if and only if $p : C' \to C$ is tamely ramified at y', or equivalently, if $G_{y'}$ has order prime to the characteristic of k [3 VI].

5. The Weil formula (compare [3 VI §4])

5.0

All schemes considered in this number will be defined over an algebraically closed base field k.

We shall use the following notation:

$$\Lambda_\nu = \mathbb{Z}/\ell^\nu\mathbb{Z},$$

where ℓ is a fixed prime number distinct from the characteristic of k.

Let $\mathbb{Z}_\ell \to \Lambda_\nu$ be the canonical homomorphism. It induces a homomorphism

$$\lambda_\nu : \mathbb{Z}_\ell[G] \longrightarrow \Lambda_\nu[G]$$

and consequently [VIII 9] a reduction homomorphism

$$\lambda_\nu^* : K^*(\mathbb{Z}_\ell[G]) \longrightarrow K^*(\Lambda_\nu[G]).$$

The images by λ_ν^* of the elements associated with the modules Sw_y, Sw'_y will be denoted by

$$Sw_y^{\Lambda_\nu}, \qquad Sw'^{\,\Lambda_\nu}_y.$$

The reduction homomorphism λ_ν^* is in fact an isomorphism [IX 1.5]. The inverse of λ_ν^* will be denoted by μ_ν.

The constant sheaf on the scheme X associated with the ring Λ_ν will be denoted by $\Lambda_{\nu,X}$.

Let V be an open subset of X, and let $j : V \to X$ be the canonical immersion. We put

$$j_!(\Lambda_{\nu,V}) = \Lambda_{\nu,V,X}.$$

Proposition 5.1. Let C' be a smooth proper algebraic curve over the algebraically closed field k, let G be a finite group acting on C', let $C = C'/G$, let $p : C' \to C$ be the canonical projection, let U be a dense open subset of C such that $U' = p^{-1}(U)$ is a principal covering of U with group G, let $Y = C - U$, let $Y' = p^{-1}(Y) = C' - U'$, and let $\mathrm{EP}(C)$ be the Euler–Poincaré characteristic of C (equal to $2 - 2g$ if C is connected of genus g). Under these conditions one has the Weil formula:

$$\mathrm{cl}_{K^*(\Lambda_\nu[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{\nu,U',C'})) = \mathrm{EP}(C)\,\mathrm{cl}_{K^*(\Lambda_\nu[G])}(\Lambda_\nu[G]) - \sum_{y\in Y} Sw_y'^{\Lambda_\nu}, \tag{5.2}$$

where the first member of (5.2) is defined using 2.1.

Proof. Denote by S_1 (resp. S_2) the first (resp. second) member of the formula to be proved. Since the reduction homomorphism λ_ν^* is in fact an isomorphism, it is enough to prove that

$$\mu_\nu(S_1) = \mu_\nu(S_2).$$

On the other hand, by (3.3.1), the trace homomorphism

$$K^*(\mathbb{Z}_\ell[G]) \xrightarrow{tr} \mathcal{F}(G, \mathbb{Z}_\ell)$$

is a monomorphism. Consequently, it is enough to prove that

$$\mathrm{tr}(\mu_\nu(S_1)) = \mathrm{tr}(\mu_\nu(S_2)).$$

The character $\sigma_2 = \mathrm{tr}(\mu_\nu(S_2))$ is plainly given by

$$\sigma_2 = \mathrm{EP}(C) r_G - \sum_{y\in Y} \sigma_y', \tag{5.2.1}$$

where σ_y' is the character of Sw_y' ((4.2.7) and (4.2.3)), and r_G is the regular representation.

Now consider the reduction homomorphisms for traces:

$$\mathcal{F}(G, \mathbb{Z}_\ell) \xrightarrow{\lambda_n^*} \mathcal{F}(G, \mathbb{Z}/\ell^n\mathbb{Z}).$$

One plainly has

$$\mathcal{F}(G, \mathbb{Z}_\ell) = \varprojlim_n \mathcal{F}(G, \mathbb{Z}/\ell^n\mathbb{Z}).$$

It is therefore enough to prove, for every n, that

$$\lambda_n^*(\mathrm{tr}(\mu_\nu(S_1))) = \lambda_n^*(\sigma_2).$$

The first member is equal to $\mathrm{tr}(\lambda_n^*(\mu_\nu(S_1)))$, while

$$\lambda_n^*(\mu_\nu(S_1)) = \mathrm{cl}_{K^*(\Lambda_n[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'})),$$

by the Künneth formula (SGA 4 XVIII)

$$\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}) \otimes^{L}_{\Lambda_n} \Lambda_\nu \simeq \mathrm{R}\Gamma_{C'}(\Lambda_{\nu,U',C'}) \quad \text{for } n \geq \nu.$$

Thus we are reduced to proving the equality

$$\mathrm{tr}\big(\mathrm{cl}_{K^*(\Lambda_n[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}))\big) = \lambda_n^*(\sigma_2).$$

Let τ denote the first member of this equality, and let $g \in G$. The element g plainly induces an endomorphism of the distinguished triangle

$$\begin{tikzcd}
& \mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'}) \arrow[dr] & \\
\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}) \arrow[rr] & & \mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})
\end{tikzcd}$$

Therefore

$$\tau(g) = \mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'})} = \mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} - \mathrm{Tr}^* g_{\mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'})}.$$

For brevity, Tr^* has been written instead of $\mathrm{Tr}^*_{\Lambda_n}$.

Applying a trivial Lefschetz formula to the discrete space Y' on which g acts gives

$$\mathrm{Tr}^* g_{\mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'})} = \sum_{\substack{y' \in Y' \\ gy' = y'}} 1.$$

On the other hand, the Lefschetz formula (III 4.7) for g acting on C' gives, if $g \neq e$,

$$\mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} = \sum_{y' \in Y'} v_{y'}(g),$$

the sum of the multiplicities of the fixed points of the map $g_{C'} : C' \to C'$. Also

$$\mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} = \mathrm{EP}(C') \cdot 1_\Lambda \quad \text{for } g = e,$$

by the explicit computation of $H^*(C', \Lambda_{n,C'})$, via (SGA 4 IX 4) and Künneth (SGA 4 XVII). We therefore obtain the following formula for τ:

$$\tau(g) = \begin{cases} \displaystyle\sum_{\substack{y' \in Y' \\ g(y') = y'}} \big(v_{y'}(g) - 1\big), & g \neq e, \\ \mathrm{EP}(C') - \mathrm{card}(Y'), & g = e. \end{cases} \tag{5.2.2}$$

Comparison with (5.2.1) gives easily $\tau = \lambda_n^*(\sigma_2)$. Indeed, if $g \neq e$, the equality between $\tau(g)$ and $\lambda_n^*(\sigma_2)$ is immediate from (4.5.2). If $g = e$, the formula to be checked is

$$\mathrm{EP}(C') - \mathrm{card}(Y') = N\,\mathrm{EP}(C) - \sum_{y \in Y} \sigma'_y(e), \qquad N = \mathrm{card}(G),$$

that is, by (4.5.2),

$$\mathrm{EP}(C') - N\,\mathrm{EP}(C) = -\deg(\mathfrak{D}_{C'/C}),$$

which is nothing other than the classical Hurwitz formula.

6. Definition of the local terms $\varepsilon_x^\Delta(F)$

6.1

Let C be a connected smooth algebraic curve over the algebraically closed field k, let A be a Λ-algebra ($\Lambda = \mathbb{Z}/\ell^\nu\mathbb{Z}$, with ℓ distinct from the characteristic of k), and let $\Delta \subset D^+(A)$ be a triangulated subcategory of $D^+(A)$ stable under direct factors. Let F be a complex of sheaves of A-modules on C satisfying the following conditions:

i) For every geometric point $\bar{x} \in C(k)$, one has $F_{\bar{x}} \in \mathrm{ob}(\Delta)$.

ii) There exists a non-empty open subset of C on which F is locally constant; equivalently, there exist a non-empty open subset U of C and a connected Galois covering $C' \xrightarrow{p} C$ of C, with group G, such that, if $F' = p^*(F)$ is the inverse image of F by p, then $F'|U'$ is constant ($U' = p^{-1}(U)$).

The sheaf $F'|U'$ is therefore defined by a complex M of A-modules on which G acts. If η (resp. η') denotes the generic point of C (resp. C'), and if $\bar{\eta}$ (resp. $\bar{\eta}'$) is a geometric point above η (resp. η'), then

$$p(\eta') = \eta, \qquad F_{\bar{\eta}} = F'_{\bar{\eta}'} = M. \tag{6.1.1}$$

As a complex of A-modules, M is isomorphic to $F_{\bar{\eta}}$ and, by condition i), is an object of Δ. Hence, by (3.6), for every $x \in C(k)$,

$$\mathrm{Hom}^\bullet_{A[G]}(Sw_x^A, M)$$

is an object of Δ, and so defines an element of $K(\Delta)$.

In fact the object of Δ defined by $\mathrm{Hom}^\bullet_{A[G]}(Sw_x^A, M)$ is independent, up to isomorphism, of the Galois covering $C' \to C$ which trivializes $F|U$. To show this it is enough to compare the elements associated with two coverings

$$C' \xrightarrow{p} C, \qquad C'' \xrightarrow{p'} C' \xrightarrow{p} C.$$

In this case, if G' is the Galois group of the covering $C'' \xrightarrow{p''} C$, then G is a quotient group of G'. Let

$$\varphi : \mathbb{Z}_\ell[G'] \longrightarrow \mathbb{Z}_\ell[G]$$

be the associated canonical epimorphism. The Swan modules are then related by

$$Sw_{G,x} \simeq Sw_{G',x} \otimes_{\mathbb{Z}_\ell[G']} \mathbb{Z}_\ell[G],$$

which follows immediately from the fact that the character $a_{G,x}$ is the character induced from G' to G ([3] VI Prop. 3).

On the other hand, there is an isomorphism of complexes of $A[G']$-modules

$$F_{\bar{\eta}'} \simeq [F_{\bar{\eta}''}]_\varphi.$$

It follows that there is a canonical isomorphism of complexes of A-modules between

$$\mathrm{Hom}^\bullet_{A[G]}(Sw_{G,x}^A, F_{\bar{\eta}}) \quad \text{and} \quad \mathrm{Hom}^\bullet_{A[G']}(Sw_{G',x}^A, F_{\bar{\eta}''})$$

(the adjunction formula).

6.2

We shall use the following notation:

$$\alpha_x^\Delta(F) = \mathrm{cl}_{K(\Delta)}\big(\mathrm{Hom}^\bullet_{A[G]}(Sw_x^A, F_{\bar\eta})\big), \tag{6.2.1}$$

$$\varepsilon_x^\Delta(F) = \alpha_x^\Delta(F) + \big(\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \mathrm{cl}_{K(\Delta)}(F_x)\big). \tag{6.2.2}$$

Of course, in formula (6.2.1), $F_{\bar\eta}$ is endowed with its $A[G]$-module structure by the isomorphism (6.1.1). Note that if $x \in C(k)$ is such that C' is tamely ramified at x, then by (4.6) $Sw_x = 0$, hence $\alpha_x^\Delta(F) = 0$, and in this case, in conformity with Ogg-Shafarevich, one has

$$\varepsilon_x^\Delta(F) = \mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \mathrm{cl}_{K(\Delta)}(F_x). \tag{6.2.3}$$

It is clear that $\varepsilon_x^\Delta(F) = 0$ if $x \notin Y = C - U$, and therefore only finitely many x satisfy $\varepsilon_x^\Delta(F) \neq 0$.

We now give examples of categories Δ and complexes of sheaves F for which the conditions above are satisfied.

a) Let $\Delta = D^b(A)_{\mathrm{coh}}$ (VIII 5), and let F be a constructible complex of sheaves of A-modules (SGA 4 XVII) with bounded cohomology. In this case $\varepsilon_x^\Delta(F) \in K_0(A)$.

b) Take $\Delta = D(A)_{\mathrm{parf}}$ (VIII 6), and let F be a constructible complex of sheaves of A-modules of finite tor-dimension. In this case $\varepsilon_x^\Delta(F) \in K^*(A)$.

c) Let A, B be two Λ-algebras and let $\varphi : A \to B$ be a homomorphism of Λ-algebras. Take $\Delta = D^b(B \,\mathrm{rel}\, A)_{\mathrm{coh}}$ (VIII 10), and let F be a complex of sheaves of B-modules, constructible as a sheaf of complexes of A-modules and with bounded cohomology. One obtains $\varepsilon_x^\Delta(F) \in K_0(B \,\mathrm{rel}\, A)$. Of course, one can vary this as above to obtain, for example, elements $\varepsilon_x^\Delta(F) \in K^*(B \,\mathrm{rel}\, A)$.

7. Euler–Poincaré formula

Theorem 7.1. Let C be a connected, smooth, proper algebraic curve over the algebraically closed field k, let η be the generic point of C, let $\Delta \subset D^+(A)$ be a triangulated subcategory of $D^+(A)$ stable under direct factors, and let F be a complex of sheaves of A-modules satisfying conditions i), ii) of 7.1. Under these conditions one has $\mathrm{R}\Gamma_C(F) \in \mathrm{ob}(\Delta)$, and the following formula (Euler–Poincaré formula) is true:

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \sum_{x \in C(k)} \varepsilon_x^\Delta(F), \tag{7.2}$$

where the $\varepsilon_x^\Delta(F)$ are defined in 6.2 (formula 6.2.2).

The proof of this theorem will be made in a few steps.

a) Let Ψ be a bounded below complex of sheaves of $\Lambda[G]$-modules on C. There is an isomorphism

$$R\Gamma^G(\mathrm{R}\Gamma_C(\Psi)) \simeq \mathrm{R}\Gamma_C(R\Gamma^G(\Psi)), \tag{7.3}$$

where Γ^G is the functor "subsheaf of G-invariant sections." Indeed, Γ_C sends an injective sheaf of $\Lambda[G]$-modules to an injective $\Lambda[G]$-module, and Γ^G sends an injective sheaf of $\Lambda[G]$-modules to an injective sheaf of Λ-modules (using, for example, the evident adjoints of Γ_C and Γ^G). Moreover

$$\Gamma^G \circ \Gamma_C = \Gamma_C \circ \Gamma^G.$$

Taking derived functors on both sides and applying [5] prop. 3.1) gives (7.3).

b) Let Φ be a bounded below complex of sheaves of Λ-modules on C, zero outside U. There is an isomorphism

$$\Phi \simeq R\Gamma^G(p_*(\Phi')), \qquad \Phi' = p^*(\Phi). \tag{7.4}$$

First note that the functors p^* and $\Gamma^G \circ p_*$ establish quasi-inverse equivalences between the category of sheaves of Λ-modules on C' on which G acts and which are zero outside $U' = p^{-1}(U)$, and the category of sheaves of Λ-modules on C which are zero outside U. Hence one has the isomorphism

$$\Phi \simeq R((\Gamma^G \circ p_*) \circ p^*)(\Phi).$$

On the other hand, one can again apply [5] Prop. 3.1), which gives, since $Rp^* = p^*$ and $Rp_* = p_*$:

$$\Phi \simeq R(\Gamma^G \circ p_*)(p^*(\Phi)) \simeq (R\Gamma^G)(p_* p^*(\Phi)),$$

which is the desired isomorphism.

c) Let Φ be a complex of sheaves as in b). One has the isomorphism

$$\mathrm{R}\Gamma_C(\Phi) = (R\Gamma^G)(\mathrm{R}\Gamma_{C'}(\Phi')), \qquad \Phi' = p^*(\Phi). \tag{7.5}$$

Indeed,

$$(R\Gamma^G)(\mathrm{R}\Gamma_{C'}(\Phi')) \simeq (R\Gamma^G)(\mathrm{R}\Gamma_C(p_*(\Phi'))) \simeq \mathrm{R}\Gamma_C(R\Gamma^G(p_*(\Phi')))$$

by (7.3), and (7.5) follows from the isomorphism (7.4).

d) Let $i' : U' \to C'$, $i : U \to C$ be the canonical inclusions and put

$$\Phi = F_{U,C} = i_!(F|U).$$

To compute $\mathrm{R}\Gamma_C(\Phi)$, one may apply formula (7.5). On the other hand, with the notation of 7.1,

$$\Phi' \simeq \Lambda_{U',C'} \otimes^L_\Lambda M,$$

and the Künneth formula (SGA 4 XVII) gives

$$\mathrm{R}\Gamma_{C'}(\Phi') \simeq \mathrm{R}\Gamma_{C'}(\Lambda_{U',C'} \otimes^L_\Lambda M) \simeq \mathrm{R}\Gamma_{C'}(\Lambda_{U',C'}) \otimes^L_\Lambda M. \tag{7.6}$$

This last isomorphism is in fact an isomorphism between the objects of $D(A[G])$ defined by the two sides. Combining (7.6), (7.5), and formula (3.9.1), which is applicable by 2.1, one obtains

$$\mathrm{R}\Gamma_C(F_{U,C}) \simeq R\Gamma^G(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'}) \otimes^L_\Lambda M) \simeq \mathrm{RHom}_{A[G]}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})^\vee, M). \tag{7.7}$$

Thus, since $M \simeq F_{\bar\eta} \in \mathrm{ob}(\Delta)$ (condition (i)) and by (3.6), we find

$$\mathrm{R}\Gamma_C(F_{U,C}) \in \mathrm{ob}(\Delta).$$

On the other hand, there is the distinguished triangle (where $Y = C - U$)

$$\begin{tikzcd} & \mathrm{R}\Gamma_Y(F|Y) \arrow[dr] & \\ \mathrm{R}\Gamma_C(F_{U,C}) \arrow[rr] & & \mathrm{R}\Gamma_C(F) \end{tikzcd} \tag{7.8}$$

where obviously

$$\mathrm{R}\Gamma_Y(F) = \sum_{y\in Y} F_y \in \mathrm{ob}(\Delta),$$

because $F_y \in \mathrm{ob}(\Delta)$ (condition (i)). It follows that $\mathrm{R}\Gamma_C(F) \in \mathrm{ob}(\Delta)$, which is the first assertion of the theorem. The distinguished triangle (7.8) furthermore gives the relation

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F_{U,C})) + \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y),$$

and, using the isomorphism 7.7,

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \mathrm{cl}_{K(\Delta)}\big(\mathrm{RHom}_{A[G]}(\mathrm{R}\Gamma_{C'}(\Lambda_{U',C'})^\vee, F_{\bar\eta})\big) + \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y).$$

To obtain the Euler–Poincaré formula it remains only to use the Weil formula (5.2). Indeed, first 4.4, (5.2), and 3.6 (in the particular case $B = A[G]$) give

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)) = \sum_{y\in Y} \mathrm{cl}_{K(\Delta)}(F_y) + \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \sum_{y\in Y} \big(\alpha_y^\Delta(F) + \mathrm{cl}_{K(\Delta)}(F_{\bar\eta})\big).$$

Since $\varepsilon_x^\Delta(F) = 0$ for $x \notin Y$, this is exactly formula (7.2). For a complex of sheaves F satisfying the conditions of 7.1, we introduce the notation

$$\mathrm{EP}_\Delta(C,F) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(F)).$$

Corollary 7.9. If F, F' are complexes of sheaves satisfying the conditions of 7.1 and if they are locally isomorphic (for the étale topology) as complexes of A-modules, then

$$\mathrm{EP}_\Delta(C,F) = \mathrm{EP}_\Delta(C,F'). \tag{7.10}$$

In particular, if F is locally constant of value $M^\bullet$, then

$$\mathrm{EP}_\Delta(C,F) = \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(M^\bullet). \tag{7.11}$$

Corollary 7.12. Let C be a connected smooth proper algebraic curve over k, let V be a non-empty open subset of C, let Δ be a subcategory of $D^+(A)$ as in 7.1, and let F be a complex of sheaves of A-modules on V such that its restriction to a suitable non-empty open subset of V is trivialized by a Galois covering of V. Suppose moreover that $F_{\bar x} \in \mathrm{ob}(\Delta)$ for every $x \in V$. Under these conditions one has

$$(*) \qquad \mathrm{R}\Gamma_!V(F),\ \mathrm{R}\Gamma_V(F) \in \mathrm{ob}(\Delta),$$

$$(**) \qquad \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_!V(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_V(F)).$$

Put $\mathrm{EP}_\Delta(V,F) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_V(F))$ and

$$\mathrm{EP}(V) = \mathrm{EP}_{D(F_\ell)_{\mathrm{coh}}}(V,F_\ell) = \mathrm{EP}(C) - \mathrm{card}(Z),$$

where $Z = C - V$ and where F_ℓ is the sheaf on C defined in dimension zero by the constant sheaf associated with $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$. Then one has the formula

$$\mathrm{EP}_\Delta(V,F) = \mathrm{EP}(V)\,\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \sum_{x\in V(k)} \varepsilon_x^\Delta(F) - \sum_{x\in Z} \alpha_x^\Delta(F), \tag{7.13}$$

where ε_x^Δ and α_x^Δ are defined by (6.2.1), (6.2.2).

We shall use the following lemma.

Lemma 7.14. Let K be the field of fractions of a strictly local discrete valuation ring R, let $W = \operatorname{Spec} K$, let A be a noetherian ring annihilated by an integer n prime to the residual characteristic p of R, and let F be a complex of sheaves of A-modules on W trivialized by a finite extension of K. If Δ is a triangulated subcategory of $D^+(A)$ stable under direct factors and if $F_{\bar\eta} \in \mathrm{ob}(\Delta)$ (η the generic point of S), then $\mathrm{R}\Gamma_W(F) \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_W(F)) = 0.$$

Proof of 7.14. Let π be the fundamental group of W. Then F corresponds to a complex of A-modules $M^\bullet$ with π as group of operators. The underlying complex of A-modules of $M^\bullet$ is isomorphic to $F_{\bar\eta}$ and consequently belongs to $\mathrm{ob}(\Delta)$. Introduce the quotient group $\bar\pi$ of π by its unique Sylow p-subgroup P [4, II 33], that is, the projective limit of discrete quotient groups of π which are of order prime to p, where p is the characteristic exponent of the residue field of R (if $p = 1$, put $\bar\pi = \pi$). Thus there is an exact sequence

$$1 \longrightarrow P \longrightarrow \pi \longrightarrow \bar\pi \longrightarrow 0,$$

and one obtains

$$\mathrm{R}\Gamma_W(F) \simeq R\Gamma^\pi(M^\bullet) \simeq R\Gamma^{\bar\pi}(R\Gamma^P(M^\bullet))$$

(Hochschild–Serre isomorphism). Now P acts on $M^\bullet$ through a finite quotient group, which is therefore a p-group, hence of order prime to n. Thus

$$R\Gamma^P(M^\bullet) \simeq \Gamma^P(M^\bullet),$$

and $\Gamma^P(M^\bullet)$ is a direct factor of $M^\bullet$, hence belongs to $\mathrm{ob}(\Delta)$. On the other hand

$$\bar\pi = \prod_{\ell \neq p} \mathbb{Z}_\ell$$

([4], II 33). We are therefore reduced to proving that if $M^\bullet \in \mathrm{ob}(\Delta)$, then $R\Gamma^{\bar\pi}(M^\bullet) \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}(R\Gamma^{\bar\pi}(M^\bullet)) = 0.$$

For this, let f be a topological generator of $\bar\pi$. We may suppose that the components M^i of M are $\bar\pi$-modules such that

$$H^j(\bar\pi, M^i) = 0 \quad \text{if } j \geq 1.$$

There is then a distinguished triangle

$$\begin{tikzcd} & M^\bullet \arrow[dr, "1-f"] & \\ R\Gamma^{\bar\pi}(M^\bullet) \arrow[rr] & & M^\bullet. \end{tikzcd} \tag{7.15}$$

Indeed, on the one hand it is clear that $R\Gamma^{\bar\pi} = \mathrm{Ker}(1-f)$. On the other hand $1 - f : M^\bullet \to M^\bullet$ is surjective. Indeed

$$H^1(\bar\pi, M) = M^\bullet/(1-f)M^\bullet$$

(see for this calculation [4] pp. 197, in a substantially analogous case), and therefore $M^\bullet = (1-f)M^\bullet$. The triangle (7.15) gives at the same time $R\Gamma^{\bar\pi}(M^\bullet) \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}(R\Gamma^{\bar\pi}(M^\bullet)) = 0,$$

which proves 7.14.

Proof of 7.12. The fact that $\mathrm{R}\Gamma_! V(F) \in \mathrm{ob}(\Delta)$ follows immediately from 7.1 applied to the complex $i_!(F)$, where $i : V \to C$; indeed

$$\mathrm{R}\Gamma_! V(F) = \mathrm{R}\Gamma_C(i_! F).$$

To prove that $\mathrm{R}\Gamma_V(F) \in \mathrm{ob}(\Delta)$, consider the distinguished triangle

$$\begin{tikzcd} & Q^\bullet \arrow[dr] & \\ Ri_!(F) \arrow[rr] & & Ri_*(F), \end{tikzcd}$$

where $Q^\bullet$ is concentrated on $Z = C - V$. Applying $\mathrm{R}\Gamma_C$ gives the distinguished triangle

$$\begin{tikzcd} & \mathrm{R}\Gamma_Z(Q^\bullet|Z) \arrow[dr] & \\ \mathrm{R}\Gamma_! V(F) \arrow[rr] & & \mathrm{R}\Gamma_V(F). \end{tikzcd} \tag{7.16}$$

Since $\mathrm{R}\Gamma_! V(F) \in \mathrm{ob}(\Delta)$, it follows that $\mathrm{R}\Gamma_V(F) \in \mathrm{ob}(\Delta)$ if and only if $\mathrm{R}\Gamma_Z(Q^\bullet|Z) \in \mathrm{ob}(\Delta)$, that is, if and only if $(Ri_*(F))_{\bar x} \in \mathrm{ob}(\Delta)$ for every $x \in Z$. Recall that to calculate $(Ri_*(F))_{\bar x}$, one considers the cartesian square

$$\begin{tikzcd} \bar V \arrow[r, "j"] \arrow[d, "u"'] & C_x \arrow[d] \\ V \arrow[r] & C \end{tikzcd}$$

where C_x is the strictly localized scheme corresponding to the geometric point x. One has

$$Ri_*(F)_{\bar x} \simeq \mathrm{R}\Gamma_{\bar V}(u^*(F)),$$

by virtue of 7.14, and therefore $(Ri_*(F))_{\bar x} \in \mathrm{ob}(\Delta)$ and

$$\mathrm{cl}_{K(\Delta)}((Ri_*(F))_{\bar x}) = 0.$$

It follows that

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_Z(Q^\bullet|Z)) = 0,$$

and the distinguished triangle (7.16) gives

$$\mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_! V(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_V(F)).$$

Finally, to establish (7.13), it is enough to use the Euler–Poincaré formula (7.2) for the complex of sheaves $i_!(F)$:

$$\mathrm{EP}_\Delta(V, F) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_! V(F)) = \mathrm{cl}_{K(\Delta)}(\mathrm{R}\Gamma_C(i_!(F))) = \mathrm{EP}(C)\,\mathrm{cl}_{K(\Delta)}(F_{\bar\eta}) - \sum_{x \in C(k)} \varepsilon_x^\Delta(i_!(F)).$$

But

$$\sum_{x\in C(k)} \varepsilon_x^{\Delta}(i_!(F)) = \sum_{x\in V(k)} \varepsilon_x^{\Delta}(F) + \sum_{x\in Z} \varepsilon_x^{\Delta}(F).$$

For $x \in Z$ one has $(i_!(F))_{\bar{x}} = 0$, and consequently, for such x,

$$\varepsilon_x^{\Delta}(i_!(F)) = \alpha_x^{\Delta}(F) + \mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}),$$

which gives formula (7.13).

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